Inattentive Network: Theory and Evidence from EDGAR*

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Abstract

This paper provides novel evidence on firms' attention allocation in a production network by measuring their browsing intensity of other firms' filings using EDGAR log files. We document that firms allocate relatively greater attention to other firms that exhibit higher volatility and are closer in network distance, and that their total browsing activity increases as the volatility of their targets rises. These empirical patterns closely match the theoretical predictions for rationally inattentive firms subject to elastic information capacity. We also underscore an expectation-management channel relevant for policy design, which emerges when policymakers internalize agents' endogenous information choices.

Keywords: Rational inattention, production network, monetary policy

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1. Introduction

How do firms optimally allocate scarce attention within complex supply-chain networks? What economic mechanisms explain the cross-sectional heterogeneity in firms' information acquisition activities? Moreover, what are the policy implications when policymakers internalize firms' endogenous information allocation decisions? Addressing these questions has become particularly pertinent in light of recent empirical findings at both the firm and industry levels.[1] Despite rapid development in recent literature that studies the macroeconomic impacts of informational frictions and bounded rationality, these questions remain largely unexplored.

This paper contributes to addressing these questions along several dimensions. Theoretically, we develop a model of rationally inattentive firms embedded within a production network, providing an analytical characterization of attention allocation and strategic information acquisition in general equilibrium. In our baseline scenario with elastic information capacity, firms optimally direct more attention toward counterparties exhibiting greater volatility and stronger network exposure. Empirically, we propose a novel measure of firms' attention allocation constructed from their browsing activity of other firms' electronic filings available on the EDGAR platform. Our analysis reveals that bilateral browsing intensity declines with increasing network distance but rises with heightened economic volatility, aligning closely with our theoretical predictions. Additionally, we document several other empirical regularities that serve to identify the appropriate information structure. On the applied dimension, integrating this attention-allocation framework into a monetary model demonstrates that optimal price stabilization policies inadvertently amplify dispersion in sectoral price rigidities. This occurs because firms' incentives for information acquisition—and consequently their pricing inertia—respond endogenously to policy interventions.

Theory. Our theoretical framework extends the standard production network model by incorporating rationally inattentive firms. Firms optimally select signals subject to information capacity constraints (Sims, 2003; Maćkowiak et al., 2023), and set prices based on these signals without perfect knowledge of underlying productivity shocks. Our baseline specification features a costly but elastic information capacity: firms can increase their information capacity to make prices more responsive to shocks—thereby enhancing price flexibility—but doing so entails higher costs associated with acquiring more precise signals. In a partial equilibrium (PE) setting, the key predictions for individual firms are that (1) the total amount of information acquired is increasing in the volatility of firms' marginal costs, and (2) they allocate more attention towards other firms that are more volatile and that are more important as suppliers.

In general equilibrium (GE), firms' information acquisition decisions interact strategically through the production network. When firms in one sector acquire more information, they not only alter their own price responsiveness but also influence the volatility of marginal costs in downstream sectors

[1] See Carvalho et al. (2021), Candia et al. (2023), and Pastén et al. (2024).

through input linkages. This, in turn, creates incentives for other firms to adjust their own information acquisition strategies. These feedback effects render the equilibrium allocation a fixed-point problem. Our general equilibrium characterization establishes conditions for the existence and uniqueness of this fixed point and derives a formula for the strategic response in firms' information acquisition. Ultimately, firms allocate more attention to others with greater network exposure—measured by the influence matrix—highlighting the role of general equilibrium forces in shaping attention patterns.

Attention measurement. Firms' attention allocations are crucial for their decision-making and for testing different theories on information acquisition, but they are not directly observable. To overcome this difficulty, we utilize the EDGAR "Logfile" which contains detailed information on the online viewing records for every SEC filing published on the EDGAR platform, including a partially masked IP address and the time of the access requested, among other information. We then uncover the browsers' complete IP addresses using the method developed in Chen et al. (2020), and map each viewer's uncovered IP address to their true identity with the service provided by ip-info.io. This procedure allows us to construct a panel of firms' browsing volume for other firms' files. The browsing intensities towards different firms, therefore, provide an account of attention allocation.

We link firms' browsing activity to their broader economic behavior and document several empirical patterns. First, firms allocate greater attention to other firms that exhibit higher volatility and are more relevant to their own payoffs. This finding is robust to different measures of payoff relevance, including input and sales shares, model-based GE exposures, as well as network distance, following Carvalho et al. (2021). It aligns with our theoretical characterization of attention allocation with limited information capacity and provides empirical support for the key prediction of Maćkowiak et al. (2023) in a network environment. Second, a firm's overall browsing activity comoves with the volatility of its targets of interest. This pattern is consistent with models featuring elastic information capacity and poses a challenge to frameworks premised on fixed information capacity. Third, a firm's absolute forecast error declines with its recent total browsing activity, after controlling for the volatility of relevant variables. This result supports the interpretation that increased information acquisition effort mitigates informational frictions and enhances forecast accuracy.

In addition, this novel measurement provides useful moments for distinguishing among alternative information technologies and signal structures beyond our theory framework. For instance, the empirically observed hazard function for information acquisition is increasing, a feature at odds with standard benchmarks such as Calvo-type updating, which implies a flat hazard function, or deterministic periodic updating, which predicts a hazard function concentrated at a single point.

Optimal monetary policy. On the applied side, the endogeneity of price flexibility arising from firms' information acquisition introduces an expectation-management channel for monetary policy. By influencing the stochastic processes of marginal costs, policy can affect firms' price responsiveness. Importantly, optimal policy design must account for the fact that firms' information choices are interconnected through the production network. Despite these complexities, we derive a closed-form

optimal policy rule in the form of a price stabilization index. This index consists of two components: an exogenous component, which holds firms' price rigidities fixed and corresponds to the formulation in La'O and Tahbaz-Salehi (2022); and an endogenous component, which allows price rigidities to adjust in response to policy and is unique to our framework. The endogenous component captures the influence of information acquisition on shaping optimal policy and highlights the strategic role of expectation management in a networked economy.

Notably, the interaction between the price stabilization index and sectoral price flexibilities gives rise to policy-induced nominal rigidity. Specifically, the policymaker is incentivized to place greater weight on more rigid sectors in order to stabilize prices where they are least responsive. However, these sectors typically exhibit lower volatilities of marginal costs and weaker incentives for information acquisition. The resulting policy-induced stabilization further suppresses volatility in these sectors, thereby reducing firms' incentives to acquire information and reinforcing rigidity. This feedback loop between optimal policy and endogenous information acquisition amplifies the dispersion of price rigidities, in contrast to models with exogenously fixed nominal rigidities.

Quantification. To quantify these mechanisms, we calibrate the model to match the distribution of the absolute forecast error of earnings per share. The model also yields the distribution of price-change frequencies similar to the data. The calibrated model is able to produce the attention allocation pattern that is consistent with the sectoral evidence in the browsing activities. In addition, the model predicts a positive correlation between the sectoral shock volatility and the sectoral price-change frequency, which resembles the pattern in the data. Such correlation is natural in our model as firms allocate more attention—and thus adjust prices more frequently—when the volatility of payoff-relevant variables is higher. Such a pattern is difficult to reconcile with models featuring ad hoc price rigidities.

We show that in the calibrated model, the optimal monetary policy rule significantly differs from that under exogenous information or with fixed information capacity. The difference in the policy rule can be almost entirely accounted for by the heterogeneity in the initial price rigidities. Due to the aforementioned policy-rigidity feedback channel, more rigid sectors will be assigned additional weight in the optimal policy with elastic information acquisition than that with exogenous information. At the same time, the sectoral price rigidities are much more dispersed once the optimal monetary policy is in place.

Related literature. Our paper contributes to the extensive literature on costly information acquisition, and in particular to the rational inattention framework, by introducing a novel and direct measurement of firms' attention allocation. While the theoretical foundations and economic implications of rational inattention have been well developed (see Maćkowiak et al. (2023) for a comprehensive review), empirical validation has yet to catch up, primarily due to the difficulty in observing attention allocation and information acquisition actions directly.

In this respect, our paper is closely related to recent work that infers attention from textual analysis of firms' reports (Song et al., 2022; Flynn and Sastry, 2023) and from media coverage (Chahrour et al.,

2021; Bui et al., 2024). In contrast, our approach offers a more comprehensive and direct measure of information acquisition by capturing firms' actual browsing behavior on EDGAR, enabling the study of bilateral attention allocation across firms. Our empirical strategy complements other approaches that explore information acquisition in different contexts, such as Chu et al. (2024) in healthcare decision-making and Dean and Neligh (2023) in controlled experimental settings. While these studies offer detailed insights into the cognitive process of acquiring information, our method has the advantage of observing firms engaged in real-time, economically meaningful decisions, and can be readily linked to a wide range of existing firm-level datasets.

Our paper also complements an expanding literature that has incorporated information frictions in the models of production networks (Chahrour et al., 2021; Angeletos and Huo, 2021; Auclert et al., 2020; La'O and Tahbaz-Salehi, 2022; Pellet and Tahbaz-Salehi, 2023; Bui et al., 2024; Nikolakoudis, 2024). These studies differ in their assumed information structures, ranging from exogenous and endogenous information to models with sticky information. Our paper differs by explicitly modeling endogenous information acquisition, which offers two key advantages: empirically, it enables direct connection to observed patterns in firms' browsing behavior; and theoretically, it enables the interaction between production network linkages and firms' strategic information choices. Afrouzi et al. (2025) further depart from the commonly used mutual information benchmark with the neighborhood-based cost function (Hébert and Woodford, 2021) and study the role of salient sectors in shaping expectation formation. Jamilov et al. (2024) study sentiment-driven aggregate fluctuations in production networks with a pre-specified two-dimensional signal structure. They find that attention is centered on downstream firms due to their role as information agglomerators.

Finally, our paper contributes to the growing literature on optimal monetary policy design in the presence of production networks.[2] Much of the existing literature has focused on models with time-dependent pricing frictions, such as Calvo pricing (Rubbo, 2023), or exogenous information frictions (La'O and Tahbaz-Salehi, 2022). Our paper instead examines shock propagation in production networks under state-dependent pricing arising from endogenous information acquisition. We show that monetary policy not only affects prices directly but also shapes the distribution of price rigidities, giving rise to an expectation-management channel. Relative to the literature on optimal policy under endogenous information frictions (Adam, 2007; Paciello and Wiederholt, 2013; Li and Wu, 2016; Angeletos et al., 2020; Ou et al., 2024), we extend the analysis to a network environment and highlight a novel feedback mechanism between firms' attention allocation and policy implementation.

2. Theory

In this section, we present the baseline model that features both input-output linkages and costly information acquisition. This framework allows us to characterize firms' attention allocation in a

[2] Earlier work has examined optimal monetary policy in horizontally or vertically structured economies, e.g., Aoki (2001), Benigno (2004), and Huang and Liu (2005).

production network setting, paving the way to connect theoretical predictions with firms' browsing activities observed in EDGAR in Section 3.

2.1 Setup

The economy is comprised of N sectors indexed by i and j, and each sector consists of a continuum of firms indexed by ι. Firms in the same sector operate the same constant returns to scale production technology

$$Y_{i,\iota,t} = Z_{it} L_{i,\iota,t}^{\alpha_i} \prod_{j=1}^{N} X_{i,\iota,j,t}^{a_{ij}},$$

where Z_{it} is the technology shock in sector i

$$z_{it} \equiv \log Z_{it} \sim \mathbb{N}(0, \sigma_i^2),$$

$L_{i,\iota,t}$ is the labor input, and $X_{i,\iota,j,t}$ is the intermediate input from sector j. The parameter α_i measures the relative importance of labor, and a_{ij} measures the relative importance of intermediate inputs. The production network is therefore captured by the input-output matrix with elements defined by a_{ij}. The variance of the sectoral TFP shock is σ_i^2, and we denote the diagonal covariance matrix of sectoral shocks as $\mathbf{\Sigma} = \operatorname{diag}\left(\sigma_i^2, \ldots, \sigma_N^2\right)$.

Within a sector, each firm ι produces a different variety. The aggregate goods in sector i is given by the standard Dixit-Stiglitz aggregator with a constant elasticity of substitution θ_i, which yields a downward-sloping demand schedule

$$Y_{it} = \left[\int Y_{i,\iota,t}^{\frac{\theta_i - 1}{\theta_i}} d\iota\right]^{\frac{\theta_i}{\theta_i - 1}}, \qquad Y_{i,\iota,t} = \left(\frac{P_{i,\iota,t}}{P_{i,t}}\right)^{-\theta_i} Y_{it}.$$

In each period, there are two stages. In the first stage, firms set their prices, potentially under incomplete information about the realization of sectoral technology shocks. They may acquire additional information beyond their prior beliefs, but doing so is costly. In the second stage, all shocks are realized and information becomes public. Firms then make their hiring and intermediate input decisions to meet demand.

In the second stage, given the wage rate W_t and the vector of sectoral goods prices $\{P_{jt}\}$, the nominal marginal cost of a firm in sector i is

$$\mathrm{MC}_{i,t} = \frac{1}{Z_{i,t}} W_t^{\alpha_i} \prod_{j=1}^{N} P_{j,t}^{a_{ij}}.$$

In the first stage, anticipating the marginal cost process and the demand schedule, firms' pricing

and information acquisition problem can be written as

$$\max_{P_{i,\iota,t},\boldsymbol{x}_{i,\iota,t}} \quad \mathbb{E}\left[\left(\frac{P_{i,\iota,t}}{P_{i,t}}\right)^{-\theta_i} Y_{i,t}\Big((1+\tau_i)P_{i,\iota,t} - \mathrm{MC}_{i,t}\Big)\,\middle|\, \boldsymbol{x}_{i,\iota,t}\right] - \mathcal{F}_i(\boldsymbol{x}_{i,\iota,t}). \tag{2.1}$$

where $\tau_i = \frac{1}{\theta_i - 1}$ is a standard subsidy rate that eliminates the price markup in the steady state. Crucially, when setting the price $P_{i,\iota,t}$, firms do not yet observe the sectoral productivity shocks and therefore they have to form expectations about the marginal cost.

In our baseline specification, firms are rationally inattentive. They can actively acquire additional signals $\boldsymbol{x}_{i,\iota,t}$ subject to some information acquisition costs indicated by $\mathcal{F}_i(\boldsymbol{x}_{i,\iota,t})$, in the spirit of Sims (2003). In Section 2.2, we will specify the details of the cost function.

Beyond the baseline specification, we will also consider comparisons with alternative information structures. For instance, signal processes may be specified exogenously, as in Woodford (2003) for a one-sector economy or La'O and Tahbaz-Salehi (2022) in a multi-sector production network. Alternatively, the overall information capacity can be determined exogenously, as in Maćkowiak and Wiederholt (2009). We also consider the implications of sticky information models (Mankiw and Reis, 2002; Alvarez et al., 2016) in Section 3, where agents update information periodically.

2.2 Endogenous Information Acquisition

Frictionless case. Consider momentarily the frictionless case. In this case, there is no information acquisition cost and firms can perfectly observe all the underlying shocks. The solution to firms' problem (2.1) then takes a particular simple form: the sectoral price is identical to the nominal marginal cost $P_{i,\iota,t} = P_{it} = \mathrm{MC}_{it}$. In terms of log-deviation from steady state, the marginal cost in sector i is given by[3]

$$\mathrm{mc}_{it} = -z_{it} + \alpha_i w_t + \sum_j a_{ij} p_{jt}, \tag{2.2}$$

and accordingly, the variations in prices satisfy

$$p_{it} = \mathrm{mc}_{it}.$$

Elastic attention. Now we specify our baseline information acquisition problem. Instead of observing the nominal marginal cost perfectly, firms need to form expectations based on imperfect information. The nominal marginal costs depend on the vector of sectoral productivity shocks $\boldsymbol{z}_t$, which requires firms to allocate their limited capacity in learning about these underlying states. In our baseline analysis, we adopt the elastic attention approach (Maćkowiak et al., 2023), where the marginal cost in reducing uncertainty measured in mutual information is constant. The following lemma first

[3]Throughout, we use small letter x_t to denote a variable X_t's log-deviation from its steady state.

transforms the original problem (2.1) into a tracking problem.

Lemma 2.1. *Under a quadratic approximation, firms' problem with elastic attention can be written as*

$$\max_{p_{i,\iota,t},\, \boldsymbol{x}_{i,\iota,t}} \quad -\frac{1}{2}\lambda_i\theta_i\mathbb{E}\left[\left(p_{i,\iota,t}-mc_{it}\right)^2\right]-\chi_i\mathbb{I}\left(mc_{it};\boldsymbol{x}_{i,\iota,t}\right) \tag{2.3}$$

subject to

$$\begin{aligned} p_{i,\iota,t} &= \mathbb{E}[mc_{it}|\boldsymbol{x}_{i,\iota,t}], \\ \mathbb{I}\left(mc_{it};\boldsymbol{x}_{i,\iota,t}\right) &= \frac{1}{2}\left(\log\mathbb{V}(mc_{it})-\log\mathbb{V}(mc_{it}|\boldsymbol{x}_{i,\iota,t})\right). \end{aligned}$$

Lemma 2.1 illustrates the basic trade-off faced by firms when choosing the information set: more accurate signals reduce the pricing error, but it incurs additional information acquisition costs. Here, λ_i is sector i's steady-state sales which determines the benefit of additional information,[4] and χ_i is the constant marginal cost when reducing the uncertainty measured by the standard mutual information $\mathbb{I}\left(mc_{it};\boldsymbol{x}_{i,\iota,t}\right)$ between the target and the signals. Here, $\mathbb{V}(mc_{it})$ and $\mathbb{V}(mc_{it}|\boldsymbol{x}_{i,\iota,t})$ denote the prior variance and posterior conditional variance of sector i's marginal cost, respectively.

Based on the technique developed in Afrouzi and Yang (2021) and Miao et al. (2022), the optimal signal structure takes a rather simple form independent of the perceived stochastic processes of the marginal cost.

Proposition 2.1. *For any linear laws of motion of the marginal cost, the optimal signal structure satisfies*

1. *When $\chi_i < \lambda_i\theta_i\mathbb{V}(mc_{it})$, firms acquire a single signal about the marginal cost*

$$x_{i,\iota,t} = mc_{it} + u_{i,\iota,t}, \qquad u_{i,\iota,t} \sim \mathbb{N}(0, v_i^2), \tag{2.4}$$

   where $\mathtt{mc}_{it}$ satisfies condition (2.2) and the variance of the private noise v_i^2 is given by

$$v_i^2 = \frac{\chi_i\mathbb{V}(mc_{it})}{\lambda_i\theta_i\mathbb{V}(mc_{it})-\chi_i}.$$

2. *When $\chi_i \geq \lambda_i\theta_i\mathbb{V}(mc_{it})$, firms acquire no new information.*

The optimal signal structure is quite intuitive: firms choose to obtain a noisy version of their ideal price change and the noise level is increasing in the marginal cost χ_i. It is important to emphasize that, in principle, the optimal information acquisition decision uniquely determines the posterior covariance matrix. Firms must evaluate how much to learn about each underlying state. In our setting, the optimal posterior covariance matrix corresponds to a tractable signal structure, as characterized by condition (2.4). Since the marginal cost mc_{it} evolves as a function of the underlying states, the signal

[4] In the envrionment in Section 4, λ_i corresponds to the steady-state Domar weight.

about marginal cost implicitly governs the firm's optimal allocation of attention across states. From now on, we focus on the case in which χ_i is sufficiently small, ensuring that firms find it optimal to actively acquire new information.

Attention allocation in partial equilibrium. What are the implied properties of attention allocation? We first discuss the total amount of attention spent in acquiring information. From Proposition 2.1, it is clear that the choice of information capacity varies with the prior uncertainty of the marginal cost process. It turns out that as the variance of the marginal cost increases, firms will acquire more information to keep track of the more volatile target.

Proposition 2.2 (Information Capacity)**.** *For any processes of prices and wages, the information capacity* $\mathbb{I}(mc_{it}; \boldsymbol{x}_{i,\iota,t})$ *increases in the volatility of firms' marginal costs* $\mathbb{V}(mc_{it})$.

The applied implication is that firms' information acquisition activities should be intensified with the level of uncertainty of their marginal cost processes, a prediction that we will test in the next section. In contrast, the information capacity will remain unchanged in an alternative environment with exogenous information or fixed information capacity.

Next, we turn to the attention allocation across different sectors. Recall that the optimal signal structure (2.4) applies to any processes of sectoral prices that appear in the marginal cost. From a partial equilibrium perspective, it is helpful to examine a special case where prices are uncorrelated both with each other and with the nominal wage. In this case, the incentive for firms' attention allocation becomes transparent.

Proposition 2.3 (Attention Allocation in PE)**.** *For firms from sector* i*, denote* γ_j *and* $\widehat{\gamma}_{ij}$ *as the prior and posterior uncertainty about* p_{jt}*, respectively. When prices are uncorrelated, the reduction of uncertainty satisfies*

$$\log\left(1 - \frac{\widehat{\gamma}_{ij}}{\gamma_j}\right) = const_i + \log \gamma_j + \log a_{ij}^2, \quad \text{for } j \neq i. \tag{2.5}$$

The left-hand side of equation (2.5) relates to the reduction of uncertainty about prices in sector j after the information acquisition, which can be viewed as a measure of amount of attention allocated to sector j. It says that comparing two sectors, firms will allocate more attention to the one that is perceived with more uncertainty (higher γ_j) and that is more relevant for firms in sector i in terms of payoff (higher a_{ij}). These implications directly generalize the insights from a one-sector environment (Maćkowiak and Wiederholt, 2009) to a network setting.

This special case is particularly relevant when firms' input prices are dominated by their corresponding specific shocks or when firms subjectively downplay the cross-sector correlation due to behavioral reasons (Gabaix, 2014; Lian, 2021). As a result, it can serve as a good approximation when depicting a single firm's belief system during information acquisition. Of course, in general equilibrium, prices will be correlated, and the exact covariance structure will be shaped by trade linkages and endogenous information acquisition, as we demonstrate next.

2.3 Equilibrium Characterization and Attention Allocation

In this subsection, we move to the general equilibrium analysis and discuss how the attention allocation and the price dynamics are jointly determined in the production network.

Implied price flexibilities. An important feature of the model is that the effects of informational frictions can be succinctly summarized by the corresponding price flexibilities in firms' pricing decision. Lemma 2.1 directly implies that firms set their prices according to the expected marginal costs

$$p_{i,\iota,t} = \mathbb{E}[\mathrm{mc}_{it}|x_{i,\iota,t}].$$

Based on Proposition 2.1, the sectoral pricing formula follows from the standard Bayesian inference after averaging out idiosyncratic noises.

$$p_{it} = \int p_{i,\iota,t} d\iota = \mu_i \, \mathrm{mc}_{it}, \quad \text{where} \quad \mu_i \equiv 1 - \frac{\chi_i}{\theta_i \lambda_i \mathbb{V}(\mathrm{mc}_{it})} \in [0,1]. \tag{2.6}$$

The variable μ_i effectively represents the signal-to-noise ratio and can be mapped one-to-one to the endogenously chosen information capacity. Simultaneously, μ_i shapes the price flexibility: a higher μ_i implies greater responsiveness to variations in nominal marginal costs and therefore smaller nominal rigidity.

In this section, we take the wage function as given denoted by $w_t = \boldsymbol{\kappa} \boldsymbol{z}_t = \sum_j \kappa_j z_{j,t}$.[5] Supposing the information capacities or the price flexibilities $\{\mu_i\}$ are already known, it is straightforward to derive from condition (2.6) that prices can be expressed in terms of the underlying productivity shocks.

Lemma 2.2. *Given the price flexibilities $\{\mu_i\}$ and the wage function $\boldsymbol{\kappa}$, the price dynamics obey*

$$\boldsymbol{p}_t = \boldsymbol{\phi} \boldsymbol{z}_t = (\mathbf{I} - \boldsymbol{\mu}\mathbf{A})^{-1} \boldsymbol{\mu}(-\mathbf{I} + \boldsymbol{\alpha}\boldsymbol{\kappa}) \boldsymbol{z}_t, \tag{2.7}$$

where $\boldsymbol{\mu} = \mathrm{diag}(\mu_1, \ldots, \mu_N)$ is a diagonal matrix.

This influence matrix $\boldsymbol{\phi}$ in (2.7) is akin to a modified Leontief inverse where the network matrix $\mathbf{A}$ is dampened by $\boldsymbol{\mu}$, which encodes the responses of sectoral prices to different sectoral shocks. It nests the frictionless benchmark by setting $\boldsymbol{\mu} = \mathbf{I}$. It is important to keep in mind that though the formula resembles that in La'O and Tahbaz-Salehi (2022), the dampening matrix $\boldsymbol{\mu}$ in our setting is endogenous to equilibrium while it is exogenous in their environment.

Attention best response and fixed point. It remains to show how to determine $\{\mu_i\}$ in equilibrium. Note that μ_i inherits the property of the precision of optimal signals. As the marginal costs are influenced by prices in other sectors, the price flexibilities and the information acquisition decisions across sectors are interdependent as well. This interdependence gives rise to a fixed-point problem in the general equilibrium, which is formalized in the following proposition.

[5] In Section 4, we derive the optimal wage function that minimizes the social welfare loss.

Proposition 2.4. *Given a nominal wage process $w_t = \boldsymbol{\kappa} \boldsymbol{z}_t = \sum_j \kappa_j z_{j,t}$, the price flexibilities $\{\mu_i\}$ solve the following fixed-point problem*

$$\mu_i = \mathcal{T}_i(\boldsymbol{\mu}, \boldsymbol{\kappa}) = 1 - \frac{\chi_i}{\theta_i \lambda_i \mathbb{V}(mc_{it})}, \qquad \text{and} \qquad \mathbb{V}(mc_{i,t}) = \left\| \mathbf{e}_i (\mathbf{I} - \mathbf{A}\boldsymbol{\mu})^{-1} (-\mathbf{I} + \boldsymbol{\alpha}\boldsymbol{\kappa}) \boldsymbol{\Sigma}^{\frac{1}{2}} \right\|^2, \tag{2.8}$$

where $\mathcal{T}_i(\boldsymbol{\mu}; \boldsymbol{\kappa})$ is the attention best response function for firms in sector i.[6]

Proposition 2.4 highlights the endogeneity of price flexibilities and information acquisition decisions. Firms' choice of μ_i not only depends on the volatilities of underlying shocks ($\boldsymbol{\Sigma}$), but also on other firms' information acquisition decision. For example, when firms in sector j acquire more information, their prices become more responsive to underlying shocks, which in turn increases the volatility of sector i's marginal costs via production network linkages. As a result, it provides incentive for firms in sector i to also acquire more information. In the end, the equilibrium corresponds to the fixed-point of the vector of best responses.

The aforementioned mechanism alludes to a form of *strategic complementarity* in information acquisition, which is related to the coordination motive in Hellwig and Veldkamp (2009) and the expectation conformity in Pavan and Tirole (2022). Although the strategic interaction is more involved in our network setting, a precise characterization of the response of μ_i to changes in μ_j still holds.

Proposition 2.5. *Given the attention best response $\mathcal{T}(\boldsymbol{\mu}, \boldsymbol{\kappa})$, the impact of changes in μ_j on μ_i is given by*

$$\frac{\partial \mathcal{T}_i(\boldsymbol{\mu}, \boldsymbol{\kappa})}{\partial \mu_j} = \frac{1 - \mu_i}{\mathbb{V}(mc_{it})} \mathrm{Cov}(mc_{it}, mc_{jt}) \left((\mathbf{I} - \mathbf{A}\boldsymbol{\mu})^{-1} \mathbf{A} \right)_{ij}. \tag{2.9}$$

This formula characterizes how firms in sector i respond to a change in sector j's information acquisition decision. The first term, $\frac{1-\mu_i}{\mathbb{V}(\mathrm{mc}_{it})}$, reflects the sensitivity of sector i's attention allocation. As μ_i approaches one, sector i's attention allocation becomes increasingly inelastic to external changes. A shift in μ_j directly alters sector j' pricing decision, and p_{jt} adjusts proportionally to mc_{jt} according to condition (2.6). At the same time, sector i's choice of μ_i depends on the volatility of its own marginal cost, mc_{it}. The extent to which μ_i responds to changes in μ_j is governed by the degree of comovement across sectors, captured by the covariance term $\mathrm{Cov}(\mathrm{mc}_{it}, \mathrm{mc}_{jt})$. Furthermore, a change in μ_j indirectly affects pricing decisions in all other sectors, even when their price flexibilities remain constant, consistent with the general equilibrium condition (2.7). These direct and indirect network effects are summarized by the (i, j)-th element of the matrix $(\mathbf{I} - \mathbf{A}\boldsymbol{\mu})^{-1}\mathbf{A}$.

As a preview, we will derive the optimal price stabilization policy in Section 4, which internalizes the impact of monetary policy on firms' endogenous information choices—a channel that is absent when information is exogenous or when information capacity is fixed. This endogenous information channel leverages the strategic interactions in information acquisition outlined in Proposition 2.5.

[6] $\mathbf{e}_i$ denotes i-th standard basis (row) vector.

Finally, we comment on the uniqueness of the equilibrium. The equilibrium system characterized by Proposition 2.4 is nonlinear and generally does not admit a closed-form solution. In Appendix D.10, we provide a sufficient condition that guarantees uniqueness of the equilibrium. The proof relies on Kellogg's fixed point Theorem in Banach spaces (Kellogg, 1976), and it may be applicable to other high-dimensional economic models.

Attention allocation in general equilibrium. Having characterized the general equilibrium, we now revisit the properties of firms' attention allocation. Different from the partial equilibrium results in Proposition 2.3, we can no longer assume that prices in different sectors are independent of each other. Generically, prices in all sectors depend on all the underlying sectoral shocks

$$p_{it} = \sum_j \phi_{ij} z_{jt},$$

where ϕ_{ij} is an element in the influence matrix shown in condition (2.7).

Parallel with our partial equilibrium discussion in Subsection 2.2, we measure the attention allocated towards a particular sector as a reduction in uncertainty in terms of sectoral fundamentals. For firms from sector i, denote $\widehat{\sigma}^2_{j|i}$ as the posterior uncertainty about z_{jt}. The reduction of uncertainty about sector j is given by

$$\omega_{ij} = \frac{\sigma_j^2 - \widehat{\sigma}^2_{j|i}}{\sigma_j^2}. \tag{2.10}$$

A higher value of ω_{ij} indicates that firms in sector i experience a greater reduction in uncertainty regarding productivity in sector j. That is, more attention is allocated to learning about sector j's conditions. To which extent their signals reveal the fundamentals about the underlying states in different sectors depends on the endogenous exposure of marginal costs to shocks in different sectors. The following proposition highlights how such attention allocation hinges on both the primitive shocks and equilibrium forces.

Proposition 2.6 (Attention Allocation in GE)**.** *The attention allocation satisfies*

$$\omega_{ij} = \mu_i \frac{\sigma_j^2 \phi_{ij}^2}{\sum_k \sigma_k^2 \phi_{ik}^2} \qquad \textit{and} \qquad \sum_j \omega_{ij} = \mu_i.$$

This proposition makes it clear that with either a higher variance of productivity shock z_{jt} or a larger dependence of p_{it} on z_{jt} will lead firms to allocate more attention towards sector j. To understand this result, recall from condition (2.6) that there is a one-to-one mapping between marginal costs and sectoral prices

$$\text{mc}_{it} = \frac{1}{\mu_i} p_{it} = \frac{1}{\mu_i} \sum_j \phi_{ij} z_{jt}.$$

Firms acquire signals regarding their marginal costs, which are influenced by sector-specific fundamentals. The informativeness of the signal about sector j's fundamental increases with the product of the volatility of z_{jt} and the equilibrium exposure measured by ϕ_{ij}. A higher volatility σ_j^2 amplifies the relative importance of sector j's fundamental within firms' signals in sector i, directing greater attention towards sector j. Simultaneously, a higher equilibrium exposure ϕ_{ij} intensifies the general equilibrium impact, endogenously increasing the prominence of sector j's component in sector i's signals.

2.4 Alternative Information Specifications

We consider two alternative specifications for the information structure. The first one is the case in which firms face a fixed information capacity constraint when collecting information (Sims, 2003; Maćkowiak and Wiederholt, 2009). This corresponds to setting $\chi_i = 0$ in equation (2.3) while requiring

$$\mathbb{I}(\boldsymbol{z}_t; \boldsymbol{x}_{i,\iota,t}) \leq \delta_i,$$

where δ_i is some exogenous parameter that determines the maximum capacity a firm can choose to reduce faced uncertainty.

The second alternative is the exogenous information approach where the signal structure and its precision are exogenously determined. Particularly, we consider the specification as in La'O and Tahbaz-Salehi (2022), in which firms in sector i observe a vector of signals about $\boldsymbol{z}_t$

$$x_{i,\iota,j,t} = z_{j,t} + u_{i,\iota,j,t}, \qquad u_{i,\iota,j,t} \sim \mathbb{N}(0, \tau_i \sigma_j^2).$$

Notice that when inferring productivities in different sectors, firms face the same signal-to-noise ratio controlled by the exogenous parameter τ_i.

It turns out that both of these two alternatives yield observationally equivalent pricing strategies to our baseline elastic attention case.

Proposition 2.7. *With alternative information structures, the optimal pricing strategy is*

$$p_{i,t} = \mu_i \, mc_{it},$$

where $\mu_i = 1 - e^{-2\delta_i}$ *with fixed information capacity and* $\mu_i = \frac{1}{1+\tau_i}$ *with exogenous information.*[7]

It is worth noting that though the pricing formula in these alternatives resembles that from the elastic attention approach, the implied price rigidity μ_i in both cases are determined only by exogenous parameters and are independent of the stochastic processes of the economy's endogenous variables. Therefore, it is also independent to the market structure or the monetary policy.

[7] In this paper we adopt natural logarithm log in computing the Shannon entropy, and the unit of information is called a "nat." The usual base for logarithm in the entropy formula is 2 , in which case the unit of information is a "bit." Adopting natural logs simplifies the algebra without changing any of our results.

Corollary 2.1. *For any processes of prices and wages, the information capacity summarized by* $\{\mu_i\}$ *is invariant to the market outcomes.*

This property is in contrast to that outlined in Proposition 2.2 under elastic attention. In the next section, we will provide further evidence that helps select the relevant approach in acquiring information.

3. Measuring Firms' Attention: Browsing Activities

We leverage a novel dataset that records firms' browsing histories on other firms' electronic filings in EDGAR to provide empirical evidence in support of our model's predictions. We begin by describing the data and our browsing-based attention measure and presenting a set of key stylized facts on firms' information acquisition behavior. We highlight that firms' information acquisition behavior is consistent with our elastic attention approach.

3.1 Data and Stylized Facts

Our empirical analysis combines three main data sources: (1) firm-to-firm browsing activity from the SEC's EDGAR platform, (2) survey-based data on firms' forecasts, and (3) input-output linkages at both the firm and industry levels. Appendix A provides additional details on data sources and construction procedures.

3.1.1 The EDGAR Browsing Data

Our primary measure of firm-to-firm attention is constructed from the EDGAR (Electronic Data Gathering, Analysis, and Retrieval) system, maintained by the U.S. Securities and Exchange Commission (SEC).[8] Since 1996, all U.S. public companies have been required to file disclosures electronically via EDGAR. These filings, freely accessible online, are widely used by corporate managers, investors, analysts, and other market participants.

Beginning in 2003, the SEC's Division of Economic and Risk Analysis (DERA) has made available detailed logs of web traffic to EDGAR filings. This dataset, known as the "EDGAR Logfile", contains detailed internet access records for all filings on the platform. This includes a partially masked IP address of the viewer, timestamp of the access, and a unique document identifier (Accession number).

We define a *browser* as the viewer of a document and a *browsee* as the owner of that document. To uncover browsers' full IP addresses, we follow the method developed in Chen et al. (2020), and then use services provided by ipinfo.io—a leading IP information provider—to map each viewer's uncovered IP address to its firm identity. These steps yield a firm-level, daily browser-browsee database

[8] EDGAR supports mandatory electronic disclosures under the Securities Act of 1933, the Securities Exchange Act of 1934, the Trust Indenture Act of 1939, and the Investment Company Act of 1940. See https://www.sec.gov/edgar/about for more information.

covering the period from 2009 to 2016.[9]

Browsing-based attention measure. We measure the *browsing intensity* of a browser i on a browsee j, denoted by B_{ijt}, as the total number of online accesses made by firm i to all EDGAR filings owned by firm j over time interval t,

$$B_{ijt} = \sum_{s \in t} a_{ijs}, \tag{3.1}$$

where $a_{ijs} = 1$ if browser i accessed a filing of browsee j at time s, and zero otherwise. The time unit s can be measured as accurately as to a second, while the interval t may refer to a day, month, quarter, or year.[10] In our empirical analysis, we define the log browsing volume as $b_{ijt} = \log\big(B_{ijt}\big)$.[11]

We argue that our browsing-based measure provides a reasonable proxy for firms' attention allocation. The core premise is that when firm i pays greater attention to firm j, this should manifest empirically through more frequent browsing from firm i to firm j's SEC filings on the EDGAR system.

This premise is supported by two key features of EDGAR filings. First, as the SEC's official platform for corporate disclosures, EDGAR offers the most timely and comprehensive access to firm-specific information, including unscheduled material events (Form 8-K) and annual financial statements (Form 10-K).[12] Second, unlike paid services such as Bloomberg or Yahoo Finance, EDGAR provides free, unfiltered access to all mandatory disclosures while preserving the original granularity of financial statements that third-party platforms often aggregate or omit. These features establish EDGAR as the primary resource for market participants seeking authoritative, up-to-date firm information. For example, General Electric's 2003 10-K received 21,987 downloads through EDGAR versus just 800 from the company's website.[13]

Our browsing-based attention measure thus complements and extends the existing text-based proxies commonly used in recent literature. First, while existing measures typically rely on the content of firms' public disclosures—which may be influenced by strategic reporting incentives—our measure captures firms' actual daily information acquisition behavior, free from such biases. Second, our data are available at a high daily frequency, in contrast to quarterly or annual frequency of most text-based proxies. Third, unlike media-based proxies that typically reflect aggregate attention, our approach enables a direct and comprehensive mapping of firm-to-firm attention linkages.[14]

[9] We begin our sample in 2009 because prior to 2009, the number of browsings was orders of magnitude less, largely due to significant changes in the disclosure requirement by the SEC and EDGAR. The SEC discontinued public release of EDGAR internet traffic data after 2017. Additional details are provided in Appendix A.1.2.

[10] For example, if t corresponds to January 2, 2015, and firm i accessed firm j's documents three times on that day, then $B_{ijt} = 3$.

[11] For industry-level analysis, the browsing volume is always positive. For firm-level analysis, we define $b_{ijt} = \log\Big(B_{ijt} + 1\Big)$ whenever $b_{ijt} = 0$.

[12] Form 10-K filings on EDGAR typically precede formal annual reports by 61 days on average. Appendix Table A.1 lists the 10 types of filed documents receiving the most browsings.

[13] The news article can be accessed on the Wall Street Journal website at https://www.wsj.com/articles/the-109-894-word-annual-report-1433203762.

[14] In Appendix A.2, we construct a text-based firm-to-firm attention measure $\widetilde{B}_{ij}$—analogous to our browsing-based measure B_{ij}—using established methods based on firms' 10-K disclosures. In 2016, the text-based measure captures only 2,265

Baseline sample. While the full dataset includes browsing activity by companies, institutions, and individuals, we restrict our baseline sample to public firms for three main reasons. First, EDGAR's disclosure mandate applies exclusively to public companies. Second, the set of IP addresses that can be reliably matched to firm identities is predominantly associated with public firms. Third, unlike private firms, public companies can be linked to standard datasets like Compustat. Our final sample comprises 7,622 public firms with 713 million unique IP addresses, capturing 52.4% of all EDGAR browsing activity during our sample period.[15] Additional details on sample construction are provided in Appendix A.1.1.

Table 1 reports summary statistics of the final baseline sample. On an annual basis, the sample typically contains 3,626 unique browsers and 6,254 unique browsees. A typical browser views filings from 180 firms each year, while a typical browsee is viewed by 105 distinct firms. The average browser–browsee pair records 60 browsing events annually. Aggregating across all interactions, the median browser initiates 11,356 views per year, while the median browsee receives 5,897 views.

Table 1: EDGAR Browsing Data: Summary Statistics

	Mean	Median	S.D.
No. Browsers per Year	3,626	3,608	251
No. Browsees per Year	6,254	6,280	321
No. Browsees viewd by a Browser per Year	180	176	25
No. Browsers viewing a Browsee per Year	105	74	115
Browsing Volume sent by a Browser per Year	11,356	9,881	7,764
Browsing Volume received by a Browsee per Year	5,897	5,389	4,869
Browsing Volume per Browser-Browsee pair per Year	60	50	34

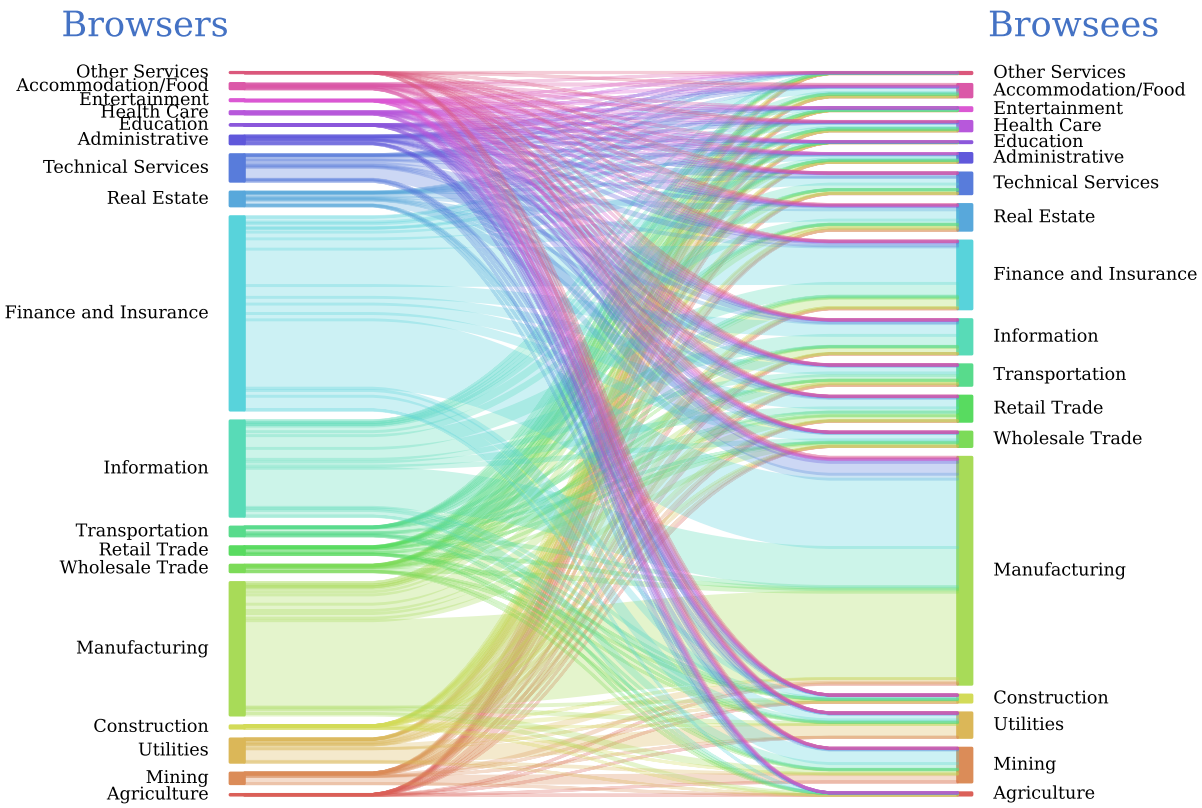
Finally, Figure 1 visualizes the bilateral browsing network at the 2-digit NAICS industry level after excluding media and financial investment firms. Firms in the finance and insurance sector exhibit the highest volume of outbound browsing activity, particularly directed toward firms in the manufacturing sector. At the same time, manufacturing firms receive the greatest overall attention from browsers across all industries.

Appendix B.1 presents additional descriptive evidence on firms' browsing behavior. First, firms with larger sales and employment tend to both browse more intensively and attract more attention from others. Second, firms allocate approximately 75% of their browsing activity to suppliers and customers, while only 12.6% is directed toward competitors.

browser–browsee pairs, whereas our browsing-based data covers 2.6 million pairs. The two measures are weakly positively correlated, with a correlation coefficient of 0.19 (standard error 0.06).

[15] Automated web crawlers account for a growing share of internet traffic in recent years. To isolate genuine information-seeking behavior, we follow the approach in Cao et al. (2023) and exclude IP addresses that viewed filings from more than 50 firms in a single day, as well as addresses that self-identify as bots via their user-agent strings. Our results are robust to the inclusion of these excluded observations.

Figure 1: Bilateral Browsing at the Industry Level



Note: This figure plots the bilateral browsing volumes between industries. The width of the bar is proportional to the browsing volumes. The media companies and financial investment firms are excluded. Firms in the finance and insurance industry have the largest browsing volume on other firms, especially those in the manufacturing industry. Firms in the manufacturing industry also receive the most attention from the browsers.

3.1.2 Other Datasets

The IBES Managerial Forecast Data. The I/B/E/S Managerial Guidance dataset extracts public companies' quantitative managerial forecasts from earnings call transcripts and press releases. Managers provide forward-looking estimates for up to 14 firm-specific variables. For our analysis, we focus on two widely reported and economically salient variables: earnings per share (EPS) and sales. These variables are the most frequently forecasted and thereby contain the largest number of observations. We match these forecasts with actual outcomes from IBES Actuals to compute forecast errors.[16]

The Factset Supply Chain Data. We obtain firm-level supply chain relationships from the FactSet Revere Supply Chain dataset accessed through WRDS. This dataset provides detailed information on publicly disclosed supply chain links between firms, identifying the supplying firm ("supplier") and the procuring firm ("customer").[17]

[16] When forecasts are reported as ranges, we take the midpoint of the upper and lower bounds as the forecasted value.

[17] Supply chain links in FactSet Revere are sourced from public disclosures, including annual and quarterly filings, investor call transcripts, capital market presentations, sell-side conferences, firm websites, and press releases. Public filings are the

We extract all reported relationships from 2009 to 2016 to align with the timeframe of the browsing data. Each observation includes firm identifiers for both the supplier and customer, as well as contract start and end dates. For approximately 11% of supplier–customer pairs, the data also report the customer's purchases as a share of the supplier's total revenue in a given year. The final sample comprises 4,260 unique suppliers and 4,518 unique customers. On average, each supplier reports 7.6 customer relationships per year, while each customer reports 7.55 supplier relationships.

The BEA input-Ouput Database and Other Data Sources. We construct industry-level input–output (IO) linkages using the supply and use tables published by the U.S. Bureau of Economic Analysis (BEA), including annual IO tables for 71 industries from 1997 to 2021 and IO tables for 405 industries for the years 2007 and 2012. To match firms in our datasets to its corresponding BEA industries, we apply a concordance between BEA industry codes and NAICS classifications.

We also draw on several additional data sources in our analysis. Firm-level characteristics, including total factor productivity (TFP), are obtained from Compustat. We use producer price index (PPI) data from the Bureau of Labor Statistics (BLS) to compute industry-level inflation rates. Industry-level measures of TFP are taken from the BEA. Summary statistics for IO tables, inflation volatility, sales growth volatility, and TFP volatility appear in Appendix Table B.5.

3.1.3 Browsing Statistics

We begin by presenting descriptive statistics on firms' browsing behavior. The hazard function of information acquisition, along with other empirical moments, provides stronger support for state-dependent models—such as our baseline specification with elastic attention—than for time-dependent alternatives.

Browsings are frequent and inattentive spells vary substantially. Consider a sample of firms over T periods. Let I_{it} be a browsing indicator, where $I_{it} = 1$ if firm i's total browsing exceeds zero during period t. The frequency of browsing for firm i, denoted f_i, is calculated as $f_i = \sum_t I_{it}/T$.

The first two rows in Table 2 report the mean and median frequencies of browsing across firms. We compute the weekly, bi-weekly, monthly, and quarterly browsing frequencies by defining periods of varying lengths, as shown in the respective columns. Firms exhibit frequent browsing behavior: on average, 57% of firms acquire information about other firms within a given week. This fraction rise to 87% over the course of a quarter.

An inattentive spell is a consecutive sequence of periods during which a firm engages in no browsing activity. The length of an inattentive spell for firm i is given by $\sum_{s=\tau}^{t} I_{is}$, where $I_{i(\tau-1)} = 1$, $I_{i(t+1)} = 1$, and $I_{is} = 0$ for $\tau - 1 < s < t + 1$. For each firm, we compute the standard deviation of the logarithm of inattentive spell lengths. The last two rows in Table 2 present the mean and median of this statistic across firms. For a given firm, the inattentive spells vary substantially during the sample period: for

primary source of these data (Culot et al., 2023).

example, the standard deviation is 1.36 weeks when periods are defined weekly and 0.47 quarters under a quarterly definition.

Table 2: Frequency of Browsing and Standard Deviation of Inattentive Spells

Statistic	Weekly	Bi-weekly	Monthly	Quarterly
Mean frequency	0.57	0.67	0.73	0.87
Median frequency	0.62	0.75	0.83	0.96
Standard deviation of inattentive spells (mean)	1.36	0.99	1.48	0.44
Standard deviation of inattentive spells (median)	1.30	0.99	1.41	0.47

At face value, these patterns appear inconsistent with theories that predict firms acquire information in a periodic manner, such as sticky information models (Mankiw and Reis, 2002; Alvarez et al., 2016). A caveat to this interpretation is that when firms produce multiple products or delegate information acquisition to multiple individuals, the aggregated activity may appear more random. Nevertheless, these empirical moments can still be informative for testing or calibrating the information technologies embedded in different theoretical frameworks.

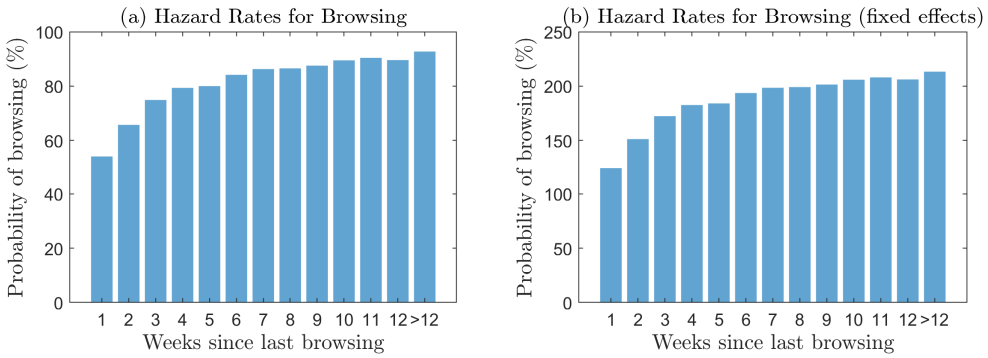
Hazard function is upward-sloping. We estimate the hazard function of information acquisition, which represents the probability that a firm acquires information as a function of the time elapsed since its last browsing event. Panel (a) of Figure 2 shows that the hazard function is upward-sloping—the likelihood of browsing increases with the duration of inattention. This pattern is consistent with state-dependent models of information acquisition, in contrast to time-dependent models, which imply a flat hazard function. The upward slope persists after controlling for frequency fixed effects to account for survivor bias, as shown in Panel (b).[18] Appendix B.2 provides details on how we estimate the hazard function.

To illustrate how the hazard function helps identify the appropriate information technology, consider first the case in which firms acquire information in a Calvo fashion. In that case, the implied hazard function will be completely flat, as the decision to update information is independent of the time since the last update. Alternatively, if firms update information in a perfectly periodic manner, the hazard function will spike at a single point corresponding to the fixed updating interval. Neither of these benchmark models aligns with the empirical hazard function.

Browsing volumes are variable. For a given firm, the standard deviation of the logrithm of its browsing volume over the sample period is 1.22, indicating considerable volatility in browsing activity over time. From the lens of models with information acquisition, this variability suggests that firms continuously adjust their information processing capacity, consistent with the elastic attention

[18] We sort firms into deciles based on their hazard rates and compute the unconditional hazard rate within each decile. This approach accounts for heterogeneity in hazard rates and avoids survivor bias; see Klenow and Kryvtsov (2008) for a detailed discussion of the latter. Note that the hazard rates reported in Panel (b) can exceed one, as they are expressed relative to fixed effects.

Figure 2: Hazard Rates for Inattention



Note: This figure displays the estimated hazard rate for browsing. Panel (b) displays results after controlling for the frequency fixed effects to account for survivor bias.

framework. Section 3.3 provides additional evidence that the information capacity co-moves with the volatility of firms' learning targets.

3.2 Network Linkages, Uncertainty and Attention Allocation

In this section, we provide both firm-level and industry-level evidence showing that firms allocate more attention to others that are more payoff-relevant, measured by network distance and input-output linkages, and those subject to greater uncertainty.

3.2.1 Firm Level Evidence

In a partial equilibrium setting, our theory predicts that firm i allocates more attention to a supplier that is more relevant—measured by input share—and that is perceived with more uncertainty, as formalized in Proposition 2.3.

Input Shares, volatility and attention allocation. We begin by testing this prediction empirically using a subsample of firms from the FactSet dataset that report transaction values with specific trading partners. Specifically, we examine how a firm's attention to a supplier varies with the supplier's importance (input share) and this supplier's perceived uncertainty, proxied by volatility in TFP or sales. Motivated by equation (2.5), we estimate the following regression:

$$b_{ijt} = \alpha_i + \gamma_t + \beta_1 \text{input_share}_{ijt} + \beta_2 \sigma_{jt} + \beta_3 X_{jt} + \varepsilon_{ijt}, \tag{3.2}$$

where b_{ijt} is the log browsing volume of firm i on firm j in year t, input_share_{ijt} denotes the share of firm i's purchases from supplier j in year t, and σ_{jt} is the standard deviation of firm j's TFP/log sales over the past-5 years prior to year t.[19] The regression includes browser-year fixed effects α_{it}, and the

[19] As it is generally difficult to obtain firm-level product price information, we therefore use volatility of log sales and TFP as a proxy for firms' perceived uncertainty. Later, we present results using inflation (price) volatility at the industry level.

vector X_{jt} contains standard time-varying firm-level covariates for the supplier. The coefficients β_1 and β_2 capture how the browsing intensity of firm i depends on its input share from supplier j and the sales/TFP volatility of supplier j, respectively.

Table 3 reports the estimation results. Column (1) shows that a 10 percentage points increase in input share is associated with a 44.3 percent increase in the buyer's browsing volume to the supplier, holding the supplier's TFP volatility constant. Conditional on input share, a one-standard-deviation increase in the supplier's TFP volatility (equal to 0.12) leads to a 14.5 percent increase in customer's browsing volume. Column (4) displays a similar magnitude when we replace suppliers' TFP volatility with sales volatility. Our results continue to hold when we add the time-varying controls for browsees and the browser and time fixed effects to the regression.

Table 3: Input Share, Volatility and Browsing Intensity

	Browsing Intensity					
	(1)	(2)	(3)	(4)	(5)	(6)
Input Share	4.43***	4.35***	6.93*	1.29***	1.43*	9.37***
	(0.83)	(0.83)	(3.57)	(0.46)	(0.73)	(2.74)
TFP Volatility	1.21***	1.14***	1.19**			
	(0.39)	(0.39)	(0.47)			
Sales Volatility				0.20**	0.29***	0.24***
				(0.080)	(0.097)	(0.074)
Controls		✓	✓		✓	✓
Browser FE			✓			✓
Time FE			✓			✓
Adjusted R^2	0.03	0.03	0.55	0.01	0.02	0.55
No. Observations	2025	2000	1298	6614	4868	3758

Note: This table shows how the browsing intensity of a firm depends on its input share from the suppliers, how volatile the suppliers' sales are, and how volatile the suppliers' TFP is. The control variables include firm size, age, leverage, fixed costs, ROA, stock returns, price-to-cost margins, cash holdings and R&D expenditures. Robust standard errors are in parentheses and clustered by browsing firms. *Significance*: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Network distance and attention allocation. The previous result relies on a subset of firms in Factset, as only about 11% of customer-supplier relationships report transaction values. To broaden the analysis to the full sample, we next proxy supplier importance using network distance and examine how a firm allocates attention along its supply chain.

We begin by constructing a network distance measure following Carvalho et al. (2021). For each firm i in year t, we classify all other firms into distinct groups based on their shortest-path distance to firm i. Firms in the "upstream distance 1" group are direct suppliers of firm i. "Upstream distance 2" firms are suppliers to firm i via one intermediary, and so on, up to a maximum distance of 4. Firms beyond distance 4 are grouped into a residual category and serve as the (omitted) base group.

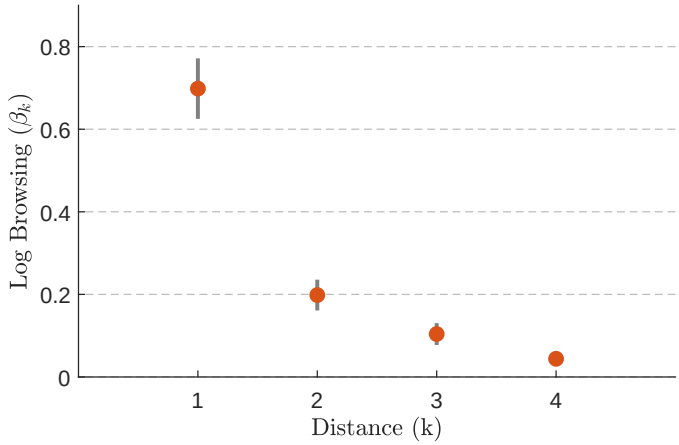
To quantify how attention allocation varies with supply chain proximity, we estimate the following

regression:

$$b_{ijt} = \alpha_{it} + \gamma_{jt} + \sum_{k=1}^{4} \beta_k \text{Upstream}_{ijt}^{(k)} + \varepsilon_{ijt}, \tag{3.3}$$

where b_{ijt} denotes the log browsing volume of firm i on firm j in year t, and $\text{Upstream}_{ijt}^{(k)}$ is a dummy variable indicating if firm j is firm i's upstream distance-k firm during year t. The regression includes browser-year fixed effects α_{it} and target browsee-year fixed effects γ_{jt}. The coefficient β_k captures the relative browsing intensity directed toward distance-k firms, compared to those in the base group.

Figure 3: Firms' Attention Hierarchy along the Supply Chain



Note: The figure depicts how the estimated coefficients β_k (red dots) vary with the suppliers' network distance. The gray bars represent the 95% confidence intervals.

Figure 3 plots the estimated coefficients $\{\beta_k\}$ (red dots) with 95% confidence intervals (gray bars). Attention declines sharply with network distance: firms pay roughly 70% more attention to direct suppliers (distance 1) and only about 20% more to distance-2 suppliers, relative to the base group. The regression results are presented in Appendix Table B.1 column (1). This result is consistent with our theoretical prediction that firms pay more attention to others that are more payoff-relevant.[20] The relatively steep decline from β_2 on further suggests that this channel is quantitatively significant.

Food for thought: event study. We wrap up our analysis with further evidence on the dynamic relationship between firms' attention allocation and input-output linkages. Leveraging the panel structure of our data, we explore an event study framework. Specifically, we ask: how does a customer adjust its attention when it starts or ends a trading relationship with a supplier? To answer this, we estimate the following regression:

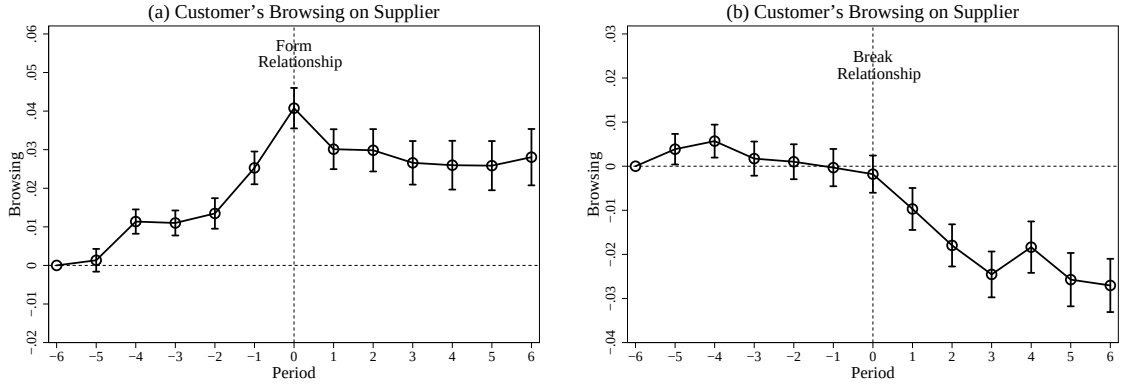
$$b_{ijt} = \sum_{\ell=-m}^{n} \beta_\ell D_{ij(t-\ell)} + \alpha_i + \alpha_j + \gamma_t + \varepsilon_{ijt}, \tag{3.4}$$

[20] To explore the role of browsees' uncertainty on browsers' attention allocation, we replace the browsee-year fixed effect using browsees' TFP or sales volatility. Results in Appendix Table B.1 column (2) - (5) show economically and statistically significant positive effect, consistent with our model prediction.

where b_{ijt} is the log browsing volume of customer i on supplier j in quarter t, and $D_{ij(t-\ell)}$ is an indicator equal to one if the pair formed or terminated a trading relationship in quarter $t-\ell$, having had no trading activity for at least four years prior (in the case of formation) or following (in the case of termination). The regression includes supplier and customer fixed effects (α_i, α_j) as well as time fixed effects (γ_t).

Figure 4 plots the estimated coefficients $\{\beta_\ell\}$.[21] Two main patterns emerge. First, attention from the customer to the supplier gradually increases in the quarters leading up to the formation of a trading relationship, peaking at the quarter of contract initiation, and remains elevated thereafter. This suggests a forward-looking channel on attention allocation: anticipating potential cooperation, firms start learning about their partners before the relationship actually takes place. In contrast, following the termination of a relationship, attention gradually declines without a pre-trend.

Figure 4: Browsing and Trading Relationship: Event Studies



Note: The figure depicts the log browsings of customers on suppliers when new trading relationships are formed or existing relationships are terminated at period 0. The frequency is quarterly and 90% confidence intervals are based on clustering at the browser level.

Comparing the medium-run effects of forming or breaking a relationship, firms allocate, on average, about 3% more attention to a partner during periods of active trading. This is consistent with our theoretical prediction that firms devote more attention to partners that are more payoff-relevant. As our model is static in nature, it cannot directly account for the dynamic anticipation effects. However, it would be interesting for future research to build on these findings.

Discussion on alternative empirical specification. Our theory predicts that, in the partial equilibrium, input shares serve as a natural measure of other firms' payoff relevance. This result depends on specific modeling assumptions, such as the form of the production function and the nature of underlying shocks. More generally, demand-side factors may also influence firms' information acquisition decisions. For example, with demand shocks instead of productivity shocks, the direction of shock

[21] Appendix Figure B.5 shows estimates for events defined without the four-year inactivity restriction. Figure B.6 shows results with 95 confidence intervals.

transmission switches from downstream to upstream (Acemoglu et al., 2016); with general CES aggregator, the distinction between upstream and downstream sectors are less clear (Carvalho et al., 2021); when firms can choose their supply functions, information on both suppliers and customers become relevant (Flynn et al., 2025).

Given these considerations, we revisit the aforementioned exercises, replacing input shares with sales shares and suppliers with customers as an alternative measure of firms' payoff relevance. The results remain similar to those in our baseline specification, and are presented in Appendix B.4.

3.2.2 Sectoral Level Evidence

We next present sector-level evidence to evaluate the model's predictions regarding attention allocation in the general equilibrium setting. Proposition 2.6 implies that, all else equal, firms in sector i allocate more attention to: (i) sectors with more volatile productivity shocks, and (ii) sectors whose shocks exert a larger general equilibrium influence on the price of sector i's output.

To empirically test these predictions, we construct four proxies for the general equilibrium exposure term ϕ_{ij} in Proposition 2.6, using two complementary approaches. The first two are heuristic: sector i's input share from sector j and its sales share to sector j.[22] The remaining two proxies are derived from our calibrated model in Section 5: the equilibrium price effect ϕ_{ij} specified in equation (2.7) and its frictionless benchmark obtained by setting $\boldsymbol{\mu} = \mathbf{I}$.

With these proxies in hand, we aggregate firm-level bilateral browsing volumes to construct annual industry-to-industry browsing flows across 71 sectors.[23] We then estimate the following regression:

$$s_{ijt} = \alpha_{it} + \beta_1 \log \sigma_{jt} + \beta_2 \log \widehat{\phi}_{ijt} + X_{jt} + \varepsilon_{ijt}, \tag{3.5}$$

where s_{ijt} denotes the log share of sector i's browsing directed toward sector j in year t, and $\log \sigma_{jt}$ is the log standard deviation of sector j's TFP growth.[24] The variable $\log \widehat{\phi}_{ijt}$ denotes one of the proxies for the GE exposure term. The specification includes browser-year fixed effects α_{it}, which absorb time-varying, sector-specific factors that may influence browsing behavior. We additionally control for a set of time-varying characteristics of the browsee sector j, such as total revenue, value added, and the share of intermediate inputs.

Table 4 presents the estimation results. In line with the model's predictions, both β_1 and β_2 are positive and statistically significant across all specifications. All four proxies for general equilibrium exposure yield economically meaningful coefficients, with the model-implied proxy producing the largest estimate. The final row of the table quantifies the effect of volatility: firms in industry i allocate 0.30–0.43 percent more attention to industry j when industry j's TFP growth volatility is 1 percent

[22] The input share is defined as the ratio of sector i's expenditures on goods from sector j to its total intermediate input purchases. The sales share is the ratio of revenues from sector j to total sales.

[23] Appendix Table B.8 shows that the results are robust to using a 405-sector IO table.

[24] Appendix Table B.7 confirms that the results continue to hold when replacing TFP growth volatility with sales or inflation volatility.

Table 4: Testing Attention Allocation (Industry level)

	Attention Allocation			
	(1)	(2)	(3)	(4)
log(input share)	0.17***			
	(0.012)			
log(sales share)		0.24***		
		(0.018)		
$\log\widehat{\phi}_{ij}$ ($\boldsymbol{\mu} = \mathbf{I}$)			3.05***	
			(0.49)	
$\log\widehat{\phi}_{ij}$ ($\boldsymbol{\mu}$ calibrated)				4.13***
				(0.75)
TFP growth volatility	0.40***	0.43***	0.30***	0.30***
	(0.033)	(0.032)	(0.030)	(0.031)
Controls	✓	✓	✓	✓
Browser-Year FE	✓	✓	✓	✓
Adjusted R^2	0.46	0.47	0.52	0.52
No. Observations	25257	25254	25280	25280

Note: This table shows regression results on firms' attention allocation in GE predicted by our model. The control variables include average firm size, age, leverage, cash holdings, return on assets(ROA), fixed costs, stock returns, price cost margin, R&D intensity, Domar weight, total value added, total inputs, and expenditure share of intermediate inputs. Robust standard errors are in parentheses and clustered by industry. *Significance*: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

higher than that of industry k, holding general equilibrium exposure constant ($\phi_{ijt} = \phi_{ikt}$).

Additional evidence using text-based attention measure. We further validate our empirical findings using a text-based measure of attention, constructed with Natural Language Processing (NLP) techniques. Specifically, we compile an industry-specific dictionary of distinctive terms for each 3-digit NAICS sector. For each firm k in industry i and year t, we count the number of times its 10-K filing references industry j, denoted $a_{ijt}(k)$, based on our dictionary. Aggregating across firms yields an annual sequence of industry-to-industry attention matrix: $a_{ijt} = \sum_{k \in i} a_{ijt}(k)$.[25] Additional details on the construction of this measure are provided in Appendix A.2.

We re-estimate our main specification (3.5) using the text-based attention data and find that the results continue to hold. Appendix Table B.6 confirms that the text-based results align closely—both qualitatively and quantitatively—with those obtained using the browsing-based measure.

3.3 Information Capacity and Uncertainty

The key distinction between information acquisition technologies with elastic information capacity and fixed (or exogenous) information capacity lies in how total information acquired responds to the underlying stochastic environment. Under elastic information capacity, total information acquisition

[25] The correlation between the text-based and the browsing-based attention measure is 0.53, with standard error of 0.005. This is computed by calculating the correlation between the two measures within each browser-year cell and then averaging across all such cells.

increases with firms' volatility of marginal costs (Proposition 2.4); while under fixed capacity, it does not (Corollary 2.1). In this section, we test these predictions by examining whether firms expand their information capacity in response to rising economic uncertainty.

To do so, we first construct the growth rate of marginal cost mc_{it} for firms in sector i,

$$\Delta \text{mc}_{it} = -\Delta z_{it} + \alpha_i \Delta w_t + \sum_j a_{ij} \Delta p_{jt},$$

where a_{ij} denotes the (i, j) entry of the input-output matrix A, and α_i is sector i's labor share. The matrix A is derived from the 71-sector annual input–output (IO) table. The sectoral quarterly inflation series Δp_{it} and wage growth Δw_t are constructed using producer price index (PPI) data and wage of private industry workers from the Bureau of Labor Statistics (BLS), while TFP growth Δz_{it} is computed at a quarterly frequency using firm-level Compustat data.

Table 5: Total Browsing and Variance of Marginal Cost

	Total Browsing			
	(1)	(2)	(3)	(4)
$\mathbb{V}(\Delta mc_{it})$	2.27***	2.84***	2.83**	0.26**
	(0.35)	(0.35)	(1.24)	(0.11)
Sector Controls		✓	✓	✓
Time FE			✓	✓
Sector FE				✓
Adjusted R^2	0.00	0.68	0.69	0.94
No. Observations	5654	5654	5654	5654

Note: This table shows how the total browsing of firms in a given sector vary with the volatility of their marginal cost. The control variables include average firm size, age, leverage, cash holdings, return on assets(ROA), fixed costs, stock returns, price cost margin, R&D intensity, Domar weight, total value added, total inputs, and expenditure share of intermediate inputs. Standard errors are in parentheses and are clustered at industry level. *Significance*: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

We then compute the variance of marginal cost growth over a rolling six-quarter window, denoted as $\mathbb{V}(\Delta \text{mc}_{it})$. To assess the relationship between information acquisition and marginal cost uncertainty, we estimate the following regression:

$$b_{it} = \lambda_i + \beta \mathbb{V}(\Delta \text{mc}_{it}) + X_{it} + \gamma_t + \varepsilon_{it},$$

where b_{it} is the log browsing volume of sector i in quarter t, λ_i are sector fixed effects, γ_t are time fixed effect, and X_{it} is a vector of time-varying sector-level controls.

The results in Table 5 support the predictions of the elastic attention framework. Firms increase their information acquisition—measured by browsing intensity—when marginal costs become more volatile. This finding is consistent with the prediction labeled "F6: Uncertainty increases responsive-

ness" in Maćkowiak et al. (2023) and, to our knowledge, provides the first direct empirical evidence in support of it. However, this does not imply that models with fixed information capacity (Maćkowiak and Wiederholt, 2009; Van Nieuwerburgh and Veldkamp, 2010) are invalid. When the objective is to analyze relative attention allocation across states, both elastic and fixed-capacity models yield similar predictions. The two approaches diverge only when the total quantity of information becomes an object of interest. Models with exogenous information capacity (La'O and Tahbaz-Salehi, 2022; Bui et al., 2024) can replicate similar cross-sectional patterns, but only if the informational frictions are parameterized to mirror the behavior implied by rational inattention.

3.4 Browsing Intensity and Forecast Accuracy

If browsing intensity serves as a valid proxy for a firm's information acquisition, then—holding other factors constant—greater browsing activity should increase the firm's information capacity and reduce forecast errors. We conclude this section by examining the relationship between firms' browsing intensity and the accuracy of their forecasts.

Consider a firm ι in sector j that forms a forecast a variable $x_{\iota T}$, whose actual value is realized on day T. Let $\widehat{x}_{\iota t}$ denote the forecast made on day $t < T$. The forecast error is defined as $y_{\iota t} = (\widehat{x}_{\iota t} - x_{\iota T})/x_{\iota T}$, and the forecast horizon is given by $T - t$. We estimate the following baseline regression,

$$y_{\iota t} = \alpha_\iota + \beta \sum_{\ell=t-n}^{t} \text{browsing}_{\iota \ell} + X_{\iota t} + \eta_{jt} + \varepsilon_{\iota t}.$$

where the key independent variable $\sum_{\ell=t-n}^{t} \text{browsing}_{i\ell}$ measures the log of total browsing over the n days preceding the forecast. In our baseline specification, we set $n = 90$, capturing approximately three months of total browsing volume. The regression includes firm fixed effects α_ι, sector-by-time fixed effects η_{jt}, and a vector of time-varying firm-level covariates $X_{\iota t}$.

Table 6 presents our baseline results. The dependent variable in columns (1)–(4) is the firm's forecast error for earning per share (EPS). Column (1) shows that a 50% increase in total browsing intensity is associated with a 2.2 percentage point reduction in forecast error. This effect is sizable relative to the median forecast error of 7.8 percent.

To address concerns that aggregate or sector-specific shocks—such as changes in uncertainty—may jointly affect forecast accuracy and browsing behavior, column (2) adds sector-by-time fixed effects. We also control for the forecast horizon and a broad set of firm-level time-varying covariates.[26] We see little changes in the estimate. Column (3) adds firm fixed effects to additionally control for unobserved time-invariant firm heterogeneity. The point estimate continues to remain economically and statistically significant. Finally, to further address the possibility that firm-level uncertainty drives both browsing and forecast accuracy, we include the standard deviation of firm-level sales and EPS

[26] The variables for firm controls include firm size, age, leverage, cash holdings, return on assets (ROA), fixed costs, stock returns, price cost margin and R&D intensity.

Table 6: Browsing Intensity and Forecast Accuracy

	Forecast Error							
	Earnings Per Share				Sales			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Browsing Intensity	-0.043***	-0.046***	-0.018***	-0.014***	-0.0020***	-0.0011**	-0.0024***	-0.0024***
	(0.0059)	(0.0098)	(0.0013)	(0.0029)	(0.00055)	(0.00046)	(0.00014)	(0.00014)
Sales Volatility				-0.25***				0.13***
				(0.055)				(0.017)
EPS Volatility				0.013***				0.0023**
				(0.0016)				(0.00081)
Controls		✓	✓	✓		✓	✓	✓
Sector-Time FE		✓	✓	✓		✓	✓	✓
Firm FE			✓	✓			✓	✓
Adjusted R^2	0.01	0.05	0.23	0.23	0.01	0.05	0.23	0.23
No. Observations	7351	7068	5960	5894	7351	7068	5960	5894

Note: Standard errors are in parentheses and are clustered at industry level. This table shows how firms' forecast accuracy is associated with firms' browsing intensity in the past 90 days prior to making the forecast. The variables for firm controls include firm size, age, leverage, cash holdings, return on assets (ROA), fixed costs, stock returns, price cost margin and R&D intensity. *Significance*: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

as additional controls in column (5).[27] Our results continue to hold. Estimates of forecast errors on sales display similar results, as shown in columns (5)-(8), but with somewhat smaller magnitude.

While the estimates should be interpreted with appropriate caution, the results are suggestive of a causal relationship: firms tend to forecast more accurately when they browse other firms more intensively in the preceding 90 days.[28] Our preferred interpretation is that exogenous changes in the marginal cost of information acquisition prompt firms to increase their browsing activity, thereby expanding their information set and reducing forecast errors.

4. Application: Optimal Monetary Policy with Endogenous Attention

Having shown that the properties of browsing activities align with the predictions of our baseline model on attention allocation in networks, we now turn to exploring the policy implications. In particular, we highlight a novel endogenous attention channel in the design of monetary policy—one that stands in contrast to results obtained under assumptions of exogenous information or fixed information capacity.

[27] We calculate the standard deviation of sales and EPS in a quarter using the previous 6 quarters (current quarter included).

[28] In Appendix Table B.4, we demonstrate that our results are robust when days prior to the forecast are set to 30 or 180.

4.1 Setup

In Section 2, the wage function $w_t = \kappa z_t$ is taken as given. Now we extend the framework to include households and a monetary authority, in which the law of motion for the wage rate κ can be effectively chosen by the policymaker.

Households. There is a representative consumer who maximizes her utility under perfect information

$$\max_{C_t, L_t} \frac{C_t^{1-\gamma}}{1-\gamma} - \frac{L_t^{1+\frac{1}{\eta}}}{1+\frac{1}{\eta}},$$

subject to

$$P_t C_t \leq W_t L_t + \Pi_t + T_t, \quad \text{and} \quad P_t C_t \leq M_t.$$

In consumers' problem, γ determines the income effects and η corresponds to the Frisch elasticity. The total expenditure of consumers must be financed by the sum of the labor income, the total profit of the firms (Π_t), and the government transfer (T_t). In addition, consumers also need to respect the Cash-in-Advance (CIA) constraint: their total nominal expenditure cannot be larger than the exogenous money supply M_t. The final consumption goods C_t and the final goods price P_t are given by the following aggregators:

$$C_t = \prod_{i=1}^{N} C_{it}^{\beta_i}, \qquad P_t = \prod_{i=1}^{N} P_{it}^{\beta_i},$$

where β_i represents the expenditure share in goods from sector i.

Monetary policy. We assume that the monetary authority is not subject to informational frictions and can commit to the following monetary policy rule

$$m_t = \log M_t = \sum_{j=1}^{N} \psi_j z_{jt}.$$

After prices have been set and all shocks have realized, the nominal wage adjusts one-to-one with the money supply m_t to satisfy the cash-in-advance (CIA) constraint. Therefore, by varying the responsiveness to sectoral shocks, the policymaker can influence the dynamics of sectoral marginal costs, which in turn affects firms' endogenous price responses in equilibrium.

Policymaker's optimization problem. When informational frictions are absent, the economy is at its efficient level. In this case, the potential output c_t* is only a function of the underlying productivity shocks, satisfying the Hulton's theorem

$$c_t^* = \frac{1+\eta}{\gamma+\eta} \sum_{i=1}^{N} \lambda_i z_{it},$$

where λ_i is the steady-state Domar weight of sector i.

The inefficiency in the model economy stems from informational frictions and firms' pricing errors. To see this point more clearly, denote the average pricing error in sector i, relative to the perfect-information benchmark, as $e_{it} \equiv p_{it} - p_{it}^*$. The output gap between actual and potential output can then be expressed as

$$c_t - c_t^* = -\frac{\eta}{1+\gamma\eta} \sum_{i=1}^{N} \beta_i e_{it}, \tag{4.1}$$

which is a weighted average of sectoral price errors.

In addition to the aggregate output gap, social welfare is further reduced by inefficiencies arising from both within-sector and cross-sector price dispersion. Building on La'O and Tahbaz-Salehi (2022), the following lemma provides the decomposition.

Lemma 4.1. *The second-order approximation to the expected welfare loss is*

$$\mathcal{W} = \frac{1}{2}\left[\left(\gamma + \frac{1}{\eta}\right)\mathbb{V}(c_t - c_t^*) + \sum_{i=0}^{N} \lambda_i \mathcal{C}_i + \sum_{i=1}^{N} \lambda_i \theta_i \mathcal{D}_i\right] \tag{4.2}$$

where the cross-sector price error dispersion is

$$\mathcal{C}_i = \mathbb{E}\left[\sum_{j=1}^{N} a_{ij} e_{jt}^2 - \left(\sum_{j=1}^{N} a_{ij} e_{jt}\right)^2\right] \qquad \mathcal{C}_0 = \mathbb{E}\left[\sum_{j=1}^{N} \beta_j e_{jt}^2 - \left(\sum_{j=1}^{N} \beta_j e_{jt}\right)^2\right],$$

and the within-sector price dispersion $\mathcal{D}_i$ is

$$\mathcal{D}_i = \mathbb{E}\left[\int_0^1 (p_{i,\iota,t} - p_{it})^2 d\iota\right] = \mu_i^2 \nu_i^2.$$

The first component of welfare loss corresponds to the variance of the output gap. This component depends only on a weighted average of sectoral pricing errors, as captured by condition (4.1). Notably, eliminating output gap volatility requires only that the average price error remains constant, i.e., $\sum_j \beta_j e_{jt} = 0$, an insight developed in Rubbo (2023).

The second component relates to the cross-sector dispersion of price errors. Even if aggregate output gap volatility is minimized, sector-specific pricing errors may persist. These errors induce inefficient relative movements across sectors, which are suboptimal from the planner's perspective.

The third component reflects within-sector price dispersion. Even when average sectoral prices align with those in a frictionless benchmark, idiosyncratic noise in firms' signals still generates inefficient within-sector price variation. The magnitude of this dispersion depends on both the variance of idiosyncratic noise, ν_i^2, and the responsiveness of firms to their signals, μ_i. In our framework, these two factors are endogenously linked through condition (2.6).

The monetary authority aims to minimize welfare loss by implementing an optimal monetary policy rule. From the firms' perspective, their pricing errors—and thus the welfare loss—depend solely on the wage function, $w_t = \boldsymbol{\kappa} \boldsymbol{z}_t$. Moreover, for any given wage function, there exists a corresponding monetary policy rule that can support it in equilibrium.[29] Therefore, the monetary authority can directly choose the wage coefficient vector $\boldsymbol{\kappa}$ and derive the appropriate monetary policy rule subsequently. This is the approach we adopt when solving for the optimal policy. Ultimately, it suffices to consider the following policy design problem.

Lemma 4.2. *The optimal policy solves the following problem*

$$\min_{\boldsymbol{\kappa}} \mathcal{W} = \frac{1}{2}\left[\left(\gamma + \frac{1}{\eta}\right)\boldsymbol{\beta}'\boldsymbol{\Sigma}_e\boldsymbol{\beta} + \boldsymbol{\lambda}' \operatorname{diag}(\boldsymbol{\Sigma}_e) - \boldsymbol{\lambda}' \operatorname{diag}\left(\mathbf{A}\boldsymbol{\Sigma}_e\mathbf{A}'\right) - \boldsymbol{\beta}'\boldsymbol{\Sigma}_e\boldsymbol{\beta} + \boldsymbol{\chi}' \operatorname{diag}\left(\boldsymbol{\mu}\right)\right],$$

where $\boldsymbol{\Sigma}_e$ is the covariance matrix of the cross-sectional sectoral price errors $\boldsymbol{e}_t = \left[e_{1t}, \ldots, e_{Nt}\right]'$

$$\boldsymbol{e}_t = (\mathbf{I} - \boldsymbol{\mu}\mathbf{A})^{-1}(\mathbf{I} - \boldsymbol{\mu})\left((\mathbf{I} - \mathbf{A})^{-1} - \mathbf{1}\boldsymbol{\kappa}\right)\boldsymbol{z}_t,$$

and $\boldsymbol{\mu}$ saitsifies the fixed-point problem in Proposition 2.4.

This optimal policy problem in our model incorporates two novel elements relative to the existing literature. First, price flexibilities $\{\mu_i\}$ are endogenous to the general equilibrium and depend on the choice of monetary policy. As a result, the policymaker gains an additional policy lever: the ability to influence equilibrium attention allocation and, consequently, price flexibilities. This opens up a channel for managing expectations—one that is absent in models with exogenous information frictions, where price rigidities are fixed and unaffected by policy choices.

Second, expectation management interacts with input-output linkages. Attention choices across sectors are interdependent due to the structure of marginal cost processes. The policymaker thus faces a non-trivial task of designing sector-specific price flexibilities while accounting for the relationship between the distribution of attention and the underlying production network.

4.2 Optimal Policy with Expectation Management

Conceptually, with elastic attention, variations in the policy instrument $\boldsymbol{\kappa}$ influence the social welfare via two channels: the first is the exogenous channel when holding firms' attentions/flexibility $\boldsymbol{\mu}$ constant, and the second is the endogenous attention channel allowing responses of $\boldsymbol{\mu}$ to policy changes. The first-order condition with respect to $\boldsymbol{\kappa}$ naturally yields this decomposition

$$\frac{d\mathcal{W}}{d\kappa_i} = \underbrace{\frac{\partial \mathcal{W}}{\partial \kappa_i}}_{\text{exogenous chanel}} + \underbrace{\sum_{j=1}^{N} \frac{\partial \mathcal{W}}{\partial \mu_j}\frac{d\mu_j}{d\kappa_i}}_{\text{attention channel}} = 0. \tag{4.3}$$

[29] See Appendix E.4 for more details.

With exogenous information, fixed information capacity, or Calvo type pricing frictions, only the exogenous channel $\frac{\partial \mathcal{W}}{\partial \kappa_i}$ is at work as the implied nominal rigidity is invariant to policy changes or general equilibrium. What is unique in our setting is that the choice of μ_i in each sector responds to both the monetary policy $\boldsymbol{\kappa}$ and to other sectors' attention choices $\{\mu_j\}$ via general equilibrium forces. In what follows, we will spell out the details of the attention response $\frac{d\mu_i}{d\kappa_j}$, the welfare exposure $\frac{\partial \mathcal{W}}{\partial \mu_i}$, and finally the formula of the optional monetary policy.

Exogenous channel. As a benchmark, consider first the case where only the exogenous channel is at work. In fact, due to the equivalence result provided in Proposition 2.7, setting $\frac{\partial \mathcal{W}}{\partial \kappa_i} = 0$ leads to the same optimal monetary policy as that in an environment with exogenous information or fixed information capacity. The following proposition provides the characterization of such optimal policy.

Proposition 4.1. *With fixed information capacity or exogenous information, the optimal monetary policy is identical and can be implemented via a price stabilization policy*

$$\sum_{i=1}^{N} \varphi_i^x p_{it} = 0, \tag{4.4}$$

where the sectoral weights are given by

$$\varphi_i^x = \Bigg[\underbrace{\frac{(1-\rho_0)}{(\gamma+1/\eta)}\lambda_i}_{\text{output gap}} + \underbrace{\sum_{j=1}^{N} \left(1-\mu_i\right)\lambda_j \rho_j (\mathbf{I}-\mathbf{A})^{-1}_{j,i} + \left(\rho_0 - \rho_i\right)\lambda_i}_{\text{cross-sector dispersion}} + \underbrace{\mu_i \lambda_i \theta_i \rho_i}_{\text{within-sector dispersion}} \Bigg] \left(\frac{1}{\mu_i} - 1\right), \tag{4.5}$$

and ρ_i for $i = 1, \ldots, N$ and ρ_0 are given by

$$\rho_i = \mathbf{e}_i (\mathbf{I} - \mathbf{A}\boldsymbol{\mu})^{-1} \boldsymbol{\alpha}, \quad \rho_0 = \sum_{i=1}^{N} \beta_i \mu_i \rho_i.$$

This optimal policy rule is identical to that in La'O and Tahbaz-Salehi (2022) with exogenous signals, and it turns out that the same rule also applies to the economy with fixed information capacity. While the signal structures in these two economies are substantially different, they are nevertheless equivalent in the eyes of the policymaker. The three terms in (4.5) addresses the three different inefficiencies in the policymaker's objective function (4.2), respectively.

The weight on each sector not only depends on the Domar weight, but also depends on its interaction between sectoral rigidities captured by ρ_i. In La'O and Tahbaz-Salehi (2022), ρ_i is interpreted as the upstream price flexibility of sector i, and this measure will also show up in the endogenous component of the policy rule when allowing the attention channel.

Attention response to policy change. Importantly, when deriving the formula (4.5), sectoral attentions or nominal rigidities $\{\mu_i\}$ are taken as exogenously determined. Now we allow them to be chosen

by firms. Recall from Proposition 2.4 and Proposition 2.5, firms' attention choices are characterized by a fixed-point problem, and their best response $\mathcal{T}(\boldsymbol{\mu}, \boldsymbol{\kappa})$ hinges on all other firms' decisions and the wage function. A change in the wage function directly translates into changes in the marginal cost processes in different sectors, which induce changes in firms' attention choices. Due to network linkages, firms' attention choices must be further adjusted in response to changes in the attention choices of other firms. The following proposition formalizes these considerations.

Proposition 4.2. *The impact of changes in monetary policy $\boldsymbol{\kappa}$ on equilibrium attention $\{\mu_i\}$ are given by an $N \times N$ matrix*

$$\left[\frac{d\mu_i}{d\kappa_j}\right]_{i,j} = \left[\mathbf{I} - \mathcal{T}_{\boldsymbol{\mu}}\right]^{-1} \mathcal{T}_{\boldsymbol{\kappa}}, \tag{4.6}$$

where the $(i,j)-$th element of $\mathcal{T}_{\boldsymbol{\kappa}}$ is given by

$$\frac{\partial \mathcal{T}_i(\boldsymbol{\mu}, \boldsymbol{\kappa})}{\partial \kappa_j} = \frac{1-\mu_i}{\mathbb{V}(mc_{it})} \mathrm{Cov}(mc_{it}, z_{jt}) \left((\mathbf{I} - \mathbf{A}\boldsymbol{\mu})^{-1} \boldsymbol{\alpha} \mathbf{1}'\right)_{ij}, \tag{4.7}$$

and the $(i,j)-$th element of $\mathcal{T}_{\boldsymbol{\mu}}$, $\frac{\partial \mathcal{T}_i(\boldsymbol{\mu}, \boldsymbol{\kappa})}{\partial \mu_j}$, is taken from Proposition 2.5:

$$\frac{\partial \mathcal{T}_i(\boldsymbol{\mu}, \boldsymbol{\kappa})}{\partial \mu_j} = \frac{1-\mu_i}{\mathbb{V}(mc_{it})} \mathrm{Cov}(mc_{it}, mc_{jt}) \left((\mathbf{I} - \mathbf{A}\boldsymbol{\mu})^{-1} \mathbf{A}\right)_{ij}. \tag{4.8}$$

In Proposition 4.2, the (i,j) elements of matrices $\mathcal{T}_k$ and $\mathcal{T}_{\boldsymbol{\mu}}$ measure the marginal effects of changing policy instrument κ_j and sector j's attention μ_j on an individual firm's attention in sector i. The matrix $\left[\mathbf{I} - \mathcal{T}_{\boldsymbol{\mu}}\right]^{-1} \mathcal{T}_{\boldsymbol{\kappa}}$ then encodes all the general equilibrium consideration, allowing firms to internalize that all other firms' attention choice will further adjust according to their attention best response functions.

In the same vein of Proposition 2.5, the formula (4.7) summarizes how a different wage response to z_{jt} affects firms' attention choice μ_i via altering the variance of the marginal cost. The first term, $\frac{1-\mu_i}{\mathbb{V}(\mathrm{mc}_{it})}$, captures the variability of sector i's attention. The second term, $\mathrm{Cov}(\mathrm{mc}_{it}, z_{jt})$, corresponds to the exposure of the marginal cost to the sectoral productivity shock. The last term, $\rho_i = \mathbf{e}_i(\mathbf{I} - \mathbf{A}\boldsymbol{\mu})^{-1}\boldsymbol{\alpha}$, measures the importance of such wage change from a network perspective. Note that even holding other firms $\boldsymbol{\mu}$ unchanged, a different wage function still implies different price dynamics in different sectors, while ρ_i in turn summarizes the consideration of these changes for a firm in sector i.

So far, the discussion of conditions (4.7) remains at the partial equilibrium level, as it holds other firms' attention choices unchanged. In general equilibrium, once sector i's attention choice changes, it will induce endogenous response of all other sectors' attention choices captured by $\frac{\partial \mu_i}{\partial \mu_j}$, which in turn will lead to another round of attention modification, and so on. This infinite order of general equilibrium reasoning can be summarized by the matrix expansion of $\left[\mathbf{I} - \mathcal{T}_{\boldsymbol{\mu}}\right]^{-1} \mathcal{T}_{\boldsymbol{\kappa}}$

$$\left[\mathbf{I} - \mathcal{T}_{\boldsymbol{\mu}}\right]^{-1} \mathcal{T}_{\boldsymbol{\kappa}} = \mathcal{T}_{\boldsymbol{\kappa}} + \mathcal{T}_{\boldsymbol{\mu}}\mathcal{T}_{\boldsymbol{\kappa}} + (\mathcal{T}_{\boldsymbol{\mu}})^2\mathcal{T}_{\boldsymbol{\kappa}} + (\mathcal{T}_{\boldsymbol{\mu}})^3\mathcal{T}_{\boldsymbol{\kappa}} + \ldots. \tag{4.9}$$

Different from the logic of the standard Leontief inverse expansion, $(\mathbf{I}-\mathbf{A})^{-1}$, condition (4.9) incorporates two novel ingredients of the model: (i) the attention choices are endogenous to both shock volatility and government policy; (ii) the attention linkages hinges on production networks due to general equilibrium feedback and strategic complementarity in information acquisition. These features not only provide an endogenous account of observed heterogeneity in sectoral browsing activities, they also point to a novel feedback between endogenous rigidities and optimal policy, which we explore in the sequel.

Lastly, the importance of the endogenous expectation channel also depends on the welfare exposure to changes in sectoral attention, $\frac{\partial \mathcal{W}}{\partial \mu_i}$, as in equation (4.3). The following lemma characterizes such exposure.

Lemma 4.3. *Let the vector* $\boldsymbol{r}=\left[\frac{\partial \mathcal{W}}{\partial \mu_1},\ldots,\frac{\partial \mathcal{W}}{\partial \mu_N}\right]$ *denote the welfare exposure to sectoral attention changes, which satisfies*

$$\boldsymbol{r}=\left(\frac{1}{\gamma+1/\eta}-1\right)(\boldsymbol{\beta}'\mathbf{M})\odot\left(\boldsymbol{\beta}'(\mathbf{I}-\boldsymbol{\mu}\mathbf{A})^{-1}\right)+\boldsymbol{\lambda}'\left[\mathbf{M}\odot(\mathbf{I}-\boldsymbol{\mu}\mathbf{A})^{-1}-(\mathbf{A}\mathbf{M})\odot\left(\mathbf{A}(\mathbf{I}-\boldsymbol{\mu}\mathbf{A})^{-1}\right)\right]+\frac{1}{2}\boldsymbol{\chi}',$$

where $\mathbf{M}=(\mathbf{I}-\mathbf{A})^{-1}(\boldsymbol{\mu}-\mathbf{I})\boldsymbol{\Sigma}_{mc}$, *and* $\boldsymbol{\Sigma}_{mc}$ *represents the covariance matrix of sectoral marginal costs.*

Here, the vector $\boldsymbol{r}$ summarizes the impact of changes in μ on welfare through output gap volatility, cross-sector price error dispersion, and within-sector price error dispersion.

Optimal monetary policy. Now we are ready to characterize the optimal monetary policy rule. Because endogenous attention is the fixed point of the best response $\mathcal{T}(\boldsymbol{\mu},\boldsymbol{\kappa})$, the solution to the first-order condition (4.3) is highly nonlinear. Despite this difficulty, the optimal policy still admits a compact price stabilization representation.

Proposition 4.3. *The optimal monetary policy can be implemented by a price-stabilization policy of the form*

$$\sum_{i=1}^{N}\varphi_i p_{it}=0,\quad \textit{with}\quad \varphi_i=\varphi_i^x+\varphi_i^e, \tag{4.10}$$

where φ_i^e *represents the weight due to endogenous attention given by*

$$\varphi_i^e=\left\{2\left[\sum_{j=1}^{N}r_j[\mathbf{I}-\mathcal{T}_{\boldsymbol{\mu}}]_{ij}^{-1}\right]-(\lambda_i\theta_i\mathbb{V}(mc_{it})-\chi_i)\right\}\frac{\rho_i}{\mu_i}\frac{1-\mu_i}{\mathbb{V}(mc_{it})}, \tag{4.11}$$

and φ_i^x *given by (4.5) represents the weight when holding the attention constant.*

The endogenous component of the policy rule (4.11) underscores the importance of expectation management and the network structure. The sectoral weight φ_i^e reflects welfare exposures to price flexibility across all sectors (captured by $\boldsymbol{r}$ in Lemma 4.3), with their relevance scaled by the entries

of the matrix $[\mathbf{I} - \mathcal{T}\boldsymbol{\mu}]^{-1}$. This formulation shows that the policymaker internalizes the strategic complementarities in firms' attention allocation within the network. Moreover, the weight is inversely related to the volatility of sectoral marginal costs, $\mathbb{V}(\text{mc}_{it})$, because price flexibility is endogenously determined rather than fixed. Finally, the term $(\lambda_i \theta_i \mathbb{V}(\text{mc}_{it}) - \chi_i)$ adjusts for the gap between elastic and fixed information capacity.

What is hidden from the formula, however, is the fact that price flexibilities $\boldsymbol{\mu}$ are themselves endogenously determined under the optimal policy. Given a policy rule $\boldsymbol{\varphi}$, the corresponding nominal wage function can be expressed as

$$\boldsymbol{\kappa} \propto \boldsymbol{\varphi}(\mathbf{I} - \boldsymbol{\mu}\mathbf{A})^{-1}\boldsymbol{\mu}, \quad \text{subject to} \quad \boldsymbol{\kappa}\boldsymbol{\alpha} = 1. \tag{4.12}$$

Recall that in the best response function, $\mu_i = \mathcal{T}_i(\boldsymbol{\mu}, \boldsymbol{\kappa})$, a firm's attention allocation depends on the perceived wage function. Under optimal monetary policy, however, the wage function is itself determined by firms' attention allocations, as shown in (4.12). This feedback loop between policy rules and firms' active information acquisition decisions generates endogenous equilibrium price rigidities—an element that is absent in environments with exogenous or fixed information capacity. In what follows, we explore the properties of these implied price flexibilities.

4.3 Policy Induced Rigidity: Understanding the Mechanism

In this subsection, we examine the implications of optimal policy design for firms' attention allocation and the resulting nominal rigidity. We conduct three exercises, each focusing on minimizing a single component of the social loss function (4.2). We show that, compared to an economy with exogenous information or fixed information capacity, the optimal policy under elastic attention can actually induce greater nominal rigidity.

Minimizing output gap volatility. We start with the case in which the central bank only attempts to resolve welfare losses associated with the inefficient output-gap volatility. Surprisingly, the optimal policy rule in this case takes a particularly simple form as presented in the following proposition.

Proposition 4.4. *When minimizing only the output gap volatility, the optimal policy weight becomes*

$$\varphi_i = \frac{(1-\rho_0)}{(\gamma + 1/\eta)} \lambda_i \left(\frac{1}{\mu_i} - 1 \right), \tag{4.13}$$

conditional on the aggregate output gap being completely stabilized, $\mathbb{V}\left(c_t - c_t^*\right) = 0$.

Notice that when output-gap volatility is fully stabilized, $\mathbb{V}\left(c_t - c_t^*\right) = 0$, the optimal policy rules under exogenous information and elastic attention coincide.[30] Achieving output-gap stabilization requires eliminating the *aggregate* pricing error, i.e., ensuring $\sum_j \beta_j e_{jt} = 0$, which in turn implies that

[30] In our quantitative analysis, we find that the optimal output-gap (OG) policy indeed satisfies this condition under our model calibration.

the endogenous component of the optimal policy rule, φ_i^e, equals zero. This envelope-type result significantly simplifies the analysis.[31]

However, this result does not imply that the equilibrium allocation is the same under elastic attention as it is under alternative information structures. Under elastic attention, μ_i in condition (4.13) is endogenously determined, whereas under fixed information capacity or exogenous information, it is externally specified.

To illustrate, consider starting from the exogenous-information case, where the attention allocation $\{\mu_i\}$ is fixed. According to formula (4.13), it is optimal to place more weight on stabilizing sectors with smaller μ_i—that is, sectors with greater nominal rigidity. Consequently, the volatility of prices p_{it} in these more rigid sectors will tend to be lower. If sector i also relies heavily on its own output as an intermediate input (as is empirically common), then the volatility of its marginal cost mc_{it} will also be lower.

Now consider our baseline environment with elastic attention. Given the lower marginal cost volatility, sector i will optimally allocate less attention, resulting in an even smaller μ_i. This, in turn, leads the policymaker—via formula (4.13)—to place even greater weight on stabilizing sector i. This feedback loop between optimal policy design and firms' best-response attention allocation creates a reinforcing channel that endogenously amplifies the dispersion of nominal rigidities—a mechanism we further quantify in Section 5. In Appendix C, we present a simple example with symmetric input linkages but asymmetric shock volatilities to illustrate the feedback between optimal policy and endogenous price flexibilities.

Minimizing within-sector dispersion. The same logic also applies if the policymaker only tries to minimize the within-sector dispersion component in the welfare loss function. With this single objective, the sum of the endogenous and exogenous component in the price stabilizing rule also leads to a much simpler formula.

Proposition 4.5. *When minimizing only the within-sector dispersion, the optimal policy weight becomes*

$$\varphi_i = \left(\sum_{j=1}^{N} \chi_j \left[\mathbf{I} - \mathcal{T}_{\boldsymbol{\mu}} \right]^{-1}_{j,i} \right) \frac{\rho_i}{\mathbb{V}(mc_{it})} \left(\frac{1}{\mu_i} - 1 \right). \tag{4.14}$$

Fixing $\mathcal{T}_{\boldsymbol{\mu}}$, the information acquisition complementarity matrix, a lower μ_i or less volatile marginal cost translates into a larger weight in sector i. More stabilized price in sector i further reduces μ_i and $\mathbb{V}(\text{mc}_{it})$ through endogenous attention choice. As a result, the policy-attention feedback points to the same implication on the implied nominal rigidities.[32]

Also recall from Proposition 4.1, with exogenous information or fixed information capacity, the policy weight is $\varphi_i = \lambda_i \theta_i \rho_i (1 - \mu_i)$. This policy rule also implies a higher weight on more rigid

[31] Note, however, that this result no longer holds when the planner is simultaneously targeting cross-sector or within-sector pricing error dispersion.

[32] In our quantitative exercise in Section 5, we verify this intuition in our calibrated model.

sectors. Differently, there is no further modification of μ_i resulting from a higher φ_i.

Minimizing cross-sector dispersion. Lastly, if the policymaker only aims to eliminate the cross-sector dispersion, the only way to do so is to encourage all firms to pay full attention, ($\mu_i = 1$), so that the sectoral pricing errors vanish. One way to achieve this goal is to set the monetary policy in a way such that the wage function exhibits extremely large response to all shocks, which results in extremely large volatility of sectoral marginal costs. Consequently, there is no longer an interior solution to the optimal policy problem, and the price stabilization rule fails to hold.

It is useful to note that though the cross-sector dispersion term alone will lead to a corner solution, simultaneously considering all the three terms in objective (4.2) permits an interior solution. As we will show in the next section, the first two terms are dominating forces quantitatively in shaping the optimal monetary policy, and most intuition can be obtained by inspecting the two special cases characterized by Proposition 4.4 and 4.5.

5. Quantification

In this section, we provide a quantitative account of the theory. We show that the calibrated model is capable of replicating salient patterns of attention allocation as in firms' browsing activities. The model also predicts a positive correlation between sectoral price flexibility and sectoral shock volatility, which helps distinguish alternative information structures. As anticipated in the previous subsection, we compare the optimal policy rule with that under exogenous information and quantify the effects of policy-induced dispersion of endogenous price rigidities.

5.1 Calibration

The model is calibrated at a quarterly frequency. The parameters calibrated externally are based on conventional values in the literature. The elasticity of intertemporal substitution γ is set to 1. The Frisch elasticity η is calibrated to be 2. To achieve an average markup of 20%, the elasticity of substitution between goods within a sector θ_i is chosen to be 6 for all i. There are also a set of parameters that can be obtained directly from data. The elements in the production network matrix a_{ij} and the final goods shares β_i are computed based on the input-output table from BEA. The volatilities of sector-level productivity shocks are calibrated using the BEA/BLS Integrated Industry-Level Production Account (ILPA).

The calibration of the information acquisition costs $\{\chi_i\}$ is more unique in our setting and is calibrated internally. The choice of this parameter set involves two challenges. First, most variables in the model, such as price flexibilities, are endogenous to policy. Therefore, internal calibration requires us to first take a stance on the monetary policy rule when bringing the model to the data. As a baseline,

Table 7: Calibrated Parameters

Param.	Value	Source	Related to
Exogenously determined parameters			
γ	1	—	income elasticity
η	2	—	Frisch elasticity
θ	6	20% markup	elasticity of substitution
a_{ij}		BEA	input-output matrix
σ_i		BEA	productivity shock volatility
Endogenously determined parameters			
δ_0	0.30	sectoral forecast error of EPS	information acquisition cost
δ_1	0.71	sectoral forecast error of EPS	information acquisition cost
δ_2	1.60	sectoral forecast error of EPS	information acquisition cost

we assume that the current monetary authority employs a CPI price stabilization index, that is,

$$\sum_i \varphi_i p_{it} = 0, \quad \text{with} \quad \varphi_i = \frac{1}{\beta_i}.$$

Note that, unlike the optimal policy, the CPI stabilization rule, $\varphi_i = \frac{1}{\beta_i}$, contains no endogenous variables, thereby excluding any interaction between the policy rule and the endogenous attention choice.

Second, a large number of parameters must be determined, as χ_i varies at the sector level. Motivated by condition (2.8), we allow χ_i to depend on λ_i and σ_i in a flexible way and impose the following parsimonious functional form:

$$\log \chi_i = \delta_0 + \delta_1 \log \lambda_i + \delta_2 \log \sigma_i.$$

We calibrate the parameters δ_0, δ_1 and δ_2 to match the distribution of *adjusted* forecast errors of earnings per share (EPS) at the sector level, which are directly related to the informational frictions. Particularly, we target the mean level, 25 percetile, and 75 perticle of the forecast error in the sectoral distribution.[33] We choose EPS as our target because it provides a significantly larger number of observations compared to other forecasted variables in the IBES, allowing us to aggregate the forecast error at the sector level with small standard errors. Table 7 lists the calibrated parameters.

[33]Specifically, we choose δ_0, δ_1 and δ_2 to minimize the following loss function,

$$\min_{\delta_0,\delta_1,\delta_2} \left(\mathrm{FE}^{\text{data}}_{i=25\%} - \mathrm{FE}^{\text{model}}_{i=25\%}\right)^2 + \left(\sum_{i=1}^{N} \frac{1}{N} \mathrm{FE}^{\text{data}}_i - \sum_{i=1}^{N} \frac{1}{N} \mathrm{FE}^{\text{model}}_i\right)^2 + \left(\mathrm{FE}^{\text{data}}_{i=75\%} - \mathrm{FE}^{\text{model}}_{i=75\%}\right)^2,$$

where FE_i is the mean absolute forecast error of sector i. However, the volatility of EPS differs in the data and in the model. To ensure that the forecast error is comparable, we normalize it by the volatility of EPS, both in the data and in the model.

Table 8: Model Fit: Forecast Error and Price-Change Frequency

Moment	Forecast Error		Frequency	
	Data	Model	Data	Model
Moments targeted				
Mean	0.28	0.27		
25th percentile	0.20	0.22		
75 percentile	0.33	0.33		
Moments not targeted				
Mean			0.23	0.24
Standard deviation	0.12	0.08	0.13	0.11
10th percentile	0.13	0.18	0.12	0.12
25th percentile			0.14	0.16
50th percentile	0.26	0.27	0.19	0.23
75 percentile			0.29	0.30
90 percentile	0.48	0.36	0.36	0.39

As shown in Table 8, the model closely matches the targeted moments related to the cross-sector distribution of forecast errors. For the non-targeted moments, note that there is a one-to-one mapping between the price flexibilities $\{\mu_i\}$ in our model and the sector-level price-change frequencies, as in Mankiw and Reis (2002) and La'O and Tahbaz-Salehi (2022). The model also replicates the distribution of implied price-change frequencies reasonably well, generating the degree of sectoral price rigidities comparable to that observed in the data across different percentiles.

5.2 Model v.s. Data: Attention, Volatility, and Price Flexibility

In this subsection, we compare various aspects of the calibrated model predictions with their data counterparts and illustrate the role of endogenous information acquisition.

Attention allocation. As documented in Section 3, a salient feature of firms' browsing activities is that they are positively associated with both the input-output linkages and sectoral shock volatilites. To see to what extent our model can replicate the empirical regularity, we leverage equation 2.10, where the attention allocated to a sector j from sector i, ω_{ij}, is measured by the reduction of posterior uncertainty of sector i's productivity shock relative to its prior. Parallel to the empirical specification in equation (3.5), we run the following regression using model-generated data.

$$\omega_{ij} = \alpha_i + \beta_1 \log \phi_{ijt} + \beta_2 \text{ TFP_volatility}_j + \varepsilon_{ij}.$$

Unlike its empirical counterpart, the model is stationary, and the independent variables are time-invariant.

Table 9: Attention Allocation: Sectoral-Level Regression

Data		Model	
$\log(\phi_{ij})$	TFP volatility	$\log(\phi_{ij})$	TFP volatility
4.13***	0.30***	1.28***	0.99***
(0.75)	(0.03)	(0.05)	(0.03)

Table 9 displays the regression coefficients. The model yields a qualitatively similar pattern as that in the data (Table 4 column (4) in Section 3.2.2): the percentage reduction of uncertainty about a sector is positively related with the sectoral TFP volatility and bilateral trade linkages. This is much anticipated based on Proposition 2.6, which states that ω_{ij} is determined by the product of equilibrium exposure of sector i's price to sector j's shock and the shock volatility in sector j. This pattern underscores the importance of the endogeneity of firms' information acquisition choice. For example, with exogenous information such as La'O and Tahbaz-Salehi (2022), though firms in different sectors face different degrees of information frictions, the attention of a firm is allocated in a uniform way across sectors, that is, $\omega_{ij} = \omega_{ik}$.

Price flexibility and shock volatility. Another important implication of the endogenous information acquisition channel is that the implied price flexibilities are directly linked to the underlying shock volatilities. This feature is absent in models with Calvo-type frictions or exogenous information frictions, where price flexibilities are exogenously imposed and unrelated to fundamentals.

Figure 5: Frequency and Volatility

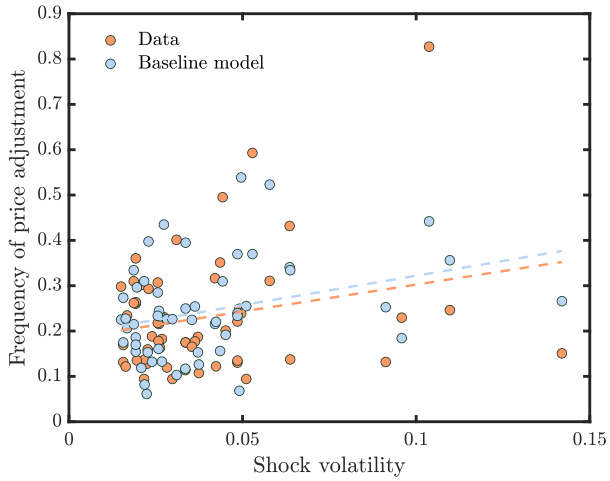


Figure 5 illustrates this relationship in both the data and our calibrated model. Each dot represents a sector's own productivity shock volatility and its frequency of price adjustment. The red dots show that, in the data, higher shock volatility is associated with greater price flexibility. The blue dots depict the same relationship in the model. It turns out that our model reproduces a similarly positive

slope. The intuition is straightforward: condition (2.6) implies that price flexibility increases with the volatility of marginal cost, and own productivity shocks play a quantitatively important role in driving marginal cost fluctuations.

5.3 Allocation under Optimal Monetary Policy

With the calibrated model, we now quantify the effects of endogenous information acquisition in shaping the optimal monetary policy and the policy-induced price flexibilities.

We start with the comparison of the monetary policy rule with elastic attention in our baseline model with that under exogenous information and fixed information capacity. To make these economies comparable with each other, we invoke Proposition 2.7 and set the maximum mutual information capacity $\{\delta_i\}$ and the variance of noise $\{\tau_i\}$ so that the implied price flexibilities $\{\mu_i\}$ are identical to that in our baseline calibration with elastic attention and CPI price stabilization. Due to Proposition 4.1, the optimal monetary policy rule under exogenous information coincides with that under fixed information capacity, and we will only refer to the exogenous information case when making comparison with our baseline model.

Figure 6 displays the two vectors of sectoral weights under the optimal price stabilization rule, both of which are normalized so that they sum to one. The beige bars represent the weights with exogenous information according to Proposition 4.1, where the informational friction is independent of the policy. The blue bars represent the weights in our baseline model with elastic attention according to Proposition 4.3, where the policy weights contains the additional component due to the additional expectation management motive. The two policy rules are significantly different, with an average percentage difference change equaling 24.3%, which suggests that the nature of the underlying informational frictions matters for the design of optimal policy rule.

The next question is: what factor drives the differences between these two policy rules? To answer this question, Panel (a) of Figure 7 plots the percentage difference of the optimal sectoral weights between the exogenous information economy and the elastic attention economies against the initial price flexibilities under CPI stabilization. Anticipated from the discussion in Section 4.3, there is a sharp pattern that the difference is amplified when the initial level of price flexibility is low, and the heterogeneity in the initial price flexibilities account for more than 85% of the policy weight difference. With exogenous information, the more rigid sectors receive a higher weight. With elastic attention, the policymaker will put an additional emphasis on these sectors due to the feedback effects between the policy and the endogenous flexibilities, which verifies the intuition developed in Proposition 4.4 and 4.5 in the full calibrated model.

Panel (b) of Figure 7 further highlights the aforementioned endogenous response of price flexibilities. The sectors that are relatively more rigid (flexible) under the CPI stabilization policy rule becomes even more so under the optimal monetary policy rule. The dispersion of price rigidities is therefore amplified. Again, this result is consistent with the theoretical findings in Section 4.3.

Figure 6: Optimal Monetary Policy Rule

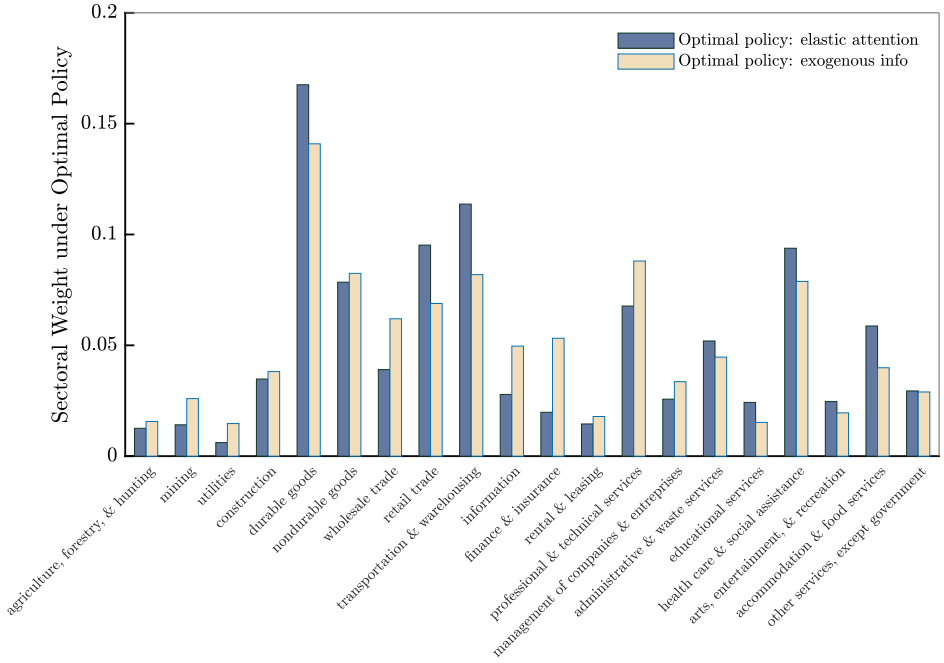
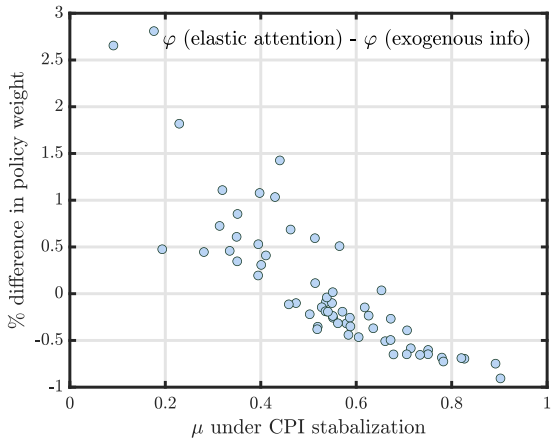
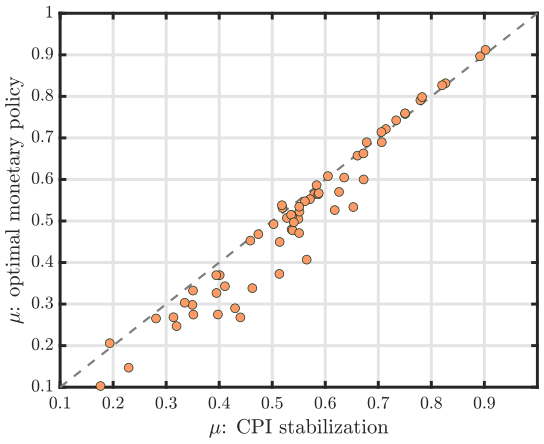


Figure 7: Optimal Policy Rule and Endogenous Price Rigidity

(a) Difference in policy rules



(b) Policy-induced price rigidity



6. Conclusion

This paper explores how firms acquire information within complex production networks. We develop a parsimonious theoretical framework that characterizes firms' information acquisition behavior with input-output linkages. Using a novel dataset on firms' browsing activities related to other firms' disclosed documents, we find that firms acquire more information about those that are more volatile, more payoff-relevant, and closer in network distance. Moreover, firms' total browsing activity increases with the volatility of their targeted prices. These empirical patterns are consistent with

our theoretical prediction under endogenous information capacity. Embedding this framework into a monetary model reveals an additional expectation management channel in the design of optimal monetary policy, giving rise to policy-induced nominal rigidities and cross-sectional dispersion.

Several promising research directions remain to be explored. First, the browsing data can be used to construct a firm-level attention network, enabling the study of shock transmission under informational frictions at a more granular level. We pursue this direction in a follow-up paper (Huo et al., 2025), where we show that the uneven attention received by firms can allow some salient firms to influence aggregate fluctuations, even if their size is relatively small. Second, our current analysis treats the network structure as exogenous. However, the evidence on browsing activity during relationship formation or dissolution in Section 3.2 suggests the need for a framework that jointly models network formation and information acquisition.

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Online Appendix

Inattentive Networks: Theory and Evidence from EDGAR

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A. Data Appendix

A.1 The EDGAR Browsing Data

A.1.1 Constructing the Browsing Data

The EDGAR log files contain the full viewing history of the filed documents published on EDGAR, including the viewers' IP address, the timestamp of the request, and the page requested. These log files were made available online by the SEC at https://www.sec.gov/data/edgar-log-file-data-set. The SEC provides the first three octets of the viewers' IP address and anonymizes the fourth octet with a 3-character string that preserves the last octet's uniqueness without revealing the IP's full identity.[34] Following the method proposed by Chen et al. (2020), we decipher the last octet and reveal the full IP address. We further map each viewer's uncovered IP address to their true identity using services provided by ip-info.io, a leading IP information provider. The final sample contains a total of 713,157,510 unique IP addresses of 7,622 public companies in our sample period, representing 52.4% of all browsings of disclosed files recorded in EDGAR. The summary statistics of the baseline sample are presented in Table 1 in the main text.

A.1.2 The Surge of EDGAR Usage

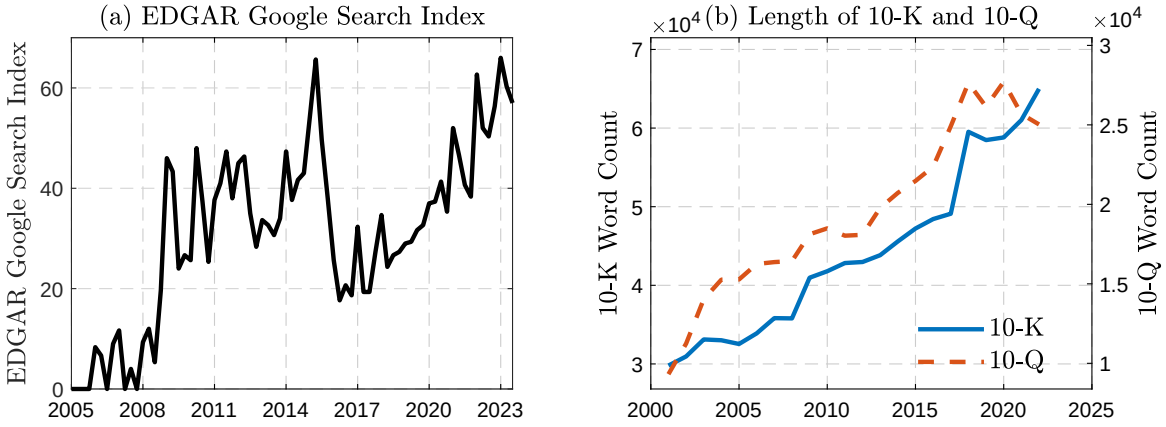
The year 2009 marks a noticeable milestone in terms of EDGAR adoption. Panel (a) of Figure A.1 shows the Google Trend for the keyword "SEC EDGAR search", directly linking to EDGAR's search page, which is the primary point of entry for human users. The number of searches surged a few months prior to 2009, followed by a stable level of searches thereafter.

Despite the general rule of new product adoption which predicts a Bass Diffusion Model growth curve (Rogers et al. (2017); Bass (1969)), the information content of disclosures may also contribute to the increasing usage of the EDGAR platform. The reason is twofold. First, due to the changing regulatory landscape, SEC requires the disclosure of more and more content over time.[35] Panel (b) of Figure A.1 illustrates the average lengths of 10-Ks and 10-Qs, two primary types of public disclosures, over time. It can be seen that the lengths are steadily increasing, suggesting that the information in disclosures gets richer over the years. Due to the benefits of reducing information asymmetry and the associated benefits, such as lowering the cost of capital(Francis et al. (2008)), mitigating the adverse selection problems (Milgrom (1981); Diamond and Verrecchia (1991)), and offsetting the negative effects of complex financial statements on the information environment (Guay et al. (2016)), firms also have incentives to disclose more information voluntarily.

[34] An octet is an integer in the range [0, 256). For example, an IP address provided by the SEC is "10.191.131.ace", where the fourth octet "ace" masks the last three digits of the IP address.

[35] For instance, SEC started mandating firms to disclose unresolved SEC staff comments (2005), the effectiveness of internal control over financial reporting (2007), risk factors (Item 1A, 2006), mining safety (Item 4, 2011), oil and gas reserves in possession of energy companies, hedging policies (2018), and payment to US federal and foreign governments by natural resource extractors (2020).

Figure A.1: The Surge of EDGAR Usage



Note: Panel (a) plots the Google Trend index for the keyword "SEC EDGAR search". Panel (b) displays the average lengths of 10Ks and 10Qs over time.

A.1.3 Most Viewed Filed Documents

Table A.1 lists the top ten types of documents filed on EDGAR in 2016 that received the highest browsing volume. These documents fall broadly into two categories: filings related to unscheduled events and periodic reports such as quarterly and annual disclosures. Both types provide timely and detailed firm-level information. The top three filings—Form 8-K (unscheduled material events), Form 10-Q (quarterly financial reports), and Form 10-K (annual financial reports)—account for a disproportionately large share of total browsing activity.

Table A.1: Most Viewed Filings

Form	Avg Browsing	Description
8-K	1,102,683,079	Unscheduled material events or corporate changes importance to the shareholders or to the SEC
10-Q	1,034,609,909	Quarterly financial report
10-K	778,512,251	Comprehensive overview of the registrant and the annual official presentation of the company's financials
424B2	272,580,533	Transaction specific information for a delayed shelf offering private issuers
497K	208,204,363	Prospectus for investment companies using Form N-1A.
6-K	207,299,363	The equivalent of the 8-K for certain foreign private issuers seasoned issuers
485	190,956,241	Annual update filed by investment companies
497	138,954,321	Prospectus for investment companies using Form N-1A
DEF14A	93,240,818	Annual meeting proxy
485APOS	66,608,633	Unscheduled material changes to the prospectus for companies

Note: This table lists the 10 forms filed on the EDGAR with the most browsings in 2016.

A.2 Constructing Text-Based Attention Measure

10-K filings. The SEC requires public companies to file Form 10-K annually. The 10-K filing is a detailed document that provides a comprehensive overview of a company's financial performance, operations, and risks, among other important information. 10-K goes beyond regular annual reports and provides a more in-depth backward-looking and forward-looking analysis of a company.

Methodology. We propose two measures of attention based on a text-based approach similar to that of Flynn and Sastry (2023) and Song et al. (2022). The first measure is firms' attention on other firms and the second measure is firms' attention on other industries. The idea is to choose a dictionary of "signal words" unique to each industry or firm and calculate the frequency counts in a company's 10-K disclosures. For example, the word "uranium" is most likely to be mentioned when a firm is discussing the "Mining (except Oil and Gas)" industry (NAICS code 212). If a company repeatedly mentions "uranium" in its corporate disclosures (high word frequency), it likely devotes greater managerial attention to NAICS 212.

We discuss in detail how we construct firms' attention measure on other industries, and similar procedure can be applied to obtain firms' attention on other firms. Based on the dictionary, for each firm k in industry i in year t, we count how many times its annual 10-K document mentions an industry j, denoted as $a_{ijt}(k)$. Aggregating the number of mentions within industry i, we obtain a measure of industry i's attention towards industry j, expressed as $a_{ijt} = \sum_{k \in i} a_{ijt}(k)$, and therefore construct an annual sequence of attention-allocation matrices across industries.

Specifically, we select the keyword list for each industry i (defined by three-digit NAICS codes) following the steps below:

(1) Prepare corpus texts pertaining to each industry from two sources. First, we extract company descriptions for all public US companies from Osiris, a Bureau van Dijk data product that contains comprehensive information on public and private companies across the globe. In Osiris, each company has a short description of its history and an overview of its business and products. By matching these descriptions with their industry codes, they jointly portray the essence of the industries that they belong to. The second corpus is the descriptive texts of NAICS codes, maintained by the NAICS Association, LLC, a company specializing in NAICS code services.

(2) Pre-processing. Two python packages, Spacy and NLTK (Natural Language Toolkit) are prerequisites. We execute two procedures before running the main program. First, we use NLTK's embedded stop word list to remove common words like "the", "an", and "at" that lack meaning. Second, we run entity_recognization.py to leverage Spacy's Named Entity Recognition (NER) algorithm and the large English language model en_core_web_lg to identify and exclude geographic and geopolitical entities. This prevents some geographically-concentrated industries from capturing words like "Texas" and "China".

(3) After pre-processing, we tokenize the texts using NLTK's pre-trained language model and select all nouns (NN), adjectives (JJ), and verbs (VB) - the three parts of speech that form sentences. We first

extract the 100 most common words across all industries (subjectively chosen but works well) and remove them, as they are unlikely to be unique to a specific industry.

(4) We employ the TF-IDF (Term Frequency-Inverse Document Frequency) algorithm, a method that is intended to reflect how relevant a word is to a document in a corpus (Qaiser and Ali, 2018), to select a dictionary of signal words Θ_j of each NAICS3 industry j. The intuition behind TF-IDF is that a term that appears frequently in a document but rarely in the rest of the corpus is more likely to be representative of the topic of the document. We vectorize the corpus documents, calculate the tf-idf score of every qualified word, and select the top 15 words with the highest tf-idf scores into the industry j's dictionary Θ_j.

(5) Using the dictionary generated in (4), We search for the word frequency of words in Θ_j for every j in firms' 10-K filings. We restrict the search to Item 1A (Risk Factors) and Item 7 (Management Discussion and Analysis), as companies are most likely to discuss an industry about their risk or strategic concerns in these two parts. Finally, we aggregate the total number of occurrences of each word in industry j's dictionary Θ_j, indexed as k. Firm i's attention to industry j in year t is calculated as:

$$\text{Attention}_{i,j,t} = \sum_{k \in \Theta_j} \text{WordFreq}_{i,k,t} \tag{A.1}$$

Similarly, firm i's attention to firm k in year t is given by

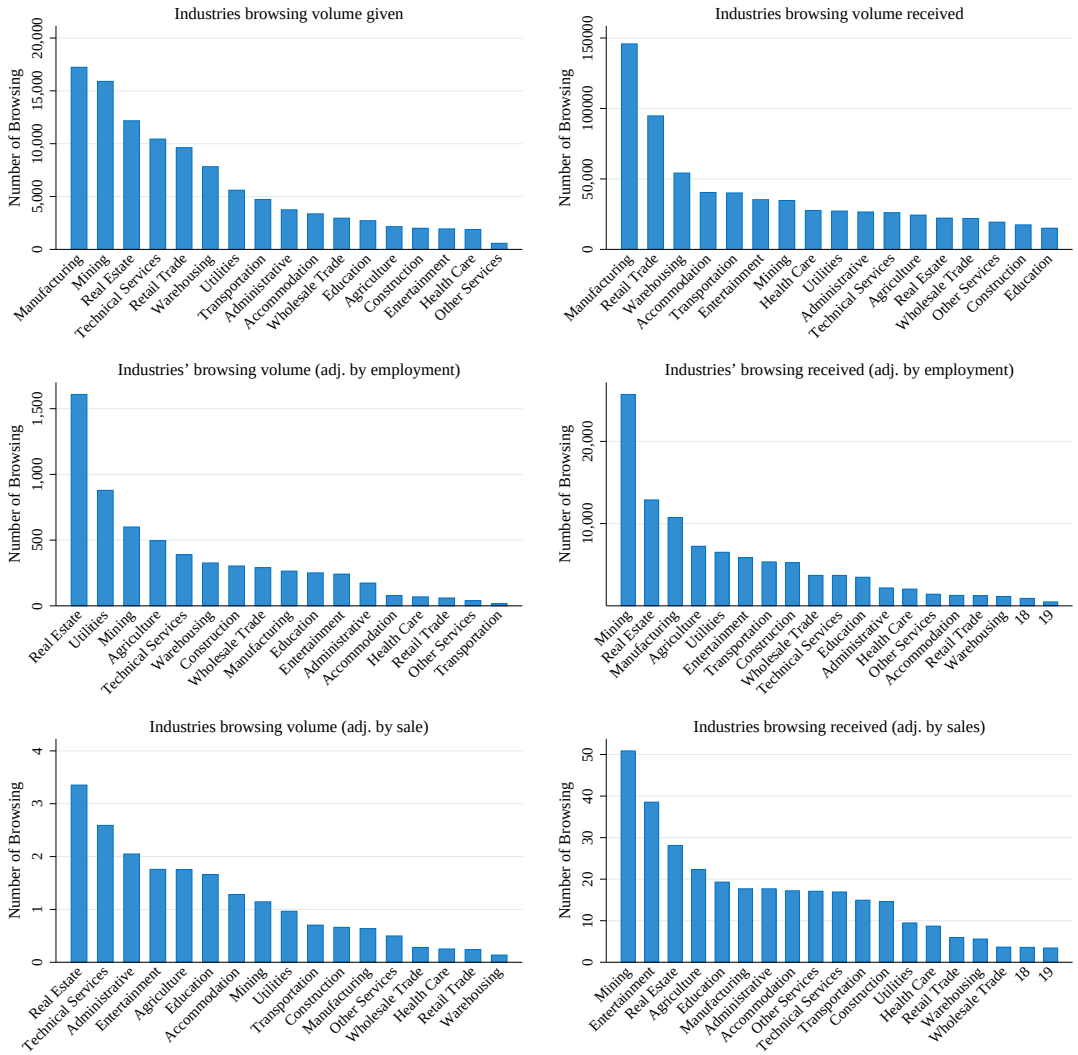
$$\text{Attention}_{i,k,t} = \sum_{\ell \in \Theta_k} \text{WordFreq}_{i,\ell,t}.$$

B. Additional Empirical Results

B.1 Additional Results on Browsing Activities

Distribution of browsing activities across industries. The first row in Figure B.1 shows the distribution of browsing activity at the industry level in 2016, where the left panel shows the browsings sent and the right panel shows the browsings received. To account for size bias, we adjust a firm's total browsing by its total employment or sales (second and third rows of Figure B.1). The browsings are distributed unequally across industries. The Manufacturing industry both send and receive the most browsings, while adjusted for employment and sales the real estate and mining industry ranks first, respectively.

Figure B.1: Distribution of Browsing Activity

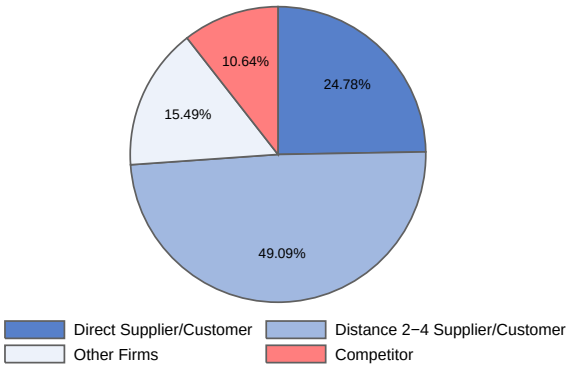


Employment, sales and browsing. Firms with larger size, measured by employment or sales, tend to send and receive more browsings. In particular, the correlation between log browsing received and log sales is 0.29 (standard error: 0.02), while the correlation between log browsing sent and log sales is 0.22 (standard error: 0.02). The correlation between log browsing received and log employment is 0.27 (standard error: 0.02), while the correlation between log browsing sent and log employment is 0.20 (standard error: 0.02).

Attention Allocated to Competitors, Suppliers, and Customers. Figure B.2 shows how firms allocate their browsing activity across three categories: direct suppliers or customers, competitors, and firms that are neither. To identify competitors, we follow the approach in Hoberg and Phillips (2016), which uses textual analysis of 10-K filings to measure product similarity across firms.

On average, a firm allocates 24.78% of its browsing activity to its direct suppliers and customers, and an additional 49.09% to firms that are distance-2, distance-3, or distance-4 in the supply chain. In contrast, only 12.6% of browsing is directed toward competitors. These estimates likely understate the true attention paid to suppliers and customers for two reasons. First, the FactSet dataset captures only a subset of actual supplier-customer relationships. Second, we classify firms as competitors even when they are also suppliers or customers. As such, the reported figures represent a lower bound on the attention allocated to firms with which a company has supply-chain links.

Figure B.2: Allocation of Browsing Activity



Note: This figure plots the allocation of a firm's browsing volume to the direct and indirect suppliers or customers, its competitors, and other firms in the Factset dataset in 2016.

B.2 Estimating the Hazard Function

At weekly frequency, we estimate hazard rates for each age of inattention. Let κ_{it} denote the current age of inattention for firm i in week t and I_{it} be a browsing indicator; $I_{it} = 1$ if $B_{it} > 0$ and $B_{i,t-1} = 0$. We estimate the hazard rate for age κ using the following expression,

$$h_\tau = \{\bar{I} | \kappa_{it} = \kappa\} = \frac{\sum_i \sum_t I_{it} I\{\kappa_{it} = \kappa\}}{\sum_i \sum_t I\{\kappa_{it} = \kappa\}}$$

To address the concern for survivor bias, we split the firms into deciles, $d = 1, 2, ..., 10$, based on their unconditional hazard rate. We then calculate the unconditional hazard rate for each decile, denoted as h^d_τ. We finally calculate the hazard rate for each age relative to the decile fixed effect,

$$\{\overline{I_{it}/h^d_\tau}|\kappa_{it} = \kappa\} \equiv \frac{\sum_i \sum_t [I_{it}/h^d_\tau] I\{\kappa_{it} = \kappa\}}{\sum_i \sum_t I\{\kappa_{it} = \kappa\}}.$$

B.3 Network Distance, Volatility and Attention Allocation

In this section, we drop the browsee-by-year fixed effects in specification (B.3) and explicitly consider the effect of browsees' sectoral TFP and sales volatility on browsers' attention allocation. Specifically, we estimate the following equation separately for $k = 1, 2, 3, 4$.

$$\log y_{ijt} = \alpha_{it} + \beta_1 \log \widehat{\sigma}_{jt} + \beta_2 \log \widehat{\phi}_{ijt} + X_{jt} + \varepsilon_{ijt}, \tag{B.1}$$

where $\text{Upstream}^{(k)}_{ijt}$ is a dummy variable indicating if firm j is firm i's upstream distance-k firm. For every regression, the control (base) group include firms whose distance to firm i is five or larger. The term σ_{jt} represents the standard deviation of firm j's TFP/log sales over the past 5-years prior to year t. Table B.1 presents the results. To interpret the magnitude of the estimated coefficients, the standard deviation of log sales is 0.48 and TFP's standard deviation is 0.11. The results show that, conditional on the network distance, firms pay more attention to those firms with greater TFP/sales volatility. A firm pays more attention to the firm that is closer to it when two firms have the same TFP/sales volatility.

B.4 Alternative Specification: Empirical Evidence

As discussed in Section 3.2.1, our model predicts that, in partial equilibrium, customers pay direct attention to suppliers, whereas suppliers do not pay direct attention to customers. In contrast, alternative frameworks in the literature (Acemoglu et al., 2016; Carvalho et al., 2021; Flynn et al., 2025) permit suppliers to allocate attention to customers. In this section, we revisit the firm-level empirical analysis from Section 3.2.1, replacing input shares with sales shares and reinterpreting suppliers as customers. We find that the results remain broadly consistent with those from our baseline specification.

Sales Shares, Volatility and Attention Allocation. Leveraging the same subsample as in Section 3.2.1 "Input shares, volatility and attention allocation", we explore how a firm's attention paid to a customer varies with the firm's sales share to that customer and the customer's sales or TFP volatility. Specifically, we run the following regression equation:

$$y_{ijt} = \alpha_{it} + \beta_1 \text{sales_share}_{ijt} + \beta_2 \sigma_{jt} + X_{jt} + \varepsilon_{ijt}, \tag{B.2}$$

Table B.1: Browsing Intensity, Upstream Distance and Volatility

	Browsing Intensity				
	(1)	(2)	(3)	(4)	(5)
Upstream distance 1	0.70***	0.91***	0.70***	0.92***	0.67***
	(0.037)	(0.02)	(0.03)	(0.01)	(0.03)
Upstream distance 2	0.20***	0.24***	0.19***	0.26***	0.18
	(0.019)	(0.004)	(0.01)	(0.002)	(0.008)
Upstream distance 3	0.10***	0.14***	0.09***	0.15***	0.10***
	(0.013)	(0.002)	(0.003)	(0.001)	(0.004)
Upstream distance 4	0.044***	0.06***	0.05***	0.07***	0.06***
	(0.0077)	(0.002)	(0.001)	(0.003)	
TFP volatility		0.23***	0.23***		
		(0.0035)	(0.015)		
Sales volatility				0.041***	0.026***
				(0.00058)	(0.0019)
Controls	✓	✓	✓	✓	✓
Browser-Year FE	✓		✓		✓
Browsee-Year FE	✓				
Adjusted R^2	0.45	0.01	0.68	0.01	0.65
No. Observations	102902727	8333741	8333741	26739980	26739980

Note: This table shows how the browsing intensity of a firm depends on its sales share from the downstream customers, how volatile the customers' sales are, and how volatile the customers' TFP is. The control variables include firm size, age, leverage, fixed costs, stock returns, price-to-cost margins, and cash holdings. Robust standard errors are in parentheses and clustered by browsing firms. *Significance*: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

where b_{ijt} is the log browsing volume of firm i on firm j in year t, sales_share$_{ijt}$ is defined as the share of firm i's sales to firm j as firm i's total revenue in year t, and σ_{jt} is the standard deviation of firm j's TFP/log sales over the past-5 years prior to year t. Here, α_{it} is the firm-year fixed effect, and ε_{ijt} is the error term. The controls consist of a broad array of time-varying covariates of firm j. The coefficients β_1 and β_2 measure how the browsing intensity of firm i depends on its sales share to firm j and the TFP/sales volatility of firm j, respectively.

Table B.2 presents the estimates. Column (1) shows that a 10 percent increase in sales share to a customer leads to a 27.6 percent increase in the supplier's browsing volume to that customer, holding the customer's sales volatility constant. Conditional on the same sales share, when a customer's sales volatility increases by one standard deviation (equal to 0.16), the supplier's browsing volume increases by 9 percent. Column (4) displays a similar magnitude when we replace customers' sales volatility with the TFP volatility. When a customer's TFP volatility increases by one standard deviation (equal to 0.12), the supplier's browsing volume increases by 14.4 percent. Our results continue to hold when we add the time-varying controls, as well as the browser fixed effect and time fixed effect to the regression.

Table B.2: Browsing Intensity, Sales Share and Volatility (Firm Level)

	Browsing Intensity					
	(1)	(2)	(3)	(4)	(5)	(6)
Sales Share	3.56***	3.50***	1.58*	2.72***	3.12***	2.00**
	(0.39)	(0.40)	(0.85)	(0.22)	(0.28)	(0.88)
TFP Volatility	1.20**	1.52**	1.14			
	(0.51)	(0.59)	(0.85)			
Sales Volatility				0.58***	0.91***	0.16
				(0.17)	(0.29)	(0.48)
Controls		✓	✓		✓	✓
Browser FE			✓			✓
Time FE			✓			✓
Adjusted R^2	0.07	0.09	0.60	0.04	0.08	0.59
No. Observations	1404	1379	681	4795	2897	1675

Note: This table shows how the browsing intensity of a customer on its supplier depends on the supplier's upstream distance, how volatile the supplier's sales are, and how volatile the supplier's TFP is. The variables for firm controls include firm size, age, leverage, cash holdings, return on assets (ROA), fixed costs, stock returns, price cost margin and R&D intensity. Robust standard errors are in parentheses and clustered by browsing firms. *Significance*: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Suppliers' Attention on Customers and Network Distance. We next examine whether a supplier's attention to customers varies systematically with the customers' network distance. Following the approach in our baseline specification, we classify, for each firm i in year t, all other firms into groups based on their shortest-path distance to firm i in the customer–supplier network. Firms at *downstream distance 1* are direct customers of firm i; those at *distance 2* are connected via one intermediary, and so on, up to a maximum of distance 4. Firms beyond distance 4 are grouped into a residual category, which serves as the omitted base group.

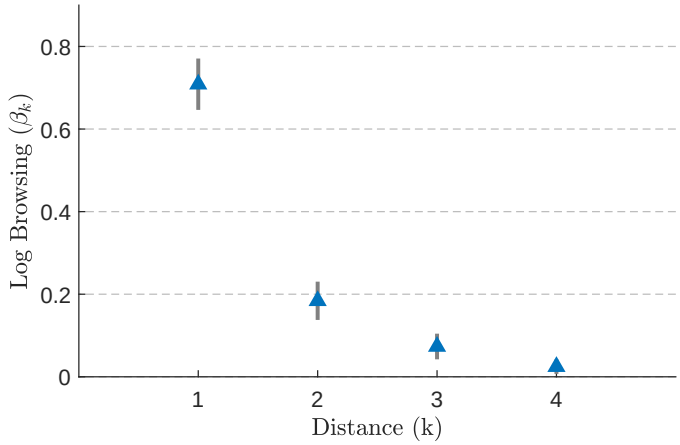
We estimate the following specification:

$$b_{ijt} = \alpha_{it} + \gamma_{jt} + \sum_{k=1}^{4} \beta_k \text{Downstream}_{ijt}^{(k)} + \varepsilon_{ijt}, \tag{B.3}$$

where b_{ijt} denotes the log browsing volume of firm i on firm j in year t, and $\text{Downstream}_{ijt}^{(k)}$ is an indicator equal to one if firm j is at downstream distance k from firm i in year t. The regression includes browser-year fixed effects (αit) and browsee-year fixed effects (γ_{jt}).

Figure B.3 plots the estimated coefficients β_k (blue dots) along with 95% confidence intervals (gray bars). As in our baseline findings, attention declines sharply with network distance: firms allocate approximately 70% more attention to direct customers (distance 1) and only about 20% more to distance-2 customers, relative to the base group. The corresponding regression estimates are reported in Table B.3, column (1). Columns (2)–(5) show that, conditional on network distance, suppliers allocate more attention to customers with greater volatility in TFP and sales.

Figure B.3: Network Distance and Attention Allocation (Suppliers on Customers)



Note: The figure depicts how the estimated coefficients β_k (blue dots) vary with the customers' network distance. The gray bars represent the 95% confidence intervals.

Table B.3: Browsing Intensity, Downstream Distance and Volatility

	Browsing Intensity				
	(1)	(2)	(3)	(4)	(5)
Downstream distance 1	0.71***	1.01***	0.90***	1.04***	0.90***
	(0.031)	(0.015)	(0.035)	(0.0095)	(0.028)
Downstream distance 2	0.18***	0.33***	0.27***	0.33***	0.27***
	(0.023)	(0.0037)	(0.012)	(0.0022)	(0.012)
Downstream distance 3	0.073***	0.18***	0.15***	0.18***	0.15***
	(0.016)	(0.0017)	(0.0065)	(0.00098)	(0.0069)
Downstream distance 4	0.025***	0.077***	0.082***	0.081***	0.083***
	(0.0088)	(0.0011)	(0.0038)	(0.00064)	(0.0040)
TFP volatility		0.22***	0.22***		
		(0.0034)	(0.014)		
Sales volatility				0.035***	0.022***
				(0.00056)	(0.0018)
Controls	✓	✓	✓	✓	✓
Browser-Year FE	✓		✓		✓
Browsee-Year FE	✓				
Adjusted R^2	0.44	0.01	0.69	0.01	0.66
No. Observations	103694229	8461039	8461039	27128093	27128093

Note: This table shows how the browsing intensity of a supplier on a customer depends on the customer's downstream distance, how volatile the customers' sales are, and how volatile the customers' TFP is. The variables for firm controls include firm size, age, leverage, cash holdings, return on assets (ROA), fixed costs, stock returns, price cost margin and R&D intensity. Robust standard errors are in parentheses and clustered by browsing firms. *Significance*: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

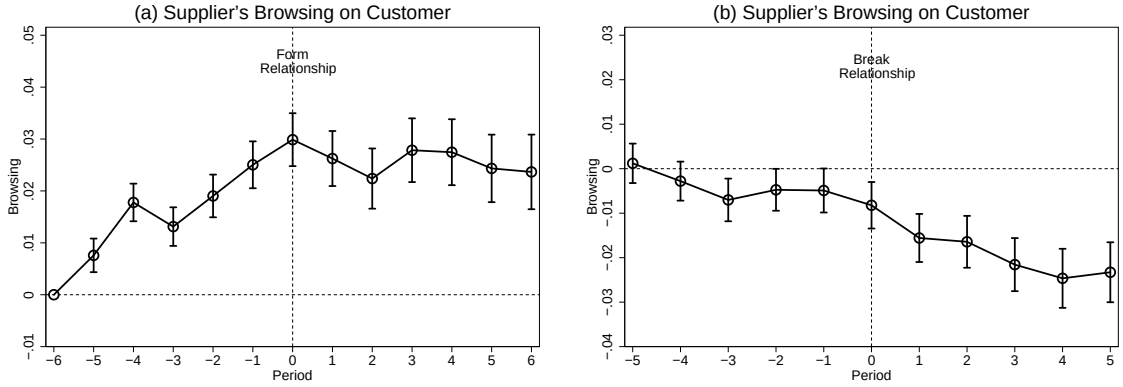
Event Study: Suppliers' Attention on Customers. To examine how a supplier adjusts its attention when initiating or terminating a trading relationship with a customer, similar to our baseline specification, we estimate the following event-study specification:

$$b_{ijt} = \sum_{\ell=-m}^{n} \beta_\ell D_{ij(t-\ell)} + \alpha_i + \alpha_j + \gamma_t + \varepsilon_{ijt}, \tag{B.4}$$

where b_{ijt} denotes the log browsing volume of supplier i on customer j in quarter t, and $D_{ij(t-\ell)}$ is an indicator equal to one if the pair either formed or ended a trading relationship in quarter $t-\ell$, having had no trading activity for at least four years prior (in the case of formation) or subsequent (in the case of termination). The regression includes supplier and customer fixed effects (α_i, α_j) as well as time fixed effects (γ_t).

Figure B.4 plots the estimated coefficients $\{\beta_\ell\}$.[36] The results closely mirror those in our baseline analysis. Attention from the supplier to the customer gradually increases in the quarters preceding the formation of a trading relationship, peaks at the time of contract initiation, and remains persistently elevated thereafter. In contrast, following the termination of a relationship, attention declines gradually, with no evidence of a pre-trend. On average, suppliers devote approximately 3% more attention to active trading partners relative to inactive ones.

Figure B.4: Browsing and Trading Relationship: Event Studies



Note: The figure plots the log browsings of customers on suppliers when new trading relationships are formed at period 0. The frequency is quarterly and 90% confidence intervals are based on clustering at the browser level.

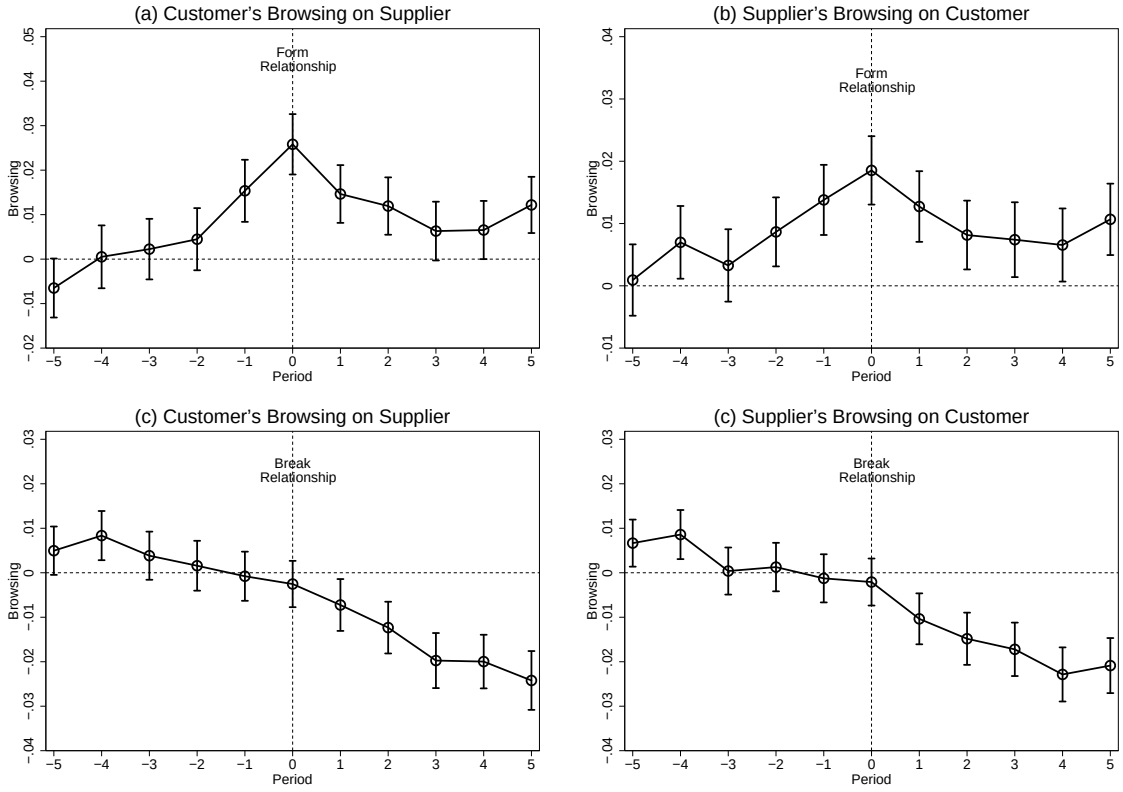
B.5 Event Study: Robustness

We show that the results estimated in our event-study analysis are robust when we relax the four-year inactivity restriction in defining the event, or when we consider 95% confidence intervals. Figure B.5 shows that the dynamic patterns remain almost unchanged when we define an event as forming or

[36] Appendix Figure B.5 presents estimates without the four-year inactivity restriction; Figure B.6 shows results with 95% confidence intervals.

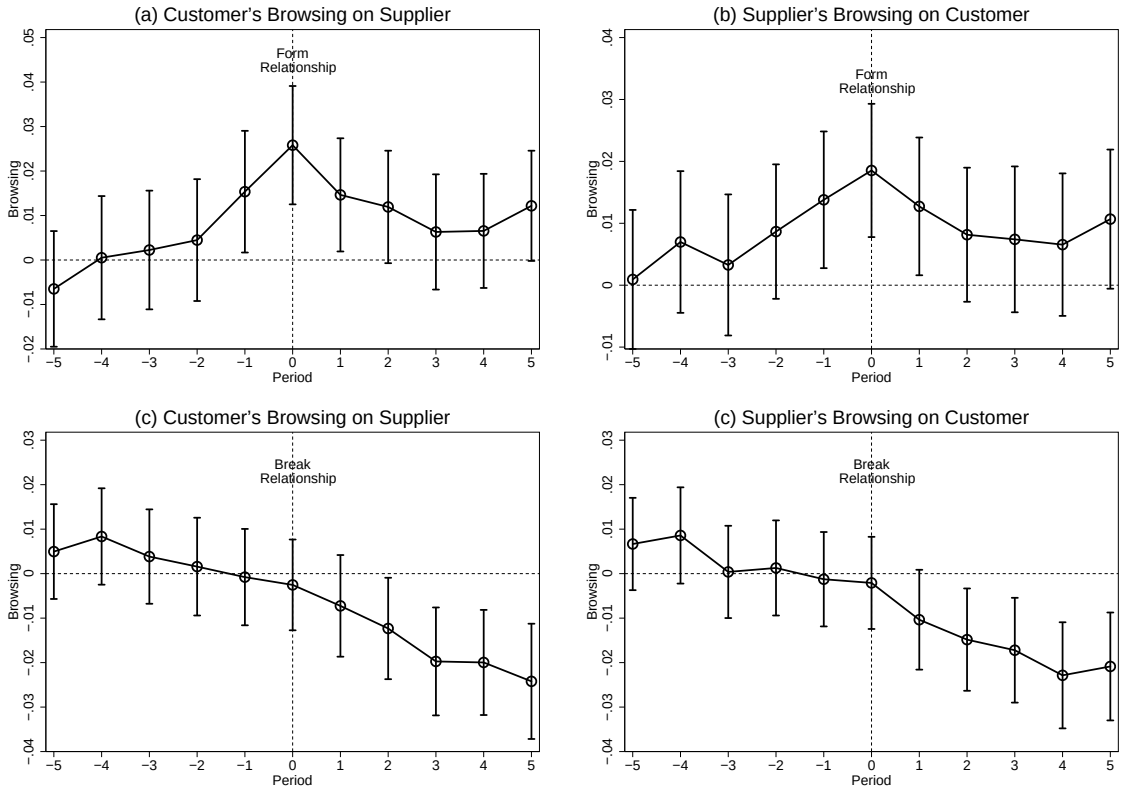
terminating a trading relationship. Figure B.6 shows that our results are robust if we apply the 95% confidence bands.

Figure B.5: Browsing and Trading Relationship: Event Studies



Note: The figure depicts the log browsings of customers on suppliers (left panel) and suppliers on customers (right panel) when new trading relationships are formed or existing relationships are terminated at period 0, relaxing the four-year inactivity restriction in defining the event in our baseline specification. The frequency is quarterly and 90% confidence intervals are based on clustering at the browser level.

Figure B.6: Browsing and Trading Relationship: Event Studies (95 CI)



Note: The figure depicts the log browsings of customers on suppliers (left panel) and suppliers on customers (right panel) when new trading relationships are formed or existing relationships are terminated at period 0. The frequency is quarterly and 95% confidence intervals are based on clustering at the browser level.

B.6 Browsing Intensity and Forecast Accuracy: Robustness

In our baseline specification, we show that a firm forecasts more accurately when it browses other firms more intensively during the 90 days preceding the forecast. In Table B.4, we demonstrate our results remain robust when the browsing window is shortened to 30 days or extended to 180 days prior to the forecast.

Table B.4: Browsing Intensity and Forecast Accuracy

	Forecast Error			
	Earnings Per Share		Sales	
	(1)	(2)	(3)	(4)
Browsing Intensity (30 days)	-0.020**		-0.0011***	
	(0.0073)		(0.00023)	
Browsing Intensity (180 days)		-0.042***		-0.0016**
		(0.0090)		(0.00056)
Controls	✓	✓	✓	✓
Sector-Time FE	✓	✓	✓	✓
Firm FE	✓	✓	✓	✓
Adjusted R^2	0.25	0.25	0.42	0.42
No. Observations	6722	6722	7607	7607

Note: Standard errors are in parentheses and are clustered at industry level. This table shows how firms' forecast accuracy is assocatiated with firms' browsing intensity in the past 30 days or 180 days before the forecast day. The variables for firm controls include firm size, age, leverage, cash holdings, return on assets (ROA), fixed costs, stock returns, price cost margin and R&D intensity. .*Significance*: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

B.7 Industry-Level Evidence: Additional Results

Summary Statistics. Table B.5 reports summary statistics for industry-level variables. Given the large number of sectors (71 and 405, respectively), the average input share of a supplier industry and the sales share of a customer industry are small but exhibit considerable variation, as reflected in their relatively large standard deviations. The average inflation volatility is 0.10 (standard deviation: 0.15), average sales growth volatility is 0.09 (standard deviation: 0.058), and average TFP volatility is 0.031 (standard deviation: 0.03).

Results using Text-based Attention Measure. We re-estimate our main specification (3.5) using the text-based attention data. Table B.6 presents the estimation results and confirms that the text-based results align closely—both qualitatively and quantitatively—with those obtained using the browsing-based measure. The model-implied proxy continues to generate the largest estimate.

Estimation using Inflation ro Sales Growth Volatility. Table B.7 demonstrates that our industry-level results remain robust when we replace TFP growth volatility in regression equation (3.5) with

either sales growth volatility or inflation volatility. Sales growth volatility for year t is calculated using annual sales growth data from years $t-5$ to t. Inflation volatility for year t is computed using monthly inflation data within the same year.

Results using 405 Sectors. Finally, we estimate specification (3.5) using data from the 405-sector input-output (IO) table. Because our quantitative model is calibrated to data moments from the 71-sector IO table, we rely solely on input shares and sales shares to proxy for general equilibrium exposure ϕ_{ij}. The results are closely aligned—both qualitatively and quantitatively—with those based on the 71-sector data.

Table B.5: Industry-level Summary Statistics

	71 Sectors			405 Sectors		
	Mean	Median	S.D.	Mean	Median	S.D.
Input Share	0.014	0.0025	0.040	0.0025	0.000029	0.014
Sales Share	0.014	0.0030	0.046	0.0025	0.000075	0.019
Inflation Volatility	0.10	0.053	0.15	0.091	0.049	0.12
Sales Growth Volatility	0.090	0.080	0.058	0.12	0.09	0.13
TFP Volatility	0.031	0.024	0.030			

Note: This table presents the summary statistics of the industry-level variables for baseline sample of the EDGAR browsing data.

Table B.6: Testing Attention Allocation (Text-based Attention Measure)

	Attention Allocation			
	(1)	(2)	(3)	(4)
log(Input share)	0.057***			
	(0.0061)			
log(Sales share)		0.085***		
		(0.0099)		
$\log\widehat{\phi}_{ij}$ ($\mu = 1$)			0.93***	
			(0.13)	
$\log\widehat{\phi}_{ij}$ (μ calibrated)				1.32***
				(0.20)
TFP growth volatility	0.28***	0.30***	0.25***	0.25***
	(0.013)	(0.014)	(0.012)	(0.012)
Controls	✓	✓	✓	✓
Industry-Year FE	✓	✓	✓	✓
Adjusted R^2	0.55	0.56	0.56	0.56
No. Observations	22178	22175	22207	22207

Note: This table shows estimation results of regression (3.5) using text-based attention measure. The control variables include average firm size, age, leverage, cash holdings, return on assets(ROA), fixed costs, stock returns, price cost margin, R&D intensity, Domar weight, total value added, total inputs, and expenditure share of intermediate inputs. Robust standard errors are in parentheses and clustered by industry. *Significance*: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table B.7: Testing Attention Allocation (Inflation/Sales Growth Volatility)

	Attention Allocation							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Input Share	0.15***				0.14***			
	(0.013)				(0.013)			
Sales Share		0.24***				0.22***		
		(0.019)				(0.022)		
$\log\widehat{\phi}_{ij}$ ($\mu = 1$)			3.02***				2.94***	
			(0.49)				(0.49)	
$\log\widehat{\phi}_{ij}$ (μ calibrated)				4.12***				3.92***
				(0.76)				(0.74)
Sales Growth Volatility	0.71***	0.79***	0.65***	0.66***				
	(0.044)	(0.043)	(0.041)	(0.042)				
Inflation Volatility					0.15***	0.15***	0.14***	0.13***
					(0.024)	(0.022)	(0.019)	(0.020)
Industry Controls	✓	✓	✓	✓	✓	✓	✓	✓
Industry-Year FE	✓	✓	✓	✓	✓	✓	✓	✓
Adjusted R^2	0.46	0.47	0.52	0.52	0.43	0.44	0.49	0.49
No. Observations	25193	25190	25216	25216	22682	22658	22682	22682

Note: This table shows that our results are robust if we use sales growth volatility or inflation volatility instead of TFP growth volatility in regression (3.5). The control variables include average firm size, age, leverage, cash holdings, return on assets(ROA), fixed costs, stock returns, price cost margin, R&D intensity, Domar weight, total value added, total inputs and expenditure share of intermediate inputs. Robust standard errors are in paraentheses and clustered by industry. *Significance*: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table B.8: Testing Attention Allocation (405 Sectors)

	Attention Allocation			
	(1)	(2)	(3)	(4)
Input Share	0.092***		0.097***	
	(0.0026)		(0.0029)	
Sales Share		0.14***		0.15***
		(0.0064)		(0.0070)
Sales Volatility	0.17***	0.20***		
	(0.011)	(0.0096)		
Inflation Volatility			0.13***	0.14***
			(0.0098)	(0.0093)
Controls	✓	✓	✓	✓
Industry-Year FE	✓	✓	✓	✓
Adjusted R^2	0.55	0.56	0.56	0.56
No. Observations	22178	22175	22207	22207

Note: This table shows estimation results of regression (3.5) using the 405-sector IO table. The control variables include average firm size, age, leverage, cash holdings, return on assets(ROA), fixed costs, stock returns, price cost margin, R&D intensity, Domar weight, total value added, total inputs, and expenditure share of intermediate inputs. Robust standard errors are in parentheses and clustered by industry. *Significance*: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

C. An Illustrating Example on Policy Induced Rigidity

To further illustrate the interaction between endogenous rigidities and optimal monetary policy, we consider an economy where all sectors are identical except for the volatility of productivity shocks, and labor is the only production input. As a result, the nominal marginal cost in sector i is

$$\mathrm{mc}_{it} = -z_{it} + w_t,$$

and firms' attention is given by $\mu_i = 1 - \frac{\chi}{\theta\lambda\mathbb{V}(\mathrm{mc}_{it})}$. If the monetary policy implies a wage rule $\boldsymbol{\kappa}$ that is independent of firms' attention choice, then the sectoral rigidity dispersion will be pinned down by the volatility of productivity shocks and any strategic interactions among firms will be muted. One example of such symmetric and exogenous policy is that $\kappa_i = \frac{1}{N}$. However, this is not the case under the optimal monetary policy rule and κ_i will indeed depend on endogenous variables.

For simplicity, we focus on the special case where the policymaker only aims to minimize the aggregate output gap. Applying Proposition 4.4, the weight in the price stabilization rule is proportional to the inverse of sectoral attention, $\varphi_i \propto \frac{1}{\mu_i} - 1$. By condition (4.12), this monetary policy rule implies the following nominal wage function that is endogenous to the sectoral rigidities, $\kappa_i = \frac{1-\mu_i}{\sum_j(1-\mu_j)}$.[37] Therefore, combining the best response of firms' attention choices and the policy minimizing the output gap leads to the following policy-induced fixed-point problem

$$\mu_i = 1 - \frac{\chi}{\theta\lambda}\frac{1}{(1-\kappa_i)^2\sigma_i^2 + \sum_{j\neq i}\kappa_j^2\sigma_j^2}, \tag{C.1}$$

$$\kappa_i = \frac{1-\mu_i}{\sum_j(1-\mu_j)}. \tag{C.2}$$

First assume momentarily that the wage function is symmetric across sectors, $\kappa_i = \frac{1}{N}$. When a sector i is with a relatively low productivity volatility (small σ_i^2), the volatility of their marginal cost is also relatively low, which implies firms in this sector have less incentive to acquire information and therefore choose a smaller μ_i according to (C.1). Next, we allow the wage function to respond to sectoral rigidities. According to (C.2), a smaller μ_i due to lower σ_i^2 now implies more policy emphasis or relatively large φ_i and κ_i, which in turn further reduces the volatility of sector i's marginal costs and incentivizes firms to pay even less attention. On the other hand, the sectors with volatility productivity shocks will end up paying even more attention with smaller φ_i and κ_i. The feedbacks between firms optimal attention allocation and the monetary policy reinforce each other and amplify the dispersion of sectoral rigidities.

To visualize the aforementioned mechanism, we divide the sectors into two groups: the volatility of productivity shocks in the first m sectors is lower than that of remaining m sectors, $\sigma_L^2 < \sigma_H^2$. With the exogenous wage function $\kappa_i = \frac{1}{N}$, the rigidities are pinned down by the shock volatility in their

[37] This is obtained by using condition (4.12) with $\mathbf{A} = 0$ and $\alpha_i = \alpha = 1$ in our example.

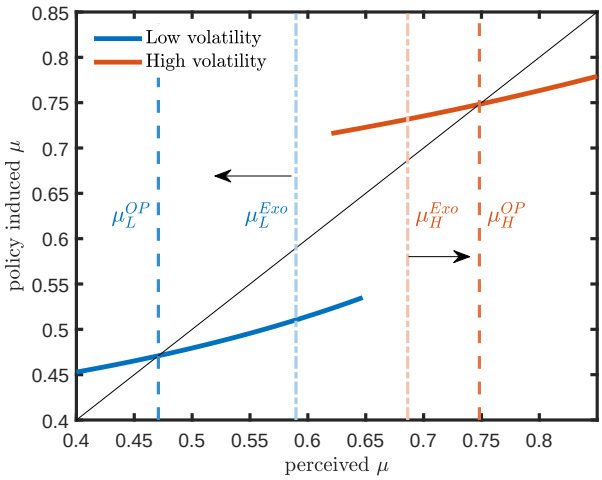


Figure C.1: Optimal Policy Induced Rigidity Dispersion

own sector. The broken blue line in Figure C.1 represents the level of attention in the low volatility sectors (μ_L^{Exo}), while the red line represents the level of attention in the high volatility sectors (μ_H^{Exo}). When switching to the output gap minimizing policy, the dashed blue and red lines represent the new equilibrium level of attention allocation (μ_L^{OP} and μ_H^{OP}). Note that $\mu_L^{OP} < \mu_L^{Exp}$ and $\mu_H^{OP} > \mu_H^{Exp}$, which underscores the additional dispersion of sectoral rigidities with optimal monetary policy.

Furthermore, the solid blue and red lines illustrate the working of the fixed-point system (C.1) and (C.2). For example, the solid blue line shows that when holding μ_H at its equilibrium level μ_H^{OP}, a perceived μ_L in the monetary policy function (C.2) maps to the actual attention choice μ_L in (C.1). The fixed-point is then located at the cross with the 45 degree line. Similarly, the red solid line represents the mapping for high volatility sectors holding $\mu_L = \mu_L^{OP}$. In either case, the fixed-point mapping pushes the endogenous rigidity further away from each other.

D. Proofs for Section 2

In this paper, we adopt the convention that two vectors or matrices have the relation $\mathbf{A} \geq \mathbf{B}$ if $a_{ij} \geq b_{ij}, \forall i, j$, $\mathbf{A} > \mathbf{B}$ if $\mathbf{A} \geq \mathbf{B}$, and $\mathbf{A} \neq \mathbf{B}$. $\mathbf{A} >> \mathbf{B}$ if $a_{ij} > b_{ij}, \forall i, j$. Therefore $\mathbf{A} > \mathbf{0}$ implies that $a_{ij} \geq 0, \ \forall i, j$ and $\mathbf{A} \neq \mathbf{0}$. In addition, vectors in this paper are column vectors by default unless specified otherwise (for example, $\boldsymbol{\kappa}$ and $\boldsymbol{\varphi}$ are row vectors). Throughout our theoretical, quantitative, and optimal policy analysis, we impose the following parameter assumption on the economy's production network.

Assumption 1. *The economy's production network is represented by a nonnegative adjacency matrix* $\mathbf{A} > \mathbf{0}$ *such that for each sector* $\sum_{j=1}^{N} a_{ij} < 1; \quad \forall i = 1, 2, ...N$. *Each sector's production displays constant return to scale, therefore the labor share vector* $\boldsymbol{\alpha} = \left(\{\alpha_i\}_{i=1}^{N}\right)'$ *satisfies* $\alpha_i = 1 - \sum_{j=1}^{N} a_{ij} > 0, \quad \forall i = 1, 2, ...N$. *We denote* $'$ *as the matrix transpose.*

In addition, in this paper we define the following matrix operator,

$$\text{diag}(\cdot) : \mathbb{R}^{N \times N} \longmapsto \mathbb{R}^{N}; \qquad \text{diag}(\cdot) : \mathbb{R}^{N} \longmapsto \mathbb{R}^{N \times N}$$

where operator $\text{diag}(\cdot)$ either extracts the diagonal vector from a given matrix as a column vector, or perform the inverse operation that transform a given vector into diagonal matrix with corresponding diagonal elements equal to the input vector. For example, consider a N-dimensional vector of interest,

$$\boldsymbol{\mu}^{v} = \left(\mu_1, \mu_2, ...\mu_N\right)' \in \mathbb{R}^{N} = \text{diag}\left(\boldsymbol{\mu}\right); \qquad \boldsymbol{\mu} = \text{diag}\left(\boldsymbol{\mu}^{v}\right).$$

Therefore, the nature of this matrix operation hinges on the input object, which is consistent with the Matlab convention.

Next, we adopt the following conventions for matrix differentiation:

1. The derivative between a scalar and a vector follows an arrangement of the same form as this vector. For example,

$$\frac{\partial \mu_i}{\partial \boldsymbol{\kappa}} = \begin{bmatrix} \frac{\partial \mu_i}{\partial \kappa_1} & \frac{\partial \mu_i}{\partial \kappa_2} & \cdots & \frac{\partial \mu_i}{\partial \kappa_N} \end{bmatrix}; \qquad \frac{\partial \boldsymbol{\mu}^{v}}{\partial \kappa_i} = \begin{bmatrix} \frac{\partial \mu_1}{\partial \kappa_i} & \frac{\partial \mu_2}{\partial \kappa_i} & \cdots & \frac{\partial \mu_N}{\partial \kappa_i} \end{bmatrix}'.$$

2. The derivative between a scalar and a matrix follows an arrangement of the same form as this matrix. For example,

$$\frac{\partial \mu_i}{\partial \mathbf{A}} = \begin{bmatrix} \frac{\partial \mu_i}{\partial a_{11}} & \frac{\partial \mu_i}{\partial a_{12}} & \cdots & \frac{\partial \mu_i}{\partial a_{1N}} \\ \frac{\partial \mu_i}{\partial a_{21}} & \frac{\partial \mu_i}{\partial a_{22}} & \cdots & \frac{\partial \mu_i}{\partial a_{2N}} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial \mu_i}{\partial a_{N1}} & \frac{\partial \mu_i}{\partial a_{N2}} & \cdots & \frac{\partial \mu_i}{\partial a_{NN}} \end{bmatrix}; \qquad \frac{\partial \mathbf{A}}{\partial \mu_i} = \begin{bmatrix} \frac{\partial a_{11}}{\partial \mu_i} & \frac{\partial a_{12}}{\partial \mu_i} & \cdots & \frac{\partial a_{1N}}{\partial \mu_i} \\ \frac{\partial a_{21}}{\partial \mu_i} & \frac{\partial a_{22}}{\partial \mu_i} & \cdots & \frac{\partial a_{2N}}{\partial \mu_i} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial a_{N1}}{\partial \mu_i} & \frac{\partial a_{N2}}{\partial \mu_i} & \cdots & \frac{\partial a_{NN}}{\partial \mu_i} \end{bmatrix}.$$

3. The derivative between two vectors: the numerator vector determines the arrangement of its derivative matrix. For example,

$$\frac{\partial \boldsymbol{\mu}^v}{\partial \boldsymbol{\kappa}} = \frac{\partial \boldsymbol{\mu}^v}{\partial \boldsymbol{\kappa}'} = \begin{bmatrix} \frac{\partial \boldsymbol{\mu}^v}{\partial \kappa_1} & \frac{\partial \boldsymbol{\mu}^v}{\partial \kappa_2} & \cdots & \frac{\partial \boldsymbol{\mu}^v}{\partial \kappa_N} \end{bmatrix} = \begin{bmatrix} \frac{\partial \mu_1}{\partial \kappa_1} & \frac{\partial \mu_1}{\partial \kappa_2} & \cdots & \frac{\partial \mu_1}{\partial \kappa_N} \\ \frac{\partial \mu_2}{\partial \kappa_1} & \frac{\partial \mu_2}{\partial \kappa_2} & \cdots & \frac{\partial \mu_2}{\partial \kappa_N} \\ \vdots & \vdots & \vdots & \vdots \\ \frac{\partial \mu_N}{\partial \kappa_1} & \frac{\partial \mu_N}{\partial \kappa_2} & \cdots & \frac{\partial \mu_N}{\partial \kappa_N} \end{bmatrix}; \quad \frac{\partial (\boldsymbol{\mu}^v)'}{\partial \boldsymbol{\kappa}} = \frac{\partial (\boldsymbol{\mu}^v)'}{\partial \boldsymbol{\kappa}'} = \begin{bmatrix} \frac{\partial \mu_1}{\partial \kappa_1} & \frac{\partial \mu_2}{\partial \kappa_1} & \cdots & \frac{\partial \mu_N}{\partial \kappa_1} \\ \frac{\partial \mu_1}{\partial \kappa_2} & \frac{\partial \mu_2}{\partial \kappa_2} & \cdots & \frac{\partial \mu_N}{\partial \kappa_2} \\ \vdots & \vdots & \vdots & \vdots \\ \frac{\partial \mu_1}{\partial \kappa_N} & \frac{\partial \mu_2}{\partial \kappa_N} & \cdots & \frac{\partial \mu_N}{\partial \kappa_N} \end{bmatrix}.$$

D.1 Proof of Lemma 2.1

Proof. We first derive a linear-quadratic-Gaussian (LQG) approximation to each firm's expected profit function. We focus on a firm ι from sector i, at the first stage, its expected profit function is given by

$$\max_{P_{i\iota t}} \mathbb{E}_{i\iota t}\left[\frac{U'(C_t)}{P_t}\left\{\left((1+\tau_i)P_{i\iota t}^{(1-\theta_i)} - P_{i\iota t}^{-\theta_i} MC_{it}\right)P_{it}^{\theta_i} Y_{it}\right\}\right],$$

where we substitute firm (i,ι)'s demand function under monoplistic competition into the objective. Since the representative household's stochastic discount factor $\frac{U'(C_t)}{P_t}$ and does not affect firms' optimal pricing decisions in our static model, they do not affect firms' optimal information choices. Therefore, we simplify the objective function as[38]

$$\max_{P_{i\iota t}} \mathbb{E}_{i\iota t}\left[\underbrace{(1+\tau_i)P_{i\iota t}^{(1-\theta_i)} P_{it}^{\theta_i} Y_{it} - P_{i\iota t}^{-\theta_i} P_{it}^{\theta_i} Y_{it} MC_{it}}_{\equiv \Pi_{i\iota t}}\right],$$

which parallels (2.1). Its marginal cost is given by $\text{MC}_{it} = \frac{W_t^{\alpha_i} \prod_{j=1}^N P_{jt}^{a_{ij}}}{Z_{it}}$. Note that the marginal cost is identical across firms within the same sector. We express the objective function in log-deviations from the steady-state,

$$\Pi_{i\iota t} = \frac{\theta_i}{(\theta_i - 1)}\lambda_i e^{\left[(1-\theta_i)p_{i\iota t} + \theta_i p_{it} + y_{it}\right]} - \lambda_i e^{\left[-\theta_i p_{i\iota t} + \theta_i p_{it} + y_{it} + \alpha_i w_t + \sum_{j=1}^N a_{ij} p_{jt} - z_{it}\right]}, \tag{D.1}$$

where $\lambda_i = P_i Y_i$ is the steady-state sales of firms. In Appendix E.1, we present the solution of perfect-information, deterministic steady state in this model, we show that in the general equilibrium λ_i is also the Domar weight of industry i such that

$$\lambda_i = P_i Y_i = \text{MC}_i Y_i.$$

[38] The term $P_{it}^{\theta_i} Y_{it}$ can also be removed from the objective, because the LQG approximation will be unique up to the steady-state constant $\frac{U'(C)}{P} P_i^{\theta_i} Y_i$, but such difference can be normalized by choosing approriate information cost χ_i.

To ease the exposition, we define a vector of state variables for firm (i, ι),

$$\boldsymbol{\Upsilon}_{it} = \begin{bmatrix} p_{1t} & p_{2t} & \dots, p_{it} & \dots & p_{Nt} & w_t & z_{it} & y_{it} \end{bmatrix}'.$$

Next, we perform a second-oder Taylor series expansion of $\Pi_{i\iota t}$ in terms of the control variable $p_{i\iota t}$ and the state vector $\boldsymbol{\Upsilon}_{it}$,

$$\Pi_{i\iota t}\left(p_{i\iota t}, \boldsymbol{\Upsilon}_{it}\right) = \Pi_{i\iota t}\left(0, \mathbf{0}\right) + \mathbf{D}_{\Pi_{i\iota}(0,\mathbf{0})}\mathcal{X}_{i\iota t} + \frac{1}{2}\mathcal{X}'_{i\iota t}\mathbf{H}_{\Pi_{i\iota}(0,\mathbf{0})}\mathcal{X}_{i\iota t} + O(\varepsilon^3). \tag{D.2}$$

where $\mathcal{X}_{i\iota t} \equiv \begin{bmatrix} \boldsymbol{\Upsilon}_{it} \\ p_{i\iota t} \end{bmatrix}$, $\mathbf{D}_{\Pi_{i\iota}(0,\mathbf{0})}$ and $\mathbf{H}_{\Pi_{i\iota}(0,\mathbf{0})}$ are Jacobian and Hessian matrix of derivatives, respectively. Using standard LQG control techniques, it is straightforward to show that only the quadratic term matters for the optimal information choice under rational inattention.[39] In the approximation, the Hessian matrix is given by $\frac{1}{2}\mathbf{H}_{\Pi_{i\iota}(0,\mathbf{0})} \equiv \begin{bmatrix} \mathbf{H}^i_{xx} & (\mathbf{H}^i_{ux})' \\ \mathbf{H}^i_{ux} & H^i_{uu} \end{bmatrix}$, where $H^i_{uu} = -\frac{1}{2}\lambda_i\theta_i$ and $\mathbf{H}^i_{ux} = \frac{1}{2}\lambda_i\theta_i \begin{bmatrix} a_{i1} & a_{i2} & \dots & a_{iN} & \alpha_i & -1 & 0 \end{bmatrix}$.

We do not spell the expression for block matrix $\mathbf{H}^i_{xx}$ because it is irrelevant for the optimal pricing decision and information choice. Then, we express the second-order approximation of the profit function under rational inattention in quadratic form

$$\max_{p_{i\iota t}} -\mathbb{E}_{i\iota t}\left[\boldsymbol{\Upsilon}'_{it}\mathbf{W}_i\boldsymbol{\Upsilon}_{it} + p'_{i\iota t}R_i p_{i\iota t} + 2\boldsymbol{\Upsilon}'_{it}\mathbf{S}_i p_{i\iota t}\right] \tag{D.3}$$

with coefficient matrices defined by $\mathbf{W}_i = -\mathbf{H}^i_{xx}$, $R_i = -H^i_{uu}$, $\mathbf{S}_i = -(\mathbf{H}^i_{ux})'$.

such that $\begin{bmatrix} \mathbf{W}_i & \mathbf{S}_i \\ \mathbf{S}'_i & R_i \end{bmatrix} \geq 0$ and $R_i > 0$. Using acquisition-filter-control separation principle and the certainty-equivalence principle of the LQG control Tanaka et al. (2017), the optimal pricing decision is given by[40]

$$p_{i\iota t} = -\mathbf{F}_i\mathbb{E}_{i\iota t}\left[\Upsilon_{it} \mid \boldsymbol{x}_{i\iota t}\right] = \mathbb{E}_{i\iota t}\left[\mathrm{mc}_{it} \mid \boldsymbol{x}_{i\iota t}\right]; \quad \mathbf{F}_i = R_i^{-1}\mathbf{S}'_i = -\begin{bmatrix} \mathbf{e}_i\mathbf{A} & \alpha_i & -1 & 0 \end{bmatrix}. \tag{D.4}$$

Agggating (D.4) over the the entire sector, the linear functions of prices and marginal costs coincides with log-linear decision rule (2.2). The equivalence between log-linear approximation and linear-quadratic approximation remains valid for ANY information structure firm chooses. Since y_{it} does not appear in firm's ideal price, the marginal cost, it has no incentive to pay attention to this state variable. Therefore, from now on we drop y_{it} and refine the state variable as $\boldsymbol{\Upsilon}_{it} = \begin{bmatrix} p_{1t} & p_{2t} & \dots, p_{it} & \dots & p_{Nt} & w_t & z_{it} \end{bmatrix}'$.

[39] To see this point, we can transform (D.2) into a pure quadratic form (D.3) by introducing an additional constant state 1, leading to a augmented tracking problem akin to (D.5). Since all firms are able to forecast the deterministic state 1 perfectly without incuring any cost, its associated mean-square errors (variance and covariances) are identically 0. Therefore, we can safely disregard the constant and first-order term in the RI problem.

[40] We denote $\mathbf{e}_i$ as the ith standard basis row vector in $\mathbb{R}^N$.

The Hessian block matrix is given by $\mathbf{H}_{ux}^{i} = \frac{1}{2}\lambda_i\theta_i \begin{bmatrix} a_{i1} & a_{i2} & \dots & a_{iN} & \alpha_i & -1 \end{bmatrix}$. The optimal decision rule is simplified to $\mathbf{F}_i = -\begin{bmatrix} \mathbf{e}_i\mathbf{A} & \alpha_i & -1 \end{bmatrix}$.[41]

Next, we employ Lemma 1 in Miao et al. (2022) to cast the profit maximization problem of (D.3) into a tracking problem of information choices,

$$\max_{\boldsymbol{x}_{i\iota t}} -\mathbb{E}\left[(\boldsymbol{\Upsilon}_{it} - \mathbb{E}\left[\boldsymbol{\Upsilon}_{it} \mid \boldsymbol{x}_{i\iota t}\right])' \boldsymbol{\Omega}_i \left(\boldsymbol{\Upsilon}_{it} - \mathbb{E}\left[\boldsymbol{\Upsilon}_{it} \mid \boldsymbol{x}_{i\iota t}\right]\right)\right], \tag{D.5}$$

where the positive-semidefinite weighting matrix is given by $\boldsymbol{\Omega}_i = \mathbf{F}_i' R_i \mathbf{F}_i \succeq 0$. Using this observation and (D.4), we obtain the tracking formulation in terms of optimal pricing decision,

$$\max_{\boldsymbol{x}_{i\iota t}} -\frac{1}{2}\lambda_i\theta_i\mathbb{E}\left[\left(p_{i\iota t} - \mathrm{mc}_{it}\right)^2\right],$$

as desired. The existence of a pure tracking problem in optimal information choice relies on the fact the state variables Υ_{it} are "exogenous" in the eyes of firm (i, ι). Since information costs χ are sector-specific, it is clear that firms' optimal information choice problems are homogeneous within each sector but heterogeneous across sectors in the network. In the tracking formulation, firm's information acquisition motive is dictated by the optimal pricing decision, which in turn hinge on network structure via the marginal cost.

Finally, in the general equilibrium we note that the pricing decision and state variables $\boldsymbol{\Upsilon}_{it}$ are both functions of sectoral productivity shocks $\boldsymbol{z}_t$. Under LQG setting with Shannon-entropy cost function $C_i(\boldsymbol{x}_{i,\iota,t})$, it is well known that the optimal information structure is Gaussian, represented by an signal structure in the form

$$\boldsymbol{x}_{i,\iota,t} = \mathbf{H}_i\boldsymbol{z}_t + \mathbf{u}_{i,l,t}, \quad \mathbf{u}_{i,\iota,t} \sim \mathbb{N}(\mathbf{0}, \mathbf{V}_i)$$

where $\mathbf{H}_i$ and $\mathbf{V}_i \succ 0$ are matrices of unknown dimensions to be determined by the RI problem. The information cost function is parameterized by the classical Gaussian (conditional) mutual information, $\mathbb{I}(\boldsymbol{z}_t; \boldsymbol{x}_{i\iota t}|\boldsymbol{\Sigma}_z) = \frac{1}{2}\left(\log\det\boldsymbol{\Sigma}_z - \log\det\boldsymbol{\Sigma}_{z|x_i}\right)$. To make the notation more clear, in the Appendix, we denote the prior covariance matrix of shocks as

$$\boldsymbol{\Sigma} = \boldsymbol{\Sigma}_z. \tag{D.6}$$

The proof is now complete. □

D.2 Proof of Proposition 2.1

Proof. We first transform the optimal information choice problem into a semidefinite progamming problem. Let $\boldsymbol{p}_t = \boldsymbol{\phi}\boldsymbol{z}_t$. $\boldsymbol{\phi}$ is the pricing function ($N \times N$ influence matrix) under rational inattention, which is determined in general equilibrium.

[41] It is straightforward to verify that such refinement doe not change any of our results.

Lemma D.1. *The RI problem for firms in sector i is represented by the following matrix optimization problem,*

$$\min_{\boldsymbol{\Sigma}_{z|x_i}} \ \operatorname{tr}\left(\boldsymbol{\Omega}_{iz}\boldsymbol{\Sigma}_{z|x_i}\right) + \chi_i\left[\frac{1}{2}\log\det\boldsymbol{\Sigma}_z - \frac{1}{2}\log\det\boldsymbol{\Sigma}_{z|x_i}\right]$$
$$s.t. \quad \mathbf{0} \preceq \boldsymbol{\Sigma}_{z|x_i} \preceq \boldsymbol{\Sigma}_z,$$

where $\boldsymbol{\Sigma}_{z|x_i}$ denotes the posterior covariance matrices of the fundamental shocks $\boldsymbol{z}_t$. The modified weighting matrix admits representation as $\boldsymbol{\Omega}_{iz} = \mathbf{G}_i'\mathbf{G}_i$, where $\mathbf{G}_i = \sqrt{R_i}\mathbf{e}_i\left(\mathbf{A}\boldsymbol{\phi} - \mathbf{I} + \boldsymbol{\alpha}\boldsymbol{\kappa}\right)$.

Proof. We write (2.2) in matrix form as

$$\mathrm{mc}_{it} = \mathbf{e}_i\left(-\mathbf{I} + \boldsymbol{\alpha}\boldsymbol{\kappa} + \mathbf{A}\boldsymbol{\phi}\right)\boldsymbol{z}_t. \tag{D.7}$$

Substite (D.4) and (D.7) into the objective function in Lemma 2.1, we express the tracking problems in terms of $\boldsymbol{z}_t$,[42]

$$\min_{\boldsymbol{x}_{it}} \mathrm{E}\left[\left(\boldsymbol{z}_t - \mathrm{E}[\boldsymbol{z}_t|\boldsymbol{x}_{it}]\right)'\boldsymbol{\Omega}_{iz}\left(\boldsymbol{z}_t - \mathrm{E}[\boldsymbol{z}_t|\boldsymbol{x}_{it}]\right)\right] + \chi_i\mathbb{I}(\boldsymbol{z}_t;\boldsymbol{x}_{it}|\Sigma_{\boldsymbol{z}}), \tag{D.8}$$

where the modified weighting matrix is given by $\boldsymbol{\Omega}_{iz} = \mathbf{G}_i'\mathbf{G}_i \succeq 0$, and $\mathbf{G}_i = \sqrt{R_i}\mathbf{e}_i\left(\mathbf{A}\boldsymbol{\phi} - \mathbf{I} + \boldsymbol{\alpha}\boldsymbol{\kappa}\right)$. Using the invariance property of the trace operator under circular shifts , the first term in problem (D.8) can be expressed as $\mathrm{E}\left[\left(\boldsymbol{z}_t - \mathrm{E}[\boldsymbol{z}_t|\boldsymbol{x}_{it}]\right)'\boldsymbol{\Omega}_{iz}\left(\boldsymbol{z}_t - \mathrm{E}[\boldsymbol{z}_t|\boldsymbol{x}_{it}]\right)\right] = \operatorname{tr}\left(\boldsymbol{\Omega}_{iz}\boldsymbol{\Sigma}_{z|x_i}\right)$. Finally, given any signal structure firms choose, they are fully rational in predicting the unobserved states. Therefore, the Bayesian updating formula holds such that

$$\mathrm{SNR}_i \equiv \mathbf{H}_i'\mathbf{V}_i^{-1}\mathbf{H}_i = \boldsymbol{\Sigma}_{z|x_i}^{-1} - \boldsymbol{\Sigma}_z^{-1} \succeq 0. \tag{D.9}$$

where SNR_i is defined as matrix-valued signal-to noise ratio. Since $\boldsymbol{\Sigma}_{z|x_i} \succeq 0$ is by construction positive-semidefinite, elementary theory of positive-definite matrices implies that $0 \preceq \boldsymbol{\Sigma}_{z|x_i} \preceq \boldsymbol{\Sigma}_z$. The underlying "no-forgetting constraint" reflects the full-rationality of firms: they cannot achieve better forecasts of some states by eliminating existing memories of other states. This completes the proof. □

Next, we derive a closed-form solution to the matrix optimization problem defined in Lemma D.1. The solution then produces the optimal signal structure as a by-product. In particular, we employ techniques developed by Miao et al. (2022), which also apply to more general models of dynamic, multivariate rational inattention.

To begin with, we note that by (D.7), $\mathrm{mc}_{it} = R_i^{-\frac{1}{2}}\mathbf{G}_i\boldsymbol{z}_t$. We define the variance of the marginal

[42] Alternatively, we can use (D.5) to derive (D.8) by defining the mapping betweem endogenous states and shocks: $\Upsilon_{it} = \boldsymbol{\Psi}_i\boldsymbol{z}_t$. $\boldsymbol{\Psi}_i = \begin{bmatrix}\boldsymbol{\phi}\\ \boldsymbol{\kappa}\\ \mathbf{e}_i\end{bmatrix}$. Two derivations are equivalent.

cost (profit-maximizing price, or ideal price) as $\mathbb{V}(\text{mc}_{it})$. As we emphasized in the main text, this marginal-cost volatility serves as the key determinant of firms' information acquisition strategy. The following relationship between $\mathbb{V}(\text{mc}_{it})$ and $\mathbf{G}_i$ holds:

$$\mathbb{V}(\text{mc}_{it}) = R_i^{-1} \parallel \boldsymbol{\Sigma}_z^{\frac{1}{2}} \mathbf{G}_i' \parallel^2, \tag{D.10}$$

where $\|\cdot\|$ denotes the Euclidean norm. Since $\boldsymbol{\Omega}_{iz}$ is symmetric and $\boldsymbol{\Sigma}_z \succ 0$, we construct a symmetric, positive-semidefinite matrix $\boldsymbol{\Sigma}_z^{\frac{1}{2}} \boldsymbol{\Omega}_{iz} \boldsymbol{\Sigma}_z^{\frac{1}{2}}$, where $\boldsymbol{\Sigma}_z^{\frac{1}{2}} = \begin{bmatrix} \sigma_1 & 0 & \dots & 0 \\ 0 & \sigma_2 & \dots & 0 \\ \vdots & \cdots & \ddots & \vdots \\ 0 & \cdots & \cdots & \sigma_N \end{bmatrix} \succ 0$ denotes the positive square root matrix of the prior covariance matrix. Next, perform an unitary eigendecomposition, $\boldsymbol{\Sigma}_z^{\frac{1}{2}} \boldsymbol{\Omega}_{iz} \boldsymbol{\Sigma}_z^{\frac{1}{2}} = \mathbf{U}_i \boldsymbol{\Omega}_d^i \mathbf{U}_i'$, where $\mathbf{U}_i \mathbf{U}_i' = I$ is unitary and $\boldsymbol{\Omega}_d^i = \text{diag}\left(\{d_k^i\}_{k=1}^N\right)$ is a diagonal matrix of non-negative eigenvalues with descending order. Note that $\mathbf{G}_i = \sqrt{R_i}\mathbf{e}_i(\mathbf{A}\boldsymbol{\phi} - \mathbf{I} + \boldsymbol{\alpha}\boldsymbol{\kappa})$ is a $1 \times N$ vector, and the rank of the outer product $\boldsymbol{\Omega}_{iz} = \mathbf{G}_i' \mathbf{G}_i$ is equal to one. Therefore, $\text{rank}\left(\boldsymbol{\Sigma}_z^{\frac{1}{2}} \boldsymbol{\Omega}_{iz} \boldsymbol{\Sigma}_z^{\frac{1}{2}}\right) = 1$ by construction. Consequently, there is only one strictly positive eigenvalue with an associated eigenvector denoted as

$$\begin{aligned} d_1^i &= \parallel \boldsymbol{\Sigma}_z^{\frac{1}{2}} \mathbf{G}_i' \parallel^2 = \frac{1}{2}\theta_i \lambda_i \mathbb{V}(\text{mc}_{it}), \\ \boldsymbol{\zeta}_1^i &= \frac{\boldsymbol{\Sigma}_z^{\frac{1}{2}} \mathbf{G}_i'}{\parallel \boldsymbol{\Sigma}_z^{\frac{1}{2}} \mathbf{G}_i' \parallel}. \end{aligned} \tag{D.11}$$

The rest of eigenvalues are equal to 0 by definition. The next lemma presents a generalized reverse water-filling solution of optimal posterior matrix $\boldsymbol{\Sigma}_{z|x_i}$,[43]

Lemma D.2. *The optimal solution to the matrix optimization problem defined in Lemma D.1 is given by*

$$\boldsymbol{\Sigma}_{z|x_i} = \boldsymbol{\Sigma}_z^{\frac{1}{2}} \mathbf{U}_i \ \text{diag}\left(\left\{\min\left(1, \frac{\chi_i}{2d_k^i}\right)\right\}_{k=1}^N\right) \mathbf{U}_i' \boldsymbol{\Sigma}_z^{\frac{1}{2}} \tag{D.12}$$

Proof. Define an instrument matrix $\widehat{\boldsymbol{\Sigma}}_i = \mathbf{U}_i' \boldsymbol{\Sigma}_z^{-\frac{1}{2}} \boldsymbol{\Sigma}_{z|x_i} \boldsymbol{\Sigma}_z^{-\frac{1}{2}} \mathbf{U}_i \succeq 0$. It is clear that

$$\boldsymbol{\Sigma}_{z|x_i} = \boldsymbol{\Sigma}_z^{\frac{1}{2}} \mathbf{U}_i \widehat{\boldsymbol{\Sigma}}_i \mathbf{U}_i' \boldsymbol{\Sigma}_z^{\frac{1}{2}}. \tag{D.13}$$

Using basic properties of the trace operator,

$$\text{tr}\left(\boldsymbol{\Omega}_{iz} \boldsymbol{\Sigma}_{z|x_i}\right) = \text{tr}\left(\boldsymbol{\Omega}_{iz} \boldsymbol{\Sigma}_z^{\frac{1}{2}} \mathbf{U}_i \widehat{\boldsymbol{\Sigma}}_i \mathbf{U}_i' \boldsymbol{\Sigma}_z^{\frac{1}{2}}\right) = \text{tr}\left(\boldsymbol{\Sigma}_z^{\frac{1}{2}} \boldsymbol{\Omega}_{iz} \boldsymbol{\Sigma}_z^{\frac{1}{2}} \mathbf{U}_i \widehat{\boldsymbol{\Sigma}}_i \mathbf{U}_i'\right) = \text{tr}\left(\mathbf{U}_i \boldsymbol{\Omega}_d^i \mathbf{U}_i' \mathbf{U}_i \widehat{\boldsymbol{\Sigma}}_i \mathbf{U}_i'\right) = \text{tr}\left(\boldsymbol{\Omega}_d^i \widehat{\boldsymbol{\Sigma}}_i\right).$$

[43] This result holds for arbitrarily-correlated prior matrix $\boldsymbol{\Sigma}_z$ and for problems with arbitrary number of optimal decisions such that $\text{rank}(\boldsymbol{\Omega}_{iz}) > 1$. See Miao et al. (2022) for more details.

Similarly, using properties of matrix determinant, $\log\det\boldsymbol{\Sigma}_z - \log\det\boldsymbol{\Sigma}_{z|x_i} = -\log\det\widehat{\boldsymbol{\Sigma}}_i$. The "no-forgeting" constraint $\boldsymbol{\Sigma}_{z|x_i} \preceq \boldsymbol{\Sigma}_z$ can also be recasted as

$$\boldsymbol{\Sigma}_{z|x_i} \preceq \boldsymbol{\Sigma}_z \Leftrightarrow \boldsymbol{\Sigma}_z^{\frac{1}{2}}\mathbf{U}_i\widehat{\boldsymbol{\Sigma}}_i\mathbf{U}_i'\boldsymbol{\Sigma}_z^{\frac{1}{2}} \preceq \boldsymbol{\Sigma}_z^{\frac{1}{2}}\mathbf{U}_i\mathbf{U}_i'\boldsymbol{\Sigma}_z^{\frac{1}{2}} \Leftrightarrow \widehat{\boldsymbol{\Sigma}}_i \preceq \mathbf{I},$$

where we use the properties of positive-semidefinite matrices. Therefore, we convert the optimization problem in terms of surrogate matrix $\widehat{\boldsymbol{\Sigma}}_i$,

$$\min_{\widehat{\boldsymbol{\Sigma}}_i} \operatorname{tr}\left(\boldsymbol{\Omega}_d^i\widehat{\boldsymbol{\Sigma}}_i\right) - \frac{1}{2}\chi_i\log\det\widehat{\boldsymbol{\Sigma}}_i, \tag{D.14}$$

subject to a semidefinite constraint,

$$\widehat{\boldsymbol{\Sigma}}_i \preceq \mathbf{I}. \tag{D.15}$$

The optimization is a well-defined convex progamming problem, because $\log\det$ function is concave and (D.15) defines a convex (positive, semidefinite) cone. Recall that $\boldsymbol{\Omega}_d^i$ is a diagonal matrix of eigenvalues, hence off-diagonal elements in $\widehat{\boldsymbol{\Sigma}}_i$ does not affect the value of $\operatorname{tr}\left(\boldsymbol{\Omega}_d^i\widehat{\boldsymbol{\Sigma}}_i\right)$. Meanwhile, the Hadamard inequality for positive definite matrices implies the determinant of $\widehat{\boldsymbol{\Sigma}}_i$ is bounded by the product of its diagonal elements, $\det\widehat{\boldsymbol{\Sigma}}_i \leq \prod_{j=1}^N \widehat{\boldsymbol{\Sigma}}_i(j,j)$. The equality holds if and only if $\widehat{\boldsymbol{\Sigma}}_i$ is diagonal. Therefore, it is straightforward to conclude that the optimal solution to (D.14) and (D.15) must be diagonal. This observation allow us to convert the optimization into an equivalent form,

$$\min_{\left\{\widehat{\boldsymbol{\Sigma}}_i(k,k)\right\}_{k=1}^N} \sum_{k=1}^N d_k^i\widehat{\boldsymbol{\Sigma}}_i(k,k) - \frac{1}{2}\chi_i\sum_{k=1}^N \log\widehat{\boldsymbol{\Sigma}}_i(k,k),$$

subject to constraints: $\widehat{\boldsymbol{\Sigma}}_i(k,k) \leq 1$ for $k = 1,2,...N$. By convexity, the Kuhn-Tucker conditions allow us to deliver the unique solution to this problem,

$$\widehat{\boldsymbol{\Sigma}}_i(k,k) = \min\left(1, \frac{\chi_i}{2d_k^i}\right); \qquad k = 1,2,...N.$$

Subtitute this expression into the original solution (D.13) yields Lemma's claim as desired. □

Now given the solution in the preceding lemma, in our rank-one model the solution reduces to

$$\boldsymbol{\Sigma}_{z|x_i} = \boldsymbol{\Sigma}_z^{\frac{1}{2}}\mathbf{U}_i\widehat{\boldsymbol{\Sigma}}_i\mathbf{U}_i'\boldsymbol{\Sigma}_z^{\frac{1}{2}} = \boldsymbol{\Sigma}_z^{\frac{1}{2}}\mathbf{U}_i\begin{bmatrix}\min\left(1, \frac{\chi_i}{2d_1^i}\right) & 0\\ 0 & \mathbf{I}_{N-1}\end{bmatrix}\mathbf{U}_i'\boldsymbol{\Sigma}_z^{\frac{1}{2}}, \tag{D.16}$$

since $d_k^i = 0; \forall k = 2,3,...N$. It follows that that when the exogenous marginal cost $\frac{\chi_i}{2d_1^i} \geq 1$ is large enough, $\widehat{\boldsymbol{\Sigma}}_i = I$ and $\boldsymbol{\Sigma}_{z|x_i} = \boldsymbol{\Sigma}_z$. That is, no information is collected (posterior equals prior) when information cost is too high. Using the definition of positive eigenvalue d_1^i, we arrives at the condition

for no information acquisition,

$$\chi_i \geq \lambda_i \theta_i \mathbb{V}\left(\mathrm{mc}_{it}\right),$$

as desired. On the other hand, the required threshold condition for processing information is written as

$$\chi_i < 2d_1^i. \tag{D.17}$$

In this case, firms would choose to collect information.[44] By the Bayesian updating formula,

$$\mathrm{SNR}_i = \mathbf{H_i}'\mathbf{V}_i^{-1}\mathbf{H_i} = \mathbf{\Sigma}_{z|x_i}^{-1} - \mathbf{\Sigma}_z^{-1} = \mathbf{\Sigma}_z^{-\frac{1}{2}}\mathbf{U}_i \begin{bmatrix} \left(\frac{2d_1^i}{\chi_i} - 1\right) & 0 \\ 0 & \mathbf{0}_{N-1} \end{bmatrix} \mathbf{U}_i'\mathbf{\Sigma}_z^{-\frac{1}{2}}.$$

By construction, the signal-to-noise ratio matrix (SNR_i) is rank-one with one strictly positive singular value. Therefore, the optimal signal dimension is 1,

$$\dim\left(\boldsymbol{x}_{iit}\right) = \mathrm{rank}\left(\mathrm{SNR}_i\right) = 1,$$

by Proposition 1 of Miao et al. (2022) and Tanaka et al. (2017). Next, we partition the unitary matrix $\mathbf{U}_i$ comfortably as $\mathbf{U}_i = \begin{bmatrix} \boldsymbol{\zeta}_1^i & \boldsymbol{\zeta}_2^i \end{bmatrix}$ where eigenvector $\boldsymbol{\zeta}_1^i$ corresponds to d_1^i as we defined before, then the matrix SNR is given by

$$\mathbf{H_i}'\mathbf{V}_i^{-1}\mathbf{H_i} = \mathbf{\Sigma}_z^{-\frac{1}{2}}\boldsymbol{\zeta}_1^i\left(\frac{2d_1^i}{\chi_i} - 1\right)(\boldsymbol{\zeta}_1^i)'\,\mathbf{\Sigma}_z^{-\frac{1}{2}}.$$

Therefore, a natural solution for the univariate optimal signal is

$$\mathbf{V}_i = \frac{\chi_i}{(2d_1^i - \chi_i)}; \qquad \mathbf{H_i} = (\boldsymbol{\zeta}_1^i)'\,\mathbf{\Sigma}_z^{-\frac{1}{2}} = \frac{G_i}{\left\|\mathbf{\Sigma}_z^{\frac{1}{2}}G_i'\right\|}.$$

Note that the optimal signal structure can also be computed by the singular-value decomposition. In our univariate case, the optimal signal is unique modulo a scalar normalization constant. When $\mathbf{H_i}$ is scaled by a constant b, $\mathbf{V}_i$ is scaled by b^2. Such normalization yields informationally-equivalent signals that produce identical posteriors, and does not affect the sectoral and aggregate equilibrium of our model.[45]

Given this observation and the fact that $\mathrm{mc}_{it} = R_i^{-\frac{1}{2}}\mathbf{G}_i\boldsymbol{z}_t$, we let $b = R_i^{-\frac{1}{2}}\left\|\mathbf{\Sigma}_z^{\frac{1}{2}}G_i'\right\|$. Accordingly, the noise variance is multiplied by $b^2 = R_i^{-1}\left\|\mathbf{\Sigma}_z^{\frac{1}{2}}G_i'\right\|^2 = \mathbb{V}\left(\mathrm{mc}_{it}\right)$. The normalized optimal signal is given

[44] Of course, d_1^i is an endogenous objective to be determined in the general equilibrium. In our proof of equilibrium existence in Proposition D.2 , we provide conditions for χ_i such that (D.17) always holds in general equilibrium.

[45] Normalization does not change the response functions to fundamental shocks, but the responses to idosyncratic noise shocks are scaled by $\frac{1}{b}$. Since idiosyncratic noises wash out at the sectoral and aggregate level, our model equilibrium is unaffected. In Appendix D.6, we provide further evidence.

by

$$x_{\iota it} = \text{mc}_{it} + u_{\iota it}, \quad u_{\iota it} \sim \mathbb{N}\left(0, v_i^2\right).$$

The endogenous noise variance is given by

$$v_i^2 = \frac{\chi_i \mathbb{V}\left(mc_{it}\right)}{\lambda_i \theta_i \mathbb{V}\left(mc_{it}\right) - \chi_i}, \tag{D.18}$$

where we use the fact that $d_1^i = \left\| \mathbf{\Sigma}_z^{\frac{1}{2}} \mathbf{G}_i' \right\|^2 = \frac{1}{2}\theta_i \lambda_i \mathbb{V}\left(mc_{it}\right)$. Note that by construction, the endogenous Gaussian noise $u_{\iota it}$ is independent of the marginal costs. The noise is also sector-specific in the sense that firms in the same sector face noisy signals drawn from the same distribution. Within-sector information heterogeneity arises ONLY from different realizations of noise $u_{\iota it}$. The underlying results in the Proposition applies to firms in ALL sectors, $i = 1, 2, ...N$. The proof is now complete. □

D.3 Proof of Proposition 2.2

Proof. Using our previous result, $\mathbb{I}\left(\text{mc}_{it}, x_{i,\iota,t}\right) = \frac{1}{2}\left(\log\det \mathbf{\Sigma}_z - \log\det \mathbf{\Sigma}_{z|x_i}\right) = -\frac{1}{2}\log\det \widehat{\mathbf{\Sigma}}_i$. The optimal solution for $\widehat{\mathbf{\Sigma}}_i$ is diagonal, $\widehat{\mathbf{\Sigma}}_i\left(k,k\right) = \min\left(1, \frac{\chi_i}{2d_k^i}\right)$ for $k = 1, 2, ...N$. Therefore, it follows that

$$\mathbb{I}\left(\text{mc}_{it}, x_{i,\iota,t}\right) = -\frac{1}{2}\log\frac{\chi_i}{2d_1^i} = -\frac{1}{2}\log\frac{\chi_i}{\theta_i\lambda_i\mathbb{V}\left(\text{mc}_{it}\right)}.$$

Note when firm collect information, $\frac{\chi_i}{\theta_i\lambda_i\mathbb{V}(\text{mc}_{it})} \in [0, 1)$, therefore the mutual information is monotonically increasing in the volatility of marginal cost. In Appendix D.6, we also establish equivalence between signal precision of sector i, μ_i, and the total attention capacity chosen by sector i. Therefore, the price-flexibility is also monotonically increasing in $\mathbb{V}\left(\text{mc}_{it}\right)$. The proof is now complete.

□

D.4 Proof of Proposition 2.3

Proof. Consider the equilibrium process for sector i's marginal cost,

$$\text{mc}_{it} = -z_{it} + \alpha_i w_t + \sum_j a_{ij} p_{jt}. \tag{D.19}$$

Suppose in the partial equilibrium, firms belives that the marginal cost consists of two parts,

$$\text{mc}_{it} = f_{it} + \sum_{j\neq i} a_{ij} p_{jt}, \quad f_{it} = -z_{it} + \alpha_i w_t + a_{ii} p_{it}.$$

By the assumption of Proposition, we let $f_{it} \sim N(0, \sigma_f^2)$ be independent of other sectors' prices, while prices in other sectors $p_{jt} \sim N(0, \gamma_j)$ are also mutually independent. Now define the diagonal prior

covariance matrix of $\left[f_{it}, \mathbf{p}_{-it}\right]$ to be $\Sigma_\Upsilon = \begin{bmatrix} \sigma_f^2 & \mathbf{0} \\ 0 & \operatorname{diag}\left(\{\gamma_j\}_{j\neq i}\right) \end{bmatrix}$. Now using the same techniques in the proof of Proposition 2.1, we define

$$\mathbf{G}_i = \sqrt{\frac{1}{2}\lambda_i\theta_i}\begin{bmatrix} 1 & \mathbf{A}_{(i,-j)} \end{bmatrix}; \qquad R_i = \frac{1}{2}\lambda_i\theta_i; \qquad d_1^i = \|\Sigma_\Upsilon^{\frac{1}{2}}\mathbf{G}_i'\|^2 \tag{D.20}$$

where $\mathbf{A}_{(i,j\neq i)}$ denotes the ith row of $\mathbf{A}$ without the diagonal element. The volatility of marginal cost is expressed as $\mathbb{V}(\mathrm{mc}_{it}) = R_i^{-1}\|\Sigma_\Upsilon^{\frac{1}{2}}\mathbf{G}_i'\|^2$. The optimal signal is still one-dimensional, $x_{iit} = \mathrm{mc}_{it} + u_{iit}$, $u_{iit} \sim \mathbb{N}\left(0, \nu_i^2\right)$. The endogenous noise variance is given by

$$\nu_i^2 = \frac{\chi_i \mathbb{V}(mc_{it})}{\lambda_i\theta_i\mathbb{V}(mc_{it}) - \chi_i}. \tag{D.21}$$

In the proof of Proposition D.1, we obtain the optimal posteior covariance matrix (MSE) matrix,

$$\mathbf{\Sigma}_{\Upsilon|x_i} = \mathbf{\Sigma}_\Upsilon - \frac{\chi_i}{2R_i d_1^i \nu_i^2}\left(\mathbf{\Sigma}_\Upsilon \mathbf{\Omega}_{i\Upsilon}\mathbf{\Sigma}_\Upsilon\right), \tag{D.22}$$

where $\mathbf{\Omega}_{i\Upsilon} = \mathbf{G}_i'\mathbf{G}_i$ is the weighting matrix. We denote $\widehat{\gamma}_{ij} = \mathbb{E}\left[\left(\mathbb{E}\left[p_{jt} \mid x_{iit}\right] - p_{jt}\right)^2 \mid x_{iit}\right]$ as the diagonal element of the posterior matrix that represents posterior uncertainty about sector j's price. Now substitute (D.20) and (D.21) into (D.22), after some algebra we obtain

$$\widehat{\gamma}_{ij} = \gamma_j\left[1 - \frac{\mu_i}{\mathbb{V}(\mathrm{mc}_{it})}\gamma_j a_{ij}^2\right].$$

Therefore, the percentage reduction in certainty is given by

$$\omega_{ij} = \frac{\gamma_j - \widehat{\gamma}_{ij}}{\gamma_j} = \frac{\mu_i}{\mathbb{V}(\mathrm{mc}_{it})}\gamma_j a_{ij}^2.$$

Now take logs to the above equation,

$$\log\left(1 - \frac{\widehat{\gamma}_{ij}}{\gamma_j}\right) = \mathrm{const}_i + \log\gamma_j + \log a_{ij}^2, \quad \text{for } j \neq i.$$

as desired. The proof is now complete. □

D.5 Derivation of Pricing Equation (2.6)

We derive pricing equation using the following Corrolary.

Corollary D.1. *At the sectoral level, the optimal pricing strategy with elastic attention is given by*

$$p_{i,t} = \int p_{i,\iota,t} d\iota = \mu_i \, mc_{it}, \quad \textit{where} \quad \mu_i \equiv 1 - \frac{\chi_i}{\theta_i \lambda_i \mathbb{V}(mc_{it})} \in [0,1]. \tag{D.23}$$

Proof. In the preceding analysis, we have derived the optimal pricing decision rules for firm (i, ι) under rational inattention,

$$p_{i\iota t} = \mathbb{E}\left[\mathrm{mc}_{it} \mid x_{i\iota t}\right].$$

Given the optimal signal structure, Bayesian forecasting formula implies that

$$p_{i\iota t} = \mu_i x_{i\iota t} = \mu_i \left(\mathrm{mc}_{it} + u_{i\iota t}\right); \qquad \mu_i = \frac{\mathbb{V}\left(\mathrm{mc}_{it}\right)}{\mathbb{V}\left(\mathrm{mc}_{it}\right) + \nu_i^2} \in [0,1], \tag{D.24}$$

given the independence between sectoral marginal cost and noises. Substitute the formula of endogenous noise variance into the above expression,

$$\mu_i = 1 - \frac{\chi_i}{\lambda_i \theta_i \mathbb{V}\left(\mathrm{mc}_{it}\right)},$$

as desired. Note that μ_i is sector specific because all firms in sector i solve an identical information choice problem. Aggregating the individual firms' decisions in sector i, we obtain

$$p_{it} = \mu_i \mathrm{mc}_{it},$$

where idiosyncratic noises wash out $\int u_{i\iota t} d\iota = 0$. The proof is now complete. □

D.6 Equivalence between Attention and Signal Precision (Sectoral Nominal Rigidity)

In this subsection, we establish equivalence between signal precision of sector i, μ_i, and the total attention capacity chosen by sector i. Let δ_i^* be the endogenous amount of information capacity chosen by firms, measured in terms of nats. By definition,

$$\delta_i^* = \frac{1}{2} \log\det \mathbf{\Sigma}_z - \frac{1}{2} \log\det \mathbf{\Sigma}_{z|x_i}.$$

The following proposition establishes a monotone, one-to-one mapping between δ_i^* and μ_i.

Proposition D.1. *The endogenous attention (total entropy reduction) δ_i^* chosen by firms in sector i is monotonically increasing with μ_i and is given by:*

$$\delta_i^* = -\frac{1}{2} \log\left(1 - \mu_i\right).$$

Proof. Using our previous result (D.16) and partition $\mathbf{U}_i = \begin{bmatrix} \boldsymbol{\zeta}_1^i & \boldsymbol{\zeta}_2^i \end{bmatrix}$ where eigenvector $\boldsymbol{\zeta}_1^i$ corresponds

to d_1^i defined in Appendix D.2,

$$\begin{aligned}
\boldsymbol{\Sigma}_{z|x_i} &= \boldsymbol{\Sigma}_z^{\frac{1}{2}} \begin{bmatrix} \boldsymbol{\zeta}_1^i & \boldsymbol{\zeta}_2^i \end{bmatrix} \begin{bmatrix} \frac{\chi_i}{2d_1^i} & 0 \\ 0 & \mathbf{I}_{N-1} \end{bmatrix} \begin{bmatrix} \boldsymbol{\zeta}_1^i & \boldsymbol{\zeta}_2^i \end{bmatrix}' \boldsymbol{\Sigma}_z^{\frac{1}{2}} \\
&= \boldsymbol{\Sigma}_z^{\frac{1}{2}} \left[\boldsymbol{\zeta}_1^i \left(\boldsymbol{\zeta}_1^i\right)' + \boldsymbol{\zeta}_2^i \left(\zeta_2^i\right)' - \left(1 - \frac{\chi_i}{2d_1^i}\right) \boldsymbol{\zeta}_1^i \left(\boldsymbol{\zeta}_1^i\right)' \right] \boldsymbol{\Sigma}_z^{\frac{1}{2}} \\
&= \boldsymbol{\Sigma}_z^{\frac{1}{2}} \left[\mathbf{I} - \left(1 - \frac{\chi_i}{2d_1^i}\right) \frac{\boldsymbol{\Sigma}_z^{\frac{1}{2}} \boldsymbol{\Omega}_{iz} \boldsymbol{\Sigma}_z^{\frac{1}{2}}}{d_1^i} \right] \boldsymbol{\Sigma}_z^{\frac{1}{2}} \\
&= \boldsymbol{\Sigma}_z - \left[\frac{\left(2d_1^i - \chi_i\right)}{2(d_1^i)^2} \right] \left(\boldsymbol{\Sigma}_z \boldsymbol{\Omega}_{iz} \boldsymbol{\Sigma}_z \right) \\
&= \boldsymbol{\Sigma}_z - \frac{\chi_i}{2R_i d_1^i \nu_i^2} \left(\boldsymbol{\Sigma}_z \boldsymbol{\Omega}_{iz} \boldsymbol{\Sigma}_z \right),
\end{aligned}$$

where we use the definition of unitary matrix $\boldsymbol{\zeta}_1^i(\boldsymbol{\zeta}_1^i)' + \boldsymbol{\zeta}_2^i(\boldsymbol{\zeta}_2^i)' = \mathbf{U}_i \mathbf{U}_i' = \mathbf{I}$ and the derivation of endogenous noise variance in Appendix D.2. Next, we compute the ratio of determinants of prior and posterior covariances,

$$\frac{\det \boldsymbol{\Sigma}_{z|x_i}}{\det \boldsymbol{\Sigma}_z} = \det \boldsymbol{\Sigma}_{z|x_i} \boldsymbol{\Sigma}_z^{-1} = \det \left(\mathbf{I} - \frac{\chi_i}{2R_i d_1^i \nu_i^2} \boldsymbol{\Sigma}_z \boldsymbol{\Omega}_{iz} \right) = \det \left(\mathbf{I} - \frac{\chi_i}{2R_i d_1^i \nu_i^2} \boldsymbol{\Sigma}_z \mathbf{G}_i' \mathbf{G}_i \right),$$

since $\boldsymbol{\Sigma}_z \mathbf{G}_i'$ is a $N \times 1$ vector while $\mathbf{G}_i$ is a $1 \times N$ vector, we employ the matrix determinant lemma for outer product $\boldsymbol{\Sigma}_z \mathbf{G}_i' \mathbf{G}_i$,

$$\det \left(\mathbf{I} - \frac{\chi_i}{2R_i d_1^i \nu_i^2} \boldsymbol{\Sigma}_z \mathbf{G}_i' \mathbf{G}_i \right) = 1 - \frac{\chi_i}{2R_i d_1^i \nu_i^2} \mathbf{G}_i \boldsymbol{\Sigma}_z \mathbf{G}_i' = 1 - \frac{\chi_i}{2d_1^i \nu_i^2} \mathbb{V}\left(\mathrm{mc}_{it}\right) = 1 - \frac{\mathbb{V}\left(\mathrm{mc}_{it}\right)}{\mathbb{V}\left(\mathrm{mc}_{it}\right) + \nu_i^2} = 1 - \mu_i,$$

where the second equality follows from $\mathbb{V}\left(\mathrm{mc}_{it}\right) = R_i^{-1} \parallel \boldsymbol{\Sigma}_z^{\frac{1}{2}} \mathbf{G}_i' \parallel^2$ in (D.10). The last two equalities follows from previous derivations of d_1^i, ν_i^2, and μ_i: (D.11), (D.18), and (D.24). Therefore, it follows that $\delta_i^* = -\frac{1}{2} \log\left(1 - \mu_i\right)$, which completes the proof. To summarize, the isomorphic relation between δ_i^* and μ_i indicates that sectoral attentions and signal precisions, and therefore nominal rigidities, are indeed equivalent. □

D.7 Proof of Lemma 2.2

Assume that the equilibrium sectoral price follows $\boldsymbol{p}_t = \boldsymbol{\phi} \boldsymbol{z}_t$. The marginal cost , or the ideal price, is given by

$$\mathbf{mc}_t = -\boldsymbol{z}_t + \boldsymbol{\alpha} w_t + \mathbf{A} \boldsymbol{p}_t,$$

which generalizes (2.2). Meanwhile, (2.6) indicates that equilibrium price vector is given by

$$\boldsymbol{p}_t = \boldsymbol{\mu}\mathbf{mc}_t,$$

where $\boldsymbol{\mu} = \text{diag}\left(\{\mu_i\}_{i=1}^N\right)$. Combining these two equation leads to

$$\boldsymbol{p}_t = (\mathbf{I} - \boldsymbol{\mu}\mathbf{A})^{-1}\boldsymbol{\mu}(-\mathbf{I} + \boldsymbol{\alpha}\boldsymbol{\kappa})\boldsymbol{z}_t; \qquad \boldsymbol{\phi} = (\mathbf{I} - \boldsymbol{\mu}\mathbf{A})^{-1}\boldsymbol{\mu}(-\mathbf{I} + \boldsymbol{\alpha}\boldsymbol{\kappa}). \tag{D.25}$$

D.8 Proof of Proposition 2.4

Proof. Given that $\boldsymbol{p}_t\boldsymbol{\phi}\boldsymbol{z}_t$, the marginal cost can be solved as

$$\mathbf{mc}_t = \left(\mathbf{I} + \mathbf{A}(\mathbf{I} - \boldsymbol{\mu}\mathbf{A})^{-1}\boldsymbol{\mu}\right)(-\mathbf{I} + \boldsymbol{\alpha}\boldsymbol{\kappa})\boldsymbol{z}_t.$$

Next, we present a technical Lemma using the theory of nonnegative matrices. This result will be applied repeatedly in subsequent analysis.

Lemma D.3. *For any vector $\boldsymbol{\mu}^v \in [0,1]^N$,*[46] *the spectral radius of the non-negative matrix $\boldsymbol{\mu}\mathbf{A}$ is strictly smaller than 1, $\rho(\boldsymbol{\mu}\mathbf{A}) < 1$. Therefore, $\mathbf{I} - \boldsymbol{\mu}\mathbf{A}$ is a M matrix.*

Proof. By definition,

$$\boldsymbol{\mu}\mathbf{A} = \begin{bmatrix} a_{11}\mu_1 & a_{12}\mu_1 & \dots & a_{1N}\mu_1 \\ a_{21}\mu_2 & a_{22}\mu_2 & \dots & a_{2N}\mu_2 \\ \vdots & \vdots & \ddots & \dots \\ a_{N1}\mu_N & a_{N2}\mu_N & \dots & a_{NN}\mu_N \end{bmatrix},$$

is nonnegative. Therefore, we employ the following Theorem in matrix analysis,

Theorem (Gershgorin Circle Theorem). *For an $N \times N$ matrix $\mathbb{L}$, each eigenvalue z of $\mathbb{L}$ satisfy the following conditions,*

$$|z - l_{ii}| \leq \sum_{j=1;j\neq i}^{N} |l_{ij}|, \quad i \in \{1, 2, \cdots, N\},$$

where l denotes element of $\mathbb{L}$.

By the Gershgorin circle theorem and the Perron-Frobenius Theorem (Berman and Plemmons (1994), Chapter 2), the absolute value of maximal eigenvalue of $\boldsymbol{\mu}\mathbf{A}$, or its spectral radius $\rho(\boldsymbol{\mu}\mathbf{A})$, is

[46] Recall that we define $\boldsymbol{\mu}^v = \text{diag}(\boldsymbol{\mu})$.

bounded above,[47]

$$\rho\left(\boldsymbol{\mu}\mathbf{A}\right) \leq \max_i \left\{ \sum_{j=1}^{N} a_{ij}\mu_j : i = 1, 2, ...N \right\} \leq \max_i \left\{ \sum_{j=1}^{N} a_{ij} : i = 1, 2, ...N \right\} = \left\|\mathbf{A}\right\|_\infty < 1,$$

where constant $\left\|\mathbf{A}\right\|_\infty \in (0, 1)$ is the matrix operator norm induced by the maximal row-sum of $\mathbf{A}$.[48]

$$\left\|\mathbf{A}\right\|_\infty = \max_i \left\{ \sum_{j=1}^{N} |a_{ij}| : i = 1, 2, ...N \right\} = \max_i \left\{ \sum_{j=1}^{N} a_{ij} : i = 1, 2, ...N \right\}. \tag{D.26}$$

By definition in Plemmons (1977), $(\mathbf{I} - \boldsymbol{\mu}\mathbf{A})$ is a M matrix, such that $(\mathbf{I} - \boldsymbol{\mu}\mathbf{A})^{-1}$ exists and is nonnegative. In other words, the matrix inverse admits series expansion as

$$(\mathbf{I} - \boldsymbol{\mu}\mathbf{A})^{-1} = \sum_{n=0}^{\infty} (\boldsymbol{\mu}\mathbf{A})^n = \mathbf{I} + (\boldsymbol{\mu}\mathbf{A}) + (\boldsymbol{\mu}\mathbf{A})^2 + ... > 0.$$

□

Given the matrix expansion, we have

$$\mathbf{I} + \mathbf{A}\left(\mathbf{I} - \boldsymbol{\mu}\mathbf{A}\right)^{-1}\boldsymbol{\mu} = \mathbf{I} + \mathbf{A}\left(\mathbf{I} + (\boldsymbol{\mu}\mathbf{A}) + (\boldsymbol{\mu}\mathbf{A})^2 + ...\right)\boldsymbol{\mu} = \mathbf{I} + (\mathbf{A}\boldsymbol{\mu}) + (\mathbf{A}\boldsymbol{\mu})^2 + ... = (\mathbf{I} - \mathbf{A}\boldsymbol{\mu})^{-1} > 0 \tag{D.27}$$

where the matrix $(\mathbf{I} - \mathbf{A}\boldsymbol{\mu})^{-1}$ is also a M matrix with spectral radius bounded by 1. We also have an important matrix identity

$$\boldsymbol{\mu}\left(\mathbf{I} - \mathbf{A}\boldsymbol{\mu}\right)^{-1} = \left(\mathbf{I} - \boldsymbol{\mu}\mathbf{A}\right)^{-1}\boldsymbol{\mu}, \tag{D.28}$$

which will be used repeatedly in subsequent analysis. Note that above results DOES NOT rely on the invertibility of $\boldsymbol{\mu}$. In addition, the volatility of marginal cost is now given by

$$\mathbb{V}\left(mc_{i,t}\right) = \left\| \mathbf{e}_i(\mathbf{I} - \mathbf{A}\boldsymbol{\mu})^{-1}(-\mathbf{I} + \boldsymbol{\alpha}\boldsymbol{\kappa})\boldsymbol{\Sigma}_z^{\frac{1}{2}} \right\|^2.$$

To summarize, the fixed-point of nominal rigidities (sectoral attentions) is given by (2.2) while influence matrix $\boldsymbol{\phi}$ is given by (D.25), as desired. The proof is now complete. □

D.9 Proof of Proposition 2.5

Proof. We postpone the proof of Proposition 2.5 in (E.48) of Appendix E.7. □

[47] If $\boldsymbol{\mu}\mathbf{A} >> 0$ is strictly positive, then the Perron-Frobenius Theorem implies that there exists a unique positive and maximal (dominant) eigenvalue that is equals to its spectral radius.

[48] Here we adopt Horn and Johnson (2012)'s notation on matrix norms.

D.10 Equilibrium Existence and Uniquness

The highly-nonlinear equilibrium system characterized in Proposition 2.4 does not admits closed-form solution in general. Additionally, multiple equilibria is also a pervasive phenomenon in this types of models with endogenous information acquisition and strategic complementarity (Hellwig and Veldkamp, 2009). Equilibrium multiplicity invalidates the use of comparative statics methods to analyze the impact of parameter perturbations on equilibrium outcomes; it also complicates welfare and policy analysis. To address this problem, we present the following proposition that guarantees equilibrium uniqueness.

Proposition D.2. *There exists a unique fixed point $\boldsymbol{\mu}$ that satisfies condition (2.8) if for each sector $i = 1, 2, ...N$,*

1. *The monetary policy accommodate a wage rule that satisfies $\kappa_i < 1$;*[49]
2. *The information cost χ_i satisfies $0 < \chi_i < \theta_i \lambda_i \min\left\{\sigma_i^2 \left(1 - \varkappa_i \kappa_i\right)^2,\ \sigma_i^4 \left(1 - \varkappa_i \kappa_i\right)^4 (2\varsigma_i)^{-1}\right\}$. Denote $\boldsymbol{\Gamma} = -\mathbf{I} + \boldsymbol{\alpha}\boldsymbol{\kappa}$, and the auxiliary parameters $\{\varkappa_i\}$ and $\{\varsigma_i\}$ are defined as*

$$\varkappa_i = \begin{cases} 1, & 0 \leq \kappa_i < 1 \\ \alpha_i, & k_i < 0 \end{cases}, \quad \textit{and} \quad \varsigma_i = \operatorname{tr}\left(\mathbf{L}\middle|\boldsymbol{\Gamma}\boldsymbol{\Sigma}_z\boldsymbol{\Gamma}'\middle|\mathbf{L}'\mathbf{e}_i'\mathbf{e}_i\mathbf{L}\mathbf{A}\right),$$

where $\mathbf{L} = (\mathbf{I} - \mathbf{A})^{-1}$ is the Leontief inverse with perfect information

Proposition D.2 provides a set of sufficient conditions that ensures equilibrium existence and uniqueness. The proof of this result utilizes the Kellogg's Fixed Point Theorem in Banach spaces (Kellogg, 1976). We believe that technical strategy we adopt to derive these results has its independent value in methodology. In particular, it can be applied to other high-dimensional models of endogenous information frictions when establishing existence and uniqueness properties.

Proof. I present the proofs of this proposition in two parts. The first part proves the equilibrium existence. The second part establishes equilibrium uniqueness.

Part I: Proof of Existence.

Recall from (2.8), the price rigidities $\left\{\mu_i\right\}_{i=1}^N$ solve the following fixed-point problem

$$\mu_i = 1 - \frac{\chi_i}{\theta_i \lambda_i \mathbb{V}(\mathrm{mc}_{it})}, \qquad \text{and} \qquad \mathbb{V}(\mathrm{mc}_{it}) = \left\|\mathbf{e}_i(\mathbf{I} - \mathbf{A}\boldsymbol{\mu})^{-1}(-\mathbf{I} + \boldsymbol{\alpha}\boldsymbol{\kappa})\boldsymbol{\Sigma}_z^{\frac{1}{2}}\right\|^2.$$

Given the N-dimensional column vector of interest, $\boldsymbol{\mu}^v = \left(\mu_1, \mu_2, ...\mu_N\right)' \in \mathbb{R}^N$, we define an N-dimensional vector-valued function $\mathcal{T}\left(\boldsymbol{\mu}^v\right) : [0,1]^N \longmapsto \mathbb{R}^N$ as

$$\mathcal{T}\left(\boldsymbol{\mu}^v\right) = \left(\mathcal{T}_1\left(\boldsymbol{\mu}\right), \mathcal{T}_2\left(\boldsymbol{\mu}\right), ...\mathcal{T}_N\left(\boldsymbol{\mu}\right)\right)'; \tag{D.29}$$

[49] In Appendix E.4, we show that for any given wage function, there exists a corresponding monetary policy rule that can support it in equilibrium. Therefore, the monetary authority can alternatively chooses the wage function and derive the monetary supply rule accordingly.

$$\mathcal{T}_i(\boldsymbol{\mu}^v) = 1 - \frac{\chi_i}{\theta_i \lambda_i \left\| \mathbf{e}_i (\mathbf{I} - \mathbf{A}\boldsymbol{\mu})^{-1} (-\mathbf{I} + \boldsymbol{\alpha}\boldsymbol{\kappa}) \boldsymbol{\Sigma}_z^{\frac{1}{2}} \right\|^2}. \tag{D.30}$$

Then the fixed-point system can be written as

$$\mathcal{T}(\boldsymbol{\mu}^v) = \boldsymbol{\mu}^v.$$

By inspection, $\mathcal{T}(\boldsymbol{\mu})$ is continuous on $[0,1]^N$, and the Cartesian cube $[0,1]^N \subseteq \mathbb{R}^N$ is compact and convex. Therefore, we invoke the Brouwer fixed-point theorem in the N-dimensional Euclidean (Banach) space $\mathbb{R}^N$. We only need to show that $\mathcal{T}(\boldsymbol{\mu})$ defines a self-map from $[0,1]^N$ onto itself.

Using Lemma D.3, it is easy to see that $(\mathbf{I} - \mathbf{A}\boldsymbol{\mu})$ is a M matrix, such that $(\mathbf{I} - \mathbf{A}\boldsymbol{\mu})^{-1}$ exists and is nonnegative. In other words, the inverse admits the series expansion as $(\mathbf{I} - \mathbf{A}\boldsymbol{\mu})^{-1} = \sum_{n=0}^{\infty} (\mathbf{A}\boldsymbol{\mu})^n = \mathbf{I} + (\mathbf{A}\boldsymbol{\mu}) + (\mathbf{A}\boldsymbol{\mu})^2 + ... > 0$.

Now define the attention-distorted Leontief matrix,

$$\boldsymbol{\Delta}_\mu = (\mathbf{I} - \mathbf{A}\boldsymbol{\mu})^{-1} > 0, \tag{D.31}$$

and the wage-related coefficient matrix $\boldsymbol{\Gamma} = (-\mathbf{I} + \boldsymbol{\alpha}\boldsymbol{\kappa})$. Then follows a proposition.

Proposition D.3. *If the monetary policy (wage) rule satisfy*

$$\kappa_i < 1; \qquad \forall i = 1, 2, ...N,$$

then each sector's ideal price (marginal cost) volatility has a positive lower bound,

$$\mathbb{V}(mc_{it}) = \left\| \mathbf{e}_i (\mathbf{I} - \mathbf{A}\boldsymbol{\mu})^{-1} (-\mathbf{I} + \boldsymbol{\alpha}\boldsymbol{\kappa}) \boldsymbol{\Sigma}_z^{\frac{1}{2}} \right\|^2 \geq \sigma_i^2 (1 - \varkappa_i \kappa_i)^2 > 0,$$

where the auxiliary parameters $\{\varkappa_i\}$ is defined as

$$\varkappa_i = \begin{cases} 1, & 0 \leq \kappa_i < 1 \\ \alpha_i, & \kappa_i < 0 \end{cases}.$$

Proof. By definition of the Euclidean norm,[50]

$$\mathbb{V}(\mathrm{mc}_{it}) = \left\| \mathbf{e}_i (\mathbf{I} - \mathbf{A}\boldsymbol{\mu})^{-1} (-\mathbf{I} + \boldsymbol{\alpha}\boldsymbol{\kappa}) \boldsymbol{\Sigma}_z^{\frac{1}{2}} \right\|^2 = \left\| \mathbf{e}_i \boldsymbol{\Delta}_\mu \boldsymbol{\Gamma} \boldsymbol{\Sigma}_z^{\frac{1}{2}} \right\|^2 \geq \left\| \left[\boldsymbol{\Delta}_\mu \boldsymbol{\Gamma} \right]_{ii} \mathbf{e}_i \boldsymbol{\Sigma}_z^{\frac{1}{2}} \right\|^2, \tag{D.32}$$

where the scalar operator $\left[\cdot \right]_i$ or $\left[\cdot \right]_{ii}$ denotes the i-th or ii-th element of the vector or matrix, and

[50] The volatility bound we establish here is obviously non-unique and can be futher relaxed, but the current condition suffices for our purpose.

we use the fact that $\mathbf{\Sigma_z} > 0$ is diagonal under our baseline assumption. Next, we observe that

$$\begin{aligned}\mathbf{\Delta}_\mu \mathbf{\Gamma} &= (\mathbf{I} - \mathbf{A}\boldsymbol{\mu})^{-1} (-\mathbf{I} + \boldsymbol{\alpha}\boldsymbol{\kappa}) \\ &= -\left[\mathbf{I} + (\mathbf{A}\boldsymbol{\mu}) + (\mathbf{A}\boldsymbol{\mu})^2 + ...\right] + \left[\mathbf{I} + (\mathbf{A}\boldsymbol{\mu}) + (\mathbf{A}\boldsymbol{\mu})^2 + ...\right](\boldsymbol{\alpha}\boldsymbol{\kappa}).\end{aligned} \tag{D.33}$$

By definition of $\mathbf{A}$ and $\boldsymbol{\mu}$, $0 \leq \left[(\mathbf{A}\boldsymbol{\mu}) + (\mathbf{A}\boldsymbol{\mu})^2 + ...\right] \leq \left[(\mathbf{A}) + (\mathbf{A})^2 + ...\right]$. It is immediate that (D.33) is bounded by

$$\mathbf{\Delta}_\mu \mathbf{\Gamma} \leq -\mathbf{I} + \left[\mathbf{I} + (\mathbf{A}\boldsymbol{\mu}) + (\mathbf{A}\boldsymbol{\mu})^2 + ...\right](\boldsymbol{\alpha}\boldsymbol{\kappa}).$$

Notice that the ith row of matrix $\left[\mathbf{I} + (\mathbf{A}\boldsymbol{\mu}) + (\mathbf{A}\boldsymbol{\mu})^2 + ...\right](\boldsymbol{\alpha}\boldsymbol{\kappa})$ is $\left[\mathbf{I} + (\mathbf{A}\boldsymbol{\mu}) + (\mathbf{A}\boldsymbol{\mu})^2 + ...\right]\boldsymbol{\alpha}\kappa_i$, for $i = 1, 2, ...N$. On one hand, when $\kappa_i < 0$,

$$\left[\mathbf{\Delta}_\mu \mathbf{\Gamma}\right]_{ii} \leq \left[-\mathbf{I} + \left[\mathbf{I} + (\mathbf{A}\boldsymbol{\mu}) + (\mathbf{A}\boldsymbol{\mu})^2 + ...\right](\boldsymbol{\alpha}\boldsymbol{\kappa})\right]_{ii} = -1 + \left[\left[\mathbf{I} + (\mathbf{A}\boldsymbol{\mu}) + (\mathbf{A}\boldsymbol{\mu})^2 + ...\right]\boldsymbol{\alpha}\kappa_i\right]_i \leq -1 + \alpha_i\kappa_i < 0. \tag{D.34}$$

On the other hand, when $0 \leq \kappa_i < 1$, the column vector $\left[\mathbf{I} + (\mathbf{A}\boldsymbol{\mu}) + (\mathbf{A}\boldsymbol{\mu})^2 + ...\right]\boldsymbol{\alpha}\kappa_i$ is bounded as

$$\left[\mathbf{I} + (\mathbf{A}\boldsymbol{\mu}) + (\mathbf{A}\boldsymbol{\mu})^2 + ...\right]\boldsymbol{\alpha}\kappa_i \leq \left[\mathbf{I} + (\mathbf{A}) + (\mathbf{A})^2 + ...\right]\boldsymbol{\alpha}\kappa_i = (\mathbf{I} - \mathbf{A})^{-1}\boldsymbol{\alpha}\kappa_i = \mathbf{1}\kappa_i,$$

where I use the matrix property of $(\mathbf{I} - \mathbf{A})^{-1}\boldsymbol{\alpha} = \mathbf{1} = (1, 1, ...1)'$. In this case,

$$\left[\mathbf{\Delta}_\mu \mathbf{\Gamma}\right]_{ii} \leq \left[-\mathbf{I} + \left[\mathbf{I} + (\mathbf{A}\boldsymbol{\mu}) + (\mathbf{A}\boldsymbol{\mu})^2 + ...\right](\boldsymbol{\alpha}\boldsymbol{\kappa})\right]_{ii} \leq -1 + [\kappa_i \mathbf{1}]_i = -1 + \kappa_i < 0, \tag{D.35}$$

where the last inequality follows from the assumption that $\kappa_i < 1$. Hence, by (D.34) and (D.35), all diagonal elements of matrix $\mathbf{\Delta}_\mu \mathbf{\Gamma}$ are negative under the assumption of $\kappa_i < 1$, that is,

$$-\left[\mathbf{\Delta}_\mu \mathbf{\Gamma}\right]_{ii} > 0; \qquad \forall i = 1, 2, ...N.$$

Therefore, using (D.32), we conclude that

$$\mathbb{V}(\mathrm{mc}_{it}) = \left\|\mathbf{e}_i \mathbf{\Delta}_\mu \mathbf{\Gamma} \mathbf{\Sigma}_z^{\frac{1}{2}}\right\|^2 \geq \left\|\left[\mathbf{\Delta}_\mu \mathbf{\Gamma}\right]_{ii} \mathbf{e}_i \mathbf{\Sigma}_z^{\frac{1}{2}}\right\|^2 \geq \begin{cases} \sigma_i^2 (1 - \kappa_i)^2, & 0 \leq \kappa_i < 1. \\ \sigma_i^2 (1 - \alpha_i \kappa_i)^2, & \kappa_i < 0. \end{cases}$$

The proof is complete. □

By Proposition D.3, when $\boldsymbol{\kappa}$ satisfies $\kappa_i < 1, \forall i$,

$$0 < \frac{\chi_i}{\theta_i \lambda_i \mathbb{V}(\mathrm{mc}_{it})} \leq \frac{\chi_i}{\theta_i \lambda_i \sigma_i^2 (1 - \varkappa_i \kappa_i)^2} < 1, \tag{D.36}$$

where the last inequality follow directly from our sufficiency condition in Proposition D.2. The first

inequality follows that $(\mathbf{I}-\mathbf{A}\boldsymbol{\mu})^{-1}$ exists and is finite, the sectoral volatilities are finite, and that all sectors face non-zero information cost, i.e. $\chi_i > 0, \forall i$.

It is then immediate that for all sector $i = 1, 2, ...N$ and all vectors $\boldsymbol{\mu}^v \in [0,1]^N$,

$$\mathcal{T}_i(\boldsymbol{\mu}^v) = 1 - \frac{\chi_i}{\theta_i \lambda_i \left\| \mathbf{e}_i(\mathbf{I}-\mathbf{A}\boldsymbol{\mu})^{-1}(-\mathbf{I}+\boldsymbol{\alpha}\boldsymbol{\kappa})\boldsymbol{\Sigma}_z^{\frac{1}{2}} \right\|^2} \in (0,1), \qquad \forall i = 1, 2, ...N; \tag{D.37}$$

therefore, the N-dimensional vector-valued function $\mathcal{T}(\boldsymbol{\mu}^v) : [0,1]^N \longmapsto [0,1]^N$ is a continuous self-map. Then by the Brouwer fixed-point theorem (Hutson et al. (2005), Theorem 8.1.1), the equilibrium system in condition (2.8) has a (interior) fixed-point.

Part II: Proof of Uniqueness.

The key novel result in Proposition D.2 is the global uniqueness of general equilibrium. In models with endogenous information frictions (e.g. RI or learning from endogenous price), unique equilibrium is hard to prove and characterize, if not impossible (Adams (2019)). Due to the nonlinearity of equilibrium operator arosen from endogenous information, the Contraction Mapping Theorem cannot be applied. Instead, we employ the following extension of the Schauder's fixed-point theorem by Kellogg (1976):

Kellogg's Fixed Point Theorem. *Let X be a real Banach space with a bounded convex open subset $D \subseteq X$. Let $\bar{D}$ be its closure and let $F : \bar{D} \to \bar{D}$ be a compact continuous map which is continuously Fréchet differentiable on D. Suppose that (a) for each $x \in D$, 1 is not an eigenvalue of Fréchet derivative $F'(x)$, and (b) for each $x \in \partial D$ on boundary, $x \neq F(x)$. Then F has a unique fixed point.*

Kellogg's Theorem applies to infinite-dimensional Banach spaces. For discussions of this theorem, see Talman (1978) and Smith and Stuart (1980). As note in Kellogg's original paper, the compactness hypothesis can be dropped in finite-dimensional problems. So in our proof, we will omit the compactness condition.

We proceed with the proof of uniqueness in three steps.

Step 1: Compute the Fréchet derivative. Suppose now $\boldsymbol{\mu}^v \in [0,1]^N$. In our finite-dimensional problem, the Fréchet derivative of the vector-valued function $\mathcal{T}(\boldsymbol{\mu}^v)$ is the $N \times N$ Jacobian matrix,

$$\mathcal{T}'(\boldsymbol{\mu}^v) \equiv \frac{\partial \mathcal{T}}{\partial \boldsymbol{\mu}^v} = \mathcal{T}_{\boldsymbol{\mu}^v} = \begin{bmatrix} \frac{\partial \mathcal{T}_1}{\partial \mu_1} & \frac{\partial \mathcal{T}_1}{\partial \mu_2} & \cdots & \frac{\partial \mathcal{T}_1}{\partial \mu_N} \\ \frac{\partial \mathcal{T}_2}{\partial \mu_1} & \frac{\partial \mathcal{T}_2}{\partial \mu_2} & \cdots & \frac{\partial \mathcal{T}_2}{\partial \mu_N} \\ \vdots & \vdots & \vdots & \vdots \\ \frac{\partial \mathcal{T}_N}{\partial \mu_1} & \frac{\partial \mathcal{T}_N}{\partial \mu_2} & \cdots & \frac{\partial \mathcal{T}_N}{\partial \mu_N} \end{bmatrix}. \tag{D.38}$$

On the other hand, we define a matrix of derivatives for the ith component of $\mathcal{T}$ with respect to a

$N \times N$ **symmetric** matrix of variable, $\boldsymbol{\mu}$,

$$\frac{\partial \mathcal{T}_i(\boldsymbol{\mu})}{\partial \boldsymbol{\mu}} = \begin{bmatrix} \frac{\partial \mathcal{T}_i}{\partial \mu_{11}} & \frac{\partial \mathcal{T}_i}{\partial \mu_{12}} & \cdots & \frac{\partial \mathcal{T}_i}{\partial \mu_{1N}} \\ \frac{\partial \mathcal{T}_i}{\partial \mu_{21}} & \frac{\partial \mathcal{T}_i}{\partial \mu_{22}} & \cdots & \frac{\partial \mathcal{T}_i}{\partial \mu_{2N}} \\ \vdots & \vdots & \vdots & \vdots \\ \frac{\partial \mathcal{T}_i}{\partial \mu_{N1}} & \frac{\partial \mathcal{T}_i}{\partial \mu_{N2}} & \cdots & \frac{\partial \mathcal{T}_i}{\partial \mu_{NN}} \end{bmatrix}; \qquad i = 1, 2, ...N \tag{D.39}$$

here the diagonal elements $\mu_{11}, \mu_{22}, ...\mu_{NN}$ coincide with $\boldsymbol{\mu}^v$. The following definition establishes the relation between the Jacobian matrix and the matrix-valued derivatives.

Definition 1. *The Jacobian matrix $\mathcal{T}_{\boldsymbol{\mu}^v}$ is obtained by extracting the diagonal elements of $\frac{\partial \mathcal{T}_i(\boldsymbol{\mu})}{\partial \boldsymbol{\mu}}$ for each sector, arranged in rows.*

$$\mathcal{T}_{\boldsymbol{\mu}^v} = \frac{\partial \mathcal{T}\left(\boldsymbol{\mu}^v\right)}{\partial \boldsymbol{\mu}^v} = \left[\operatorname{diag}\left(\frac{\partial \mathcal{T}_1(\boldsymbol{\mu})}{\partial \boldsymbol{\mu}}\right) \quad \operatorname{diag}\left(\frac{\partial \mathcal{T}_2(\boldsymbol{\mu})}{\partial \boldsymbol{\mu}}\right) \quad \cdots \quad \operatorname{diag}\left(\frac{\partial \mathcal{T}_N(\boldsymbol{\mu})}{\partial \boldsymbol{\mu}}\right) \right]'. \tag{D.40}$$

Relation (D.40) holds either we treat $\boldsymbol{\mu}$ as a general $N \times N$ symmetric matrix of variables or as a diagonal matrix with off-diagonal restrictions of constant 0. Therefore, in the first step we treat $\boldsymbol{\mu}$ as a general $N \times N$ variable matrix, and exploit its diagonal (symmetric) property whenever is necessary.

It suffices to calculate the matrix of derivatives of the scalar function, $\frac{\partial \mathcal{T}_i(\boldsymbol{\mu})}{\partial \boldsymbol{\mu}}$. Recall that $\mathcal{T}_i(\boldsymbol{\mu}) = 1 - \frac{\chi_i}{\theta_i \lambda_i \mathbb{V}(\mathrm{mc}_{it})}$, using the chain rule for matrix derivatives (Magnus and Neudecker (2007)),

$$\frac{\partial \mathcal{T}_i(\boldsymbol{\mu})}{\partial \boldsymbol{\mu}} = \frac{\chi_i}{\theta_i \lambda_i \mathbb{V}(\mathrm{mc}_{it})^2} \frac{\partial \mathbb{V}(\mathrm{mc}_{it})}{\partial \boldsymbol{\mu}}, \tag{D.41}$$

where the volatility of marginal cost admits matrix representation,

$$\mathbb{V}(\mathrm{mc}_{it}) = \left\| \mathbf{e}_i (\mathbf{I} - \mathbf{A}\boldsymbol{\mu})^{-1} (-\mathbf{I} + \boldsymbol{\alpha}\boldsymbol{\kappa}) \boldsymbol{\Sigma}_z^{\frac{1}{2}} \right\|^2 = \left\| \mathbf{e}_i \boldsymbol{\Delta}_\mu \boldsymbol{\Gamma} \boldsymbol{\Sigma}_z^{\frac{1}{2}} \right\|^2 = \mathbf{e}_i \boldsymbol{\Delta}_\mu \boldsymbol{\Gamma} \boldsymbol{\Sigma}_z \boldsymbol{\Gamma}' \boldsymbol{\Delta}_\mu' \mathbf{e}_i'.$$

Recall that $\boldsymbol{\Delta}_\mu = (\mathbf{I} - \mathbf{A}\boldsymbol{\mu})^{-1}$ from (D.31). To compute the matrix derivative $\frac{\partial \mathbb{V}(\mathrm{mc}_{it})}{\partial \boldsymbol{\mu}}$, we start with matrix differentials and present 3 sets of basic results using matrix calculus.

Lemma D.4. *Suppose g is a scalar function of $N \times N$ matrix of variable $\boldsymbol{\mu}$, its differential follows,*

$$dg(\boldsymbol{\mu}) = \operatorname{tr}\left(\frac{\partial g(\boldsymbol{\mu})}{\partial \boldsymbol{\mu}'} d\boldsymbol{\mu} \right). \tag{D.42}$$

Proof. By definition,

$$dg(\boldsymbol{\mu}) = \sum_{i=1}^{N} \sum_{j=1}^{N} \frac{\partial g}{\partial \mu_{ij}} d\mu_{ij}$$

$$= \operatorname{tr}\left(\begin{bmatrix} \frac{\partial g}{\partial \mu_{11}} & \frac{\partial g}{\partial \mu_{21}} & \cdots & \frac{\partial g}{\partial \mu_{N1}} \\ \frac{\partial g}{\partial \mu_{12}} & \frac{\partial g}{\partial \mu_{22}} & \cdots & \frac{\partial g}{\partial \mu_{N2}} \\ & \cdots & & \\ \frac{\partial g}{\partial \mu_{1N}} & \frac{\partial g}{\partial \mu_{2N}} & \cdots & \frac{\partial g}{\partial \mu_{NN}} \end{bmatrix} \begin{bmatrix} d\mu_{11} & d\mu_{12} & \cdots & d\mu_{1N} \\ d\mu_{21} & d\mu_{22} & \cdots & d\mu_{2N} \\ & \cdots & & \\ d\mu_{N1} & d\mu_{N2} & \cdots & d\mu_{NN} \end{bmatrix} \right)$$

$$= \operatorname{tr}\left(\frac{\partial g(\boldsymbol{\mu})}{\partial \boldsymbol{\mu}'} d\boldsymbol{\mu} \right),$$

where $\operatorname{tr}(\cdot)$ is the trace operator. □

The trace operator is linear and has basic properties $\operatorname{tr}(\boldsymbol{XY}) = \operatorname{tr}(\boldsymbol{YX})$ and $\operatorname{tr}(\boldsymbol{X}) = \operatorname{tr}(\boldsymbol{X}')$ for any $N \times N$ matrices $\boldsymbol{X}$ and $\boldsymbol{Y}$.

Lemma D.5. *For matrix of variables $\boldsymbol{X}$, $\boldsymbol{Y}$, and constant matrices $\mathbf{B}$ and $\mathbf{C}$ of conformable dimensions, we have following differential results,*

$$d(\mathbf{BXC}) = \mathbf{B}d(\boldsymbol{X})\mathbf{C}; \qquad d\left(\boldsymbol{X}^{-1}\right) = -\boldsymbol{X}^{-1}d(\boldsymbol{X})\boldsymbol{X}^{-1};$$
$$d(\operatorname{tr}(\boldsymbol{X})) = \operatorname{tr}(d(\boldsymbol{X})); \qquad d(\boldsymbol{XY}) = d(\boldsymbol{X})\boldsymbol{Y} + \boldsymbol{X}d(\boldsymbol{Y}).$$

Proof. See Magnus and Neudecker (2007), p.160, p.208; Vetter (1973), p.357 □

Lemma D.6. *Given constant N- dimensional column vectors $\mathbf{b}$ and $\mathbf{c}$, a constant $N \times N$ matrix $\mathbf{B}$, and a matrix of variables $\boldsymbol{X}$, the scalar function $\mathbf{b}'\boldsymbol{X}\mathbf{B}\boldsymbol{X}'\mathbf{c}$ of variable matrix $\boldsymbol{X}$ admits differential*

$$d\left(\mathbf{b}'\boldsymbol{X}\mathbf{B}\boldsymbol{X}'\mathbf{c}\right) = \operatorname{tr}\left[\mathbf{b}'d(\boldsymbol{X})\mathbf{B}\boldsymbol{X}'\mathbf{c} + \mathbf{c}'d(\boldsymbol{X})\mathbf{B}'\boldsymbol{X}'\mathbf{b}\right]. \tag{D.43}$$

Proof. Using formulas from the last lemma and the the basic properties of trace operator we deduce that

$$\begin{aligned} d\left(\mathbf{b}'\boldsymbol{X}\mathbf{B}\boldsymbol{X}'\mathbf{c}\right) &= d\left(\operatorname{tr}\left(\mathbf{b}'\boldsymbol{X}\mathbf{B}\boldsymbol{X}'\mathbf{c}\right)\right) \\ &= \operatorname{tr}\left(d\left(\mathbf{b}'\boldsymbol{X}\mathbf{B}\boldsymbol{X}'\mathbf{c}\right)\right) \\ &= \operatorname{tr}\left(\mathbf{b}'d\left(\boldsymbol{X}\mathbf{B}\boldsymbol{X}'\right)\mathbf{c}\right) \\ &= \operatorname{tr}\left[\mathbf{b}'\left(d(\boldsymbol{X})\mathbf{B}\boldsymbol{X}' + \boldsymbol{X}\mathbf{B}d\left(\boldsymbol{X}'\right)\right)\mathbf{c}\right] \\ &= \operatorname{tr}\left[\mathbf{b}'d(\boldsymbol{X})\mathbf{B}\boldsymbol{X}'\mathbf{c}\right] + \operatorname{tr}\left[\mathbf{b}'\boldsymbol{X}\mathbf{B}d\left(\boldsymbol{X}'\right)\mathbf{c}\right] \\ &= \operatorname{tr}\left[\mathbf{b}'d(\boldsymbol{X})\mathbf{B}\boldsymbol{X}'\mathbf{c}\right] + \operatorname{tr}\left[\mathbf{c}'d(\boldsymbol{X})\mathbf{B}'\boldsymbol{X}'\mathbf{b}\right] \\ &= \operatorname{tr}\left[\mathbf{b}'d(\boldsymbol{X})\mathbf{B}\boldsymbol{X}'\mathbf{c} + \mathbf{c}'d(\boldsymbol{X})\mathbf{B}'\boldsymbol{X}'\mathbf{b}\right]. \end{aligned}$$

□

Combining results from Lemma D.4 - D.6, we compute the differential of modified Leontief inverse

$\boldsymbol{\Delta}_\mu = (\mathbf{I} - \mathbf{A}\boldsymbol{\mu})^{-1}$ as $d\left(\boldsymbol{\Delta}_\mu\right) = -\boldsymbol{\Delta}_\mu d\left(\boldsymbol{\Delta}_\mu^{-1}\right)\boldsymbol{\Delta}_\mu = \boldsymbol{\Delta}_\mu \mathbf{A} d(\boldsymbol{\mu})\boldsymbol{\Delta}_\mu$. Similarly,

$$d\left(\boldsymbol{\Delta}_\mu'\right) = \left(d\left(\boldsymbol{\Delta}_\mu\right)\right)' = \boldsymbol{\Delta}_\mu' \left(d(\boldsymbol{\mu})\right)' \mathbf{A}'\boldsymbol{\Delta}_\mu' = \boldsymbol{\Delta}_\mu' d(\boldsymbol{\mu})\mathbf{A}'\boldsymbol{\Delta}_\mu',$$

where we exploit the diagonal (symmetric) property of $\boldsymbol{\mu}$ in the last equality. Next, we derive the differential for the volatility of marginal cost,

Lemma D.7. *The volatility of marginal cost, $\mathbb{V}(mc_{it})$ admits differential,*

$$d\left[\mathbb{V}(mc_{it})\right] = 2\,\mathrm{tr}\left[\boldsymbol{\Delta}_\mu \boldsymbol{\Gamma}\boldsymbol{\Sigma}_z \boldsymbol{\Gamma}'\boldsymbol{\Delta}_\mu' \mathbf{e}_i' \mathbf{e}_i \boldsymbol{\Delta}_\mu \mathbf{A} d(\boldsymbol{\mu})\right]. \tag{D.44}$$

Proof. Combining results from Lemma D.4-D.6,

$$\begin{aligned}
d\left[\mathbb{V}(\mathrm{mc}_{it})\right] &= d\left(\mathbf{e}_i \boldsymbol{\Delta}_\mu \boldsymbol{\Gamma}\boldsymbol{\Sigma}_z \boldsymbol{\Gamma}'\boldsymbol{\Delta}_\mu' \mathbf{e}_i'\right) \\
&= \mathrm{tr}\left[\mathbf{e}_i \left(d(\boldsymbol{\Delta}_\mu)\boldsymbol{\Gamma}\boldsymbol{\Sigma}_z \boldsymbol{\Gamma}'\boldsymbol{\Delta}_\mu' + \boldsymbol{\Delta}_\mu \boldsymbol{\Gamma}\boldsymbol{\Sigma}_z \boldsymbol{\Gamma}' d(\boldsymbol{\Delta}_\mu')\right)\mathbf{e}_i'\right] \\
&= \mathrm{tr}\left[\mathbf{e}_i d(\boldsymbol{\Delta}_\mu)\boldsymbol{\Gamma}\boldsymbol{\Sigma}_z \boldsymbol{\Gamma}'\boldsymbol{\Delta}_\mu' \mathbf{e}_i' + \mathbf{e}_i \boldsymbol{\Delta}_\mu \boldsymbol{\Gamma}\boldsymbol{\Sigma}_z \boldsymbol{\Gamma}' d(\boldsymbol{\Delta}_\mu')\mathbf{e}_i'\right] \\
&= 2\,\mathrm{tr}\left[\mathbf{e}_i d(\boldsymbol{\Delta}_\mu)\boldsymbol{\Gamma}\boldsymbol{\Sigma}_z \boldsymbol{\Gamma}'\boldsymbol{\Delta}_\mu' \mathbf{e}_i'\right] \\
&= 2\,\mathrm{tr}\left[\mathbf{e}_i \boldsymbol{\Delta}_\mu \mathbf{A} d(\boldsymbol{\mu})\boldsymbol{\Delta}_\mu \boldsymbol{\Gamma}\boldsymbol{\Sigma}_z \boldsymbol{\Gamma}'\boldsymbol{\Delta}_\mu' \mathbf{e}_i'\right] \\
&= 2\,\mathrm{tr}\left[\boldsymbol{\Delta}_\mu \boldsymbol{\Gamma}\boldsymbol{\Sigma}_z \boldsymbol{\Gamma}'\boldsymbol{\Delta}_\mu' \mathbf{e}_i' \mathbf{e}_i \boldsymbol{\Delta}_\mu \mathbf{A} d(\boldsymbol{\mu})\right],
\end{aligned}$$

where we use the rotational property of the trace operator. □

Note that by equation (D.42) in Lemma D.4 and equation (D.44) in Lemma D.7,

$$d\left[\mathbb{V}(\mathrm{mc}_{it})\right] = \mathrm{tr}\left(\frac{\partial \mathbb{V}(\mathrm{mc}_{it})}{\partial \boldsymbol{\mu}'} d\boldsymbol{\mu}\right) = \mathrm{tr}\left(\frac{\partial \mathbb{V}(\mathrm{mc}_{it})}{\partial \boldsymbol{\mu}} d\boldsymbol{\mu}\right) = 2\,\mathrm{tr}\left[\boldsymbol{\Delta}_\mu \boldsymbol{\Gamma}\boldsymbol{\Sigma}_z \boldsymbol{\Gamma}'\boldsymbol{\Delta}_\mu' \mathbf{e}_i' \mathbf{e}_i \boldsymbol{\Delta}_\mu \mathbf{A} d(\boldsymbol{\mu})\right], \tag{D.45}$$

because $\boldsymbol{\mu} = \boldsymbol{\mu}'$. Since $d(\boldsymbol{\mu})$ is a matrix of differentials for variable $\boldsymbol{\mu}$, in order for (D.45) to hold, we must have

$$\frac{\partial \mathbb{V}(\mathrm{mc}_{it})}{\partial \boldsymbol{\mu}} = 2\boldsymbol{\Delta}_\mu \boldsymbol{\Gamma}\boldsymbol{\Sigma}_z \boldsymbol{\Gamma}'\boldsymbol{\Delta}_\mu' \mathbf{e}_i' \mathbf{e}_i \boldsymbol{\Delta}_\mu \mathbf{A}, \tag{D.46}$$

by the method of undetermined coefficients.[51]

Note that equation (D.46) holds for general symmetric matrix of variables $\boldsymbol{\mu}$. In our special case

[51] This is indeed the First Identification Theorem that links matrix differentials to Jacobian matrices of derivatives (Magnus and Neudecker (2007); Theorem 5.6, 5.11 and Chapter 9). We also verify the correctness of formula (D.46) on the matrix calculus website: www.matrixcalculus.org. In the univariate case $N = 1$ and $\kappa = 0$, (D.46) reduces to the simple derivative $\left(\frac{\partial \mathbb{V}(\mathrm{mc}_{it})}{\partial \mu}\right) = \frac{2a\sigma^2}{(1-a\mu)^3}$.

in which $\boldsymbol{\mu}$ is diagonal, it is clear that $\frac{\partial \mathbb{V}(\mathrm{mc}_{it})}{\partial \boldsymbol{\mu}}$ is also diagonal, and

$$\frac{\partial \mathbb{V}(\mathrm{mc}_{it})}{\partial \boldsymbol{\mu}} = 2\left[\boldsymbol{\Delta}_{\mu}\boldsymbol{\Gamma}\boldsymbol{\Sigma}_{z}\boldsymbol{\Gamma}'\boldsymbol{\Delta}_{\mu}'\mathbf{e}_{i}'\mathbf{e}_{i}\boldsymbol{\Delta}_{\mu}\mathbf{A}\right]_{d+}, \tag{D.47}$$

where we define a linear annihilation operator $\left[\,\cdot\,\right]_{d+} : \mathbb{R}^{N\times N} \longmapsto \mathbb{R}^{N\times N}$ that restricts all off-diagonal elements to 0 while keeps the diagonal elements unchanged. Consequently,

$$\left(\frac{\partial \mathbb{V}(\mathrm{mc}_{it})}{\partial \boldsymbol{\mu}}\right)_{nn} = 2\left(\boldsymbol{\Delta}_{\mu}\boldsymbol{\Gamma}\boldsymbol{\Sigma}_{z}\boldsymbol{\Gamma}'\boldsymbol{\Delta}_{\mu}'\mathbf{e}_{i}'\mathbf{e}_{i}\boldsymbol{\Delta}_{\mu}\mathbf{A}\right)_{nn}; \qquad n = 1, 2, ...N,$$

holds for every diagonal element.

Now, we are ready to conclude Step 1 by summarizing the properties of the Fréchet derivative for the general equilibrium system.

Proposition D.4. *The Fréchet derivative for the general equilibrium system (D.29) and (D.30) in terms of attention vector $\boldsymbol{\mu}^{v} = diag(\boldsymbol{\mu}) \in [0,1]^{N}$ exists and is given by*

$$\mathcal{T}'(\boldsymbol{\mu}^{v}) \equiv \frac{\partial \mathcal{T}}{\partial \boldsymbol{\mu}^{v}} = \mathcal{T}_{\boldsymbol{\mu}^{v}} = \left[\mathrm{diag}\left(\frac{\partial \mathcal{T}_{1}(\boldsymbol{\mu})}{\partial \boldsymbol{\mu}}\right) \quad \mathrm{diag}\left(\frac{\partial \mathcal{T}_{2}(\boldsymbol{\mu})}{\partial \boldsymbol{\mu}}\right) \quad \cdots \quad \mathrm{diag}\left(\frac{\partial \mathcal{T}_{N}(\boldsymbol{\mu})}{\partial \boldsymbol{\mu}}\right)\right]'.$$

For each sector $i = 1, 2, ...N$,

$$\frac{\partial \mathcal{T}_{i}(\boldsymbol{\mu})}{\partial \boldsymbol{\mu}} = \frac{\chi_{i}}{\theta_{i}\lambda_{i}\mathbb{V}(mc_{it})^{2}}\frac{\partial \mathbb{V}(mc_{it})}{\partial \boldsymbol{\mu}}; \qquad \frac{\partial \mathbb{V}(mc_{it})}{\partial \boldsymbol{\mu}} = 2\left[\boldsymbol{\Delta}_{\mu}\boldsymbol{\Gamma}\boldsymbol{\Sigma}_{z}\boldsymbol{\Gamma}'\boldsymbol{\Delta}_{\mu}'\mathbf{e}_{i}'\mathbf{e}_{i}\boldsymbol{\Delta}_{\mu}\mathbf{A}\right]_{d+},$$

where $\boldsymbol{\Delta}_{\mu} = (\mathbf{I} - \mathbf{A}\boldsymbol{\mu})^{-1}$ is the attention-distorted Leontief matrix. $\left[\,\cdot\,\right]_{d+} : \mathbb{R}^{N\times N} \longmapsto \mathbb{R}^{N\times N}$ is a linear annihilation operator that restricts all off-diagonal elements to 0 while keep the diagonal elements unchanged.[52]

Step 2: Establishing Eigenvalue Bound. By the property of spectral radius and the Gershgorin Circle Theorem,

$$\rho\left(\mathcal{T}_{\boldsymbol{\mu}^{v}}\right) \leq \left\|\mathcal{T}_{\boldsymbol{\mu}^{v}}\right\|_{\infty} = \max_{i}\sum_{j=1}^{N}\left|\left(\mathcal{T}_{\boldsymbol{\mu}^{v}}\right)_{ij}\right|,$$

even if $\mathcal{T}_{\boldsymbol{\mu}^{v}}$ is not an non-negative matrix. By Definition 1, each row of the Jocabian matrix is simply the diagonals of $\left(\frac{\partial \mathcal{T}_{i}(\boldsymbol{\mu})}{\partial \boldsymbol{\mu}}\right)$; that is,

$$\sum_{j=1}^{N}\left|\left(\mathcal{T}_{\boldsymbol{\mu}^{v}}\right)_{ij}\right| = \mathrm{tr}\left(\left|\frac{\partial \mathcal{T}_{i}(\boldsymbol{\mu})}{\partial \boldsymbol{\mu}}\right|\right); \qquad i = 1, 2, ...N,$$

where the operator $\left|\cdot\right| : \mathbb{R}^{N\times N} \longmapsto \mathbb{R}^{N\times N}$ transforms all elements of the matrix to their absolute values.

[52] If $\boldsymbol{\kappa} = 0$, it is the case that $\mathcal{T}_{\boldsymbol{\mu}^{v}} \geq 0$.

Next, we show that for each sector $i = 1, 2, ...N$, the trace $\operatorname{tr}\left(\left|\frac{\partial \mathcal{T}_i(\boldsymbol{\mu})}{\partial \boldsymbol{\mu}}\right|\right)$ is bounded above by 1, under our sufficiency condition in Proposition D.2.

We start with a lemma for the trace bound.

Lemma D.8. *For each sector $i = 1, 2, ...N$, the trace for the matrix derivatives after taking absolute values, $\operatorname{tr}\left(\left|\frac{\partial \mathbb{V}(mc_{it})}{\partial \boldsymbol{\mu}}\right|\right)$, is bounded above by*

$$\operatorname{tr}\left(\left|\frac{\partial \mathbb{V}(mc_{it})}{\partial \boldsymbol{\mu}}\right|\right) \leq 2\varsigma_i; \qquad i = 1, 2...N, \tag{D.48}$$

where $\varsigma_i = \operatorname{tr}\left(\mathbf{L}\left|\boldsymbol{\Gamma}\boldsymbol{\Sigma}_z\boldsymbol{\Gamma}'\right|\mathbf{L}'\mathbf{e}_i'\mathbf{e}_i\mathbf{L}\mathbf{A}\right)$, $i = 1, 2, ...N$, and $\mathbf{L} = (\mathbf{I} - \mathbf{A})^{-1}$.

Proof. For any $N \times N$ matrices $\boldsymbol{X}$ and $\boldsymbol{Y}$, it is straightforward to verify that $\boldsymbol{XY} \leq \left|\boldsymbol{XY}\right| \leq \left|\boldsymbol{X}\right| \cdot \left|\boldsymbol{Y}\right|$. Combined with the definition of Leontief inverse $\mathbf{L} = (\mathbf{I} - \mathbf{A})^{-1} \geq \boldsymbol{\Delta}_\mu$, we derive

$$\mathbf{L}\left|\boldsymbol{\Gamma}\boldsymbol{\Sigma}_z\boldsymbol{\Gamma}'\right|\mathbf{L}'\mathbf{e}_i'\mathbf{e}_i\mathbf{L}\mathbf{A} \geq \boldsymbol{\Delta}_\mu\left|\boldsymbol{\Gamma}\boldsymbol{\Sigma}_z\boldsymbol{\Gamma}'\right|\boldsymbol{\Delta}_\mu'\mathbf{e}_i'\mathbf{e}_i\boldsymbol{\Delta}_\mu\mathbf{A} \geq \left|\boldsymbol{\Delta}_\mu\boldsymbol{\Gamma}\boldsymbol{\Sigma}_z\boldsymbol{\Gamma}'\boldsymbol{\Delta}_\mu'\mathbf{e}_i'\mathbf{e}_i\boldsymbol{\Delta}_\mu\mathbf{A}\right| \geq 0, \tag{D.49}$$

since $\mathbf{L}$, $\boldsymbol{\Delta}_\mu$, $\boldsymbol{\Sigma}_z$, $\mathbf{e}_i$ and $\mathbf{A}$ are all nonnegative. Then, by equations (D.47) and (D.49),

$$\begin{aligned}
\operatorname{tr}\left(\left|\frac{\partial \mathbb{V}(\mathrm{mc}_{it})}{\partial \boldsymbol{\mu}}\right|\right) &= \operatorname{tr}\left(2\left|\left[\boldsymbol{\Delta}_\mu\boldsymbol{\Gamma}\boldsymbol{\Sigma}_z\boldsymbol{\Gamma}'\boldsymbol{\Delta}_\mu'\mathbf{e}_i'\mathbf{e}_i\boldsymbol{\Delta}_\mu\mathbf{A}\right]_{d+}\right|\right) \\
&= 2\operatorname{tr}\left(\left|\boldsymbol{\Delta}_\mu\boldsymbol{\Gamma}\boldsymbol{\Sigma}_z\boldsymbol{\Gamma}'\boldsymbol{\Delta}_\mu'\mathbf{e}_i'\mathbf{e}_i\boldsymbol{\Delta}_\mu\mathbf{A}\right|\right) \\
&\leq 2\operatorname{tr}\left(\boldsymbol{\Delta}_\mu\left|\boldsymbol{\Gamma}\boldsymbol{\Sigma}_z\boldsymbol{\Gamma}'\right|\boldsymbol{\Delta}_\mu'\mathbf{e}_i'\mathbf{e}_i\boldsymbol{\Delta}_\mu\mathbf{A}\right) \\
&\leq 2\operatorname{tr}\left(\mathbf{L}\left|\boldsymbol{\Gamma}\boldsymbol{\Sigma}_z\boldsymbol{\Gamma}'\right|\mathbf{L}'\mathbf{e}_i'\mathbf{e}_i\mathbf{L}\mathbf{A}\right).
\end{aligned}$$

To ease notation, we define a column vector $\varsigma = \left(\{\varsigma_i\}_{i=1}^N\right)$, where $\varsigma_i = \operatorname{tr}\left(\mathbf{L}\left|\boldsymbol{\Gamma}\boldsymbol{\Sigma}_z\boldsymbol{\Gamma}'\right|\mathbf{L}'\mathbf{e}_i'\mathbf{e}_i\mathbf{L}\mathbf{A}\right)$. Then the trace bound for the absolute value of matrix derivatives follows

$$\operatorname{tr}\left(\left|\frac{\partial \mathbb{V}(\mathrm{mc}_{it})}{\partial \boldsymbol{\mu}}\right|\right) \leq 2\varsigma_i; \qquad i = 1, 2...N,$$

as desired. □

Next, we deliver an upper bound for the eigenvalues of Jacobian matrix $\mathcal{T}_{\boldsymbol{\mu}^v}$.

Proposition D.5. *Suppose for each sector $i = 1, 2, ...N$, we have following parameter restrictions,*

$$\kappa_i < 1, \quad \textit{and} \quad \chi_i < \theta_i\lambda_i\frac{\sigma_i^4\left(1 - \varkappa_i\kappa_i\right)^4}{2\varsigma_i}, \tag{D.50}$$

where $\varkappa_i$ is defined in Proposition D.3 and $\varsigma_i = \operatorname{tr}\left(\mathbf{L}\left|\boldsymbol{\Gamma}\boldsymbol{\Sigma}_z\boldsymbol{\Gamma}'\right|\mathbf{L}'\mathbf{e}_i'\mathbf{e}_i\mathbf{L}\mathbf{A}\right)$ for each sector. Then the eigenvalues of

the Jacobian matrix is bounded above by 1; that is

$$\rho\left(\mathcal{T}_{\boldsymbol{\mu}^v}\right) < 1. \tag{D.51}$$

Proof. Recall that each row of the Jacobian matrix is simply the diagonals of matrix of derivatives $\left(\frac{\partial \mathcal{T}_i(\boldsymbol{\mu})}{\partial \boldsymbol{\mu}}\right)$; that is, $\sum_{j=1}^{N}\left|\left(\mathcal{T}_{\boldsymbol{\mu}^v}\right)_{ij}\right| = \operatorname{tr}\left(\left|\frac{\partial \mathcal{T}_i(\boldsymbol{\mu})}{\partial \boldsymbol{\mu}}\right|\right)$, $i = 1, 2, ...N$. Then, from equation (D.41),

$$\begin{aligned}
\operatorname{tr}\left(\left|\frac{\partial \mathcal{T}_i(\boldsymbol{\mu})}{\partial \boldsymbol{\mu}}\right|\right) &= \frac{\chi_i}{\theta_i \lambda_i \mathbb{V}(\mathrm{mc}_{it})^2} \operatorname{tr}\left(\left|\frac{\partial \mathbb{V}(\mathrm{mc}_{it})}{\partial \boldsymbol{\mu}}\right|\right) \\
&\leq \frac{\chi_i}{\theta_i \lambda_i \left[\sigma_i^2 \left(1 - \varkappa_i k_i\right)^2\right]^2} \operatorname{tr}\left(\left|\frac{\partial \mathbb{V}(\mathrm{mc}_{it})}{\partial \boldsymbol{\mu}}\right|\right) \\
&\leq \frac{\chi_i}{\theta_i \lambda_i \sigma_i^4 \left(1 - \varkappa_i k_i\right)^4} \cdot 2\varsigma_i \\
&< 1
\end{aligned}$$

holds for each sector $i = 1, 2....N$. The first inequality follows from (D.36) in Proposition D.3, the second inequality follows from (D.48), and the last inequality follows from our assumption (D.50).

Therefore, each row sum of $\mathcal{T}_{\boldsymbol{\mu}^v}$ is strictly smaller than 1, implying that

$$\left\|\mathcal{T}_{\boldsymbol{\mu}^v}\right\|_\infty = \max_i \sum_{j=1}^{N}\left|\left(\mathcal{T}_{\boldsymbol{\mu}^v}\right)_{ij}\right| < 1$$

with finitely many sectors. Now by the Gishgorin Circle Theorem

$$\rho\left(\mathcal{T}_{\boldsymbol{\mu}^v}\right) \leq \left\|\mathcal{T}_{\boldsymbol{\mu}^v}\right\|_\infty < 1$$

as desired.[53] □

Now we are ready to proceed to the final step of the proof for Proposition D.2.

Step 3: Verify Conditions in Kellogg's Fixed Point Theorem. Let $X = \mathbb{R}^N$, which is a finite-dimensional Banach space. Let $D = (0,1)^N$, it is clear that D is bounded open convex open subset of X. Let $\bar{D} = [0,1]^N$ be its closure (which is also compact) and $x = \boldsymbol{\mu}^v \in \bar{D}$. The function $F(x) = \mathcal{T}\left(\boldsymbol{\mu}^v\right)$ is defined in equation (D.30). By Proposition D.5, the Fréchet derivative $F'(x) = \mathcal{T}'(\boldsymbol{\mu}^v)$ exists and is continuous. For any $\boldsymbol{\mu}^v \in D = (0,1)^N$, Proposition D.5 states that

$$\rho\left(F'(x)\right) = \rho\left(\mathcal{T}_{\boldsymbol{\mu}^v}\right) < 1 \tag{D.52}$$

[53] If $\boldsymbol{\kappa} = 0$, then $\mathcal{T}_{\boldsymbol{\mu}^v} \geq 0$ and the Perron-Frobenious theorem leads to the same result.

if for each sector $i = 1, 2, ...N$,

$$\kappa_i < 1, \quad \text{and} \quad \chi_i < \theta_i \lambda_i \frac{\sigma_i^4 (1 - \varkappa_i \kappa_i)^4}{2\varsigma_i}.$$

Hence 1 cannot be an eigenvalue under the assumption of Proposition D.2. Finally, by condition (D.36) and (D.37), if

$$0 < \frac{\chi_i}{\theta_i \lambda_i \mathbb{V}(\text{mc}_{it})} \leq \frac{\chi_i}{\theta_i \lambda_i \sigma_i^2 (1 - \varkappa_i \kappa_i)^2} < 1,$$

then $\mathcal{T}_i(\boldsymbol{\mu}^v) \in (0, 1)$, for all sector $i = 1, 2, ...N$ and all vectors $\boldsymbol{\mu}^v \in [0, 1]^N$. Therefore, for each $\boldsymbol{\mu}^v \in \partial D$ on boundary,

$$\boldsymbol{\mu}^v \neq \mathcal{T}(\boldsymbol{\mu}^v), \quad \forall \boldsymbol{\mu}^v \in \partial D.$$

Now all conditions of Kellogg's Fixed Point Theorem have been verified. It is clear that under the assumption of Proposition D.2, the general equilibrium system (2.8) has an unique fixed-point $\boldsymbol{\mu}^{v*} \in (0, 1)^N$ in the interior such that

$$\boldsymbol{\mu}^{v*} = \mathcal{T}(\boldsymbol{\mu}^{v*}).$$

The entire proof of equilibrium existence and uniqueness is now complete. □

Remark: Proposition D.2 provides a sufficient condition that ensures equilibrium uniqueness. The condition can be further refined and relaxed in the following corollary, utilizing the analytical form of $\mathcal{T}_{\boldsymbol{\mu}^v}$ obtained in Proposition D.4.

Corollary D.2. *Let ρ_{max} be the maximal spectral radius of $\mathcal{T}_{\boldsymbol{\mu}^v}$ in Proposition D.4 in the domain of $\boldsymbol{\mu}^v$. That is, consider an optimization problem:*

$$\rho_{max} = \max_{\boldsymbol{\mu}^v} \rho\left(\mathcal{T}_{\boldsymbol{\mu}^v}\right),$$

subject to $\mu_i \in [0, 1], \forall i = 1, 2, ...N$. If $\rho_{max} < 1$ and

$$\kappa_i < 1; \qquad 0 < \frac{\chi_i}{\theta_i \lambda_i} < \sigma_i^2 (1 - \varkappa_i \kappa_i)^2; \qquad \forall i = 1, 2, ...N,$$

where

$$\varkappa_i = \begin{cases} 1, & 0 \leq \kappa_i < 1 \\ \alpha_i, & \kappa_i < 0 \end{cases}; \qquad i = 1, 2, ...N,$$

then the general equilibrium system (2.8) has an unique fixed-point $\boldsymbol{\mu}^{v*} \in (0, 1)^N$ in the interior.

Proof. Since $\rho\left(\mathcal{T}_{\boldsymbol{\mu}^v}\right) \leq \rho_{max} < 1$, (D.52) holds and 1 cannot be an eigenvalue. By condition (D.36) and (D.37), if $\kappa_i < 1$ and $0 < \frac{\chi_i}{\theta_i \lambda_i} < \sigma_i^2 (1 - \varkappa_i \kappa_i)^2$, $\forall i = 1, 2, ...N$, which implies that $\mathcal{T}_i(\boldsymbol{\mu}^v) \in (0, 1)$,

for all sector $i = 1, 2, ...N$ and all vectors $\boldsymbol{\mu}^v \in [0,1]^N$. Therefore, for each $\boldsymbol{\mu}^v \in \partial D$ on boundary, $\boldsymbol{\mu}^v \neq \mathcal{T}(\boldsymbol{\mu}^v)$. It is clear that all conditions of Kellogg's Fixed Point Theorem are satisfied, the general equilibrium system (2.8) has an unique fixed-point $\boldsymbol{\mu}^{v*} \in (0,1)^N$. □

Remark: The optimization problem of spectral radius and attention response $\mathcal{T}$ in Corollary D.2 are independent of the equilibrium fixed-point system and can be numerically verified by Matlab. In our quantification, we verify that the calibrated parameters satisfy Corollary D.2. Therefore, the calibrated model and the model under optimal monetary policy both possess unique fixed-point in the interior.

D.11 Proof of Proposition 2.6

Proof. In the proof of Proposition D.1, we obtain the optimal posteior covariance matrix (MSE) matrix, $\boldsymbol{\Sigma}_{z|x_i} = \boldsymbol{\Sigma}_z - \frac{\chi_i}{2R_i d_1^i \nu_i^2}\left(\boldsymbol{\Sigma}_z \boldsymbol{\Omega}_{iz} \boldsymbol{\Sigma}_z\right)$, and by definition $\widehat{\sigma}^2_{j|i} = \mathbb{E}\left[\left(\mathbb{E}\left[z_{jt} \mid x_{it}\right] - z_{jt}\right)^2 \mid x_{it}\right]$ is the diagonal element of the posterior matrix. If all sectors have positive level of attention such that $\mu_i > 0, \forall i = 1, 2, ...N$, we have following relations,

$$\text{mc}_{it} = R_i^{-\frac{1}{2}} \mathbf{G}_i \boldsymbol{z}_t = \frac{1}{\mu_i} p_{it}.$$

Therefore, when $\boldsymbol{\Sigma}_z$ is diagonal, the diagonal element of posterior is given by

$$\widehat{\sigma}^2_{j|i} = \sigma_j^2 - \frac{\chi_i}{2d_1^i \nu_i^2} \frac{1}{\mu_i^2} \phi_{ij}^2 \sigma_j^4,$$

where we use the definition of weighting matrix $\boldsymbol{\Omega}_{iz}$. Applying our previous results, it is easy to deduce that $\frac{\chi_i}{2d_1^i \nu_i^2} = \frac{1}{\mathbb{V}(mc_{it}) + \nu_i^2}$. It then follows that

$$\widehat{\sigma}^2_{j|i} = \sigma_j^2 - \frac{\phi_{ij}^2 \sigma_j^4}{\mu_i^2 \left(\mathbb{V}\left(mc_{it}\right) + \nu_i^2\right)} = \sigma_j^2 - \frac{\mu_i \phi_{ij}^2 \sigma_j^4}{\sum_{k=1}^N \sigma_k^2 \phi_{ik}^2},$$

where I use the fact that $\mu_i = \frac{\mathbb{V}(mc_{it})}{\mathbb{V}(mc_{it}) + \nu_i^2}$ and $\mathbb{V}\left(p_{it}\right) = \mu_i^2 \mathbb{V}\left(mc_{it}\right)$. Finally, the relative attention allocation measure is given by

$$\omega_{ij} = \frac{\sigma_j^2 - \widehat{\sigma}^2_{j|i}}{\sigma_j^2} = \mu_i \frac{\phi_{ij}^2 \sigma_j^2}{\sum_{k=1}^N \sigma_k^2 \phi_{ik}^2}.$$

By construction, the fraction $\frac{\phi_{ij}^2 \sigma_j^2}{\sum_{k=1}^N \sigma_k^2 \phi_{ik}^2}$ is a well-defined probability measure,

$$\frac{\phi_{ij}^2 \sigma_j^2}{\sum_{k=1}^N \sigma_k^2 \phi_{ik}^2} \in [0,1]; \qquad \sum_{j=1}^N \frac{\phi_{ij}^2 \sigma_j^2}{\sum_{k=1}^N \sigma_k^2 \phi_{ik}^2} = 1; \qquad \forall i = 1, 2, ..N.$$

Therefore, we have $\sum_{j=1}^{N} \omega_{ij} = \mu_i$ as desired. The proof is complete. □

D.12 Proof of Proposition 2.7

Proof. We divide the proof in two parts.

Fixed-capacity case. Using similar approach of Lemma D.1, under fixed-capacity the optimal information choice is given by the following matrix optimization,

$$\begin{aligned}
&\min_{\boldsymbol{\Sigma}_{z|x_i}} \operatorname{tr}\left(\boldsymbol{\Omega}_{iz}\boldsymbol{\Sigma}_{z|x_i}\right) \\
&\text{s.t } 0 \prec \boldsymbol{\Sigma}_{z|x_i} \preceq \boldsymbol{\Sigma}_z, \\
&\log\det \boldsymbol{\Sigma}_z - \log\det \boldsymbol{\Sigma}_{z|x_i} \leq 2\delta_i.
\end{aligned}$$

In information theory, the above optimization leads to a classical distortion rate function (DRF),

$$D^*(\delta_i) = \min_{\boldsymbol{\Sigma}_{z|x_i}} \operatorname{tr}\left(\boldsymbol{\Omega}_{iz}\boldsymbol{\Sigma}_{z|x_i}\right).$$

The problem can solved using the same techniques in Appendix D.2.[54] The optimal distortion is given by

$$D^* = e^{-2\delta_i} \| \boldsymbol{\Sigma}_z^{\frac{1}{2}} \mathbf{G}_i' \|^2. \tag{D.53}$$

The optimal signal is still one-dimensional with form $x_{ilt} = \mathrm{mc}_{it} + u_{ilt}$, and $u_{ilt} \sim \mathbb{N}\left(0, \nu_i^2\right)$. The signal noise variance is given by

$$\nu_i^2 = R_i^{-1} \frac{D^* \| \boldsymbol{\Sigma}_z^{\frac{1}{2}} \mathbf{G}_i' \|^2}{\| \boldsymbol{\Sigma}_z^{\frac{1}{2}} \mathbf{G}_i' \|^2 - D^*}.$$

Therefore, we can compute the signal precision (attention) as

$$\mu_i = \frac{1/\nu_i^2}{1/\nu_i^2 + 1/\mathbb{V}\left(\mathrm{mc}_{it}\right)} = \frac{\| \boldsymbol{\Sigma}_z^{\frac{1}{2}} \mathbf{G}_i' \|^2 - D^*}{\| \boldsymbol{\Sigma}_z^{\frac{1}{2}} \mathbf{G}_i' \|^2} \in (0, 1),$$

[54] In particular, we solve the following rate distortion function problem,

$$\begin{aligned}
&\delta^*(D) = \min_{\boldsymbol{\Sigma}_{z|x_i}} \log\det \boldsymbol{\Sigma}_z - \log\det \boldsymbol{\Sigma}_{z|x_i} \\
&\text{s.t } 0 \prec \boldsymbol{\Sigma}_{z|x_i} \preceq \boldsymbol{\Sigma}_z, \\
&\operatorname{tr}\left(\boldsymbol{\Omega}_{iz}\boldsymbol{\Sigma}_{z|x_i}\right) \leq D
\end{aligned}$$

which leads to a rate distortion function (RDF). The two problems are equivalent in the sense that $\delta^*(D)$ and $D^*(\delta_i)$ are inverse functions of each other that define a one-to-one, monotone, continous mapping. The solution technique is identical to the elastic attention model, except that the marginal cost of information (Lagrange multiplier) is endogenously determined. The details of solution is available upon request.

by recalling that $\mathbb{V}(\text{mc}_{it}) = R_i^{-1}\left\|\Sigma_0^{\frac{1}{2}}\mathbf{G}_i'\right\|^2$. Using (D.53), we obtain,

$$\mu_i = 1 - e^{-2\delta_i},$$

as desired.

Exogenous Information Case: For each firm ι in sector i, it observes N independent signals about $\boldsymbol{z}_t$,

$$x_{i\iota jt} = z_{j,t} + u_{i\iota jt}, \quad u_{i\iota jt} \sim \mathbb{N}\left(0, \tau_i \sigma_j^2\right).$$

This implies firm's forecats about sector shocks are given by

$$\mathbb{E}_{i\iota t}\left[z_{jt} | x_{i\iota jt}\right] = \frac{\sigma_j^2}{\sigma_j^2 + \tau_i \sigma_j^2} x_{i\iota jt} = \frac{1}{1+\tau_i} x_{i\iota jt}; \qquad j = 1, 2, ...N. \tag{D.54}$$

Therefore, the signal precisions for each sector shock are identical. In Appendix D.1, we show that the pricing rule (D.4) holds for any information structure, with marginal cost function given by (2.2). Therefore, individual firms set price according to $p_{i\iota t} = \mathbb{E}_{i\iota t}\left[\text{mc}_{it} | \left\{x_{i\iota jt}\right\}_{j=1}^N\right]$. Without loss of generality, assume the equilibium function of marginal cost is given by $\text{mc}_{it} = \boldsymbol{m}'\boldsymbol{z}_t = \sum_{j=1}^N m_j z_{jt}$. Then the pricing function follows

$$p_{i\iota t} = \mathbb{E}_{i\iota t}\left[\sum_{j=1}^N m_j z_{jt} | \left\{x_{i\iota jt}\right\}_{j=1}^N\right] = \sum_{j=1}^N m_j \mathbb{E}_{i\iota t}\left[z_{jt} | \left\{x_{i\iota jt}\right\}_{j=1}^N\right] = \sum_{j=1}^N m_j \frac{1}{1+\tau_i} x_{i\iota jt} = \frac{1}{1+\tau_i}\sum_{j=1}^N m_j\left(z_{j,t} + u_{i\iota jt}\right). \tag{D.55}$$

Integrate over all firms in the sector, idiosyncratic noises wash out. Hence,

$$p_{it} = \int \frac{1}{1+\tau_i}\sum_{j=1}^N m_j\left(z_{j,t} + u_{i\iota jt}\right) d\iota = \frac{1}{1+\tau_i}\sum_{j=1}^N m_j z_{jt} = \frac{1}{1+\tau_i}\text{mc}_{it}, \tag{D.56}$$

where the last equality follows from our initial conjecture. The proof is therefore complete. □

D.13 Proof of Corrolary 2.1

Proof. By our previous results with fixed-capacity, $\mu_i = 1 - e^{-2\delta_i}$, since δ_i is exogenous, the information capacity is independent of market outcomes. □

E. Proofs for Section 4

E.1 Characterization of General Equilibrium with Full Information

We characterize the full-information general equilibrium using the following Proposition.

Proposition E.1. *In absence of information acquisition cost,*

1. *The equilibrium price responses relies on the standard Leontief inverse matrix*

$$\boldsymbol{p}_t^* = \mathbf{L}(-\boldsymbol{z}_t + \boldsymbol{\alpha} w_t), \quad \textit{with} \quad \mathbf{L} = (\mathbf{I} - \mathbf{A})^{-1}, \tag{E.1}$$

   where the (i,j)-th element of $\mathbf{A}$ is a_{ij} and $\boldsymbol{\alpha} \equiv \left[\alpha_1, \ldots, \alpha_N\right]'$.

2. *The equilibrium output change is independent of money supply*

$$c_t^* = \frac{1+\eta}{\gamma+\eta} \sum_i \lambda_i z_{i,t},$$

   where $\lambda_i = \sum_j L_{ij}\beta_j$ is the steady-state Domar weight of sector i.

Proof. In model equilibrium, households maximize their expected utility subject to their budget constraints. The optimal consumption demand and the intratemporal Euler condition satisfy

$$P_{i,t}C_{i,t} = \beta_i P_t C_t; \qquad \frac{W_t}{P_t} C_t^{-\gamma} = L_t^{\frac{1}{\eta}}.$$

With perfect information, firms' profit optimization problem reads

$$\max_{\left\{P_{i\iota t}, L_{i\iota t}, \left\{X_{i\iota jt}\right\}_{j=1}^N\right\}} \Pi_{i\iota t} = (1+\tau_i) P_{i\iota t} Y_{i\iota t} - W_t L_{i\iota t} - \sum_{j=1}^N P_{jt} X_{i\iota jt} - T_{i\iota t}$$

where $\tau_i = \frac{1}{(\theta_i - 1)}$ is the subsidy to remove monopolistic distortion, $T_{i\iota t}$ is a lump-sum tax that finances the subsidy. In equilibrium, $T_{i\iota t} = \tau_i P_{i\iota t} Y_{i\iota t}$. A canonical cost minimization problem implies that the optimal input and labor demand satisfy

$$X_{i,\iota,j,t} = a_{ij} \mathrm{MC}_{i,t} \frac{Y_{i,\iota,t}}{P_{j,t}}, \quad \text{and} \quad L_{i,t} = \alpha_i \mathrm{MC}_{i,t} \frac{Y_{i,\iota,t}}{W_t}, \tag{E.2}$$

where the marginal cost of firms in sector i is $\mathrm{MC}_{it} = Z_{it}^{-1} W_t^{\alpha_i} \prod_{j=1}^N P_{jt}^{a_{ij}}$. The market-clearing condition is given by

$$Y_{i,t} = C_{i,t} + \sum_{j=1}^N \int_0^1 X_{j,\iota,i,t} d\iota.$$

Next, we characterize the full-information equilibrium.

Frictionless benchmark. In absence of information acquisition cost, the sectoral price is identical to the nominal marginal cost,

$$P_{i,\iota,t} = P_{i,t} = \text{MC}_{i,t} = \frac{1}{Z_{i,t}} W_t^{\alpha_i} \prod_{j=1}^{N} P_{j,t}^{a_{ij}}, \tag{E.3}$$

As a result, the after-tax (net) profit for each firm equals to zero and the households' budget balance is given by $P_t C_t = W_t L_t$. And the sectoral inputs follow

$$X_{j,i,t} = X_{j,\iota,i,t} = a_{ji} \frac{P_{j,t} Y_{j,\iota,t}}{P_{i,t}}. \tag{E.4}$$

Then the goods market clearing condition for sector i follows

$$P_{i,t} Y_{i,t} = \beta_i P_t C_t + \sum_{j=1}^{N} a_{ji} P_{j,t} Y_{j,t}. \tag{E.5}$$

Let λ_{it} denote the sales-to-GDP ratio (the Domar weight), thus $\lambda_{i,t} = \beta_i + \sum_{j=1}^{N} a_{ji} \lambda_{j,t}$. Consequently, with perfect information, the Domar weight is constant. In matrix form, $\boldsymbol{\lambda} = \boldsymbol{\beta} + \mathbf{A}'\boldsymbol{\lambda}$. Then the Domar weight is given by

$$\boldsymbol{\lambda}' = \boldsymbol{\beta}'(\mathbf{I} - \mathbf{A})^{-1}. \tag{E.6}$$

Next, we provide the steady-state solution of the frictionless benchmark.

The determinant steady-state. The subscript t of a time-varying variable is removed to denote the steady-state value. We normalize the steady-state money supply M to be 1, which implies that the GDP is equal to 1 as $PC = M = 1$. In the perfect-information economy, $P_{i,\iota} = P_i = \text{MC}_i$, so the steady-state value of the prices and wage rate in (E.3) satisfy

$$P_i = \text{MC}_i = \frac{1}{Z_i} W^{\alpha_i} \prod_{j=1}^{N} P_j^{a_{ij}} = W^{\alpha_i} \prod_{j=1}^{N} P_j^{a_{ij}}, \tag{E.7}$$

where the steady-state of sectoral productivity shock $Z_i = 1$, $i = 1, 2, ...N$. Meanwhile, each firm's after-tax profit is zero, $\Pi_{i,\iota} = 0$. The steady-state household budget constraint is given by $PC = WL = M$, and the optimal labor supply condition follows $\frac{W}{P} C^{-\gamma} = L^{\frac{1}{\eta}}$. Combing these two equations.

$$W = M^{\frac{1+\eta\gamma}{1+\eta}} P^{\frac{\eta(1-\gamma)}{1+\eta}} = M^{\frac{1+\eta\gamma}{1+\eta}} \left(\prod_{j=1}^{N} P_j \right)^{\frac{\beta_j \eta(1-\gamma)}{1+\eta}}, \tag{E.8}$$

where we use $P = \prod_{i=1}^{N} P_i^{\beta_i}$ and rewrite the sector's subscript to j. Given equation (E.7) and (E.8), the steady-state of equilibrium price system is given by

$$P_i = M^{\frac{\alpha_i(1+\eta\gamma)}{1+\eta}} \left(\prod_{j=1}^{N} P_j \right)^{\frac{\alpha_i \beta_j \eta(1-\gamma)}{1+\eta} + a_{ij}},$$

which implies that the steady-state of sectors' prices are determined by the exogenous money supply $M = 1$. Explicitly, $P_i = 1$ for $i = 1, 2, ...N$. Thus, the final goods price P and the final consumption goods C can be obtained in turn:

$$P = \prod_{i=1}^{N} P_i^{\beta_i} = 1, \quad \text{and} \quad C = \frac{M}{P} = 1.$$

Furthermore, the steady-state wage rate and the aggregated labor is given by $W = 1, \quad \text{and} \quad L = \frac{M}{W} = 1$. Next, since $P_i C_i = \beta_i PC$, the sectoral consumption follows

$$C_i = \frac{\beta_i PC}{P_i} = \beta_i.$$

Using (E.2) under perfect information, firms' inputs in sector i inputs are given by

$$L_i = \alpha_i \frac{P_i Y_i}{W} = \alpha_i \lambda_i, \quad \text{and} \quad X_{i,j} = X_{i,\iota,j} = a_{ij} \frac{P_i Y_i}{P_j} = a_{ij} \lambda_i.$$

Finally, sector i's steady-state output is given by

$$Y_i = \frac{\lambda_i M}{P_i} = \lambda_i.$$

Next, we log-linearize the equilibrium system under perfect information.

Linearized system under perfect information. The log-linearized goods market clearing condition (E.5) is given by

$$P_i Y_i \left(p_{it} + y_{it} \right) = \beta_i M m_t + \sum_{j=1}^{N} a_{ji} P_j Y_j \left(p_{jt} + y_{jt} \right)$$

where we use $P_t C_t = M_t$ and $P_{it} = MC_{it}$. , In matrix form:

$$\boldsymbol{p}_t + \boldsymbol{y}_t = \operatorname{diag}(\boldsymbol{\lambda})^{-1} (\mathbf{I} - \mathbf{A}')^{-1} \boldsymbol{\beta} m_t \tag{E.9}$$

Next, the sectoral price in (E.3) can be log-linearized as

$$p_{it} = \mathrm{mc}_{it} = -z_{it} + \alpha_i w_t + \sum_j a_{ij} p_{jt}.$$

To facilitate comparison with the *Rational-Inattention equilibrium*, we denote $\boldsymbol{p}_t^*$ and $\boldsymbol{c}_t^*$ as the price-response vector and consumption vector in the *Full-Information Equilibrium*, respectively. The matrix form of the above equation can be expressed as $\boldsymbol{p}_t^* = -\boldsymbol{z}_t + \boldsymbol{\alpha} w_t + \mathbf{A}\boldsymbol{p}_t^*$, which leads to the price-response vector under perfect information,

$$\boldsymbol{p}_t^* = (\mathbf{I} - \mathbf{A})^{-1}(-\boldsymbol{z}_t + \boldsymbol{\alpha} w_t). \tag{E.10}$$

We define p_t^f as the log-deviation from the final goods price P_t , then $p_t^f = \sum_{i=1}^N \beta_i p_{it} = \boldsymbol{\beta}' \boldsymbol{p}_t^*$. Multiplying both sides of the equation (E.10) by $\boldsymbol{\beta}'$ yields

$$p_t^f = -\boldsymbol{\beta}'(\mathbf{I} - \mathbf{A})^{-1}\boldsymbol{z}_t + \boldsymbol{\beta}'(\mathbf{I} - \mathbf{A})^{-1}\boldsymbol{\alpha} w_t = -\boldsymbol{\lambda}'\boldsymbol{z}_t + w_t, \tag{E.11}$$

where we use the matrix property of $\boldsymbol{\beta}'(\mathbf{I} - \mathbf{A})^{-1}\boldsymbol{\alpha} = 1$ and $\boldsymbol{\beta}'(\mathbf{I} - \mathbf{A})^{-1} = \boldsymbol{\lambda}'$. We then log-linearize the households' budget constraint under perfect information and the Euler condition, yielding

$$p_t^f + c_t^* = w_t + \ell_t = m_t, \quad \text{and} \quad w_t^f - p_t^f - \gamma c_t^* = \frac{1}{\eta}\ell_t.$$

Eliminating ℓ_t by combining the above equations provides the linearized output as

$$c_t^* = \frac{1+\eta}{1+\gamma\eta}\left(w_t - p_t^f\right) = \frac{1+\eta}{1+\gamma\eta}\boldsymbol{\lambda}'\boldsymbol{z}_t,$$

where the second identity follows from the equation (E.11). □

E.2 Derivation of Equation (4.1)

We derive this equation using the following Corrolary

Corollary E.1. *The output gap is given by*

$$c_t - c_t^* = -\frac{\eta}{1+\gamma\eta}\sum_i \beta_i e_{i,t}, \tag{E.12}$$

where $e_{it} \equiv p_{i,t} - p_{i,t}^*$ is the average pricing error in sector i relative to the perfect-information benchmark

$$\boldsymbol{e}_t = \mathbf{Q}\left(\mathbf{L} - \mathbf{1}\boldsymbol{\kappa}\right)\boldsymbol{z}_t, \tag{E.13}$$

where $\mathbf{Q} = (\mathbf{I} - \boldsymbol{\mu}\mathbf{A})^{-1}(\mathbf{I} - \boldsymbol{\mu})$, and $\mathbf{L} = (\mathbf{I} - \mathbf{A})^{-1}$ is the Leontief inverse with perfect information

Proof. First, we derive consumption deviations. In rationally inattentive equilibrium, the nominal GDP implied by household's budget constraint follow

$$P_t^f C_t = W_t L_t + \sum_i P_{it} Y_{it}\left(1 - \mathcal{E}_{it}\right) = W_t L_t \left(1 - \sum_{i=1}^N \Lambda_{it}\left(1 - \mathcal{E}_{it}\right)\right)^{-1}, \tag{E.14}$$

where sectoral markup deviations are defined as $\mathcal{E}_{it} \equiv \frac{\mathrm{MC}_{it}}{P_{it}} \int \left(\frac{P_{itt}}{P_{it}}\right)^{-\theta_i} d\iota$ and sectoral Domar weights are given by $\Lambda_{it} = \frac{P_{it}Y_{it}}{P_t^f C_t}$ under rational inattention. Standard log-linearization leads to $1-\sum_{i=1}^N \Lambda_{it}\left(1-\mathcal{E}_{it}\right) \approx 1+\sum_{i=1}^N \lambda_i \varepsilon_{it}$. Meanwhile, log-linearized sectoral markups are given by

$$\varepsilon_{it} = mc_{it} - p_{it} - \theta_i \int \left(p_{\iota t} - p_{it}\right) = mc_{it} - p_{it}.$$

We immediately have

$$\sum_{i=1}^N \lambda_i \varepsilon_{it} = \sum_{i=1}^N \lambda_i \left(-z_{it} + \alpha_i w_t + \sum_{j=1}^N a_{ij} p_{jt} - p_{it}\right) = -p_t^f + \sum_{i=1}^N \lambda_i \left(-z_{it} + \alpha_i w_t\right) = w_t - p_t^f - \sum_{i=1}^N \lambda_i z_{it}, \tag{E.15}$$

where we use the fact that $\sum_{i=1}^N \lambda_i \alpha_i = 1$. Under rational inattention, the log-linearized budget constraint reads

$$c_t = w_t - p_t^f + \ell_t - \sum_{i=1}^N \lambda_i \varepsilon_{it}$$

subustitue (E.15) and the Euler equation into this equation, we get

$$c_t = \frac{\eta}{1+\gamma\eta}\left(w_t - p_t^f\right) + \frac{1}{1+\gamma\eta}\sum_i \lambda_i z_{it}.$$

In the proof of Proposition E.1, we show that the equilibrium consumption is given by $c_t^* = \frac{1+\eta}{1+\gamma\eta}\left(w_t^* - p_t^{f,*}\right) = \frac{1+\eta}{1+\gamma\eta}\boldsymbol{\lambda}' \boldsymbol{z}_t$. In FI flexible-price economy, without monetary policy only the relative price $\left(w_t^* - p_t^{f,*}\right)$ is determiate. Introducing CIA constraint pins down wage and final good price index separately; however, money neutrality implies that the relative price and the consumption (and labor) are independent of monetary policy. Without loss of generality, we let $w_t^* = w_t$, then the consumption deviation becomes,

$$c_t - c_t^* = -\frac{\eta}{1+\gamma\eta}\left(p_t^f - p_t^{f,*}\right) = -\frac{\eta}{1+\gamma\eta}\sum_{i=1}^N \beta_i e_{it},$$

as desired. Note the consumption deviation is independent of the choice of wage function in the FI economy. We will expolit this property in the subsequent welfare analysis. Finally, we combine (E.10) and(D.25) and simplify the algebra, we obtain that

$$\begin{aligned}
\mathbf{e}_t &= \left((\mathbf{I}-\boldsymbol{\mu}\mathbf{A})^{-1}\boldsymbol{\mu}(\mathbf{I}-\mathbf{A})(-\mathbf{L}+\mathbf{1}\boldsymbol{\kappa}) - (\mathbf{I}-\mathbf{A})^{-1}(-\mathbf{I}+\boldsymbol{\alpha}\boldsymbol{\kappa})\right)\boldsymbol{z}_t \\
&= \left((\mathbf{L}-\mathbf{1}\boldsymbol{\kappa}) - (\mathbf{I}-\boldsymbol{\mu}\mathbf{A})^{-1}\boldsymbol{\mu}(\mathbf{I}-\mathbf{A})(\mathbf{L}-\mathbf{1}\boldsymbol{\kappa})\right)\boldsymbol{z}_t \\
&= \left(\mathbf{I} - (\mathbf{I}-\boldsymbol{\mu}\mathbf{A})^{-1}\boldsymbol{\mu}(\mathbf{I}-\mathbf{A})\right)(\mathbf{L}-\mathbf{1}\boldsymbol{\kappa})\boldsymbol{z}_t
\end{aligned}$$

$$
\begin{aligned}
&= \Big((\mathbf{I} - \boldsymbol{\mu}\mathbf{A})^{-1} (\mathbf{I} - \boldsymbol{\mu}\mathbf{A} - \boldsymbol{\mu}(\mathbf{I} - \mathbf{A})) \Big) (\mathbf{L} - \mathbf{1}\boldsymbol{\kappa}) \boldsymbol{z}_t \\
&= \Big((\mathbf{I} - \boldsymbol{\mu}\mathbf{A})^{-1} (\mathbf{I} - \boldsymbol{\mu}) \Big) (\mathbf{L} - \mathbf{1}\boldsymbol{\kappa}) \boldsymbol{z}_t \\
&= \mathbf{Q} \left(\mathbf{L} - \mathbf{1}\boldsymbol{\kappa} \right) \boldsymbol{z}_t .
\end{aligned}
$$

The proof is now complete. □

E.3 Proof of Lemma 4.1

Proof. To study the optimal policy problem, we define such loss as difference of the representative household's utilities in the RI economy and in a reference economy with perfect information. In what follows, variables in the full-information (FI) economy are denoted with *.

Following La'O and Tahbaz-Salehi (2022), we first derive the second-order approximation of the welfare function as

$$
\mathcal{W} = \mathbb{E}\left[U_t - U_t^* \right] \approx -\frac{1}{2} \left[\sum_{i=1}^{N} \lambda_i \theta_i \mathcal{D}_i + (\gamma + 1/\eta) \mathbb{V}\left(c_t - c_t^* \right) + \sum_{i=0}^{N} \lambda_i C_i \right], \tag{E.16}
$$

where $\{C_i, \mathcal{D}_i\}_{i=1}^N$ are defined in the Lemma. The output-gap volatility is defined as,

$$
\mathbb{V}\left(c_t - c_t^* \right) = \frac{1}{(\gamma + 1/\eta)^2} \mathbb{E} \left(\sum_{j=1}^{n} \beta_j \bar{e}_{jt} \right)^2 . \tag{E.17}
$$

U_t and U_t^* are the utilities in the RI and FI economies, respectively.[55] The expectation operator in (4.2) is defined with respect to the central bank (CB)'s information set whose information is assumed to be perfect. For brevity, we omit the algebraic details of this second-order approximation.

Using the equilibrium optimal signal structure we derive in Section 2, the within-sector pricing error variation in sector i is given by

$$
\mathcal{D}_i = \int_0^1 \left(p_{i\iota t} - p_{it} \right)^2 d\iota = \mu_i^2 \mathbb{E} u_{i\iota t}^2 = \mu_i^2 \nu_i^2 = \frac{\chi_i}{\theta_i \lambda_i} \mu_i . \tag{E.18}
$$

where the second and third equality follows from the fact that firms within sector i solve a identical information acquisition problem with endogenous noises drawn from the same distribution: $u_{i\iota t}^2 \sim N(0, \nu_i^2)$. The last equality follows from the optimal solution for noise variance,

$$
\nu_i^2 = \frac{\frac{\chi_i}{\theta_i \lambda_i} \mathbb{V}(\mathrm{mc}_{it})}{\mathbb{V}(\mathrm{mc}_{it}) - \frac{\chi_i}{\theta_i \lambda_i}} = \frac{1}{\mu_i} \frac{\chi_i}{\theta_i \lambda_i} . \tag{E.19}
$$

[55] Both economies are approximated around the same deterministic steady-state. In the flexible-price economy, price levels are indeterminate. We normalize wage such that $W_t^* = W_t$, where W_t in the RI economy is uniquely pinned down by the monetary policy.

(E.18) implies the variations of within-sector pricing errors are determined by the nominal rigidity μ_i and the variance of endogenous noise v_i^2. The term $\mathcal{D}_i$ vanishes when information is perfect ($v_i^2 = 0$), or when information is completely rigid ($\mu_i = 0$). If firms collect information subject to capacity constraint, price flexibility and noise variance move in the opposite direction – higher attentiveness is accompanied by lower variances, with their product equals to a constant $\frac{\chi_i}{\theta_i \lambda_i} > 0$. If the information cost $\chi_i > 0$ is strictly positive, the minimally attainable noise is achieved at the maximum level of price flexibility $\mu_i = 1$ as the volatility of profit-maximizing price converges to infinity,

$$\lim_{\mathbb{V}(\mathrm{mc}_{it}) \to \infty} v_i^2 = \lim_{\mu_i \to 1} v_i^2 = \frac{\chi_i}{\theta_i \lambda_i} > 0. \tag{E.20}$$

That is, even with fully-flexible price, full-information equilibrium cannot be restored given the positive level of noise in RI equilibrium. Meanwhile, (E.18) and (E.19) indicate the price flexibility μ_i plays a dominant role in driving within-sector pricing-error dispersion. □

E.4 Equilibrium Wage Rule and Monetary Policy

In this section, we provide the condition that relates the wage rate with the money supply.

Proposition E.2. *In an equilibrium, the monetary supply and the wage rate jointly satisfy*

$$m_t = \frac{\eta}{1+\gamma\eta} w_t + \left(1 - \frac{\eta}{1+\gamma\eta}\right) \sum_i \beta_i p_{i,t} + \frac{1}{1+\gamma\eta} \sum_i \lambda_i z_{i,t}.$$

That is, in an equilibrium, the monetary authority can alternatively chooses the wage function $\boldsymbol{\kappa}$, obtain the price function via the mapping (2.8), and derive the corresponding monetary policy rule afterwards. This is the strategy we follow when solving for the optimal policy.

Proof. Recall that the households' intratemporal Euler condition, $\frac{W_t}{P_t} C_t^{-\gamma} = L_t^{\frac{1}{\eta}}$, admits the log-linearized form as

$$\ell_t = \eta(w_t - p_t^f) - \eta\gamma c_t. \tag{E.21}$$

Using the derivation on nominal GDP in (E.14), consumers' budget constraint can be written as

$$c_t = w_t - p_t^f + \ell_t - \sum_{i=1}^N \lambda_i \varepsilon_{it} = \eta(w_t - p_t^f) - \gamma\eta c_t + \sum_{i=1}^N \lambda_i z_{it} = \frac{\eta}{1+\gamma\eta}(w_t - p_t^f) + \frac{1}{1+\gamma\eta} \sum_{i=1}^N \lambda_i z_{it}, \tag{E.22}$$

where the second identity follows from (E.21) and (E.15), and the last identity from collapsing the same terms of c_t. Recall that $p_t^f = \boldsymbol{\beta}' \boldsymbol{p}_t$ from the log-linearized form of the final price, then the log-

linearized money supply constraint can be written as

$$m_t = p_t^f + c_t = \frac{\eta}{1+\gamma\eta} w_t + \left(1 - \frac{\eta}{1+\gamma\eta}\right) p_t^f + \frac{1}{1+\gamma\eta} \sum_{i=1}^{N} \lambda_i z_{it} = \frac{\eta}{1+\gamma\eta} w_t + \left(1 - \frac{\eta}{1+\gamma\eta}\right) \sum_i \beta_i p_{i,t} + \frac{1}{1+\gamma\eta} \sum_{i=1}^{N} \lambda_i z_{i,t},$$

as desired. □

E.5 Proof of Lemma 4.2

Proof. Consider the objective function in (4.2), we express the objective in matrix form,

$$\mathcal{W} = \mathbb{E}\left[U_t - U_t^*\right] = -\frac{1}{2}\left[\boldsymbol{\chi}^T \boldsymbol{\mu}^v + \frac{1}{(\gamma + 1/\eta)} \boldsymbol{\beta}^T \boldsymbol{\Sigma}_e \boldsymbol{\beta} + \boldsymbol{\lambda}^T \operatorname{diag}\left(\boldsymbol{\Sigma}_e\right) - \boldsymbol{\lambda}^T \operatorname{diag}\left(A \boldsymbol{\Sigma}_e A^T\right) - \boldsymbol{\beta}^T \boldsymbol{\Sigma}_e \boldsymbol{\beta}\right], \tag{E.23}$$

where $\boldsymbol{\mu}^v = \operatorname{diag}(\boldsymbol{\mu})$ is the digaonal vector defined before. We use the expression in (E.13) to define the covariance matrix of the pricing errors as

$$\boldsymbol{\Sigma}_e = \mathbf{Q}\left(\mathbf{L} - \mathbf{1}\boldsymbol{\kappa}\right) \boldsymbol{\Sigma}_z \left(\mathbf{L} - \mathbf{1}\boldsymbol{\kappa}\right)' \mathbf{Q}' \geq 0,$$

where we adopt the notation $\boldsymbol{\Sigma} = \boldsymbol{\Sigma}_z$ as in (D.6). The central bank's goal is to maximize the expected welfare under rational inattention, or equivalently, minimize the expected welfare loss due to RI. Therefore, we define the CB's optimal policy problem formally as

Definition 2. *The central bank designs optimal monetary policy by solving the following constrained optimization problem*

$$\min_{\{\boldsymbol{\mu}^v, \boldsymbol{\kappa}\}} \mathcal{W} = \frac{1}{2}\left[\boldsymbol{\chi}^T \boldsymbol{\mu}^v + \frac{1}{(\gamma + 1/\eta)} \boldsymbol{\beta}^T \boldsymbol{\Sigma}_e \boldsymbol{\beta} + \boldsymbol{\lambda}^T \operatorname{diag}\left(\boldsymbol{\Sigma}_e\right) - \boldsymbol{\lambda}^T \operatorname{diag}\left(\mathbf{A} \boldsymbol{\Sigma}_e \mathbf{A}^T\right) - \boldsymbol{\beta}^T \boldsymbol{\Sigma}_e \boldsymbol{\beta}\right]$$

subject to the equilibrium fixed-point for each sector $i = 1, 2...N$,

$$\mu_i = \max\left\{0, 1 - \frac{\chi_i}{\theta_i \lambda_i \mathbb{V}(mc_{it})}\right\}; \quad \mathbb{V}(mc_{it}) = \left\| \mathbf{e}_i \left[\left(\mathbf{I} - \mathbf{A}\boldsymbol{\mu}\right)^{-1} \left(-\mathbf{I} + \boldsymbol{\alpha}\boldsymbol{\kappa}\right)\right] \boldsymbol{\Sigma}_z^{1/2} \right\|^2. \tag{E.24}$$

The proof is now complete. □

E.6 Proof of Proposition 4.1

Proof. By Proposition 2.7, the optimal signal structure under fixed-capacity is identical to the elastic attention model. The sectoral nominal rigidities/attentions are exogenously given by $\mu_i = 1 - e^{-2\delta_i} = \frac{1}{1+\tau_i}$, $\forall i = 1, 2, ...N$. Under welfare objective function (E.23), we only need to update the expression for the first term of welfare loss. By (E.18), the within-sector cross-section dispersion of pricing errors

are given by

$$\mathcal{D}_i = \mathbb{E}\left[\int_0^1 \left(p_{i\iota t} - p_{it}\right)^2 d\iota\right] = \mu_i^2 \mathbb{E} u_{i\iota t}^2 = \mu_i^2 \nu_i^2.$$

Using the results in the proof of Proposition 2.7, we note that an important feature of fixed-capacity model is that the variance of endogenous noise is proportional to the volatility of marginal cost:

$$\nu_i^2 = R_i^{-1} \frac{e^{-2\kappa_i}}{1 - e^{-2\kappa_i}} \left\| \Sigma_{\mathbf{z}}^{\frac{1}{2}} \mathbf{G}_i' \right\|^2 = \frac{e^{-2\kappa_i}}{1 - e^{-2\kappa_i}} \mathbb{V}\left(\text{mc}_{it}\right). \tag{E.25}$$

It then follows from the above equation

$$\mathcal{D}_i = \mu_i (1 - \mu_i) \mathbb{V}\left(\text{mc}_{it}\right). \tag{E.26}$$

On the other hand, it follows from (D.55) and (D.56) that under exogenous information, the within-sector cross-section dispersion of pricing errors are given by

$$\mathcal{D}_i = \frac{1}{(1+\tau_i)^2} \int \sum_{j=1}^N m_j^2 u_{i\iota jt}^2 d\iota = \frac{\tau_i}{(1+\tau_i)^2} \sum_{j=1}^N m_j^2 \sigma_j^2 = \mu_i (1 - \mu_i) \mathbb{V}\left(\text{mc}_{it}\right). \tag{E.27}$$

Therefore, (E.27) and (E.26) coincides. More generally, given exogenous $\boldsymbol{\mu}$, the welfare loss functions under two cases are identical, given by (4.2). Since the attention channel of monetary policy is muted in these two cases, the unconstrained optimal monetary policy coincides in these two economies. The solution to the optimal policy is straightforward. The only updated component is given by the derivative of the within-sector welfare loss with respect to sectoral wage rule κ_s,

$$\frac{\partial \mathcal{W}^{within}}{\partial \kappa_s} = \frac{d\mathbb{E}\left[\sum_{i=1}^N \lambda_i \theta_i \vartheta_{it}\right]}{d\kappa_s} = \sum_{i=1}^N \lambda_i \theta_i \mu_i \left(1 - \mu_i\right) \frac{dV_i}{d\kappa_s}, \tag{E.28}$$

where we define $V_i \equiv \mathbb{V}\left(\text{mc}_{it}\right)$ to simplify notation. In the Proof of Proposition 4.2, (E.54), we express the following matrix derivative as

$$\begin{bmatrix} \frac{\partial V_1(\boldsymbol{\kappa})}{\partial \boldsymbol{\kappa}} \\ \frac{\partial V_2(\boldsymbol{\kappa})}{\partial \boldsymbol{\kappa}} \\ \vdots \\ \frac{\partial V_N(\boldsymbol{\kappa})}{\partial \boldsymbol{\kappa}} \end{bmatrix} = 2 \operatorname{diag}\left(\boldsymbol{\rho}\right) \boldsymbol{\mu}^{-1} \left(\mathbf{I} - \mathbf{Q}\right) \left(-\mathbf{L} + \mathbf{1}\boldsymbol{\kappa}\right) \boldsymbol{\Sigma}_{\mathbf{z}}. \tag{E.29}$$

Using the techniques developed in the proof of Lemma 4.3, Proposition E.4, and Proposition 4.3, we write the first order condition w.r.t $\boldsymbol{\kappa}$ in row:

$$\left[\boldsymbol{\lambda}' \operatorname{diag}\left(\left\{\theta_i \rho_i \left(1 - \mu_i\right)\right\}_{i=1}^N\right) \left(\mathbf{I} - \mathbf{Q}\right) + \left\{\left(\frac{1}{(\gamma + 1/\eta)} - 1\right) \left(\boldsymbol{\beta}' \mathbf{Q} \mathbf{1}\right) \boldsymbol{\beta}' + \boldsymbol{\lambda}' \left(\operatorname{diag}\left(\mathbf{Q}\mathbf{1}\right) - \operatorname{diag}\left(\mathbf{A}\mathbf{Q}\mathbf{1}\right) \mathbf{A}\right)\right\} \mathbf{Q}\right] \left(\mathbf{L} - \mathbf{1}\boldsymbol{\kappa}\right) = \mathbf{0}'.$$

Then we obtain the optimal policy weight as

$$\boldsymbol{\varphi} = \boldsymbol{\lambda}' \operatorname{diag}\left(\left\{\theta_i \rho_i \left(1-\mu_i\right)\right\}_{i=1}^{N}\right) + \left\{\left(\frac{1}{(\gamma+1/\eta)} - 1\right)\left(\boldsymbol{\beta}'\mathbf{Q1}\right)\boldsymbol{\beta}' + \boldsymbol{\lambda}'\left(\operatorname{diag}\left(\mathbf{Q1}\right) - \operatorname{diag}\left(\mathbf{AQ1}\right)\mathbf{A}\right)\right\}\mathbf{L}\left(\boldsymbol{\mu}^{-1} - \mathbf{I}\right), \tag{E.30}$$

where we use the fact that $\boldsymbol{\phi} = (\mathbf{I}-\mathbf{Q})(\mathbf{L}-\mathbf{1}\boldsymbol{\kappa})$. Written in scalars,

$$\varphi_i^{exo} = \varphi_i^{fix} = \left[\mu_i\lambda_i\theta_i\rho_i + \frac{(1-\rho_0)}{(\gamma+1/\eta)}\lambda_i + \sum_{j=1}^{N}\left(1-\mu_i\right)\lambda_j\rho_j l_{ji} + \left(\rho_0-\rho_i\right)\lambda_i\right]\left(\frac{1}{\mu_i}-1\right), \tag{E.31}$$

for each sector $i = 1, 2, ..N$. $l_{ji} = (\mathbf{I}-\mathbf{A})^{-1}_{ji}$ is jith element of the Leontief inverse. We refer readers to the proof of Lemma 4.3, Proposition E.4, and Proposition 4.3 for the transformation of the price-stabilization and the algebra details that lead to (E.31). The proof is now complete. □

E.7 Proof of Proposition 4.2

Proof. In the fixed-point system (2.8), the implicit functions of $\boldsymbol{\mu}$ and $\boldsymbol{\kappa}$ can be expressed as

$$\mu_i = \mathcal{T}_i(\boldsymbol{\mu}, \boldsymbol{\kappa}); \qquad i = 1, 2, ...N, \tag{E.32}$$

where $\mathcal{T}_i(\boldsymbol{\mu}, \boldsymbol{\kappa})$ is a scalar best response function of $\boldsymbol{\mu}$ and $\boldsymbol{\kappa}$,

$$\mathcal{T}_i(\boldsymbol{\mu}, \boldsymbol{\kappa}) = 1 - \frac{\chi_i}{\theta_i\lambda_i V_i(\boldsymbol{\mu}, \boldsymbol{\kappa})}; \qquad V_i(\boldsymbol{\mu}, \boldsymbol{\kappa}) = \mathbb{V}(\mathrm{mc}_{it}); \qquad i = 1, 2, ...N. \tag{E.33}$$

To establish a consistent notation with (D.38) and (D.39) in Proposition D.4, we define

$$\mathcal{T}_{\boldsymbol{\mu}} = \frac{\partial \mathcal{T}}{\partial \boldsymbol{\mu}} \equiv \mathcal{T}_{\boldsymbol{\mu}^v} = \frac{\partial \mathcal{T}}{\partial \boldsymbol{\mu}^v}, \tag{E.34}$$

where $\frac{\partial \mathcal{T}}{\partial \boldsymbol{\mu}}$ is defined in the main text. In addition, the $N \times N$ matrix of derivatives in terms of individual component $\mathcal{T}_i$ with respect to matrix $\boldsymbol{\mu}$ is defined in (D.39).

Take differential on both sides of (E.32),

$$d\mathcal{T}_i(\boldsymbol{\mu}, \boldsymbol{\kappa}) = d\mu_i; \qquad i = 1, 2, ...N.$$

We write $\mathcal{T}_i(\boldsymbol{\mu}, \boldsymbol{\kappa})$ briefly as $\mathcal{T}_i$, then $\frac{d\mathcal{T}_i}{d\boldsymbol{\kappa}} = \frac{d\mu_i}{d\boldsymbol{\kappa}}$; $i = 1, 2, ...N$.. In matrix form,

$$\begin{bmatrix} \frac{d\mathcal{T}_1}{d\boldsymbol{\kappa}'} & \frac{d\mathcal{T}_2}{d\boldsymbol{\kappa}'} & \cdots & \frac{d\mathcal{T}_N}{d\boldsymbol{\kappa}'} \end{bmatrix} = \begin{bmatrix} \frac{d\mu_1}{d\boldsymbol{\kappa}'} & \frac{d\mu_2}{d\boldsymbol{\kappa}'} & \cdots & \frac{d\mu_N}{d\boldsymbol{\kappa}'} \end{bmatrix}. \tag{E.35}$$

In the best response function (E.32), given $\boldsymbol{\kappa}$, the differential of μ_i can be written as $d\mu_i = \sum_{j=1}^{N} \frac{\partial \mu_i}{\partial \kappa_j} d\kappa_j$,

for $i = 1, 2, ...N$, which implies that

$$\frac{d\mu_i}{d\kappa_s} = \frac{\partial \mu_i}{\partial \kappa_s}; \qquad i, s = 1, 2, ...N. \tag{E.36}$$

Now consider differential of the best response function $\mathcal{T}_i$,

$$d\mathcal{T}_i = \left(\frac{\partial \mathcal{T}_i}{\partial \mu_1} d\mu_1 + \frac{\partial \mathcal{T}_i}{\partial \mu_2} d\mu_2 + \cdots \frac{\partial \mathcal{T}_i}{\partial \mu_N} d\mu_N\right) + \left(\frac{\partial \mathcal{T}_i}{\partial \kappa_1} d\kappa_1 + \frac{\partial \mathcal{T}_i}{\partial \kappa_2} d\kappa_2 + \cdots \frac{\partial \mathcal{T}_i}{\partial \kappa_N} d\kappa_N\right); \qquad i = 1, 2, ...N. \tag{E.37}$$

Dividing both sides of equation (E.37) by each sector's κ_i, respectively,

$$\begin{aligned}
\frac{d\mathcal{T}_i}{d\kappa_1} &= \left(\frac{\partial \mathcal{T}_i}{\partial \mu_1}\frac{d\mu_1}{d\kappa_1} + \frac{\partial \mathcal{T}_i}{\partial \mu_2}\frac{d\mu_2}{d\kappa_1} + \cdots \frac{\partial \mathcal{T}_i}{\partial \mu_N}\frac{d\mu_N}{d\kappa_1}\right) + \left(\frac{\partial \mathcal{T}_i}{\partial \kappa_1}\right) \\
\frac{d\mathcal{T}_i}{d\kappa_2} &= \left(\frac{\partial \mathcal{T}_i}{\partial \mu_1}\frac{d\mu_1}{d\kappa_2} + \frac{\partial \mathcal{T}_i}{\partial \mu_2}\frac{d\mu_2}{d\kappa_2} + \cdots \frac{\partial \mathcal{T}_i}{\partial \mu_N}\frac{d\mu_N}{d\kappa_2}\right) + \left(\frac{\partial \mathcal{T}_i}{\partial \kappa_2}\right) \\
&\cdots \\
\frac{d\mathcal{T}_i}{d\kappa_N} &= \left(\frac{\partial \mathcal{T}_i}{\partial \mu_1}\frac{d\mu_1}{d\kappa_N} + \frac{\partial \mathcal{T}_i}{\partial \mu_2}\frac{d\mu_2}{d\kappa_N} + \cdots \frac{\partial \mathcal{T}_i}{\partial \mu_N}\frac{d\mu_N}{d\kappa_N}\right) + \left(\frac{\partial \mathcal{T}_i}{\partial \kappa_N}\right).
\end{aligned} \tag{E.38}$$

Consolidating (E.38) into matrix form and combining (E.36), we get

$$\frac{d\mathcal{T}_i}{d\boldsymbol{\kappa}'} = \begin{bmatrix} \frac{\partial \mu_1}{\partial \kappa_1} & \frac{\partial \mu_2}{\partial \kappa_1} & \cdots & \frac{\partial \mu_N}{\partial \kappa_1} \\ \frac{\partial \mu_1}{\partial \kappa_2} & \frac{\partial \mu_2}{\partial \kappa_2} & \cdots & \frac{\partial \mu_N}{\partial \kappa_2} \\ \vdots & \vdots & \vdots & \vdots \\ \frac{\partial \mu_1}{\partial \kappa_N} & \frac{\partial \mu_2}{\partial \kappa_N} & \cdots & \frac{\partial \mu_N}{\partial \kappa_N} \end{bmatrix} \begin{bmatrix} \frac{\partial \mathcal{T}_i}{\partial \mu_1} \\ \frac{\partial \mathcal{T}_i}{\partial \mu_2} \\ \vdots \\ \frac{\partial \mathcal{T}_i}{\partial \mu_N} \end{bmatrix} + \begin{bmatrix} \frac{\partial \mathcal{T}_i}{\partial \kappa_1} \\ \frac{\partial \mathcal{T}_i}{\partial \kappa_2} \\ \vdots \\ \frac{\partial \mathcal{T}_i}{\partial \kappa_N} \end{bmatrix},$$

More specifically,

$$\begin{bmatrix} \frac{d\mathcal{T}_1}{d\boldsymbol{\kappa}'} & \frac{d\mathcal{T}_2}{d\boldsymbol{\kappa}'} & \cdots & \frac{d\mathcal{T}_N}{d\boldsymbol{\kappa}'} \end{bmatrix} = \begin{bmatrix} \frac{\partial \mu_1}{\partial \kappa_1} & \frac{\partial \mu_2}{\partial \kappa_1} & \cdots & \frac{\partial \mu_N}{\partial \kappa_1} \\ \frac{\partial \mu_1}{\partial \kappa_2} & \frac{\partial \mu_2}{\partial \kappa_2} & \cdots & \frac{\partial \mu_N}{\partial \kappa_2} \\ \vdots & \vdots & \vdots & \vdots \\ \frac{\partial \mu_1}{\partial \kappa_N} & \frac{\partial \mu_2}{\partial \kappa_N} & \cdots & \frac{\partial \mu_N}{\partial \kappa_N} \end{bmatrix} \begin{bmatrix} \frac{\partial \mathcal{T}_1}{\partial \mu_1} & \frac{\partial \mathcal{T}_2}{\partial \mu_1} & \cdots & \frac{\partial \mathcal{T}_N}{\partial \mu_1} \\ \frac{\partial \mathcal{T}_1}{\partial \mu_2} & \frac{\partial \mathcal{T}_2}{\partial \mu_2} & \cdots & \frac{\partial \mathcal{T}_N}{\partial \mu_2} \\ \vdots & \vdots & \vdots & \vdots \\ \frac{\partial \mathcal{T}_1}{\partial \mu_N} & \frac{\partial \mathcal{T}_2}{\partial \mu_N} & \cdots & \frac{\partial \mathcal{T}_N}{\partial \mu_N} \end{bmatrix} + \begin{bmatrix} \frac{\partial \mathcal{T}_1}{\partial \kappa_1} & \frac{\partial \mathcal{T}_2}{\partial \kappa_1} & \cdots & \frac{\partial \mathcal{T}_N}{\partial \kappa_1} \\ \frac{\partial \mathcal{T}_1}{\partial \kappa_2} & \frac{\partial \mathcal{T}_2}{\partial \kappa_2} & \cdots & \frac{\partial \mathcal{T}_N}{\partial \kappa_2} \\ \vdots & \vdots & \vdots & \vdots \\ \frac{\partial \mathcal{T}_1}{\partial \kappa_N} & \frac{\partial \mathcal{T}_2}{\partial \kappa_N} & \cdots & \frac{\partial \mathcal{T}_N}{\partial \kappa_N} \end{bmatrix}. \tag{E.39}$$

We define two canonical Jacobian matrices of $\mathcal{T}_i$ with respect to $\boldsymbol{\mu}$ and $\boldsymbol{\kappa}$, respectively as

$$\mathcal{T}_{\boldsymbol{\mu}^v} \equiv \frac{\partial \mathcal{T}}{\partial \boldsymbol{\mu}^v} = \begin{bmatrix} \frac{\partial \mathcal{T}_1}{\partial \mu_1} & \frac{\partial \mathcal{T}_1}{\partial \mu_2} & \cdots & \frac{\partial \mathcal{T}_1}{\partial \mu_N} \\ \frac{\partial \mathcal{T}_2}{\partial \mu_1} & \frac{\partial \mathcal{T}_2}{\partial \mu_2} & \cdots & \frac{\partial \mathcal{T}_2}{\partial \mu_N} \\ \vdots & \vdots & \vdots & \vdots \\ \frac{\partial \mathcal{T}_N}{\partial \mu_1} & \frac{\partial \mathcal{T}_N}{\partial \mu_2} & \cdots & \frac{\partial \mathcal{T}_N}{\partial \mu_N} \end{bmatrix}, \quad \text{and} \quad \mathcal{T}_{\boldsymbol{\kappa}} \equiv \frac{\partial \mathcal{T}}{\partial \boldsymbol{\kappa}} = \begin{bmatrix} \frac{\partial \mathcal{T}_1}{\partial \kappa_1} & \frac{\partial \mathcal{T}_1}{\partial \kappa_2} & \cdots & \frac{\partial \mathcal{T}_1}{\partial \kappa_N} \\ \frac{\partial \mathcal{T}_2}{\partial \kappa_1} & \frac{\partial \mathcal{T}_2}{\partial \kappa_2} & \cdots & \frac{\partial \mathcal{T}_2}{\partial \kappa_N} \\ \vdots & \vdots & \vdots & \vdots \\ \frac{\partial \mathcal{T}_N}{\partial \kappa_1} & \frac{\partial \mathcal{T}_N}{\partial \kappa_2} & \cdots & \frac{\partial \mathcal{T}_N}{\partial \kappa_N} \end{bmatrix}, \tag{E.40}$$

where $\mathcal{T}_{\boldsymbol{\mu}^v}$ is identical to the definition in euqation (D.38). Similarly, we define the canonical Jacobian matrix of $\boldsymbol{\mu}^v$ with respect to $\boldsymbol{\kappa}$ as,

$$J_{\boldsymbol{\mu}^v}(\boldsymbol{\kappa}) \equiv \frac{d\boldsymbol{\mu}^v}{d\boldsymbol{\kappa}} = \begin{bmatrix} \frac{\partial \mu_1}{\partial \kappa_1} & \frac{\partial \mu_1}{\partial \kappa_2} & \cdots & \frac{\partial \mu_1}{\partial \kappa_N} \\ \frac{\partial \mu_2}{\partial \kappa_1} & \frac{\partial \mu_2}{\partial \kappa_2} & \cdots & \frac{\partial \mu_2}{\partial \kappa_N} \\ \vdots & \vdots & \vdots & \vdots \\ \frac{\partial \mu_N}{\partial \kappa_1} & \frac{\partial \mu_2}{\partial \kappa_2} & \cdots & \frac{\partial \mu_N}{\partial \kappa_N} \end{bmatrix}, \tag{E.41}$$

which is defined implicitly by the equilibrium fixed-point system. Note the above equation contains all the information regarding the dependence of the diagonal matrix $\boldsymbol{\mu}$ on $\boldsymbol{\kappa}$. Using the above notations, we express equation (E.39) as

$$\begin{bmatrix} \frac{d\mathcal{T}_1}{d\boldsymbol{\kappa}'} & \frac{d\mathcal{T}_2}{d\boldsymbol{\kappa}'} & \cdots & \frac{d\mathcal{T}_N}{d\boldsymbol{\kappa}'} \end{bmatrix} = J_{\boldsymbol{\mu}^v}(\boldsymbol{\kappa})' \mathcal{T}'_{\boldsymbol{\mu}^v} + \mathcal{T}'_{\boldsymbol{\kappa}}.$$

On the other hand, we rearrange equation (E.36) into matrix form,

$$\begin{bmatrix} \frac{d\mu_1}{d\boldsymbol{\kappa}'} & \frac{d\mu_2}{d\boldsymbol{\kappa}'} & \cdots & \frac{d\mu_N}{d\boldsymbol{\kappa}'} \end{bmatrix} = \begin{bmatrix} \frac{\partial \mu_1}{\partial \kappa_1} & \frac{\partial \mu_2}{\partial \kappa_1} & \cdots & \frac{\partial \mu_N}{\partial \kappa_1} \\ \frac{\partial \mu_1}{\partial \kappa_2} & \frac{\partial \mu_2}{\partial \kappa_2} & \cdots & \frac{\partial \mu_N}{\partial \kappa_2} \\ \vdots & \vdots & \vdots & \vdots \\ \frac{\partial \mu_1}{\partial \kappa_N} & \frac{\partial \mu_2}{\partial \kappa_N} & \cdots & \frac{\partial \mu_N}{\partial \kappa_N} \end{bmatrix} = J_{\boldsymbol{\mu}^v}(\boldsymbol{\kappa})'. \tag{E.42}$$

Substituting (E.41) and (E.42) into (E.35) yields $J_{\boldsymbol{\mu}^v}(\boldsymbol{\kappa})' \mathcal{T}'_{\boldsymbol{\mu}^v} + \mathcal{T}'_{\boldsymbol{\kappa}} = J_{\boldsymbol{\mu}^v}(\boldsymbol{\kappa})'$. Therefore,

$$J_{\boldsymbol{\mu}^v}(\boldsymbol{\kappa}) \equiv \frac{d\boldsymbol{\mu}^v}{d\boldsymbol{\kappa}} = \left[\mathbf{I} - \mathcal{T}_{\boldsymbol{\mu}^v}\right]^{-1} \mathcal{T}_{\boldsymbol{\kappa}}, \tag{E.43}$$

As desired. Next, we proceed to solve the analytical form of two Jacobian matrices $\mathcal{T}_{\boldsymbol{\mu}^v}$ and $\mathcal{T}_{\boldsymbol{\kappa}}$. Define an operator
$\left[\cdot\right]_{(i,j)} : \mathbb{R}^{m\times n} \longmapsto \mathbb{R}$ that extracts the ijth element of any $m \times n$ matrix—-a notation similar to computer programming. Similarly, $\left[\ \cdot\ \right]_{(i,:)} : \mathbb{R}^{m\times n} \longmapsto \mathbb{R}^{1\times n}$ extracts the i row, and $\left[\ \cdot\ \right]_{(:,j)} : \mathbb{R}^{m\times n} \longmapsto \mathbb{R}^{m\times 1}$ extracts the j column of matrices. The next Lemma presents a matrix transformation result.

Lemma E.1. *Given a N-dimensional row vector $\mathbf{v}$ and two $N \times N$ matrices $\mathbf{X}$ and $\mathbf{Y}$, it follows that*

$$\begin{bmatrix} \mathbf{v}\mathbf{e}_1'\mathbf{e}_1\mathbf{X} \\ \mathbf{v}\mathbf{e}_2'\mathbf{e}_2\mathbf{X} \\ \vdots \\ \mathbf{v}\mathbf{e}_N'\mathbf{e}_N\mathbf{X} \end{bmatrix} = \text{diag}(\mathbf{v}')\mathbf{X}; \quad \textit{and} \quad \begin{bmatrix} \left(\text{diag}\left(\mathbf{X}\mathbf{e}_1'\mathbf{e}_1\mathbf{Y}\right)\right)' \\ \left(\text{diag}\left(\mathbf{X}\mathbf{e}_2'\mathbf{e}_2\mathbf{Y}\right)\right)' \\ \vdots \\ \left(\text{diag}\left(\mathbf{X}\mathbf{e}_N'\mathbf{e}_N\mathbf{Y}\right)\right)' \end{bmatrix} = \mathbf{X}' \odot \mathbf{Y},$$

where $\odot$ denotes the Hadamard product, and $\mathbf{e}_i$ denotes the ith standard basis (row) vector.

Proof. By definition,

$$\begin{bmatrix} \mathbf{v}\mathbf{e}_1'\mathbf{e}_1\mathbf{X} \\ \mathbf{v}\mathbf{e}_2'\mathbf{e}_2\mathbf{X} \\ \vdots \\ \mathbf{v}\mathbf{e}_N'\mathbf{e}_N\mathbf{X} \end{bmatrix} = \begin{bmatrix} \mathbf{v}\mathbf{e}_1'\mathbf{e}_1 \\ \mathbf{v}\mathbf{e}_2'\mathbf{e}_2 \\ \vdots \\ \mathbf{v}\mathbf{e}_N'\mathbf{e}_N \end{bmatrix} \mathbf{X} = \text{diag}(\mathbf{v}')\mathbf{X}.$$

Next, we observe that

$$\left(\text{diag}\left(\mathbf{X}\mathbf{e}_i'\mathbf{e}_i\mathbf{Y}\right)\right)' = \left([\mathbf{X}]_{(:,i)}\right)' \odot [\mathbf{Y}]_{(i,:)} = [\mathbf{X}']_{(i,:)} \odot [\mathbf{Y}]_{(i,:)}; \qquad \forall i = 1, 2, ...N,$$

leading to

$$\begin{bmatrix} \left(\text{diag}\left(\mathbf{X}\mathbf{e}_1'\mathbf{e}_1\mathbf{Y}\right)\right)' \\ \left(\text{diag}\left(\mathbf{X}\mathbf{e}_2'\mathbf{e}_2\mathbf{Y}\right)\right)' \\ \vdots \\ \left(\text{diag}\left(\mathbf{X}\mathbf{e}_N'\mathbf{e}_N\mathbf{Y}\right)\right)' \end{bmatrix} = \begin{bmatrix} [\mathbf{X}']_{(1,:)} \odot [\mathbf{Y}]_{(1,:)} \\ [\mathbf{X}']_{(2,:)} \odot [\mathbf{Y}]_{(2,:)} \\ \vdots \\ [\mathbf{X}']_{(N,:)} \odot [\mathbf{Y}]_{(N,:)} \end{bmatrix} = \mathbf{X}' \odot \mathbf{Y}.$$

□

Now recall from Proposition D.4, the Jacobian matirx $\mathcal{T}_{\boldsymbol{\mu}^v}$ is given by

$$\mathcal{T}_{\boldsymbol{\mu}^v} = \left[\text{diag}\left(\frac{\partial \mathcal{T}_1(\boldsymbol{\mu})}{\partial \boldsymbol{\mu}}\right) \quad \text{diag}\left(\frac{\partial \mathcal{T}_2(\boldsymbol{\mu})}{\partial \boldsymbol{\mu}}\right) \quad \cdots \quad \text{diag}\left(\frac{\partial \mathcal{T}_N(\boldsymbol{\mu})}{\partial \boldsymbol{\mu}}\right)\right]'. \tag{E.44}$$

when we treat $\boldsymbol{\kappa}$ as a coefficient row vector. For each sector $i = 1, 2, ...N$,

$$\frac{\partial \mathcal{T}_i(\boldsymbol{\mu})}{\partial \boldsymbol{\mu}} = \frac{\chi_i}{\theta_i \lambda_i \mathbb{V}(\text{mc}_{it})^2} \frac{\partial \mathbb{V}(\text{mc}_{it})}{\partial \boldsymbol{\mu}}; \qquad \frac{\partial \mathbb{V}(\text{mc}_{it})}{\partial \boldsymbol{\mu}} = 2\left[\boldsymbol{\Delta}_{\boldsymbol{\mu}}\boldsymbol{\Gamma}\boldsymbol{\Sigma}_z\boldsymbol{\Gamma}'\boldsymbol{\Delta}_{\boldsymbol{\mu}}'\mathbf{e}_i'\mathbf{e}_i\boldsymbol{\Delta}_{\boldsymbol{\mu}}\mathbf{A}\right]_{d+}.$$

where $\frac{\partial \mathcal{T}_i(\boldsymbol{\mu})}{\partial \boldsymbol{\mu}}$ is defined in (D.39). We transform (E.44) using Lemma E.1 as

$$\begin{aligned}\mathcal{T}_{\boldsymbol{\mu}^v} &= \begin{bmatrix} \left[\operatorname{diag}\left(\frac{\partial \mathcal{T}_1(\boldsymbol{\mu})}{\partial \boldsymbol{\mu}}\right)\right]' \\ \left[\operatorname{diag}\left(\frac{\partial \mathcal{T}_2(\boldsymbol{\mu})}{\partial \boldsymbol{\mu}}\right)\right]' \\ \vdots \\ \left[\operatorname{diag}\left(\frac{\partial \mathcal{T}_N(\boldsymbol{\mu})}{\partial \boldsymbol{\mu}}\right)\right]' \end{bmatrix} = 2\operatorname{diag}\left(\left\{\frac{\chi_i}{\theta_i\lambda_i V_i^2}\right\}_{i=1}^N\right)\begin{bmatrix} \left(\operatorname{diag}\left(\boldsymbol{\Delta}_\mu\boldsymbol{\Gamma}\boldsymbol{\Sigma}_z\boldsymbol{\Gamma}'\boldsymbol{\Delta}_\mu'\mathbf{e}_1'\mathbf{e}_1\boldsymbol{\Delta}_\mu\mathbf{A}\right)\right)' \\ \left(\operatorname{diag}\left(\boldsymbol{\Delta}_\mu\boldsymbol{\Gamma}\boldsymbol{\Sigma}_z\boldsymbol{\Gamma}'\boldsymbol{\Delta}_\mu'\mathbf{e}_2'\mathbf{e}_2\boldsymbol{\Delta}_\mu\mathbf{A}\right)\right)' \\ \vdots \\ \left(\operatorname{diag}\left(\boldsymbol{\Delta}_\mu\boldsymbol{\Gamma}\boldsymbol{\Sigma}_z\boldsymbol{\Gamma}'\boldsymbol{\Delta}_\mu'\mathbf{e}_N'\mathbf{e}_N\boldsymbol{\Delta}_\mu\mathbf{A}\right)\right)' \end{bmatrix} \\ &= 2\operatorname{diag}\left(\left\{\frac{\chi_i}{\theta_i\lambda_i V_i^2}\right\}_{i=1}^N\right)\left[\left(\boldsymbol{\Delta}_\mu\boldsymbol{\Gamma}\boldsymbol{\Sigma}_z\boldsymbol{\Gamma}'\boldsymbol{\Delta}_\mu'\right)\odot\left(\boldsymbol{\Delta}_\mu\mathbf{A}\right)\right],\end{aligned} \tag{E.45}$$

where we simplify the notation $V_i = \mathbb{V}(\mathrm{mc}_{it})$. By equation (D.28) and the definition of $\boldsymbol{\phi} = (\mathbf{I} - \boldsymbol{\mu}\mathbf{A})^{-1}\boldsymbol{\mu}(-\mathbf{I} + \boldsymbol{\alpha}\boldsymbol{\kappa})$, we obtain a matrix identity

$$\boldsymbol{\Delta}_\mu\boldsymbol{\Gamma} = (\mathbf{I} - \mathbf{A}\boldsymbol{\mu})^{-1}(-\mathbf{I} + \boldsymbol{\alpha}\boldsymbol{\kappa}) = \boldsymbol{\mu}^{-1}(\mathbf{I} - \boldsymbol{\mu}\mathbf{A})^{-1}\boldsymbol{\mu}(-\mathbf{I} + \boldsymbol{\alpha}\boldsymbol{\kappa}) = \boldsymbol{\mu}^{-1}\boldsymbol{\phi}. \tag{E.46}$$

Combining the above identities with the definition of covariance matrix of the vector of sectoral marginal cost,

$$\boldsymbol{\Sigma}_{\mathbf{mc}} = \boldsymbol{\mu}^{-1}\boldsymbol{\phi}\boldsymbol{\Sigma}_z\boldsymbol{\phi}'\boldsymbol{\mu}^{-1} = \mathbb{COV}\left(\mathbf{mc}_t, \mathbf{mc}_t\right), \tag{E.47}$$

and $\boldsymbol{\Delta}_{\boldsymbol{\mu}} = (\mathbf{I} - \mathbf{A}\boldsymbol{\mu})^{-1}$, (E.45) now simplifies to

$$\mathcal{T}_{\boldsymbol{\mu}^v} = 2\operatorname{diag}\left(\left\{\frac{1-\mu_i}{\mathbb{V}(\mathrm{mc}_{it})}\right\}_{i=1}^N\right)\left[\mathbb{COV}\left(\mathbf{mc}_t, \mathbf{mc}_t\right)\odot\left((\mathbf{I} - \mathbf{A}\boldsymbol{\mu})^{-1}\mathbf{A}\right)\right], \tag{E.48}$$

where the first term derives from the fixed-point system (2.8) that $\frac{\chi_i}{\theta_i\lambda_i V_i} = 1 - \mu_i$.

For the solution of $\mathcal{T}_{\boldsymbol{\kappa}}$, we follow a similar approach as in Lemma D.4 - D.7, and elaborate on the following proposition.

Proposition E.3. *In the best response function (E.33), the Jacobian matrix $\mathcal{T}_{\boldsymbol{\kappa}}$ in terms of $\boldsymbol{\kappa}$ is given by*

$$\mathcal{T}_{\boldsymbol{\kappa}} = \begin{bmatrix} \frac{\partial \mathcal{T}_1(\boldsymbol{\kappa})}{\partial \boldsymbol{\kappa}'} & \frac{\partial \mathcal{T}_2(\boldsymbol{\kappa})}{\partial \boldsymbol{\kappa}'} & \cdots & \frac{\partial \mathcal{T}_N(\boldsymbol{\kappa})}{\partial \boldsymbol{\kappa}'} \end{bmatrix}'. \tag{E.49}$$

For each sector $i = 1, 2, ...N$,

$$\frac{\partial \mathcal{T}_i(\boldsymbol{\kappa})}{\partial \boldsymbol{\kappa}'} = \frac{\chi_i}{\theta_i\lambda_i V_i(\boldsymbol{\kappa})^2}\frac{\partial V_i(\boldsymbol{\kappa})}{\partial \boldsymbol{\kappa}'}; \qquad \frac{\partial V_i(\boldsymbol{\kappa})}{\partial \boldsymbol{\kappa}'} = 2\boldsymbol{\Sigma}_z\boldsymbol{\Gamma}'\boldsymbol{\Delta}_\mu'\mathbf{e}_i'\mathbf{e}_i\boldsymbol{\Delta}_\mu\boldsymbol{\alpha}. \tag{E.50}$$

Proof. Notice that the defferential of $\boldsymbol{\Gamma} = (-\mathbf{I} + \boldsymbol{\alpha}\boldsymbol{\kappa})$ is given by $d(\boldsymbol{\Gamma}) = \boldsymbol{\alpha}d(\boldsymbol{\kappa})$. Under this circumstance, only $\boldsymbol{\kappa}$ is the independent variable in function $\mathcal{T}_i(\boldsymbol{\kappa})$ and $V_i(\boldsymbol{\kappa})$ with given $\boldsymbol{\mu}$. Therefore, using Lemma

D.4-D.6, we deduce that

$$\begin{aligned} d\left[V_i(\boldsymbol{\kappa})\right] &= d\left(\mathbf{e}_i \boldsymbol{\Delta}_\mu \boldsymbol{\Gamma}\boldsymbol{\Sigma}_z \boldsymbol{\Gamma}' \boldsymbol{\Delta}_\mu' \mathbf{e}_i'\right) \\ &= \operatorname{tr}\left[\mathbf{e}_i \left(\boldsymbol{\Delta}_\mu d(\boldsymbol{\Gamma})\boldsymbol{\Sigma}_z \boldsymbol{\Gamma}' \boldsymbol{\Delta}_\mu' + \boldsymbol{\Delta}_\mu \boldsymbol{\Gamma}\boldsymbol{\Sigma}_z d(\boldsymbol{\Gamma}')\boldsymbol{\Delta}_\mu'\right)\mathbf{e}_i'\right] \\ &= \operatorname{tr}\left[\mathbf{e}_i \boldsymbol{\Delta}_\mu d(\boldsymbol{\Gamma})\boldsymbol{\Sigma}_z \boldsymbol{\Gamma}' \boldsymbol{\Delta}_\mu' \mathbf{e}_i'\right] + \operatorname{tr}\left[\mathbf{e}_i \boldsymbol{\Delta}_\mu \boldsymbol{\Gamma}\boldsymbol{\Sigma}_z d(\boldsymbol{\Gamma}')\boldsymbol{\Delta}_\mu' \mathbf{e}_i'\right] \\ &= 2\operatorname{tr}\left[\mathbf{e}_i \boldsymbol{\Delta}_\mu d(\boldsymbol{\Gamma})\boldsymbol{\Sigma}_z \boldsymbol{\Gamma}' \boldsymbol{\Delta}_\mu' \mathbf{e}_i'\right] \\ &= 2\operatorname{tr}\left[\mathbf{e}_i \boldsymbol{\Delta}_\mu \boldsymbol{\alpha} d(\boldsymbol{\kappa})\boldsymbol{\Sigma}_z \boldsymbol{\Gamma}' \boldsymbol{\Delta}_\mu' \mathbf{e}_i'\right] \\ &= 2\operatorname{tr}\left[\boldsymbol{\Sigma}_z \boldsymbol{\Gamma}' \boldsymbol{\Delta}_\mu' \mathbf{e}_i' \mathbf{e}_i \boldsymbol{\Delta}_\mu \boldsymbol{\alpha} d(\boldsymbol{\kappa})\right]. \end{aligned} \tag{E.51}$$

By Lemma D.4,

$$d\left[V_i(\boldsymbol{\kappa})\right] = \operatorname{tr}\left[\frac{\partial V_i(\boldsymbol{\kappa})}{\partial \boldsymbol{\kappa}'} d\boldsymbol{\kappa}\right], \tag{E.52}$$

where $\frac{\partial V_i(\boldsymbol{\kappa})}{\partial \boldsymbol{\kappa}'} = \left[\frac{\partial V_i(\boldsymbol{\kappa})}{\partial \kappa_1} \;\; \frac{\partial V_i(\boldsymbol{\kappa})}{\partial \kappa_2} \;\; \cdots \;\; \frac{\partial V_i(\boldsymbol{\kappa})}{\partial \kappa_N}\right]'$ serves as a column vector. The juxtaposition of equations (E.51) and (E.52) yields

$$\frac{\partial V_i(\boldsymbol{\kappa})}{\partial \boldsymbol{\kappa}'} = 2\boldsymbol{\Sigma}_z \boldsymbol{\Gamma}' \boldsymbol{\Delta}_\mu' \mathbf{e}_i' \mathbf{e}_i \boldsymbol{\Delta}_\mu \boldsymbol{\alpha}; \quad \text{and} \quad \frac{\partial V_i(\boldsymbol{\kappa})}{\partial \boldsymbol{\kappa}} = 2\boldsymbol{\alpha}' \boldsymbol{\Delta}_\mu' \mathbf{e}_i' \mathbf{e}_i \boldsymbol{\Delta}_\mu \boldsymbol{\Gamma}\boldsymbol{\Sigma}_z.$$

By definition of $\mathcal{T}_{\boldsymbol{\kappa}}$ in (E.40), $\mathcal{T}_{\boldsymbol{\kappa}} = \left[\frac{\partial \mathcal{T}_1(\boldsymbol{\kappa})}{\partial \boldsymbol{\kappa}'} \;\; \frac{\partial \mathcal{T}_2(\boldsymbol{\kappa})}{\partial \boldsymbol{\kappa}'} \;\; \cdots \;\; \frac{\partial \mathcal{T}_N(\boldsymbol{\kappa})}{\partial \boldsymbol{\kappa}'}\right]'$. Similar to (D.41), we use the chain rule for matrix derivatives. For each sector $i = 1, 2, ...N$,

$$\frac{\partial \mathcal{T}_i(\boldsymbol{\kappa})}{\partial \boldsymbol{\kappa}'} = \frac{\chi_i}{\theta_i \lambda_i V_i(\boldsymbol{\kappa})^2} \frac{\partial V_i(\boldsymbol{\kappa})}{\partial \boldsymbol{\kappa}'}; \qquad \frac{\partial V_i(\boldsymbol{\kappa})}{\partial \boldsymbol{\kappa}'} = 2\boldsymbol{\Sigma}_z \boldsymbol{\Gamma}' \boldsymbol{\Delta}_\mu' \mathbf{e}_i' \mathbf{e}_i \boldsymbol{\Delta}_\mu \boldsymbol{\alpha}, \tag{E.53}$$

which completes the proof of Proposition E.3. □

Given the above Proposition, we transform equations (E.49) and (E.50) using Lemma E.1 as

$$
\begin{aligned}
\mathcal{T}_{\boldsymbol{\kappa}} &= \begin{bmatrix} \frac{\partial \mathcal{T}_1(\boldsymbol{\kappa})}{\partial \boldsymbol{\kappa}} \\ \frac{\partial \mathcal{T}_2(\boldsymbol{\kappa})}{\partial \boldsymbol{\kappa}} \\ \vdots \\ \frac{\partial \mathcal{T}_N(\boldsymbol{\kappa})}{\partial \boldsymbol{\kappa}} \end{bmatrix} = \operatorname{diag}\left(\left\{ \frac{\chi_i}{\theta_i \lambda_i V_i^2} \right\}_{i=1}^N \right) \begin{bmatrix} \frac{\partial V_1(\boldsymbol{\kappa})}{\partial \boldsymbol{\kappa}} \\ \frac{\partial V_2(\boldsymbol{\kappa})}{\partial \boldsymbol{\kappa}} \\ \vdots \\ \frac{\partial V_N(\boldsymbol{\kappa})}{\partial \boldsymbol{\kappa}} \end{bmatrix} = 2 \operatorname{diag}\left(\left\{ \frac{\chi_i}{\theta_i \lambda_i V_i^2} \right\}_{i=1}^N \right) \begin{bmatrix} \boldsymbol{\alpha}' \boldsymbol{\Delta}_\mu' \mathbf{e}_1' \mathbf{e}_1 \boldsymbol{\Delta}_\mu \boldsymbol{\Gamma} \boldsymbol{\Sigma}_z \\ \boldsymbol{\alpha}' \boldsymbol{\Delta}_\mu' \mathbf{e}_2' \mathbf{e}_2 \boldsymbol{\Delta}_\mu \boldsymbol{\Gamma} \boldsymbol{\Sigma}_z \\ \vdots \\ \boldsymbol{\alpha}' \boldsymbol{\Delta}_\mu' \mathbf{e}_N' \mathbf{e}_N \boldsymbol{\Delta}_\mu \boldsymbol{\Gamma} \boldsymbol{\Sigma}_z \end{bmatrix} \\
&= 2 \operatorname{diag}\left(\left\{ \frac{\chi_i}{\theta_i \lambda_i V_i^2} \right\}_{i=1}^N \right) \operatorname{diag}\left(\boldsymbol{\Delta}_\mu \boldsymbol{\alpha} \right) \boldsymbol{\Delta}_\mu \boldsymbol{\Gamma} \boldsymbol{\Sigma}_z \\
&= 2 \operatorname{diag}\left(\left\{ \frac{1-\mu_i}{V_i} \right\}_{i=1}^N \right) \operatorname{diag}\left(\boldsymbol{\rho} \right) \boldsymbol{\mu}^{-1} \boldsymbol{\phi} \boldsymbol{\Sigma}_z \\
&= 2 \operatorname{diag}\left(\left\{ \frac{1-\mu_i}{V_i} \right\}_{i=1}^N \right) \left[\boldsymbol{\mu}^{-1} \boldsymbol{\phi} \mathbb{E}\left[\boldsymbol{z}_t \boldsymbol{z}_t' \right] \odot (\mathbf{I} - \mathbf{A}\boldsymbol{\mu})^{-1} \boldsymbol{\alpha} \mathbf{1}' \right] \\
&= 2 \operatorname{diag}\left(\left\{ \frac{1-\mu_i}{\mathbb{V}(\mathrm{mc}_{it})} \right\}_{i=1}^N \right) \left[\mathbb{COV}(\mathbf{mc}_t, \boldsymbol{z}_t) \odot (\mathbf{I} - \mathbf{A}\boldsymbol{\mu})^{-1} \boldsymbol{\alpha} \mathbf{1}' \right],
\end{aligned}
\tag{E.54}
$$

where we define $\boldsymbol{\rho} = (\rho_1, \rho_2, ...\rho_N)' = \boldsymbol{\Delta}_{\boldsymbol{\mu}} \boldsymbol{\alpha} = (\mathbf{I} - \mathbf{A}\boldsymbol{\mu})^{-1} \boldsymbol{\alpha}$ as in the main text, and implement matrix identity (E.46). The proof of Proposition 4.2 is now complete, according to (E.43), (E.48), and (E.54).

□

E.8 Proof of Lemma 4.3

Proof. We trace the direct and indirect impact of monetary policy $\boldsymbol{\kappa}$ on expected welfare loss $\mathcal{W} \equiv \mathbb{E}\left[U_t - U_t^*\right]$ through the lens of functional relations as 4 layers of composite matrix derivatives,

$$
\mathcal{W} = \mathbb{E}\left[U_t - U_t^*\right] \xrightarrow{\frac{\partial L}{\partial \bar{e}_{it}}} \bar{e}_{it} \xrightarrow{\frac{\partial \bar{e}_{it}}{\partial \mathbf{Q}}} \mathbf{Q} \xrightarrow{\frac{\partial \mathbf{Q}}{\partial \boldsymbol{\mu}^v}} \boldsymbol{\mu}^v \xrightarrow{J_{\boldsymbol{\mu}^v}(\boldsymbol{\kappa})} \boldsymbol{\kappa}. \tag{E.55}
$$

Step 1: Computing Derivatives of Average Pricing Errors $\bar{e}_{it}$.
Recall from the definition of sectoral pricing errors

$$
\bar{e}_{it} = \mathbf{e}_i \mathbf{Q} (\mathbf{L} - \mathbf{1}\boldsymbol{\kappa}) \boldsymbol{z}_t, \tag{E.56}
$$

with matrix $\mathbf{Q}$ defined as $\mathbf{Q} = (\mathbf{I} - \boldsymbol{\mu}\mathbf{A})^{-1}(\mathbf{I} - \boldsymbol{\mu})$.

We start with univariate derivative of a single sector, by chain rule,

$$
\frac{d\bar{e}_{it}}{d\kappa_s} = \sum_{r=1}^N \sum_{j=1}^N \frac{\partial \bar{e}_{it}}{\partial Q_{rj}} \frac{\partial Q_{rj}}{\partial \kappa_s} + \frac{\partial \bar{e}_{it}}{\partial \kappa_s} = \operatorname{tr}\left(J_{\bar{e}_{it}}(\mathbf{Q})' J_{\mathbf{Q}}(\kappa_s) \right) + \frac{\partial \bar{e}_{it}}{\partial \kappa_s}, \tag{E.57}
$$

where we define two Jocabian matrices of composite derivatives as

$$J_{\bar{e}_{it}}(\mathbf{Q}) = \begin{bmatrix} \frac{\partial \bar{e}_{it}}{\partial Q_{11}} & \frac{\partial \bar{e}_{it}}{\partial Q_{12}} & \cdots & \frac{\partial \bar{e}_{it}}{\partial Q_{1N}} \\ \frac{\partial \bar{e}_{it}}{\partial Q_{21}} & \frac{\partial \bar{e}_{it}}{\partial Q_{22}} & \cdots & \frac{\partial \bar{e}_{it}}{\partial Q_{2N}} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial \bar{e}_{it}}{\partial Q_{N1}} & \frac{\partial \bar{e}_{it}}{\partial Q_{N2}} & \cdots & \frac{\partial \bar{e}_{it}}{\partial Q_{NN}} \end{bmatrix}; \qquad J_{\mathbf{Q}}(\kappa_s) = \begin{bmatrix} \frac{\partial Q_{11}}{\partial \kappa_s} & \frac{\partial Q_{12}}{\partial \kappa_s} & \cdots & \frac{\partial Q_{1N}}{\partial \kappa_s} \\ \frac{\partial Q_{21}}{\partial \kappa_s} & \frac{\partial Q_{22}}{\partial \kappa_s} & \cdots & \frac{\partial Q_{2N}}{\partial \kappa_s} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial Q_{N1}}{\partial \kappa_s} & \frac{\partial Q_{N2}}{\partial \kappa_s} & \cdots & \frac{\partial Q_{NN}}{\partial \kappa_s} \end{bmatrix}$$

To solve $J_{\bar{e}_{it}}(\mathbf{Q})$, we treat $\boldsymbol{\kappa}$ as a coefficient vector and construct the function of $\bar{e}_{it}(\mathbf{Q})$ only with respect to the variable $\mathbf{Q}$. Under such circumstances, the differential of $\bar{e}_{it}(\mathbf{Q})$ becomes

$$\begin{aligned} d\left[\bar{e}_{it}(\mathbf{Q})\right] &= d\left(\mathbf{e}_i \mathbf{Q}\left(\mathbf{L} - \mathbf{1}\boldsymbol{\kappa}\right)\boldsymbol{z}_t\right) \\ &= \operatorname{tr}\left[\mathbf{e}_i d(\mathbf{Q})\left(\mathbf{L} - \mathbf{1}\boldsymbol{\kappa}\right)\boldsymbol{z}_t\right] \\ &= \operatorname{tr}\left[\left(\mathbf{L} - \mathbf{1}\boldsymbol{\kappa}\right)\boldsymbol{z}_t \mathbf{e}_i d(\mathbf{Q})\right], \end{aligned} \tag{E.58}$$

where we apply results from Lemma D.4-D.6. By definition of matrix differential,

$$d\left[\bar{e}_{it}(\mathbf{Q})\right] = \operatorname{tr}\left[\frac{\partial\left[\bar{e}_{it}(\mathbf{Q})\right]}{\partial \mathbf{Q}'} d(\mathbf{Q})\right] = \operatorname{tr}\left[J_{\bar{e}_{it}}(\mathbf{Q})' d(\mathbf{Q})\right]. \tag{E.59}$$

Since $d(\mathbf{Q})$ is a matrix of differentials for variable $\mathbf{Q}$, in order to equalize (E.58) and (E.59), we must have

$$J_{\bar{e}_{it}}(\mathbf{Q})' = \frac{\partial\left[\bar{e}_{it}(\mathbf{Q})\right]}{\partial \mathbf{Q}'} = \left(\mathbf{L} - \mathbf{1}\boldsymbol{\kappa}\right)\boldsymbol{z}_t \mathbf{e}_i. \tag{E.60}$$

By inspection, the othornomal unit row vector $\mathbf{e}_i$ determines that the elements of $J_{\bar{e}_{it}}(\mathbf{Q})'$ are 0 except for the ith column:

$$J_{\bar{e}_{it}}(\mathbf{Q})' = \begin{bmatrix} 0 & \cdots & \frac{\partial \bar{e}_{it}}{\partial Q_{i1}} & \cdots & 0 \\ 0 & \cdots & \frac{\partial \bar{e}_{it}}{\partial Q_{i2}} & \cdots & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & \cdots & \frac{\partial \bar{e}_{it}}{\partial Q_{iN}} & \cdots & 0 \end{bmatrix}; \qquad \left[J_{\bar{e}_{it}}(\mathbf{Q})'\right]_{(:,i)} = \left(\mathbf{L} - \mathbf{1}\boldsymbol{\kappa}\right)\boldsymbol{z}_t,$$

and $\left[J_{\bar{e}_{it}}(\mathbf{Q})\right]_{(i,:)} = \boldsymbol{z}_t'\left(\mathbf{L} - \mathbf{1}\boldsymbol{\kappa}\right)'$, It follows that (E.57) simplifies to

$$\frac{d\bar{e}_{it}}{d\kappa_s} = \left[J_{\bar{e}_{it}}(\mathbf{Q})\right]_{(i,:)}\left[J_{\mathbf{Q}}(\kappa_s)'\right]_{(:,i)} + \frac{\partial \bar{e}_{it}}{\partial \kappa_s} = \boldsymbol{z}_t'\left(\mathbf{L} - \mathbf{1}\boldsymbol{\kappa}\right)'\left[J_{\mathbf{Q}}(\kappa_s)'\right]_{(:,i)} + \frac{\partial \bar{e}_{it}}{\partial \kappa_s}, \tag{E.61}$$

where the first equality follows from definition of trace operator, and $\left[J_{\mathbf{Q}}(\kappa_s)'\right]_{(:,i)} = \begin{bmatrix} \frac{\partial Q_{i1}}{\partial \kappa_s} & \frac{\partial Q_{i2}}{\partial \kappa_s} & \cdots & \frac{\partial Q_{iN}}{\partial \kappa_s} \end{bmatrix}'$. Using the same matrix calculus techniques from (E.58) - (E.60),

$$\frac{\partial \bar{e}_{it}}{\partial \boldsymbol{\kappa}} = -\boldsymbol{z}_t \mathbf{e}_i \mathbf{Q}\mathbf{1} = -[\mathbf{Q}\mathbf{1}]_{(i)}\boldsymbol{z}_t; \qquad \frac{\partial \bar{e}_{it}}{\partial \kappa_s} = -[\mathbf{Q}\mathbf{1}]_{(i)} z_{st}; \qquad i, s = 1, 2, \ldots N,$$

where $[\mathbf{Q1}]_{(i)} = \mathbf{e}_i\mathbf{Q1}$ is scalar. Therefore, the univariate derivative of average pricing errors in (E.61) admits the form,

$$\frac{d\bar{e}_{it}}{d\kappa_s} = \boldsymbol{z}_t' \left(\mathbf{L} - \mathbf{1}\boldsymbol{\kappa}\right)' \left[J_{\mathbf{Q}}\left(\kappa_s\right)'\right]_{(:,i)} - [\mathbf{Q1}]_{(i)} z_{st}. \tag{E.62}$$

Next, we characterize the key derivative $\left[J_{\mathbf{Q}}\left(\kappa_s\right)'\right]_{(:,i)}$ in (E.57). By definition and chain rule,

$$\left[J_{\mathbf{Q}}\left(\kappa_s\right)'\right]_{(:,i)} = \begin{bmatrix} \frac{\partial Q_{i1}}{\partial \kappa_s} \\ \frac{\partial Q_{i2}}{\partial \kappa_s} \\ \vdots \\ \frac{\partial Q_{iN}}{\partial \kappa_s} \end{bmatrix} = \begin{bmatrix} \frac{\partial Q_{i1}}{\partial (\boldsymbol{\mu}^v)'} \\ \frac{\partial Q_{i2}}{\partial (\boldsymbol{\mu}^v)'} \\ \vdots \\ \frac{\partial Q_{iN}}{\partial (\boldsymbol{\mu}^v)'} \end{bmatrix} \left[J_{\boldsymbol{\mu}^v}(\boldsymbol{\kappa})\right]_{(:,s)}. \tag{E.63}$$

In particular, $\frac{\partial Q_{ij}}{\partial (\boldsymbol{\mu}^v)'} = \begin{bmatrix} \frac{\partial Q_{ij}}{\partial \mu_1} & \frac{\partial Q_{ij}}{\partial \mu_2} & \cdots & \frac{\partial Q_{ij}}{\partial \mu_N} \end{bmatrix}$, $\forall i, j = 1, 2, \ldots..N$, and $J_{\boldsymbol{\mu}^v}(\boldsymbol{\kappa})$ is constructed in (E.41), following the expression in Proposition 4.2.

To solve for $\frac{\partial Q_{ij}}{\partial (\boldsymbol{\mu}^v)'}$ in (E.63), we note that the differential of $\mathbf{Q} = \left(\mathbf{I} - \boldsymbol{\mu}\mathbf{A}\right)^{-1}\left(\mathbf{I} - \boldsymbol{\mu}\right)$ is given by

$$\begin{aligned} d\mathbf{Q} &= d\left(\mathbf{I} - \boldsymbol{\mu}\mathbf{A}\right)^{-1}\left(\mathbf{I} - \boldsymbol{\mu}\right) + \left(\mathbf{I} - \boldsymbol{\mu}\mathbf{A}\right)^{-1} d\left(\mathbf{I} - \boldsymbol{\mu}\right) \\ &= -\left(\mathbf{I} - \boldsymbol{\mu}\mathbf{A}\right)^{-1} d\left(\mathbf{I} - \boldsymbol{\mu}\mathbf{A}\right)\left(\mathbf{I} - \boldsymbol{\mu}\mathbf{A}\right)^{-1}\left(\mathbf{I} - \boldsymbol{\mu}\right) + \left(\mathbf{I} - \boldsymbol{\mu}\mathbf{A}\right)^{-1} d\left(\mathbf{I} - \boldsymbol{\mu}\right) \\ &= \left(\mathbf{I} - \boldsymbol{\mu}\mathbf{A}\right)^{-1} d\boldsymbol{\mu}\mathbf{A}\left(\mathbf{I} - \boldsymbol{\mu}\mathbf{A}\right)^{-1}\left(\mathbf{I} - \boldsymbol{\mu}\right) - \left(\mathbf{I} - \boldsymbol{\mu}\mathbf{A}\right)^{-1} d\boldsymbol{\mu} \\ &= \left(\mathbf{I} - \boldsymbol{\mu}\mathbf{A}\right)^{-1} d\boldsymbol{\mu}\left[\mathbf{A}\left(\mathbf{I} - \boldsymbol{\mu}\mathbf{A}\right)^{-1}\left(\mathbf{I} - \boldsymbol{\mu}\right) - \mathbf{I}\right] \\ &= \left(\mathbf{I} - \boldsymbol{\mu}\mathbf{A}\right)^{-1} d\boldsymbol{\mu}\left(\mathbf{A}\mathbf{Q} - \mathbf{I}\right). \end{aligned} \tag{E.64}$$

For each element of $d\mathbf{Q} \in \mathbb{R}^{N\times N}$,

$$[d\mathbf{Q}]_{(i,j)} = \sum_{l=1}^{N} \left[\left(\mathbf{I} - \boldsymbol{\mu}\mathbf{A}\right)^{-1}\right]_{(i,l)} \left[\left(\mathbf{A}\mathbf{Q} - \mathbf{I}\right)\right]_{(l,j)} d\mu_l; \qquad \forall i, j = 1, 2, \ldots..N,$$

where I use the diagonal property of $\boldsymbol{\mu}$. Hence,

$$\frac{\partial [\boldsymbol{Q}]_{(i,j)}}{\partial \mu_l} = \frac{[d\boldsymbol{Q}]_{(i,j)}}{d\mu_l} = \left[\left(\mathbf{I} - \boldsymbol{\mu}\mathbf{A}\right)^{-1}\right]_{(i,l)} \left[\left(\mathbf{A}\mathbf{Q} - \mathbf{I}\right)\right]_{(l,j)}; \qquad \forall i, j, l = 1, 2, \ldots..N.$$

Stack the above equations in vector form,

$$\frac{\partial [\mathbf{Q}]_{(i,j)}}{\partial \boldsymbol{\mu}^v} = \left\{\left[\left(\mathbf{I} - \boldsymbol{\mu}\mathbf{A}\right)^{-1}\right]_{(i,:)}\right\}' \odot \left[\left(\mathbf{A}\mathbf{Q} - \mathbf{I}\right)\right]_{(:,j)}; \qquad \forall i, j = 1, 2, \ldots..N,$$

which can be transposed to

$$\frac{\partial [\mathbf{Q}]_{(i,j)}}{\partial \left(\boldsymbol{\mu}^v\right)'} = \left(\frac{\partial [\mathbf{Q}]_{(i,j)}}{\partial \boldsymbol{\mu}^v}\right)' = \left[\left(\mathbf{I} - \boldsymbol{\mu}\mathbf{A}\right)^{-1}\right]_{(i,:)} \odot \left[\left(\mathbf{A}\mathbf{Q} - \mathbf{I}\right)'\right]_{(j,:)}; \qquad \forall i, j = 1, 2, \ldots..N. \tag{E.65}$$

If we use the following matrix identities, $\mathbf{AQ}-\mathbf{I}=-\boldsymbol{\mu}^{-1}(\mathbf{I}-\boldsymbol{\mu}\mathbf{A})^{-1}\boldsymbol{\mu}(\mathbf{I}-\mathbf{A})=-(\mathbf{I}-\mathbf{A}\boldsymbol{\mu})^{-1}(\mathbf{I}-\mathbf{A})$, we can stack the vector in (E.65) into matrix form as

$$\begin{bmatrix} \frac{\partial Q_{i1}}{\partial(\boldsymbol{\mu}^v)'} \\ \frac{\partial Q_{i2}}{\partial(\boldsymbol{\mu}^v)'} \\ \vdots \\ \frac{\partial Q_{iN}}{\partial(\boldsymbol{\mu}^v)'} \end{bmatrix} = (\mathbf{AQ}-\mathbf{I})' \operatorname{diag}\left(\left[(\mathbf{I}-\boldsymbol{\mu}\mathbf{A})^{-1}\right]_{(i,:)}\right) = -\left[(\mathbf{I}-\mathbf{A}\boldsymbol{\mu})^{-1}(\mathbf{I}-\mathbf{A})\right]' \operatorname{diag}\left(\left[(\mathbf{I}-\boldsymbol{\mu}\mathbf{A})^{-1}\right]_{(i,:)}\right).$$

The first equality follows from the property of Hadamard product. As such, we obtain a representation of (E.63),

$$\left[J_{\mathbf{Q}}(\kappa_s)'\right]_{(:,i)} = -\left[(\mathbf{I}-\mathbf{A}\boldsymbol{\mu})^{-1}(\mathbf{I}-\mathbf{A})\right]' \operatorname{diag}\left(\left[(\mathbf{I}-\boldsymbol{\mu}\mathbf{A})^{-1}\right]_{(i,:)}\right)\left[J_{\boldsymbol{\mu}^v}(\boldsymbol{\kappa})\right]_{(:,s)}. \tag{E.66}$$

Before we move on to the next step, we express the univariate derivative by substituting (E.66) into (E.62),

$$\begin{aligned} \frac{d\bar{e}_{it}}{d\kappa_s} &= \left[(\mathbf{I}-\mathbf{A}\boldsymbol{\mu})^{-1}(\mathbf{I}-\mathbf{A})(\mathbf{1}\boldsymbol{\kappa}-\mathbf{L})\boldsymbol{z}_t\right]' \operatorname{diag}\left(\left[(\mathbf{I}-\boldsymbol{\mu}\mathbf{A})^{-1}\right]_{(i,:)}\right)\left[J_{\boldsymbol{\mu}^v}(\boldsymbol{\kappa})\right]_{(:,s)} - [\mathbf{Q1}]_{(i)}z_{st} \\ &= \boldsymbol{z}_t'\left[\boldsymbol{\mu}^{-1}(\mathbf{I}-\boldsymbol{\mu}\mathbf{A})^{-1}\boldsymbol{\mu}(-\mathbf{I}+\boldsymbol{\alpha}\boldsymbol{\kappa})\right]' \operatorname{diag}\left(\left[(\mathbf{I}-\boldsymbol{\mu}\mathbf{A})^{-1}\right]_{(i,:)}\right)\left[J_{\boldsymbol{\mu}^v}(\boldsymbol{\kappa})\right]_{(:,s)} - [\mathbf{Q1}]_{(i)}z_{st} \\ &= \boldsymbol{z}_t'\boldsymbol{\phi}'\boldsymbol{\mu}^{-1}\operatorname{diag}\left([\mathbf{H}]_{(i,:)}\right)\left[J_{\boldsymbol{\mu}^v}(\boldsymbol{\kappa})\right]_{(:,s)} - [\mathbf{Q1}]_{(i)}z_{st}. \end{aligned} \tag{E.67}$$

where the second equality follows from the matrix identities (D.28) and $(\mathbf{1}\boldsymbol{\kappa}-\mathbf{L})=\mathbf{L}(-\mathbf{I}+\boldsymbol{\alpha}\boldsymbol{\kappa})$, and the last equality from the definition of $\boldsymbol{\phi}=(\mathbf{I}-\boldsymbol{\mu}\mathbf{A})^{-1}\boldsymbol{\mu}(-\mathbf{I}+\boldsymbol{\alpha}\boldsymbol{\kappa})$ and $\mathbf{H}=(\mathbf{I}-\boldsymbol{\mu}\mathbf{A})^{-1}$. Equation (E.67) measures the total marginal impact of wage rule $\boldsymbol{\kappa}$ on the average pricing errors $\bar{e}_{it}$. We now turn to the step 2.

Step 2: Computing Derivatives of Expected Welfare Loss.

The second-order approximation of the expected welfare loss (4.2) can be expressed as

$$\begin{aligned} \mathcal{W} &= \mathcal{W}^{within} + \mathcal{W}^{OG} + \mathcal{W}^{across} \\ &= \frac{1}{2}\mathbb{E}\left[\sum_{i=1}^{N}\chi_i\mu_i + \frac{1}{(\gamma+1/\eta)}\left(\sum_{j=1}^{n}\beta_j\bar{e}_{jt}\right)^2 + \sum_{i=1}^{N}\lambda_i\sum_{j=1}^{n}a_{ij}\bar{e}_{jt}^2 - \sum_{i=1}^{N}\lambda_i\left(\sum_{j=1}^{n}a_{ij}\bar{e}_{jt}\right)^2 + \sum_{j=1}^{n}\beta_j\bar{e}_{jt}^2 - \left(\sum_{j=1}^{n}\beta_j\bar{e}_{jt}\right)^2\right] \\ &= \frac{1}{2}\mathbb{E}\left[\sum_{i=1}^{N}\chi_i\mu_i + \left(\frac{1}{(\gamma+1/\eta)}-1\right)\left(\sum_{j=1}^{n}\beta_j\bar{e}_{jt}\right)^2 + \sum_{i=1}^{N}\lambda_i\sum_{j=1}^{n}a_{ij}\bar{e}_{jt}^2 - \sum_{i=1}^{N}\lambda_i\left(\sum_{j=1}^{n}a_{ij}\bar{e}_{jt}\right)^2 + \sum_{j=1}^{n}\beta_j\bar{e}_{jt}^2\right], \end{aligned} \tag{E.68}$$

with first-order condition with respect to $\kappa_s, s=1,2,...N$ given by

$$\underbrace{\frac{1}{2}\sum_{i=1}^{N}\mathbb{E}\left[\chi_i\frac{d\mu_i}{d\kappa_s}\right]}_{\text{Term (1)}}+\underbrace{\left(\frac{1}{(\gamma+1/\eta)}-1\right)\sum_{i=1}^{N}\sum_{j=1}^{N}\beta_i\beta_j\mathbb{E}\left[\bar{e}_{it}\frac{d\bar{e}_{jt}}{d\kappa_s}\right]}_{\text{Term (2)}}+\underbrace{\sum_{i=1}^{N}\lambda_i\sum_{j=1}^{N}a_{ij}\mathbb{E}\left[\bar{e}_{jt}\frac{d\bar{e}_{jt}}{d\kappa_s}\right]}_{\text{Term (3)}}-\underbrace{\sum_{i=1}^{N}\lambda_i\sum_{j=1}^{N}\sum_{r=1}^{N}a_{ij}a_{ir}\mathbb{E}\left[\bar{e}_{rt}\frac{d\bar{e}_{jt}}{d\kappa_s}\right]}_{\text{Term (4)}}+\underbrace{\sum_{j=1}^{N}\beta_j\mathbb{E}\left[\bar{e}_{jt}\frac{d\bar{e}_{jt}}{d\kappa_s}\right]}_{\text{Term (5)}}=0. \tag{E.69}$$

Next, we compute term-by-term the expected derivatives in (E.69) using results we have derived above. In deriving these matrix expressions, we need several matrix identities involving Hadamard product and diagonal matrices, summarized in the following lemma.

Lemma E.2. *Let $\mathbf{X}_a$ and $\mathbf{X}_b$ denote two N-dimensional column vectors, then*

$$\mathbf{X}_a'\,\mathrm{diag}\,(X_b)=\mathbf{X}_a'\odot\mathbf{X}_b'.$$

Let $\mathbf{X}_A$ and $\mathbf{X}_B$ denote two $N\times N$ matrices, with elements of $\mathbf{X}_A$ denoted by $[\mathbf{X}_A]_{ij}=x_{ij}^A$. Then

$$\sum_{j=1}^{N}x_{ij}^A\mathbf{e}_j\mathbf{X}_B=[\mathbf{X}_A]_{(i,:)}\,\mathbf{X}_B.$$

Combining these two results together with the definition of Hadamard product,

$$\begin{bmatrix}\mathbf{e}_1\mathbf{X}_A\ \mathrm{diag}\left([\mathbf{X}_B]_{(1,:)}\right)\\ \mathbf{e}_2\mathbf{X}_A\ \mathrm{diag}\left([\mathbf{X}_B]_{(2,:)}\right)\\ \vdots\\ \mathbf{e}_N\mathbf{X}_A\ \mathrm{diag}\left([\mathbf{X}_B]_{(3,:)}\right)\end{bmatrix}=\begin{bmatrix}[\mathbf{X}_A]_{(1,:)}\odot[\mathbf{X}_B]_{(1,:)}\\ [\mathbf{X}_A]_{(2,:)}\odot[\mathbf{X}_B]_{(2,:)}\\ \vdots\\ [\mathbf{X}_A]_{(N,:)}\odot[\mathbf{X}_B]_{(N,:)}\end{bmatrix}=\mathbf{X}_A\odot\mathbf{X}_B.$$

Finally, suppose $\mathbf{X}_C$ is a $N\times N$ matrix,

$$\begin{bmatrix}\left([\mathbf{X}_C]_{(1,:)}\,\mathbf{X}_A\right)\odot\left([\mathbf{X}_C]_{(1,:)}\,\mathbf{X}_B\right)\\ \left([\mathbf{X}_C]_{(2,:)}\,\mathbf{X}_A\right)\odot\left([\mathbf{X}_C]_{(2,:)}\,\mathbf{X}_B\right)\\ \vdots\\ \left([\mathbf{X}_C]_{(N,:)}\,\mathbf{X}_A\right)\odot\left([\mathbf{X}_C]_{(N,:)}\,\mathbf{X}_B\right)\end{bmatrix}=\begin{bmatrix}\left([\mathbf{X}_C]_{(1,:)}\,\mathbf{X}_A\right)\\ \left([\mathbf{X}_C]_{(2,:)}\,\mathbf{X}_A\right)\\ \vdots\\ \left([\mathbf{X}_C]_{(N,:)}\,\mathbf{X}_A\right)\end{bmatrix}\odot\begin{bmatrix}\left([\mathbf{X}_C]_{(1,:)}\,\mathbf{X}_B\right)\\ \left([\mathbf{X}_C]_{(2,:)}\,\mathbf{X}_B\right)\\ \vdots\\ \left([\mathbf{X}_C]_{(N,:)}\,\mathbf{X}_B\right)\end{bmatrix}=\left(\mathbf{X}_C\mathbf{X}_A\right)\odot\left(\mathbf{X}_C\mathbf{X}_B\right).$$

The proof of Lemma E.2 is straightforward and hence omitted. Using this lemma, we carry out the derivation of (E.69).

Solution of Term (1). The first term of (E.69) follows

$$\text{Term (1)}=\sum_{i=1}^{N}\mathbb{E}\left[\chi_i\frac{d\mu_i}{d\kappa_s}\right]=\sum_{i=1}^{N}\chi_i\frac{d\mu_i}{d\kappa_s}=\boldsymbol{\chi}'\frac{d\boldsymbol{\mu}^v}{d\kappa_s}=\boldsymbol{\chi}'\left[J_{\boldsymbol{\mu}^v}(\boldsymbol{\kappa})\right]_{(:,s)}, \tag{E.70}$$

where $J_{\boldsymbol{\mu}^v}(\boldsymbol{\kappa})$ is given in the proof of Proposition 4.2.

Solution of Term (2). To begin with, we notice that a matrix identity holds as

$$\mathbf{Q}\left(\mathbf{L}-\mathbf{1}\boldsymbol{\kappa}\right)=\left[\mathbf{I}-\mathbf{L}(\boldsymbol{\mu}^{-1}-\mathbf{A})\right]\boldsymbol{\phi}=\mathbf{L}(\mathbf{I}-\boldsymbol{\mu}^{-1})\boldsymbol{\phi}, \tag{E.71}$$

where I use the matrix identity $\mathbf{L}\boldsymbol{\alpha}=\mathbf{1}$, and adopt the definition of $\boldsymbol{\phi}=(\mathbf{I}-\boldsymbol{\mu}\mathbf{A})^{-1}\boldsymbol{\mu}(-\mathbf{I}+\boldsymbol{\alpha}\boldsymbol{\kappa})$ and $\mathbf{Q}=(\mathbf{I}-\boldsymbol{\mu}\mathbf{A})^{-1}(\mathbf{I}-\boldsymbol{\mu})$. With equation (E.71) and the definition of $\boldsymbol{\Sigma}_{\mathbf{mc}}=\boldsymbol{\mu}^{-1}\boldsymbol{\phi}\boldsymbol{\Sigma}_z\boldsymbol{\phi}'\boldsymbol{\mu}^{-1}$ in (E.47), we obtain

$$\mathbf{Q}\left(\mathbf{L}-\mathbf{1}\boldsymbol{\kappa}\right)\boldsymbol{\Sigma}_z\boldsymbol{\phi}'\boldsymbol{\mu}^{-1}=\mathbf{L}(\boldsymbol{\mu}-\mathbf{I})\boldsymbol{\Sigma}_{\mathbf{mc}}. \tag{E.72}$$

Next, we use equations (E.56) and (E.67) to derive

$$\begin{aligned}
\mathbb{E}\left[\bar{e}_{it}\frac{d\bar{e}_{jt}}{d\kappa_s}\right] &= \mathbb{E}\Big\{\mathbf{e}_i\mathbf{Q}\left(\mathbf{L}-\mathbf{1}\boldsymbol{\kappa}\right)\boldsymbol{z}_t\left[\boldsymbol{z}_t'\boldsymbol{\phi}'\boldsymbol{\mu}^{-1}\operatorname{diag}\left([\mathbf{H}]_{(j,:)}\right)\left[J_{\boldsymbol{\mu}^v}(\boldsymbol{\kappa})\right]_{(:,s)}-[\mathbf{Q}\mathbf{1}]_{(j)}z_{st}\right]\Big\} \\
&= \mathbf{e}_i\mathbf{Q}\left(\mathbf{L}-\mathbf{1}\boldsymbol{\kappa}\right)\boldsymbol{\Sigma}_z\boldsymbol{\phi}'\boldsymbol{\mu}^{-1}\operatorname{diag}\left([\mathbf{H}]_{(j,:)}\right)\left[J_{\boldsymbol{\mu}^v}(\boldsymbol{\kappa})\right]_{(:,s)}-[\mathbf{Q}\mathbf{1}]_{(j)}\mathbf{e}_i\mathbf{Q}\left(\mathbf{L}-\mathbf{1}\boldsymbol{\kappa}\right)[\boldsymbol{\Sigma}_z]_{(:,s)} \\
&= \mathbf{e}_i\mathbf{L}(\boldsymbol{\mu}-\mathbf{I})\boldsymbol{\Sigma}_{\mathbf{mc}}\operatorname{diag}\left([\mathbf{H}]_{(j,:)}\right)\left[J_{\boldsymbol{\mu}^v}(\boldsymbol{\kappa})\right]_{(:,s)}-[\mathbf{Q}\mathbf{1}]_{(j)}\mathbf{e}_i\mathbf{Q}\left(\mathbf{L}-\mathbf{1}\boldsymbol{\kappa}\right)[\boldsymbol{\Sigma}_z]_{(:,s)},
\end{aligned}$$

where the second identity follows from the definition that $\mathbb{E}\left[\boldsymbol{z}_t\boldsymbol{z}_t'\right]=\boldsymbol{\Sigma}_z$ and $\mathbb{E}\left[z_tz_s\right]=[\boldsymbol{\Sigma}_z]_{(:,s)}$, and the last identity follows the matrix identity of (E.72).

The second term of (E.69) now follows

$$\begin{aligned}
\text{Term (2)} &= \sum_{i=1}^{N}\sum_{j=1}^{N}\beta_i\beta_j\mathbb{E}\left[\bar{e}_{it}\frac{d\bar{e}_{jt}}{d\kappa_s}\right] \\
&= \sum_{i=1}^{N}\sum_{j=1}^{N}\beta_i\beta_j\left[\mathbf{e}_i\mathbf{L}(\boldsymbol{\mu}-\mathbf{I})\boldsymbol{\Sigma}_{\mathbf{mc}}\operatorname{diag}\left([\mathbf{H}]_{(j,:)}\right)\left[J_{\boldsymbol{\mu}^v}(\boldsymbol{\kappa})\right]_{(:,s)}-[\mathbf{Q}\mathbf{1}]_{(j)}\mathbf{e}_i\mathbf{Q}\left(\mathbf{L}-\mathbf{1}\boldsymbol{\kappa}\right)[\boldsymbol{\Sigma}_z]_{(:,s)}\right] \\
&= \sum_{i=1}^{N}\sum_{j=1}^{N}\left(\beta_i\mathbf{e}_i\mathbf{L}(\boldsymbol{\mu}-\mathbf{I})\boldsymbol{\Sigma}_{\mathbf{mc}}\right)\left[\beta_j\operatorname{diag}\left([\mathbf{H}]_{(j,:)}\right)\right]\left[J_{\boldsymbol{\mu}^v}(\boldsymbol{\kappa})\right]_{(:,s)}-\sum_{j=1}^{N}\beta_j[\mathbf{Q}\mathbf{1}]_{(j)}\sum_{i=1}^{N}\beta_i\mathbf{e}_i\mathbf{Q}\left(\mathbf{L}-\mathbf{1}\boldsymbol{\kappa}\right)[\boldsymbol{\Sigma}_z]_{(:,s)} \\
&= \left[\left(\boldsymbol{\beta}'\mathbf{L}\left(\boldsymbol{\mu}-\mathbf{I}\right)\boldsymbol{\Sigma}_{\mathbf{mc}}\right)\odot\left(\boldsymbol{\beta}'\mathbf{H}\right)\right]\left[J_{\boldsymbol{\mu}^v}(\boldsymbol{\kappa})\right]_{(:,s)}-\left(\boldsymbol{\beta}'\mathbf{Q}\mathbf{1}\right)\boldsymbol{\beta}'\mathbf{Q}\left(\mathbf{L}-\mathbf{1}\boldsymbol{\kappa}\right)[\boldsymbol{\Sigma}_z]_{(:,s)} \\
&= \mathbf{r}^{\boldsymbol{\beta}'}\left[J_{\boldsymbol{\mu}^v}(\boldsymbol{\kappa})\right]_{(:,s)}-\left(\boldsymbol{\beta}'\mathbf{Q}\mathbf{1}\right)\boldsymbol{\beta}'\mathbf{Q}\left(\mathbf{L}-\mathbf{1}\boldsymbol{\kappa}\right)[\boldsymbol{\Sigma}_z]_{(:,s)},
\end{aligned} \tag{E.73}$$

where we exploit the property of Hadamard product in Lemma E.2 and define $\mathbf{r}^{\boldsymbol{\beta}'}=\left[\left(\boldsymbol{\beta}'\mathbf{L}\left(\boldsymbol{\mu}-\mathbf{I}\right)\boldsymbol{\Sigma}_{\mathbf{mc}}\right)\odot\left(\boldsymbol{\beta}'\mathbf{H}\right)\right]$, where $\mathbf{H}=(\mathbf{I}-\boldsymbol{\mu}\mathbf{A})^{-1}$.

Solution of Term (3). The third term in (E.69) follows

$$\begin{aligned}
\text{Term (3)} &= \sum_{i=1}^{N} \lambda_i \sum_{j=1}^{N} a_{ij} \mathbb{E}\left[\bar{e}_{jt} \frac{d\bar{e}_{jt}}{d\kappa_s}\right] \\
&= \sum_{i=1}^{N} \lambda_i \sum_{j=1}^{N} a_{ij} \left[\mathbf{e}_j \mathbf{L}(\boldsymbol{\mu} - \mathbf{I})\boldsymbol{\Sigma}_{\mathbf{mc}} \operatorname{diag}\left([\mathbf{H}]_{(j,:)}\right) \left[J_{\mu^v}(\boldsymbol{\kappa})\right]_{(:,s)} - [\mathbf{Q1}]_{(j)} \mathbf{e}_j \mathbf{Q} \left(\mathbf{L} - \mathbf{1}\boldsymbol{\kappa}\right) [\boldsymbol{\Sigma}_z]_{(:,s)}\right] \\
&= \sum_{i=1}^{N} \lambda_i \left[[\mathbf{A}]_{(i,:)} \left[\left(\mathbf{L}\left(\boldsymbol{\mu} - \mathbf{I}\right)\boldsymbol{\Sigma}_{\mathbf{mc}}\right) \odot \mathbf{H}\right] \left[J_{\mu^v}(\boldsymbol{\kappa})\right]_{(:,s)} - [\mathbf{A}]_{(i,:)} \operatorname{diag}\left(\mathbf{Q1}\right) \mathbf{Q} \left(\mathbf{L} - \mathbf{1}\boldsymbol{\kappa}\right) [\boldsymbol{\Sigma}_z]_{(:,s)} \right] \\
&= \boldsymbol{\lambda}' \mathbf{A} \mathbf{R}^{\mathbf{I}} \left[J_{\mu^v}(\boldsymbol{\kappa})\right]_{(:,s)} - \boldsymbol{\lambda}' \mathbf{A} \operatorname{diag}\left(\mathbf{Q1}\right) \mathbf{Q} \left(\mathbf{L} - \mathbf{1}\boldsymbol{\kappa}\right) [\boldsymbol{\Sigma}_z]_{(:,s)} ,
\end{aligned} \tag{E.74}$$

where we define $\mathbf{R}^I = [(\mathbf{L}\left(\boldsymbol{\mu} - \mathbf{I}\right)\boldsymbol{\Sigma}_{\mathbf{mc}}) \odot \mathbf{H}]$.

Solution of Term (4). Similarly, the fourth term of (E.69) follows

$$\begin{aligned}
\text{Term (4)} &= \sum_{i=1}^{N} \lambda_i \sum_{j=1}^{N} \sum_{r=1}^{N} a_{ij} a_{ir} \mathbb{E}\left[\bar{e}_{rt} \frac{d\bar{e}_{jt}}{d\kappa_s}\right] \\
&= \sigma_s^2 \sum_{i=1}^{N} \lambda_i \sum_{j=1}^{N} \sum_{r=1}^{N} a_{ij} a_{ir} \left[\mathbf{e}_r \mathbf{L}(\boldsymbol{\mu} - \mathbf{I})\boldsymbol{\Sigma}_{\mathbf{mc}} \operatorname{diag}\left([\mathbf{H}]_{(j,:)}\right) \left[J_{\mu^v}(\boldsymbol{\kappa})\right]_{(:,s)} - [\mathbf{Q1}]_{(j)} \mathbf{e}_r \mathbf{Q} \left(\mathbf{L} - \mathbf{1}\boldsymbol{\kappa}\right) [\boldsymbol{\Sigma}_z]_{(:,s)}\right] \\
&= \sum_{i=1}^{N} \lambda_i \left\{ \sum_{r=1}^{N} \sum_{j=1}^{N} (a_{ir} e_r \mathbf{L}(\boldsymbol{\mu} - \mathbf{I})\boldsymbol{\Sigma}_{\mathbf{mc}}) \left[a_{ij} \operatorname{diag}\left([\mathbf{H}]_{(j,:)}\right)\right] \left[J_{\mu^v}(\boldsymbol{\kappa})\right]_{(:,s)} - \sum_{j=1}^{N} a_{ij} [\mathbf{Q1}]_{(j)} \sum_{r=1}^{N} a_{ir} e_r \mathbf{Q} \left(\mathbf{L} - \mathbf{1}\boldsymbol{\kappa}\right) [\boldsymbol{\Sigma}_z]_{(:,s)} \right\} \\
&= \sum_{i=1}^{N} \lambda_i \left\{ \left[\left([\mathbf{A}]_{(i,:)} \mathbf{L}(\boldsymbol{\mu} - \mathbf{I})\boldsymbol{\Sigma}_{\mathbf{mc}}\right) \odot \left([\mathbf{A}]_{(i,:)} \mathbf{H}\right)\right] \left[J_{\mu^v}(\boldsymbol{\kappa})\right]_{(:,s)} - \left([\mathbf{A}]_{(i,:)} \mathbf{Q1}\right) \left([\mathbf{A}]_{(i,:)} \mathbf{Q}\right) \left(\mathbf{L} - \mathbf{1}\boldsymbol{\kappa}\right) [\boldsymbol{\Sigma}_z]_{(:,s)} \right\} \\
&= \boldsymbol{\lambda}' \left[(\mathbf{AL}(\boldsymbol{\mu} - \mathbf{I})\boldsymbol{\Sigma}_{\mathbf{mc}}) \odot (\mathbf{AH})\right] \left[J_{\mu^v}(\boldsymbol{\kappa})\right]_{(:,s)} - \boldsymbol{\lambda}' \operatorname{diag}\left(\mathbf{AQ1}\right) \mathbf{AQ} \left(\mathbf{L} - \mathbf{1}\boldsymbol{\kappa}\right) [\boldsymbol{\Sigma}_z]_{(:,s)} \\
&= \boldsymbol{\lambda}' \mathbf{R}^{\mathbf{A}} \left[J_{\mu^v}(\boldsymbol{\kappa})\right]_{(:,s)} - \boldsymbol{\lambda}' \operatorname{diag}\left(\mathbf{AQ1}\right) \mathbf{AQ} \left(\mathbf{L} - \mathbf{1}\boldsymbol{\kappa}\right) [\boldsymbol{\Sigma}_z]_{(:,s)} ,
\end{aligned} \tag{E.75}$$

where we define $\mathbf{R}^A = [(\mathbf{AL}\left(\boldsymbol{\mu} - \mathbf{I}\right)\boldsymbol{\Sigma}_{\mathbf{mc}}) \odot (\mathbf{AH})]$.

Solution of Term (5). Finally, the fifth term of (E.69) follows

$$\begin{aligned}
\text{Term (5)} &= \sum_{j=1}^{N} \beta_j \mathbb{E}\left[\bar{e}_{jt} \frac{d\bar{e}_{jt}}{d\kappa_s}\right] \\
&= \sum_{j=1}^{N} \beta_j \left[\mathbf{e}_j \mathbf{L}(\boldsymbol{\mu} - \mathbf{I})\boldsymbol{\Sigma}_{\mathbf{mc}} \operatorname{diag}\left([\mathbf{H}]_{(j,:)}\right) \left[J_{\mu^v}(\boldsymbol{\kappa})\right]_{(:,s)} - [\mathbf{Q1}]_{(j)} \mathbf{e}_j \mathbf{Q} \left(\mathbf{L} - \mathbf{1}\boldsymbol{\kappa}\right) [\boldsymbol{\Sigma}_z]_{(:,s)}\right] \\
&= \boldsymbol{\beta}' \left[\left(\mathbf{L}\left(\boldsymbol{\mu} - \mathbf{I}\right)\boldsymbol{\Sigma}_{\mathbf{mc}}\right) \odot \mathbf{H}\right] \left[J_{\mu^v}(\boldsymbol{\kappa})\right]_{(:,s)} - \boldsymbol{\beta}' \operatorname{diag}\left(\mathbf{Q1}\right) \mathbf{Q} \left(\mathbf{L} - \mathbf{1}\boldsymbol{\kappa}\right) [\boldsymbol{\Sigma}_z]_{(:,s)} \\
&= \boldsymbol{\beta}' \mathbf{R}^{\mathbf{I}} \left[J_{\mu^v}(\boldsymbol{\kappa})\right]_{(:,s)} - \boldsymbol{\beta}' \operatorname{diag}\left(\mathbf{Q1}\right) \mathbf{Q} \left(\mathbf{L} - \mathbf{1}\boldsymbol{\kappa}\right) [\boldsymbol{\Sigma}_z]_{(:,s)} .
\end{aligned} \tag{E.76}$$

To summarize, we have obtained the derivative expressions of the five terms in the first-order condition (E.69): (E.70), (E.73), (E.74), (E.75) and (E.76). We collect these five expressions according to the three welfare terms expressed in (E.68), and arrange them in row for all $\kappa_s, s = 1, 2, ...N$. The functional relations between attention and welfare loss is characterized in (E.55) . Therefore, using (E.67) and (E.69), it is straightforward to show that the welfare exposure to sectoral attention change is

$$\frac{\partial \mathcal{W}}{\partial (\boldsymbol{\mu}^v)'} = \boldsymbol{r} \tag{E.77}$$

In particular, the row vector $\boldsymbol{r}$ is associated with the Jocabian matrix $J_{\boldsymbol{\mu}^v}(\boldsymbol{\kappa})$,

$$\boldsymbol{r} = \left(\frac{1}{\gamma + 1/\eta} - 1\right)(\boldsymbol{\beta}'\mathbf{M}) \odot \left(\boldsymbol{\beta}'(\mathbf{I} - \boldsymbol{\mu}\mathbf{A})^{-1}\right) + \boldsymbol{\lambda}'\left[\mathbf{M} \odot (\mathbf{I} - \boldsymbol{\mu}\mathbf{A})^{-1} - (\mathbf{A}\mathbf{M}) \odot \left(\mathbf{A}(\mathbf{I} - \boldsymbol{\mu}\mathbf{A})^{-1}\right)\right] + \frac{1}{2}\boldsymbol{\chi}', \tag{E.78}$$

where $\mathbf{M} = \mathbf{L}(\boldsymbol{\mu} - \mathbf{I})\boldsymbol{\Sigma}_{\mathbf{mc}}$, $\mathbf{L} = (\mathbf{I} - \mathbf{A})^{-1}$, and $\mathbf{H} = (\mathbf{I} - \boldsymbol{\mu}\mathbf{A})^{-1}$. Note that we have combined (E.74) and (E.76) as $\boldsymbol{\lambda}'\mathbf{A}\mathbf{R}^{\mathbf{I}} + \boldsymbol{\beta}'\mathbf{R}^{\mathbf{I}} = \boldsymbol{\lambda}'\mathbf{R}^{\mathbf{I}}$ and $\boldsymbol{\lambda}'\mathbf{A} = \boldsymbol{\lambda}' - \boldsymbol{\beta}'$. The proof is now complete. □

E.9 Proof of Proposition 4.3

Proof. We assume that an interior optimal monetary policy exists such that $\mathbf{0} \ll \boldsymbol{\mu}^v \ll \mathbf{1}$. We proceed with the proof in three steps.

Step 1: FOC of Optimal Wage Rule as Matrix Fixed-Point.

In contrast to exogenous-information models, the vector-valued FOC equation (4.3) for $\boldsymbol{\kappa}$ alone cannot pin down the optimal monetary policy. To see this point, we note that Definition 2 suggests the optimal policy relies on sectoral attentions $\boldsymbol{\mu}$, but such choice of optimal policy in turn determines the equilibrium distributions of attentions via (E.24). Motivated by this observation, we characterize the solution of optimal policy in our RI equilibrium as a joint fixed-point between policy and attention.

Proposition E.4. *If an interior optimal monetary policy exists such that $\mathbf{0} \ll \boldsymbol{\mu}^v \ll \mathbf{1}$, it admits the following characterization:*

(i) Conditional on equilibrium sectoral attention allocation $\boldsymbol{\mu}$ and the covariance matrix of the equilibrium marginal costs $\boldsymbol{\Sigma}_{mc}$, the optimal monetary policy (wage) rule $\boldsymbol{\kappa}$ is determined by the fixed-point equation:

$$\boldsymbol{\kappa} = \frac{\boldsymbol{\Phi}(\boldsymbol{\kappa})}{\boldsymbol{\Phi}(\boldsymbol{\kappa})\boldsymbol{\alpha}}. \tag{E.79}$$

The $1 \times N$ vector $\boldsymbol{\Phi}(\boldsymbol{\kappa})$ is given by

$$\boldsymbol{\Phi} = \left\{\left(\frac{1}{(\gamma + 1/\eta)} - 1\right)\left(\boldsymbol{\beta}'\mathbf{Q}\mathbf{1}\right)\boldsymbol{\beta}' + \boldsymbol{\lambda}'\left(\operatorname{diag}(\mathbf{Q}\mathbf{1}) - \operatorname{diag}(\mathbf{A}\mathbf{Q}\mathbf{1})\,\mathbf{A}\right)\right\}\mathbf{Q}\mathbf{L} + \boldsymbol{r}\mathcal{R}\operatorname{diag}\left(\left\{\frac{2(1-\mu_i)\rho_i}{\mu_i V_i}\right\}_{i=1}^{N}\right)(\mathbf{I} - \mathbf{Q})\mathbf{L}, \tag{E.80}$$

where we define $\mathcal{R} \equiv \left[I - \mathcal{T}_{\boldsymbol{\mu}^v}\right]^{-1}$*, which measures the general equilibrium feedback due to strategic complementarity in information acquisition.*

(ii). Conditional on optimal policy rule $\boldsymbol{\kappa}$ *determined by (E.79), the equilibrium attention* $\boldsymbol{\mu}$ *and the covariance structure of marginal costs,* $\boldsymbol{\Sigma_{mc}}$*, are endogenously determined via the fixed-point system (E.24), which implicitly define the mapping from optimal* $\boldsymbol{\kappa}$ *to* $\boldsymbol{\mu}$ *as* $\boldsymbol{\mu} = \mathcal{G}(\boldsymbol{\kappa})$*. Therefore, the optimal wage rule* $\boldsymbol{\kappa}$ *and the endogenous degree of nominal rigidities* $\boldsymbol{\mu}$ *are determined jointly by fixed-point system (E.24) and (E.79).*

Proof. Using expressions in (E.70), (E.73), (E.74), (E.75) and (E.76), combining terms, and empolying the matrix identity $\boldsymbol{\lambda}'\mathbf{A} = \boldsymbol{\lambda}' - \boldsymbol{\beta}'$ from (E.6), the FOC (E.69) now becomes

$$\begin{aligned}&\left\{\frac{1}{2}\boldsymbol{\chi}' + \left(\frac{1}{(\gamma + 1/\eta)} - 1\right)\boldsymbol{r}^{\boldsymbol{\beta}'} + \boldsymbol{\lambda}'\left(\mathbf{R}^{\mathbf{I}} - \mathbf{R}^{\mathbf{A}}\right)\right\}\left[J_{\boldsymbol{\mu}^v}(\boldsymbol{\kappa})\right]_{(:,s)} + \\ &\left\{-\left(\frac{1}{(\gamma + 1/\eta)} - 1\right)(\boldsymbol{\beta}'\mathbf{Q1})\,\boldsymbol{\beta}' - \boldsymbol{\lambda}'\Big(\operatorname{diag}(\mathbf{Q1}) - \operatorname{diag}(\mathbf{AQ1})\,\mathbf{A}\Big)\right\}\mathbf{Q}\,(\mathbf{L} - \mathbf{1}\boldsymbol{\kappa})\,[\boldsymbol{\Sigma}_z]_{(:,s)} = 0.\end{aligned} \tag{E.81}$$

Note that the FOC holds for all sectors $s = 1, 2, ...N$, thus we can arrange equation (E.81) in a row vector for each sector s, and employ the notation in Lemma 4.3:

$$\boldsymbol{r}J_{\boldsymbol{\mu}^v}(\boldsymbol{\kappa}) + \left\{-\left(\frac{1}{(\gamma + 1/\eta)} - 1\right)(\boldsymbol{\beta}'\mathbf{Q1})\,\boldsymbol{\beta}' - \boldsymbol{\lambda}'\Big(\operatorname{diag}(\mathbf{Q1}) - \operatorname{diag}(\mathbf{AQ1})\,\mathbf{A}\Big)\right\}\mathbf{Q}\,(\mathbf{L} - \mathbf{1}\boldsymbol{\kappa})\,\boldsymbol{\Sigma}_z = \mathbf{0}'. \tag{E.82}$$

Now recall $J_{\boldsymbol{\mu}^v}(\boldsymbol{\kappa})$ from Proposition 4.2,

$$J_{\boldsymbol{\mu}^v}(\boldsymbol{\kappa}) = \left[\mathbf{I} - \mathcal{T}_{\boldsymbol{\mu}^v}\right]^{-1}\mathcal{T}_{\boldsymbol{\kappa}} = \mathcal{R}\operatorname{diag}\left(\left\{\frac{2(1-\mu_i)\rho_i}{\mu_i V_i}\right\}_{i=1}^{N}\right)\boldsymbol{\phi}\boldsymbol{\Sigma}_{\mathbf{z}}. \tag{E.83}$$

Since $\mathbf{Q} = (\mathbf{I} - \boldsymbol{\mu}\mathbf{A})^{-1}(\mathbf{I} - \boldsymbol{\mu})$, we obtain

$$\boldsymbol{\phi} = (\mathbf{I} - \boldsymbol{\mu}\mathbf{A})^{-1}\boldsymbol{\mu}(-\mathbf{I} + \boldsymbol{\alpha}\boldsymbol{\kappa}) = (\mathbf{I} - \boldsymbol{\mu}\mathbf{A})^{-1}\left((\mathbf{I} - \boldsymbol{\mu}\mathbf{A}) - (\mathbf{I} - \boldsymbol{\mu})\right)\mathbf{L}(-\mathbf{I} + \boldsymbol{\alpha}\boldsymbol{\kappa}) = (\mathbf{I} - \mathbf{Q})\mathbf{L}(-\mathbf{I} + \boldsymbol{\alpha}\boldsymbol{\kappa}). \tag{E.84}$$

(E.84) serves as an important alternative representation of the pricing function. Using (E.83) and (E.84), equation (E.82) then simplifies to

$$\boldsymbol{r}\mathcal{R}\operatorname{diag}\left(\left\{\frac{2(1-\mu_i)\rho_i}{\mu_i V_i}\right\}_{i=1}^{N}\right)(\mathbf{I} - \mathbf{Q}) + \left\{\left(\frac{1}{(\gamma + 1/\eta)} - 1\right)(\boldsymbol{\beta}'\mathbf{Q1})\,\boldsymbol{\beta}' + \boldsymbol{\lambda}'\Big(\operatorname{diag}(\mathbf{Q1}) - \operatorname{diag}(\mathbf{AQ1})\,\mathbf{A}\Big)\right\}\mathbf{Q}\Big]\mathbf{L}(-\mathbf{I} + \boldsymbol{\alpha}\boldsymbol{\kappa})\boldsymbol{\Sigma}_z = \mathbf{0}', \tag{E.85}$$

where we apply matrix identities $\mathbf{L}\boldsymbol{\alpha} = \mathbf{1}$ and $(\mathbf{L} - \mathbf{1}\boldsymbol{\kappa}) = \mathbf{L}(\mathbf{I} - \boldsymbol{\alpha}\boldsymbol{\kappa})$.

Note that in (E.85), $\boldsymbol{\Sigma}_z$ is invertible with diagonal terms $\sigma_i^2 > 0, \forall i = 1, 2, ...N$, thus we must have

$$\underbrace{r\mathcal{R}\,\mathrm{diag}\left(\left\{\frac{2(1-\mu_i)\rho_i}{\mu_i V_i}\right\}_{i=1}^{N}\right)(\mathbf{I}-\mathbf{Q})+\left\{\left(\frac{1}{(\gamma+1/\eta)}-1\right)(\boldsymbol{\beta}'\mathbf{Q1})\,\boldsymbol{\beta}'+\boldsymbol{\lambda}'\left(\mathrm{diag}\,(\mathbf{Q1})-\mathrm{diag}\,(\mathbf{AQ1})\,\mathbf{A}\right)\right\}\mathbf{Q}}_{\equiv\boldsymbol{\Phi}}\mathbf{L}(-\mathbf{I}+\boldsymbol{\alpha}\boldsymbol{\kappa})=\mathbf{0}', \tag{E.86}$$

where we define $\boldsymbol{\Phi}$ as in (E.80). Therefore, we write the above FOC as $\boldsymbol{\Phi} - \boldsymbol{\Phi}\boldsymbol{\alpha}\boldsymbol{\kappa} = \mathbf{0}'$. Finally, the optimal monetary policy rule is given by the fixed-point

$$\boldsymbol{\kappa} = \frac{\boldsymbol{\Phi}(\boldsymbol{\kappa})}{\boldsymbol{\Phi}(\boldsymbol{\kappa})\,\boldsymbol{\alpha}}, \tag{E.87}$$

as desired. □

Step 2: Transformation to Price-Stabilization

We begin the proof of the second step by a lemma, which provides necessary and sufficient conditions on the existence of price-stabilization policy.

Lemma E.3. *The monetary policy rule under Proposition E.2, which is supported by a nominal wage rule* $w_t = \sum_{i=1}^{N} \kappa_i z_{it}$*, can be implemented by a price-stabilization policy of the form*

$$\sum_{i=1}^{N} \varphi_i p_{it} = 0; \qquad \boldsymbol{\varphi} = (\varphi_1, \ldots, \varphi_N) \in \mathbb{R}^N,$$

if and only if row vector $\boldsymbol{\kappa}$ *satisfies* $\boldsymbol{\kappa}\boldsymbol{\alpha} = 1$.

Proof. Note the existence of price stabilization is equivalent to a linear matrix equation, $\boldsymbol{\varphi}\boldsymbol{\phi} = \mathbf{0}$, which holds for all realizations of shocks $\boldsymbol{z}_t$. Simple linear algebra then implies such stabilization exists if and only if the row and columns of $\boldsymbol{\phi}$ are linear dependent ($\boldsymbol{\phi}$ is non-invertible). By Proposition 2.4,

$$\boldsymbol{\phi} = (\mathbf{I} - \boldsymbol{\mu}\mathbf{A})^{-1}\,\boldsymbol{\mu}\,(-\mathbf{I} + \boldsymbol{\alpha}\boldsymbol{\kappa})$$

is non-invertible in the interior solution of $\boldsymbol{\mu}$ if and only if $\det(-\mathbf{I} + \boldsymbol{\alpha}\boldsymbol{\kappa}) = 0$. Now apply the matrix determinant lemma,

$$\det(-\mathbf{I} + \boldsymbol{\alpha}\boldsymbol{\kappa}) = (1 - \boldsymbol{\kappa}\boldsymbol{\alpha})\det(-\mathbf{I}) = -(1 - \boldsymbol{\kappa}\boldsymbol{\alpha}).$$

Therefore, a price-stabilization policy exists if and only if $\boldsymbol{\kappa}\boldsymbol{\alpha} = 1$, as desired. □

Obviously, the optimal montery policy rule in equation (E.79) and (E.87) satisfies the condition $\boldsymbol{\kappa}\boldsymbol{\alpha} = 1$, thus the optimal monetary policy can be achieved through a price stabilization policy. By equations (E.71) and (E.84), we notice that two matrix identities hold:

$$\boldsymbol{\phi} = (\mathbf{I} - \mathbf{Q})\mathbf{L}(-\mathbf{I} + \boldsymbol{\alpha}\boldsymbol{\kappa}); \qquad \mathbf{Q}\,(\mathbf{1}\boldsymbol{\kappa} - \mathbf{L}) = \mathbf{L}\left(\boldsymbol{\mu}^{-1} - \mathbf{I}\right)\boldsymbol{\phi},$$

where we use the relation $(\mathbf{I}-\mathbf{A})\mathbf{1}=\boldsymbol{\alpha}$. Using these two identities in (E.86), we obtain

$$\left[\boldsymbol{r}\mathcal{R}\,\mathrm{diag}\left(\left\{\frac{2(1-\mu_i)\rho_i}{\mu_i V_i}\right\}_{i=1}^{N}\right)+\left\{\left(\frac{1}{(\gamma+1/\eta)}-1\right)(\boldsymbol{\beta}'\mathbf{Q1})\,\boldsymbol{\beta}'+\boldsymbol{\lambda}'\Big(\mathrm{diag}\,(\mathbf{Q1})-\mathrm{diag}\,(\mathbf{AQ1})\,\mathbf{A}\Big)\right\}\mathbf{L}\left(\boldsymbol{\mu}^{-1}-\mathbf{I}\right)\right]\boldsymbol{\phi}=\mathbf{0}'. \tag{E.88}$$

Therefore, in the spirit of Lemma E.3, the price-stabilization weight is given by

$$\boldsymbol{\varphi}=\left\{\left(\frac{1}{(\gamma+1/\eta)}-1\right)(\boldsymbol{\beta}'\mathbf{Q1})\,\boldsymbol{\beta}'+\boldsymbol{\lambda}'\Big(\mathrm{diag}\,(\mathbf{Q1})-\mathrm{diag}\,(\mathbf{AQ1})\,\mathbf{A}\Big)\right\}\mathbf{L}(\boldsymbol{\mu}^{-1}-\mathbf{I})+\boldsymbol{r}\mathcal{R}\,\mathrm{diag}\left(\left\{\frac{2\rho_i}{V_i}\right\}_{i=1}^{N}\right)(\boldsymbol{\mu}^{-1}-\mathbf{I}).$$

Recall from Proposition E.4 that we define the matrix of general equilibrium propagation (feedbacks) in attentions as $\mathcal{R}\equiv\left[\mathbf{I}-\mathcal{T}_{\boldsymbol{\mu}^v}\right]^{-1}$. Next, we decompose the price-stabilization policy vector $\boldsymbol{\varphi}$ into two components: $\boldsymbol{\varphi}^e=(\varphi_1^e,\varphi_2^e,...\varphi_N^e)$ represents policy response due to endogenous changes in attention (price flexibilities), and $\boldsymbol{\varphi}^x=(\varphi_1^x,\varphi_2^x,...\varphi_N^x)$ represents policy response holding the nominal rigidities constant; that is. $\boldsymbol{\varphi}=\boldsymbol{\varphi}^e+\boldsymbol{\varphi}^x$. The two components then follows

$$\boldsymbol{\varphi}^e=\boldsymbol{r}\mathcal{R}\,\mathrm{diag}\left(\left\{\frac{2\rho_i}{V_i}\right\}_{i=1}^{N}\right)(\boldsymbol{\mu}^{-1}-\mathbf{I}); \tag{E.89}$$

$$\boldsymbol{\varphi}^x=\left\{\left(\frac{1}{(\gamma+1/\eta)}-1\right)(\boldsymbol{\beta}'\mathbf{Q1})\,\boldsymbol{\beta}'+\boldsymbol{\lambda}'\Big(\mathrm{diag}\,(\mathbf{Q1})-\mathrm{diag}\,(\mathbf{AQ1})\,\mathbf{A}\Big)\right\}\mathbf{L}(\boldsymbol{\mu}^{-1}-\mathbf{I}). \tag{E.90}$$

Step 3: Algebraic Details to Scalar Representation (4.5) and (4.11)

First, we write the endogenous components (E.89) in scalar form as

$$\varphi_i^e=2\left(\sum_{j=1}^{N}r_j\left[\mathbf{I}-\mathcal{T}_{\boldsymbol{\mu}^v}\right]_{ji}^{-1}\right)\left(\frac{1}{\mu_i}-1\right)\frac{\rho_i}{\mathbb{V}\,(\mathrm{mc}_{it})}, \tag{E.91}$$

where r_j and $\left[\mathbf{I}-\mathcal{T}_{\boldsymbol{\mu}^v}\right]_{ji}^{-1}$ are elements of $\boldsymbol{r}$ and $\mathcal{R}$, respectively.

Next, we work on the exogenous component (E.90). By definition, $\boldsymbol{\rho}=(\mathbf{I}-\mathbf{A}\boldsymbol{\mu})^{-1}\boldsymbol{\alpha}$ and $\mathbf{Q}=(\mathbf{I}-\boldsymbol{\mu}\mathbf{A})^{-1}(\mathbf{I}-\boldsymbol{\mu})$. Consequently,

$$\mathbf{Q1}=(\mathbf{I}-\boldsymbol{\mu}\mathbf{A})^{-1}(\mathbf{I}-\boldsymbol{\mu})\mathbf{L}\boldsymbol{\alpha}=\left(\mathbf{I}-(\mathbf{I}-\boldsymbol{\mu}\mathbf{A})^{-1}\boldsymbol{\mu}(\mathbf{I}-\mathbf{A})\right)\mathbf{L}\boldsymbol{\alpha}=\left(\mathbf{L}-\boldsymbol{\mu}(\mathbf{I}-\mathbf{A}\boldsymbol{\mu})^{-1}\right)\boldsymbol{\alpha}=\mathbf{1}-\boldsymbol{\mu}\boldsymbol{\rho}, \tag{E.92}$$

where I use the matrix identities $\mathbf{L}\boldsymbol{\alpha}=\mathbf{1}$ and (D.28). We define $\rho_0=\boldsymbol{\beta}'\boldsymbol{\mu}\boldsymbol{\rho}$, which implies that $\boldsymbol{\beta}'\mathbf{Q1}=1-\rho_0$ as $\sum_{i=1}^{N}\beta_i=1$.

Therefore, the component of the policy response holding the nominal rigidities constant can be further simplified as

$$\boldsymbol{\varphi}^x=\boldsymbol{\lambda}'\left\{\left(\frac{1}{(\gamma+1/\eta)}-1\right)(1-\rho_0)\,\mathbf{I}+\Big(\mathbf{I}-\boldsymbol{\mu}\,\mathrm{diag}\,(\boldsymbol{\rho})-\mathrm{diag}\,(\mathbf{A1}-\mathbf{A}\boldsymbol{\mu}\boldsymbol{\rho})\,\mathbf{A}\Big)\mathbf{L}\right\}(\boldsymbol{\mu}^{-1}-\mathbf{I}),$$

where I use $\boldsymbol{\beta}'\mathbf{L} = \boldsymbol{\lambda}'$ from (E.6). Next, we derive a matrix identity $\mathbf{A}\boldsymbol{\mu}\boldsymbol{\rho} = (\mathbf{I} - (\mathbf{I} - \mathbf{A}\boldsymbol{\mu}))(\mathbf{I} - \mathbf{A}\boldsymbol{\mu})^{-1}\boldsymbol{\alpha} = (\mathbf{I} - \mathbf{A}\boldsymbol{\mu})^{-1}\boldsymbol{\alpha} - \boldsymbol{\alpha} = \boldsymbol{\rho} - (\mathbf{I} - \mathbf{A})\mathbf{1}$, where $\boldsymbol{\alpha} = (\mathbf{I} - \mathbf{A})\mathbf{1}$. We then obtain

$$\begin{aligned}\Big(\mathbf{I} - \boldsymbol{\mu}\,\mathrm{diag}\,(\boldsymbol{\rho}) - \mathrm{diag}\,(\mathbf{A}\mathbf{1} - \mathbf{A}\boldsymbol{\mu}\boldsymbol{\rho})\,\mathbf{A}\Big)\mathbf{L} &= \Big(\mathbf{I} - \boldsymbol{\mu}\,\mathrm{diag}\,(\boldsymbol{\rho}) - \mathbf{A} + \mathrm{diag}\,(\boldsymbol{\rho})\,\mathbf{A}\Big)\mathbf{L} \\ &= \Big((\mathbf{I} - \mathbf{A}) + (\mathbf{I} - \boldsymbol{\mu})\,\mathrm{diag}\,(\boldsymbol{\rho}) - \mathrm{diag}\,(\boldsymbol{\rho})\,(\mathbf{I} - \mathbf{A})\Big)\mathbf{L} \\ &= \mathbf{I} + (\mathbf{I} - \boldsymbol{\mu})\,\mathrm{diag}\,(\boldsymbol{\rho})\,\mathbf{L} - \mathrm{diag}\,(\boldsymbol{\rho})\,.\end{aligned}$$

Consequently, the component of the policy response holding the nominal rigidities constant simplifies to

$$\boldsymbol{\varphi}^x = \boldsymbol{\lambda}'\left\{\left(\frac{(1-\rho_0)}{(\gamma + 1/\eta)} + \rho_0\right)\mathbf{I} + (\mathbf{I} - \boldsymbol{\mu})\,\mathrm{diag}\,(\boldsymbol{\rho})\,\mathbf{L} - \mathrm{diag}\,(\boldsymbol{\rho})\right\}(\boldsymbol{\mu}^{-1} - \mathbf{I}). \tag{E.93}$$

It follows the exogenous component of weight in scalar form is given by

$$\varphi_i^x = \left[\frac{(1-\rho_0)}{(\gamma + 1/\eta)}\lambda_i + \sum_{j=1}^{N}(1-\mu_i)\,\lambda_j\rho_j l_{ji} + (\rho_0 - \rho_i)\,\lambda_i\right]\left(\frac{1}{\mu_i} - 1\right). \tag{E.94}$$

Next, we modify the formula (E.91) and (E.94) by adding and subtracting the exogenous component associated with the within-sector price dispersion under the exogenous-information (fixed-capacity) model, that is, the last component in (4.5): $\mu_i\lambda_i\theta_i\rho_i\left(\frac{1}{\mu_i} - 1\right)$. In this case, the exogenous component exactly align with its counterpart under exogenous-information:

$$\varphi_i^x = \left[\frac{(1-\rho_0)}{(\gamma + 1/\eta)}\lambda_i + \sum_{j=1}^{N}(1-\mu_i)\,\lambda_j\rho_j l_{ji} + (\rho_0 - \rho_i)\,\lambda_i + \mu_i\lambda_i\theta_i\rho_i\right]\left(\frac{1}{\mu_i} - 1\right). \tag{E.95}$$

Now (E.95) and (4.5) coincide. The modified endogenous component then follows,

$$\begin{aligned}\varphi_i^e &= 2\left(\sum_{j=1}^{N}\boldsymbol{r}_j\left[\mathbf{I} - \mathcal{T}_{\boldsymbol{\mu}^v}\right]_{ji}^{-1}\right)\left(\frac{1}{\mu_i} - 1\right)\frac{\rho_i}{\mathbb{V}\,(\mathrm{mc}_{it})} - \mu_i\lambda_i\theta_i\rho_i\left(\frac{1}{\mu_i} - 1\right) \\ &= \left\{2\left(\sum_{j=1}^{N}\boldsymbol{r}_j\left[\mathbf{I} - \mathcal{T}_{\boldsymbol{\mu}^v}\right]_{ji}^{-1}\right) - (\lambda_i\theta_i\mathbb{V}\,(\mathrm{mc}_{it}) - \chi_i)\right\}\frac{\rho_i}{\mu_i}\frac{1-\mu_i}{\mathbb{V}\,(\mathrm{mc}_{it})}.\end{aligned}$$

where the last equality follows from the equilibrium fixed-point (2.8). Notice that we have unified the notation in (E.34) such that $\mathcal{T}_{\boldsymbol{\mu}} \equiv \mathcal{T}_{\boldsymbol{\mu}^v}$ and $\boldsymbol{r}$ is defined in (E.77) and (E.78). The proof is now complete. □

E.10 Proof of Proposition 4.4

Proof. Recall that the output-gap volatility in terms of cross-sectional average pricing errors is given by (E.17). We first state the optimal output-gap volatility optimization problem in the following definition.

Definition 3. *The central bank designs optimal OG policy by solving the following constrained optimization problem*

$$\min_{\{\boldsymbol{\mu}^v,\boldsymbol{\kappa}\}} \mathbb{V}\left(c_t - c_t^*\right) = \mathbb{E}\left[\frac{1}{(\gamma+1/\eta)^2}\left(\sum_{j=1}^{n}\beta_j\bar{e}_{jt}\right)^2\right],$$

subject to the equilibrium fixed-point for each sector $i = 1, 2...N$,

$$\mu_i = \max\left\{0, 1 - \frac{\chi_i}{\theta_i\lambda_i\mathbb{V}(mc_{it})}\right\}; \quad \mathbb{V}(mc_{it}) = \left\|\mathbf{e}_i\left[(\mathbf{I}-\mathbf{A}\boldsymbol{\mu})^{-1}(-\mathbf{I}+\boldsymbol{\alpha}\boldsymbol{\kappa})\right]\boldsymbol{\Sigma}_z^{1/2}\right\|^2.$$

In order to obtain the optimal OG policy, we need to solve the first-order condition of the expected output-gap volatility on the monetary policy rule,

$$\frac{d\mathbb{E}\left[\Sigma_t\right]}{d\kappa_s} = \frac{1}{(\gamma+1/\eta)^2}\sum_{i=1}^{N}\sum_{j=1}^{N}\beta_i\beta_j\mathbb{E}\left[\bar{e}_{it}\frac{d\bar{e}_{jt}}{d\kappa_s}\right] = 0; \Leftrightarrow \mathbf{r}^{\beta'}\left[J_{\boldsymbol{\mu}^v}(\boldsymbol{\kappa})\right]_{(:,s)} - (\boldsymbol{\beta}'\mathbf{Q}\mathbf{1})\,\boldsymbol{\beta}'\mathbf{Q}\,(\mathbf{L}-\mathbf{1}\boldsymbol{\kappa})\left[\boldsymbol{\Sigma}_z\right]_{(:,s)} = 0,$$

where the second identity follows from (E.73), and $\mathbf{r}^{\beta'} = [(\boldsymbol{\beta}'\mathbf{L}\,(\boldsymbol{\mu}-\mathbf{I})\,\boldsymbol{\Sigma}_{\mathbf{mc}}) \odot (\boldsymbol{\beta}'\mathbf{H})]$. Applying the same methods we employed to derive (E.86), the above FOC condition can be simplified to

$$\left[\mathbf{r}^{\beta'}\mathcal{R}\,\mathrm{diag}\left(\left\{\frac{2(1-\mu_i)\rho_i}{\mu_i V_i}\right\}_{i=1}^{N}\right)(\mathbf{I}-\mathbf{Q}) + (\boldsymbol{\beta}'\mathbf{Q}\mathbf{1})\,\boldsymbol{\beta}'\mathbf{Q}\right]\mathbf{L}(-\mathbf{I}+\boldsymbol{\alpha}\boldsymbol{\kappa})\boldsymbol{\Sigma}_z = \mathbf{0}'. \tag{E.96}$$

Therefore, the optimal OG policy rule is given by the fixed-point $\boldsymbol{\kappa}^{OG} = \frac{\boldsymbol{\Phi}^{OG}}{\boldsymbol{\Phi}^{OG}\boldsymbol{\alpha}}$, where we define

$$\boldsymbol{\Phi}^{OG} = \mathbf{r}^{\beta'}\mathcal{R}\,\mathrm{diag}\left(\left\{\frac{2(1-\mu_i)\rho_i}{\mu_i V_i}\right\}_{i=1}^{N}\right)(\mathbf{I}-\mathbf{Q})\mathbf{L} + (\boldsymbol{\beta}'\mathbf{Q}\mathbf{1})\,\boldsymbol{\beta}'\mathbf{Q}\mathbf{L}.$$

Next, recall from Lemma E.3, the optimal OG policy can be implemented through a price-stabilization policy as

$$\begin{aligned}\boldsymbol{\varphi}^{OG} &= \left[\mathbf{r}^{\beta'}\mathcal{R}\,\mathrm{diag}\left(\left\{\frac{2(1-\mu_i)\rho_i}{\mu_i V_i}\right\}_{i=1}^{N}\right)(\mathbf{I}-\mathbf{Q})\mathbf{L} + (\boldsymbol{\beta}'\mathbf{Q}\mathbf{1})\,\boldsymbol{\beta}'\mathbf{Q}\mathbf{L}\right]\boldsymbol{\mu}^{-1}(\mathbf{I}-\boldsymbol{\mu}\mathbf{A}) \\ &= \underbrace{\mathbf{r}^{\beta'}\mathcal{R}\,\mathrm{diag}\left(\left\{\frac{2(1-\mu_i)\rho_i}{\mu_i V_i}\right\}_{i=1}^{N}\right)}_{\boldsymbol{\varphi}^{e,OG}} + \underbrace{(1-\rho_0)\,\boldsymbol{\lambda}'\left(\boldsymbol{\mu}^{-1}-\mathbf{I}\right)}_{\boldsymbol{\varphi}^{x,OG}},\end{aligned} \tag{E.97}$$

where the last identity follows from matrix identities[56] $\mathbf{QL} = \mathbf{L}\left(\boldsymbol{\mu}^{-1} - \mathbf{I}\right)(\mathbf{I} - \boldsymbol{\mu}\mathbf{A})^{-1}\boldsymbol{\mu}$, $(\mathbf{I} - \mathbf{Q})\mathbf{L} = (\mathbf{I} - \boldsymbol{\mu}\mathbf{A})^{-1}\boldsymbol{\mu}$, $\boldsymbol{\beta}'\mathbf{Q}\mathbf{1} = 1 - \rho_0$, and $\boldsymbol{\beta}'\mathbf{L} = \boldsymbol{\lambda}'$. Equation (E.97) shows the price-stabilization policy can be splited into two components under the OG optimization problem, which is similar to our approach in Proposition 4.3. The next proposition characterizes a condition under which the optimal OG policy satisfies $\boldsymbol{\varphi}^{e,OG} \equiv \mathbf{0}$.

Proposition E.5. *If the interior optimal OG policy completely stabilizes output-gap volatility ($\mathbb{V}\left(c_t - c_t^*\right) = 0$), at the optimum the price-stabilization policy in (E.97) does not respond to endogenous changes in attention (price flexibilities); that is, $\boldsymbol{\varphi}^{e,OG} \equiv \mathbf{0}$.*

Proof. In matrix form, the expected output-gap volatility can be expressed as

$$\mathbb{V}\left(c_t - c_t^*\right) = \frac{1}{(\gamma + 1/\eta)^2}\boldsymbol{\beta}'\boldsymbol{\Sigma}_e\boldsymbol{\beta} = \left\|\boldsymbol{\beta}'\mathbf{Q}\left(\mathbf{L} - \mathbf{1}\boldsymbol{\kappa}\right)\boldsymbol{\Sigma}_z^{1/2}\right\|^2 \geq 0,$$

where $\boldsymbol{\Sigma}_e = \mathbf{Q}\left(\mathbf{L} - \mathbf{1}\boldsymbol{\kappa}\right)\boldsymbol{\Sigma}_z\left(\mathbf{L} - \mathbf{1}\boldsymbol{\kappa}\right)'\mathbf{Q}'$ is defined as the covariance matrix of the cross-sectional average of sectoral pricing errors. Under the interior optimal OG policy, if the output-gap volatility vanishes, then $\left(\boldsymbol{\beta}'\mathbf{Q}\left(\mathbf{L} - \mathbf{1}\boldsymbol{\kappa}\right)\boldsymbol{\Sigma}_z^{1/2}\right) = \mathbf{0}'$. Then at the optimum, since $\boldsymbol{\Sigma}_z^{1/2} > 0$ is invertible,

$$\mathbf{r}^{\beta'} = \left[\left(\boldsymbol{\beta}'\mathbf{L}\left(\boldsymbol{\mu} - \mathbf{I}\right)\boldsymbol{\Sigma}_{\mathbf{mc}}\right) \odot \left(\boldsymbol{\beta}'\mathbf{H}\right)\right] = \left[\left(\left(\boldsymbol{\beta}'\mathbf{Q}\left(\mathbf{L} - \mathbf{1}\boldsymbol{\kappa}\right)\boldsymbol{\Sigma}_z^{1/2}\right)\boldsymbol{\Sigma}_z^{1/2}\boldsymbol{\phi}'\boldsymbol{\mu}^{-1}\right) \odot \left(\boldsymbol{\beta}'\mathbf{H}\right)\right] \equiv \mathbf{0}.$$

by matrix identity (E.72). Therefore, by (E.96) the optimal OG policy rule reduces to $\boldsymbol{\kappa}^{OG} = \frac{\boldsymbol{\beta}'\mathbf{QL}}{\boldsymbol{\beta}'\mathbf{Q}\mathbf{1}}$.

As a result, $\boldsymbol{\varphi}^{e,OG} \equiv \mathbf{0}$ and the endogenous part of the price-stabilizaiton condition in (E.97) vanishes. □

Given the above property of the optimal OG policy, the price-stabilization now simplifies to

$$\boldsymbol{\varphi}^{OG} = \boldsymbol{\varphi}^{x,OG} = \left(1 - \rho_0\right)\boldsymbol{\lambda}'\left(\boldsymbol{\mu}^{-1} - \mathbf{I}\right); \qquad \boldsymbol{\varphi}^{e,OG} \equiv \mathbf{0},$$

which can be expressed as scalar form,

$$\varphi_i = \varphi_i^{x,OG} = \frac{\left(1 - \rho_0\right)}{(\gamma + 1/\eta)}\lambda_i\left(\frac{1}{\mu_i} - 1\right); \qquad \varphi_i^{e,OG} \equiv 0; \qquad i = 1, 2, ...N.$$

The proof is now complete. □

[56] Specifically, we notice that by (E.71) $\mathbf{QL} = \mathbf{L}(\boldsymbol{\mu}^{-1} - \mathbf{I})(\mathbf{I} - \boldsymbol{\mu}\mathbf{A})^{-1}\boldsymbol{\mu}$. Combining this equation with (D.28) yields

$$(\mathbf{I} - \mathbf{Q})\mathbf{L} = \mathbf{L} - \mathbf{L}(\boldsymbol{\mu}^{-1} - \mathbf{I})(\mathbf{I} - \boldsymbol{\mu}\mathbf{A})^{-1}\boldsymbol{\mu} = \mathbf{L}\left(\boldsymbol{\mu}^{-1}(\mathbf{I} - \boldsymbol{\mu}\mathbf{A}) - (\boldsymbol{\mu}^{-1} - \mathbf{I})\right)(\mathbf{I} - \boldsymbol{\mu}\mathbf{A})^{-1}\boldsymbol{\mu} = (\mathbf{I} - \boldsymbol{\mu}\mathbf{A})^{-1}\boldsymbol{\mu}.$$

The matrix identities $\boldsymbol{\beta}'\mathbf{Q}\mathbf{1} = 1 - \rho_0$ follows from (E.92) and $\boldsymbol{\beta}'\mathbf{L} = \boldsymbol{\lambda}'$ from (E.6).

E.11 Proof of Proposition 4.5

Proof. Under rational inattention, when only minimizing welfare loss associated with the within-sector price dispersion. The objective function is simply

$$\mathcal{W}^{within} = \frac{1}{2}\boldsymbol{\chi}'\boldsymbol{\mu}^{v}$$

Therefore, optimal monetary policy only operate through the endogenous-attention channel. The mathematics is identical to our previous derivations, and the optimal policy weight is simply

$$\varphi_i^{within} = \left(\sum_{j=1}^{N} \chi_j \mathcal{R}_{ji}\right) \frac{\rho_i}{\mathbb{V}(\text{mc}_{it})} \left(\frac{1}{\mu_i} - 1\right). \tag{E.98}$$

Again we notice that by definition $\mathcal{R} \equiv \left[\mathbf{I} - \mathcal{T}_{\boldsymbol{\mu}^v}\right]^{-1}$ and $\mathcal{T}_{\boldsymbol{\mu}^v} = \mathcal{T}_{\boldsymbol{\mu}}$ in the main text notation. Therefore,

$$\varphi_i^{within} = \left(\sum_{j=1}^{N} \chi_j \left[\mathbf{I} - \mathcal{T}_{\boldsymbol{\mu}}\right]_{ji}^{-1}\right) \frac{\rho_i}{\mathbb{V}(\text{mc}_{it})} \left(\frac{1}{\mu_i} - 1\right). \tag{E.99}$$

The proof is complete.

□

F. Calibration and Quantitative Analysis

F.1 Calibration of Information Cost

We use indirect inference to calibrate the marginal information cost χ_i in different sectors. In our models with monetary policy, any model-implied moment is endogeous/variant to the choice of policy. Therefore, we first fix a policy environment that is consistent with the conduct of monetary policy in our data sample period.

CPI policy environment. We focus on a model with monetary policy that targets CPI stability (inflation targeting). This is consistent with the emprical evidence that within our sample period, the CPI is largely constant. We consider the following price stabilization

$$\text{CPI} = \sum_{i=1}^{N} \varphi_i p_{it} = 0; \qquad \varphi_i = \beta_i; \qquad \forall i = 1, 2, ...N \tag{F.1}$$

where $\boldsymbol{\beta}$ is calibrated to match the share of final uses of each industry's output.

Monetary Policy Rule. Price stabilization (F.1) is an implication and feature of the monetary policy, but not the policy itself. So next, we convert the price stabilization into the corresponding monetary policy rule. Using the equilibrium pricing function (2.7), we find that

$$\sum_{i=1}^{N} \varphi_i p_{it} = \boldsymbol{\varphi}(\mathbf{I} - \boldsymbol{\mu}\mathbf{A})^{-1}\boldsymbol{\mu}(-\mathbf{I} + \boldsymbol{\alpha}\boldsymbol{\kappa})\boldsymbol{z}_t = 0, \tag{F.2}$$

which holds for any productivity shock vector $\boldsymbol{z}_t$. Thus, $\boldsymbol{\varphi}(\mathbf{I} - \boldsymbol{\mu}\mathbf{A})^{-1}\boldsymbol{\mu}(-\mathbf{I} + \boldsymbol{\alpha}\boldsymbol{\kappa}) \equiv 0$ such that

$$\boldsymbol{\kappa} = \frac{1}{\boldsymbol{\varphi}'(\mathbf{I} - \boldsymbol{\mu}\mathbf{A})^{-1}\boldsymbol{\mu}\boldsymbol{\alpha}} \boldsymbol{\varphi}'(\mathbf{I} - \boldsymbol{\mu}\mathbf{A})^{-1}\boldsymbol{\mu}, \tag{F.3}$$

which provides a correspondence between the monetary policy (wage) rule $\boldsymbol{\kappa}$ and the price stabilization policy weight $\boldsymbol{\varphi}$. As Lemma E.3 predicts, any price-stabilization policy satisfies the condition $\boldsymbol{\kappa}\boldsymbol{\alpha} = 1$.

Equilibrium under CPI stability. Under CPI policy environment, the general equilibrium fixed-point is determined by the equation system (2.8), which implicitly define the mapping from CPI-based $\boldsymbol{\kappa}^{CPI}$ to $\boldsymbol{\mu}^{CPI}$ as

$$\boldsymbol{\mu}^{CPI} = \mathcal{F}\left(\boldsymbol{\kappa}^{CPI}\right). \tag{F.4}$$

Meanwhile, to achieve CPI stability, the monetary policy rule is also endogenous, given by (F.3) and $\boldsymbol{\varphi} = \boldsymbol{\beta}$, we have

$$\left(\boldsymbol{\kappa}^{CPI}\right)' = \frac{1}{\boldsymbol{\beta}'(\mathbf{I} - \boldsymbol{\mu}^{CPI}\mathbf{A})^{-1}\boldsymbol{\mu}^{CPI}\boldsymbol{\alpha}} \boldsymbol{\beta}'(\mathbf{I} - \boldsymbol{\mu}^{CPI}\mathbf{A})^{-1}\boldsymbol{\mu}^{CPI}. \tag{F.5}$$

Therefore, the fixed point system (F.4) and (F.5) jointly determines equilibrium under CPI stability.

Forecast error in model. We first derive the expression for firms' revenue. Recall that The output of firm ι in sector i is determined by $Y_{i\iota t} = (P_{i\iota t}/P_{it})^{-\theta_i} Y_{it}$. The demand of firms in sector j for intermediate goods produced in sector i is given by

$$\begin{aligned} X_{jit} &= \int_0^1 X_{j\iota it} d\iota \\ &= a_{ji} \frac{P_{jt} Y_{jt}}{P_{it}} \gamma_{jt}, \end{aligned}$$

where $\gamma_{jt} = MC_{jt} P_{jt}^{\theta_j - 1} \int_0^1 P_{j\iota t}^{-\theta_j} d\iota$ is a function of prices.

The market clearing condition for sector good i is given by $Y_{it} = C_{it} + \sum_{j=1}^N \int_0^1 X_{j\iota it} d\iota$, which implies that $P_{it} Y_{it} = \beta_i P_t C_t + \sum_{j=1}^N a_{ji} P_{jt} Y_{jt} \gamma_{jt}$. Because the aggregate nominal spending is equal to the money supply $P_t C_t = M_t$, we have $P_{it} Y_{it} = \beta_i M_t + \sum_{j=1}^N a_{ji} P_{jt} Y_{jt} \gamma_{jt}$. Log-linearizing this equation around the steady state yields

$$\lambda_i (p_{it} + y_{it}) = \beta_i m_t + \sum_{j=1}^N a_{ji} \lambda_j (y_{jt} + \sum_{k=1}^N a_{jk} p_{kt} - z_{jt}).$$

We derive the expression for sectoral outputs in vector form,

$$\boldsymbol{y}_t = \left[(\mathbf{I} - \mathbf{A}') \operatorname{diag}(\boldsymbol{\lambda}) \right]^{-1} \left[\boldsymbol{\beta} m_t + \left(\mathbf{A}' \operatorname{diag}(\boldsymbol{\lambda}) \mathbf{A} - \operatorname{diag}(\boldsymbol{\lambda}) \right) \boldsymbol{p}_t - \mathbf{A}' \operatorname{diag}(\boldsymbol{\lambda}) \boldsymbol{z}_t \right], \tag{F.6}$$

where the matrix form of monetary supply m_t is given by Proposition E.2,

$$m_t = \boldsymbol{\psi}' \boldsymbol{z}_t = \frac{\eta}{1+\gamma\eta} \boldsymbol{\kappa} \boldsymbol{z}_t + \left(1 - \frac{\eta}{1+\gamma\eta} \right) \boldsymbol{\beta}' (\mathbf{I} - \boldsymbol{\mu}\mathbf{A})^{-1} \boldsymbol{\mu} (-\mathbf{I} + \boldsymbol{\alpha}\boldsymbol{\kappa}) \boldsymbol{z}_t + \frac{1}{1+\gamma\eta} \boldsymbol{\lambda}' \boldsymbol{z}_t.$$

$\boldsymbol{\psi}$ is $N \times 1$ column vector and $\boldsymbol{p}_t = \boldsymbol{\phi} \boldsymbol{z}_t = (\mathbf{I} - \boldsymbol{\mu}\mathbf{A})^{-1} \boldsymbol{\mu} (-\mathbf{I} + \boldsymbol{\alpha}\boldsymbol{\kappa}) \boldsymbol{z}_t$. Substituting the expression of $\boldsymbol{p}_t$ and m_t into equation (F.6) we obtain the sectoral revenue as a function of fundamental shocks $\boldsymbol{z}_t$, $\boldsymbol{r}_t = \boldsymbol{p}_t + \boldsymbol{y}_t = \boldsymbol{\Psi}_r \boldsymbol{z}_t$, where

$$\boldsymbol{\Psi}_r = \left[\mathbf{I} + \left[(I - \mathbf{A}') \operatorname{diag}(\boldsymbol{\lambda}) \right]^{-1} \left(\mathbf{A}' \operatorname{diag}(\boldsymbol{\lambda}) \mathbf{A} - \operatorname{diag}(\boldsymbol{\lambda}) \right) \right] \boldsymbol{\phi} + \left[(I - \mathbf{A}') \operatorname{diag}(\boldsymbol{\lambda}) \right]^{-1} \left[\boldsymbol{\beta} \boldsymbol{\psi}' - \mathbf{A}' \operatorname{diag}(\boldsymbol{\lambda}) \right]. \tag{F.7}$$

The revenue forecast error of i-sector is given by

$$\mathrm{FE}_{i,t}^{rev} = \frac{\left| \mathbf{e}_i \left[\boldsymbol{\Psi}_r \left(\mathbf{I} - (\mathbf{I} - \boldsymbol{\Sigma}_{z|x_i} \boldsymbol{\Sigma}_z^{-1}) \right) \boldsymbol{z}_t \right] \right|}{\left| \mathbf{e}_i \boldsymbol{\Psi}_r \boldsymbol{z}_t + 1 \right|} = \frac{\left| \mathbf{e}_i \boldsymbol{\Psi}_r \boldsymbol{\Sigma}_{z|x_i} \boldsymbol{\Sigma}_z^{-1} \boldsymbol{z}_t \right|}{\left| \mathbf{e}_i \boldsymbol{\Psi}_r \boldsymbol{z}_t + 1 \right|}; \qquad \forall i = 1, 2, \ldots N,$$

where $\boldsymbol{\Sigma}_{z|x_i}$ is the posterior covariance matrices of the fundamental shocks $\boldsymbol{z}_t$ from Lemma D.1.

The firm ι's profit is given by

$$\Pi_{i\iota t} = (1+\tau_i)P_{i\iota t}Y_{i\iota t} - W_t L_{i\iota t} - \sum_{j=1}^{N} P_j X_{i\iota jt} - T_{i\iota t} = (1+\tau_i)P_{i\iota t}Y_{i\iota t} - \mathrm{MC}_{it}Y_{i\iota t} - T_{i\iota t}, \tag{F.8}$$

Next, the sectoral level profit before tax becomes

$$\begin{aligned}\Pi^b_{it} = \int (\Pi_{i\iota t} + T_{i\iota t}) &= \int \left[(1+\tau_i)\, P_{i\iota t} - \mathrm{MC}_{it}\right] Y_{i\iota t} \\ &= P_{it}Y_{it}(1+\tau_i - \mathcal{E}_{it}),\end{aligned} \tag{F.9}$$

where $\mathcal{E}_{it} = \frac{\mathrm{MC}_{it}}{P_{it}} \int \left(\frac{P_{i\iota t}}{P_{it}}\right)^{-\theta_i}$ and its log-linearized version is

$$\varepsilon_{it} = \mathrm{mc}_{it} - p_{it} - \theta_i \int (p_{i\iota t} - p_{it}) = \mathrm{mc}_{it} - p_{it}.$$

Also, we have $\mathrm{mc}_{it} = \mu_{it}^{-1} p_{it}$. Thus the log-linearized version of Π^b_{it} is

$$\pi^b_{it} = p_{it} + y_{it} + \frac{p_{it}\left(1-\mu_{it}^{-1}\right)}{\tau_i}. \tag{F.10}$$

In matrix form,

$$\boldsymbol{\pi}^b_t = \boldsymbol{p}_t + \boldsymbol{y}_t + \mathrm{diag}\left(\boldsymbol{\tau}\right)^{-1}\left(I - \boldsymbol{\mu}^{-1}\right)\boldsymbol{p}_t = \left[\boldsymbol{\Psi}_r + \mathrm{diag}\left(\boldsymbol{\tau}\right)^{-1}\left(I - \boldsymbol{\mu}^{-1}\right)\boldsymbol{\Psi}_p\right]\boldsymbol{z}_t = \boldsymbol{\Psi}_{eps}\boldsymbol{z}_t. \tag{F.11}$$

Similarly, the EPS forecast error of each sector is given by $\mathrm{FE}^{eps}_{i,t} = \dfrac{\left|\mathbf{e}_i \boldsymbol{\Psi}_{eps} \boldsymbol{\Sigma}_{z|x_i} \boldsymbol{\Sigma}_z^{-1} \boldsymbol{z}_t\right|}{\left|\mathbf{e}_i \boldsymbol{\Psi}_{eps} \boldsymbol{z}_{t+1}\right|}$, $\forall i = 1, 2, \ldots N$.

Calibration strategy. Given the equilibrium system system (F.4) and (F.5) under CPI stabilization policy, we simulate different EPS forecast error when we calibrate the information cost $\boldsymbol{\chi}$. In particular, we impose the following functional form

$$\log \chi_i = \delta_0 + \delta_1 \log \lambda_i + \delta_2 \log \sigma_i.$$

We calibrate the parameters δ_0, δ_1 and δ_2 to match the distribution of *adjusted* forecast errors of earnings per share (EPS) at the sector level, which are directly related to the informational frictions. Specifically, we choose δ_0, δ_1 and δ_2 to minimize the following loss function,

$$\min_{\delta_0,\delta_1,\delta_2} \left(\mathrm{FE}^{eps,data}_{i=25\%} - \mathrm{FE}^{eps,model}_{i=25\%}\right)^2 + \left(\sum_{i=1}^{N} \frac{1}{N}\mathrm{FE}^{eps,data}_{i} - \sum_{i=1}^{N} \frac{1}{N}\mathrm{FE}^{eps,model}_{i}\right)^2 + \left(\mathrm{FE}^{eps,data}_{i=75\%} - \mathrm{FE}^{eps,model}_{i=75\%}\right)^2,$$

where FE_i is the mean absolute forecast error of sector i . We target the mean level of the forecast error, the 25 percentile and 75 percentile of the forecast error in the sectoral distribution. However, the volatility of EPS differs in the data and in the model. To ensure that the forecast error is comparable, we normalize it by the volatility of EPS, both in the data and in the model.

F.2 Computation

Numerical Algorithms. We provide two algorithms for solving the optimal policy problem defined in Definition 2. The first method is based on the first-oder condition (FOC). The second method utilizes constrained nonlinear programming.

FOC Method. We can solve the optimal monetary policy problem by the FOC condition: $\frac{d\mathcal{W}}{d\boldsymbol{\kappa}} = 0$, which finally yields the equilibrium system in Proposition E.4; that is, the jointly fixed-point system (E.24) and (E.79) with respect to variables $\boldsymbol{\mu}^v$ and $\boldsymbol{\kappa}$. We employ Matlab's toolbox 'fsolve' to solve this static and nonlinear system. When the unique solution to this system exists, the optimal monetary policy achieves price stabilization simultaneously by Proposition 4.3. It is worth noting that $\mu_i > 0$ for all sectors $i = 1, 2, ...N$ is implicit in the RI condition (D.17), and the solution of $\boldsymbol{\mu}^v$ to FOC must be an interior solution. The FOC method does not apply when the solution of the optimization problem is on the zero boundary of $\boldsymbol{\mu}^v$. In this case of corner solution, we can awalys use the following constrained nonlinear programming algorithm to solve the problem.

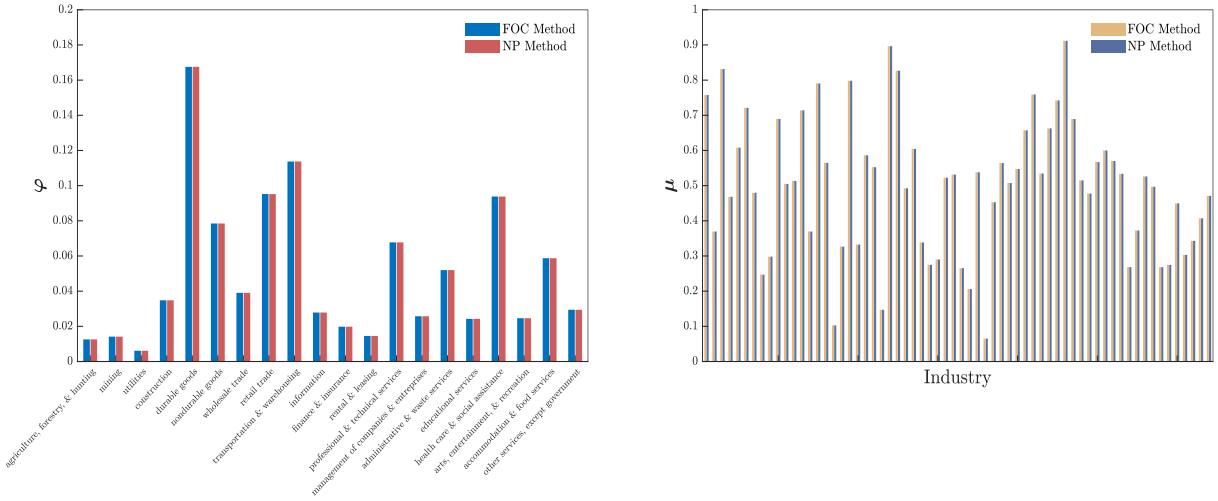
Nonlinear Constrained Programming Method. The optimal monetary policy problem defined in Definition 2 can be transformed into a nonlinear programming problem and restated as

$$\max_{\varrho} \quad \mathcal{W} = \mathcal{W}^{within} + \mathcal{W}^{OG} + \mathcal{W}^{across},$$

subject to $\varrho = \begin{bmatrix} \boldsymbol{\mu}^v \\ \boldsymbol{\kappa} \end{bmatrix}$ and nonlinear constraint (E.24). The nonlinear constrained programming (NP) method nests the FOC method as a special case. It handles both the interior solution and the corner solution. That is, the NP algorithm allows for the corner solution case where $\exists \mu_i = 0$, $i = 1, 2, ...N$. When an optimal solution under the FOC structure exists, the NP algorithm will also deliver this interior solution that coincides with the FOC solution. Numerically, we use Matlab's toolbox 'fmincon' to solve the NP problem.

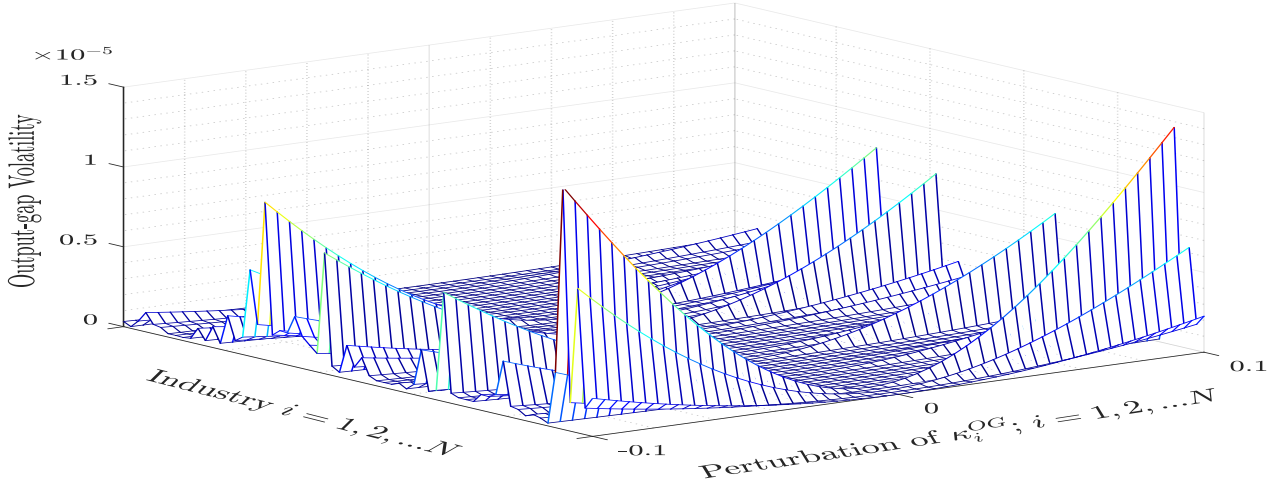
Accuracy of Solution. Utilizing the NP algorithm, we test the accuracy and correctness of our FOC-based characterization of the optimal monetary policy in Section E. Specifically, under the calibration in Table 7, the FOC-based optimal policy solution and the solution based on NP algorithm coincide. Therefore, under our model calibration, an interior optimal policy exists and is given by our characterization in Section E. Figure F.1 plots the optimal solution in terms of $\boldsymbol{\mu}$ and $\boldsymbol{\varphi}$ using two algorithms. The numerical difference of the two solutions in terms of welfare loss is smaller than $1e^{-10}$.

Figure F.1: Accuracy of Solution for Optimal Monetary Policy



Finally, we find that under model calibration, the OG optimal policy satisfies the condition imposed in Proposition 4.4, that is , $\mathbb{V}\left(c_t - c_t^*\right) = 0$. Figure F.2 plots the output-gap volatility under perturbation around different sectors' optimal OG policy, which shows that the output-gap volatility reduces to 0 at the optimum.

Figure F.2: Completely Stabilized OG Volatility



Appendix References

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